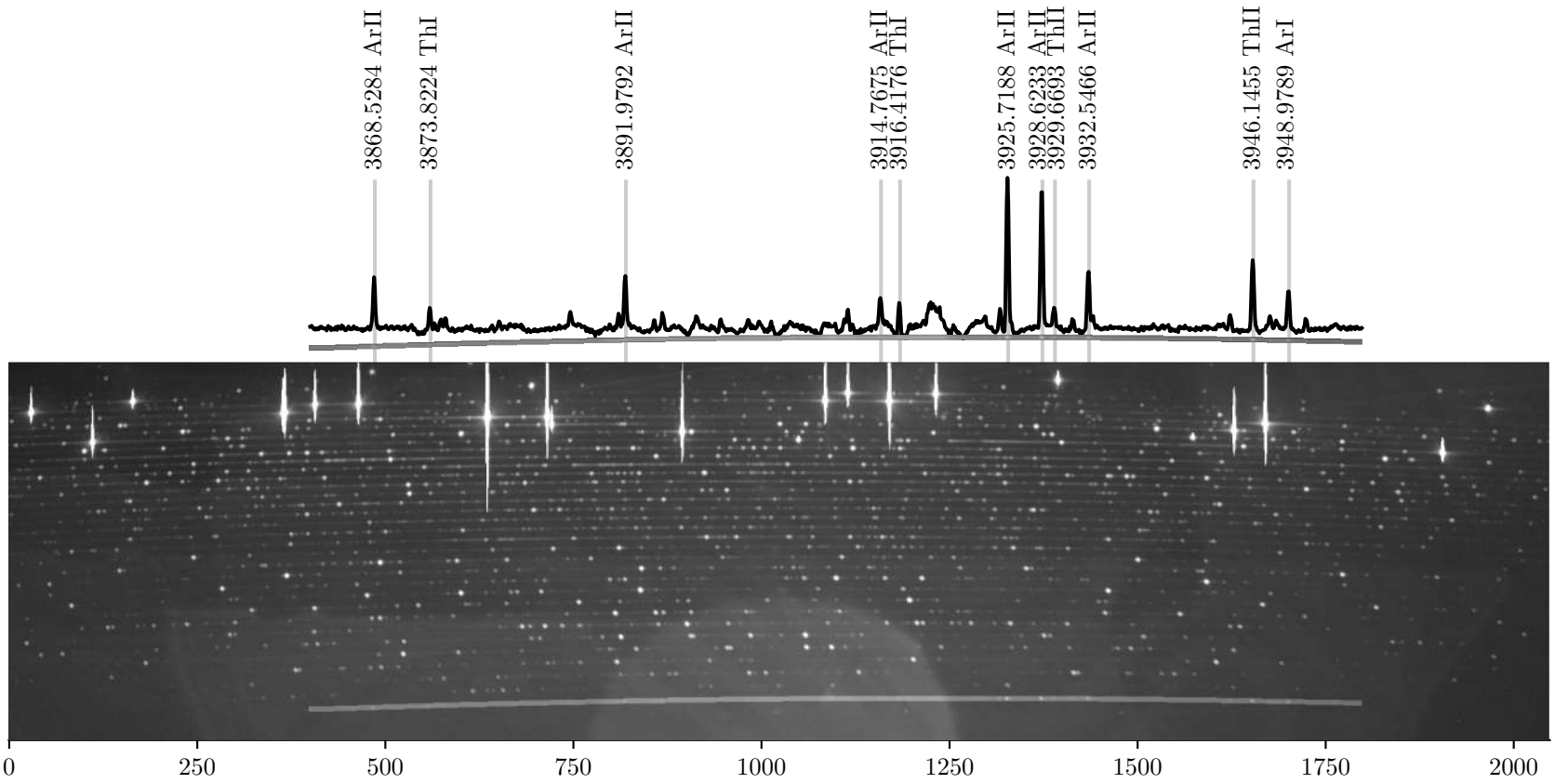
Guide to High Resolution ThAr Spectra

Echelle Spectrograph for the Nanshan Observatory

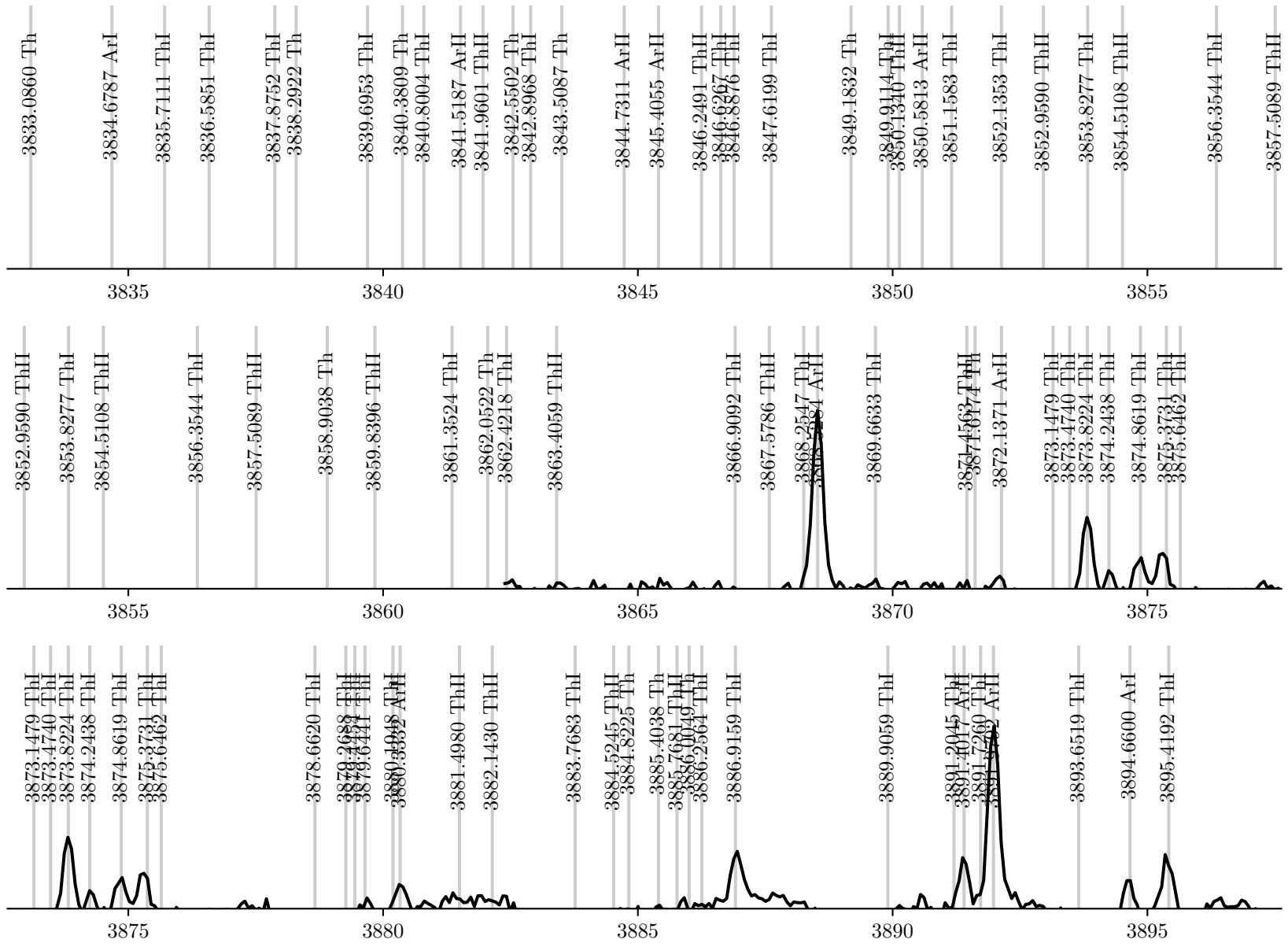
Astronomical Consultants & Equipment, Inc.

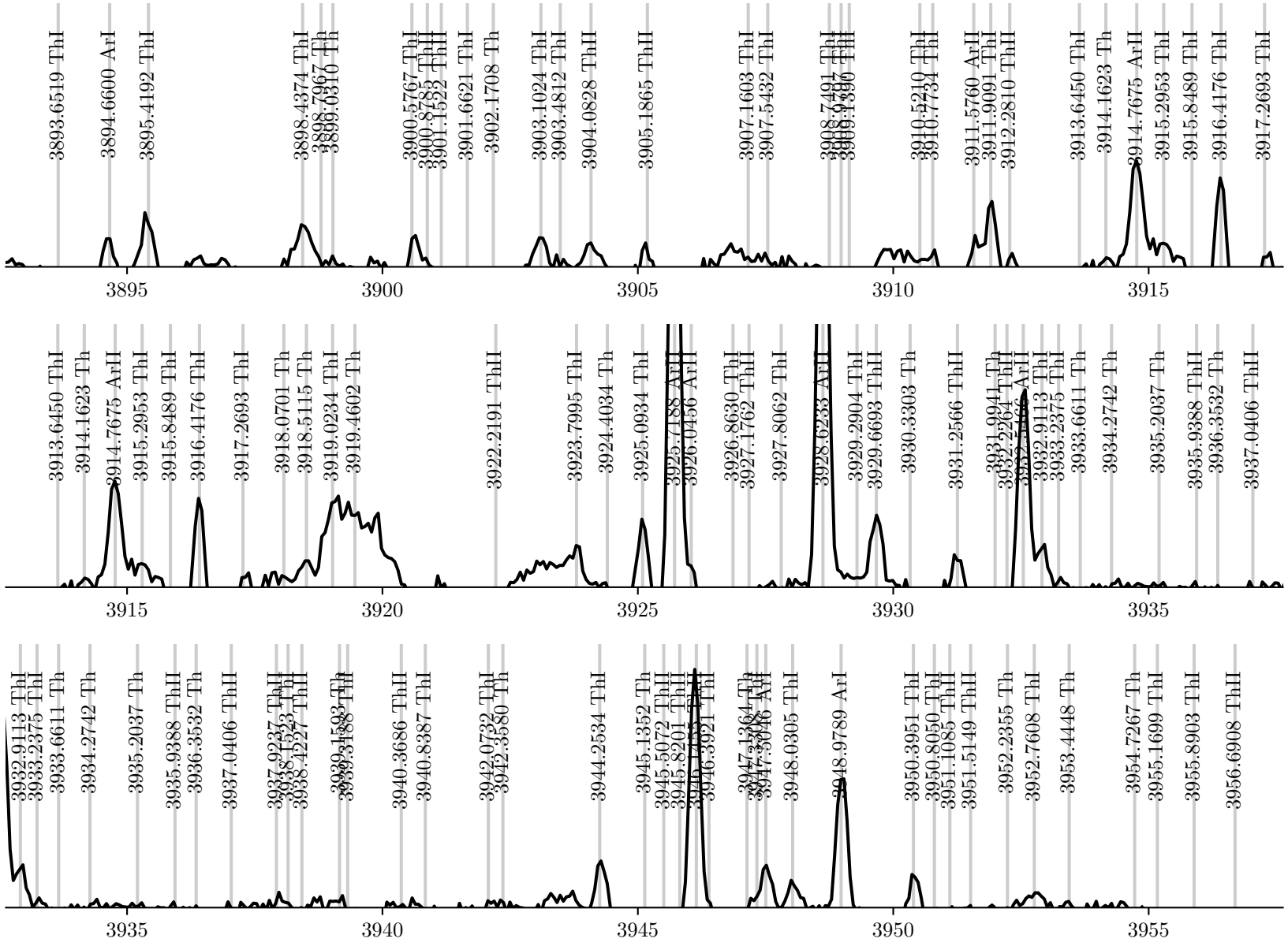
August 8, 2020

Order 58

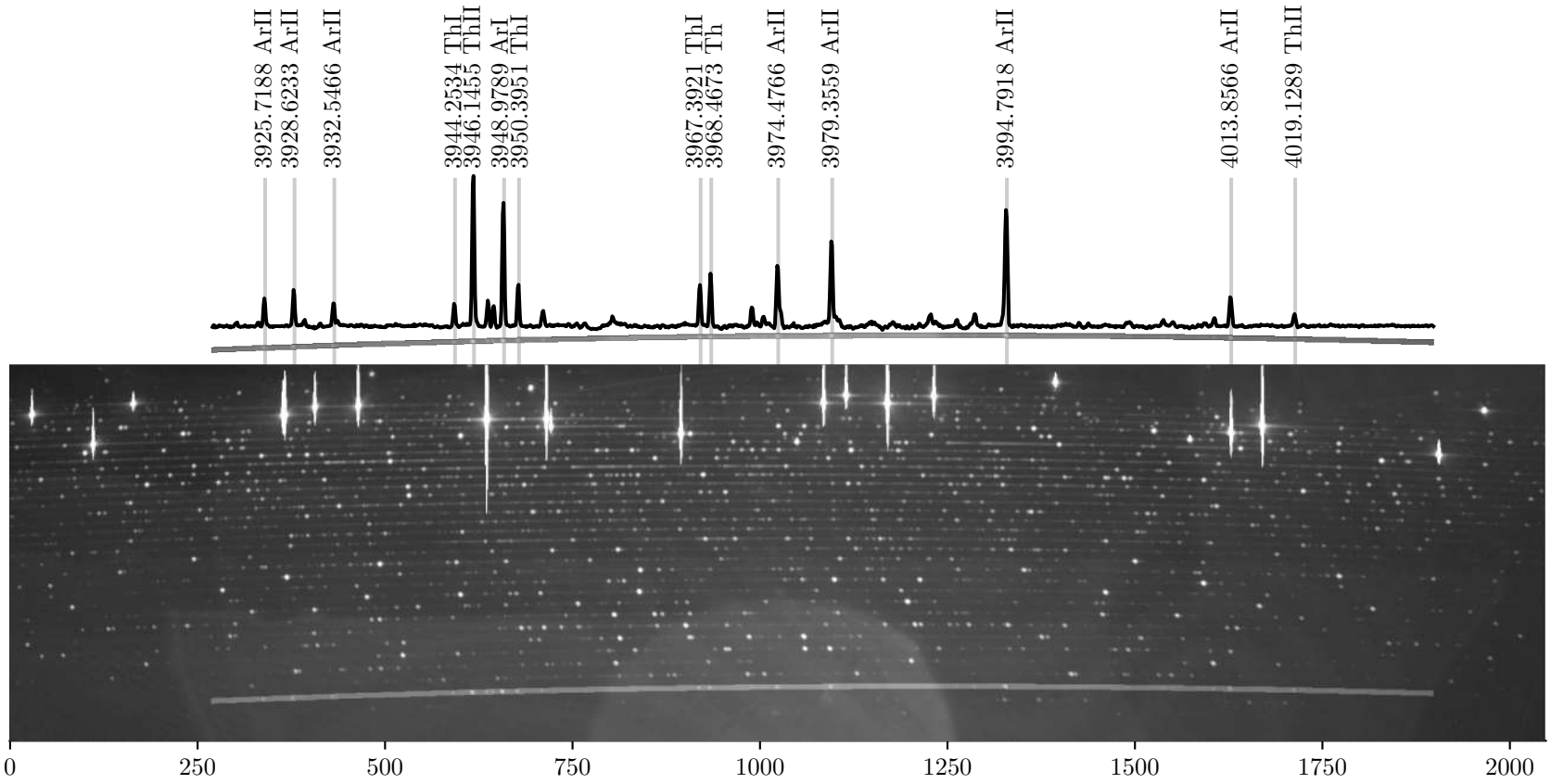


Approximate resolution:	16100
Dispersion:	0.067 Å pix^{-1}
Central wavelength:	3906 Å
Wavelength range:	3833 Å – 3969 Å

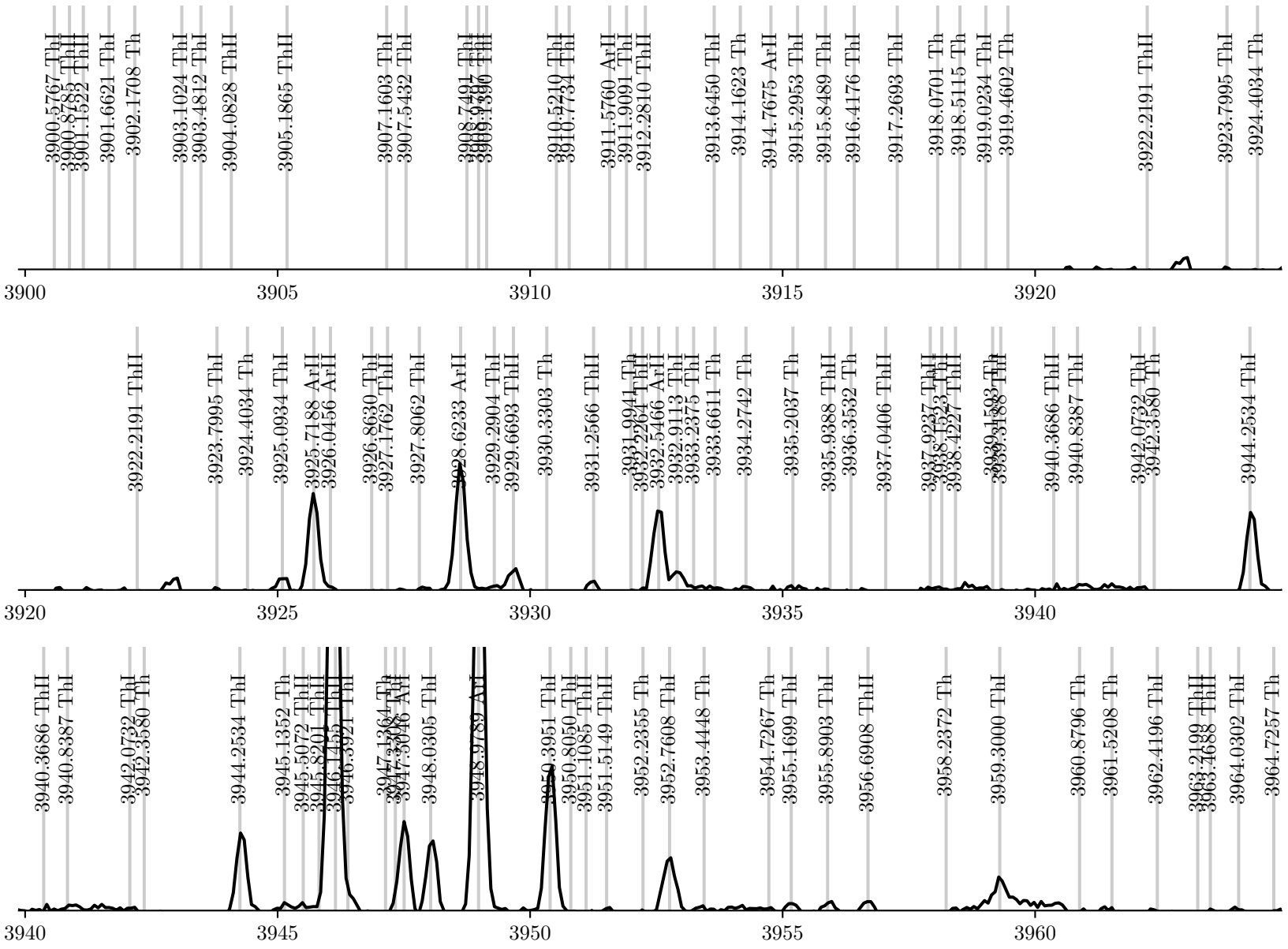

3833.0860 Th
3834.6787 ArI
3835.7111 ThI
3836.5851 ThI
3837.8752 ThI
3838.2922 Th
3839.6953 ThI
3840.3809 Th
3840.8004 ThI
3841.5187 ArII
3841.9601 ThII
3842.5502 Th
3842.8968 ThI
3843.5087 Th
3844.7311 ArII
3845.4055 ArII
3846.2491 ThII
3846.6267 ThI
3846.8876 ThI
3847.6199 ThI
3849.1832 Th
3849.9114 ThI
3850.1340 ThII
3850.5813 ArII
3851.1583 ThI
3852.1353 ThI
3852.9590 ThII
3853.8277 ThI
3854.5108 ThII
3856.3544 ThI
3857.5089 ThII
3835
3840
3845
3850
3855
3852.9590 ThII
3853.8277 ThI
3854.5108 ThII
3856.3544 ThI
3857.5089 ThII
3858.9038 Th
3859.8396 ThII
3861.3524 ThI
3862.0522 Th
3862.4218 ThI
3863.4059 ThII
3866.9092 ThI
3867.5786 ThII
3868.2547 ThI
3869.6633 ThI
3871.4563 ThII
3872.1371 ArII
3873.1479 ThI
3873.4740 ThI
3873.8224 ThI
3874.2438 ThI
3874.8619 ThI
3875.3731 ThI
3875.6462 ThI
3855
3860
3865
3870
3875
3873.1479 ThI
3873.4740 ThI
3873.8224 ThI
3874.2438 ThI
3874.8619 ThI
3875.3731 ThI
3875.6462 ThI
3878.6620 ThI
3881.4980 ThII
3882.1430 ThII
3883.7683 ThI
3884.5245 ThII
3884.8225 Th
3885.4038 Th
3886.2564 ThI
3886.9159 ThI
3889.9059 ThI
3893.6519 ThI
3894.6600 ArI
3895.4192 ThI
3875
3880
3885
3890
3895

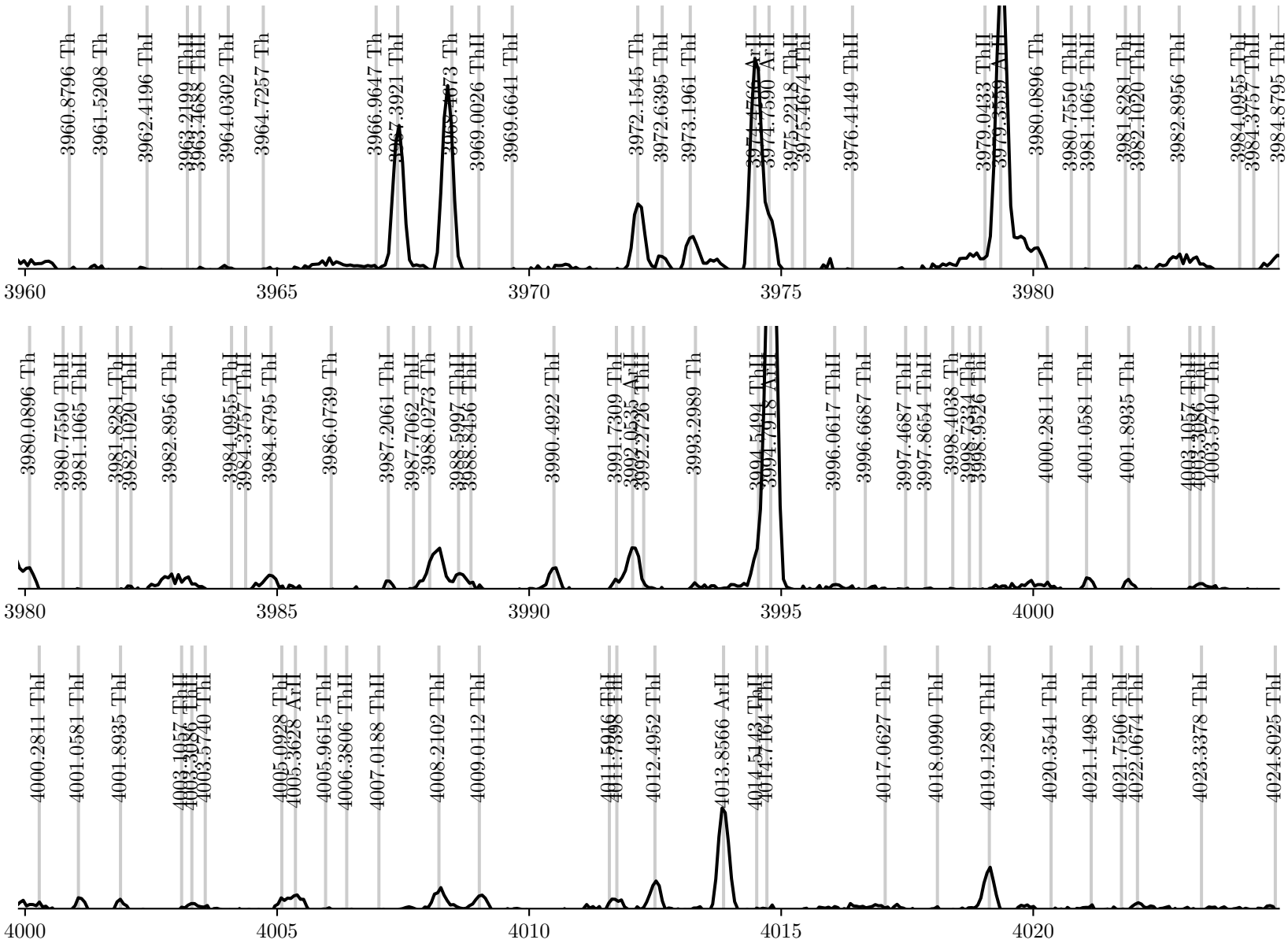

3893.6519 ThI
3894.6600 ArI
3895.4192 ThI
3898.4374 ThI
3898.7967 Th
3899.0310 Th
3900.5767 ThI
3900.8785 ThII
3901.1522 ThII
3901.6621 ThI
3902.1708 Th
3903.1024 ThI
3903.4812 ThI
3904.0828 ThII
3905.1865 ThII
3907.1603 ThI
3907.5432 ThI
3910.5210 ThI
3910.7734 ThI
3911.5760 ArII
3911.9091 ThI
3912.2810 ThII
3913.6450 ThI
3914.1623 Th
3914.7675 ArII
3915.2953 ThI
3915.8489 ThI
3916.4176 ThI
3917.2693 ThI
3895
3900
3905
3910
3915
3913.6450 ThI
3914.1623 Th
3914.7675 ArII
3915.2953 ThI
3915.8489 ThI
3916.4176 ThI
3917.2693 ThI
3918.0701 Th
3918.5115 Th
3919.0234 ThI
3919.4602 Th
3922.2191 ThII
3923.7995 ThI
3924.4034 Th
3925.0934 ThI
3925.7188 ArII
3926.0456 ArII
3926.8630 ThI
3927.1762 ThII
3927.8062 ThI
3928.6233 ArII
3929.2904 ThI
3929.6693 ThII
3930.3303 Th
3931.2566 ThII
3931.9941 Th
3932.2264 ThII
3932.5466 ArII
3932.9113 ThI
3933.2375 ThI
3933.6611 Th
3934.2742 Th
3935.2037 Th
3935.9388 ThII
3936.3532 Th
3937.0406 ThII
3915
3920
3925
3930
3935
3932.9113 ThI
3933.2375 ThI
3933.6611 Th
3934.2742 Th
3935.2037 Th
3935.9388 ThII
3936.3532 Th
3937.0406 ThII
3940.3686 ThII
3940.8387 ThI
3942.0732 ThI
3942.3580 Th
3944.2534 ThI
3945.1352 Th
3945.5072 ThII
3946.3921 ThI
3948.0305 ThI
3948.9789 ArI
3950.3951 ThI
3950.8050 ThI
3951.1085 ThII
3951.5149 ThII
3952.2355 Th
3952.7608 ThI
3953.4448 Th
3954.7267 Th
3955.1699 ThI
3955.8903 ThI
3956.6908 ThII
3935
3940
3945
3950
3955

Order 57

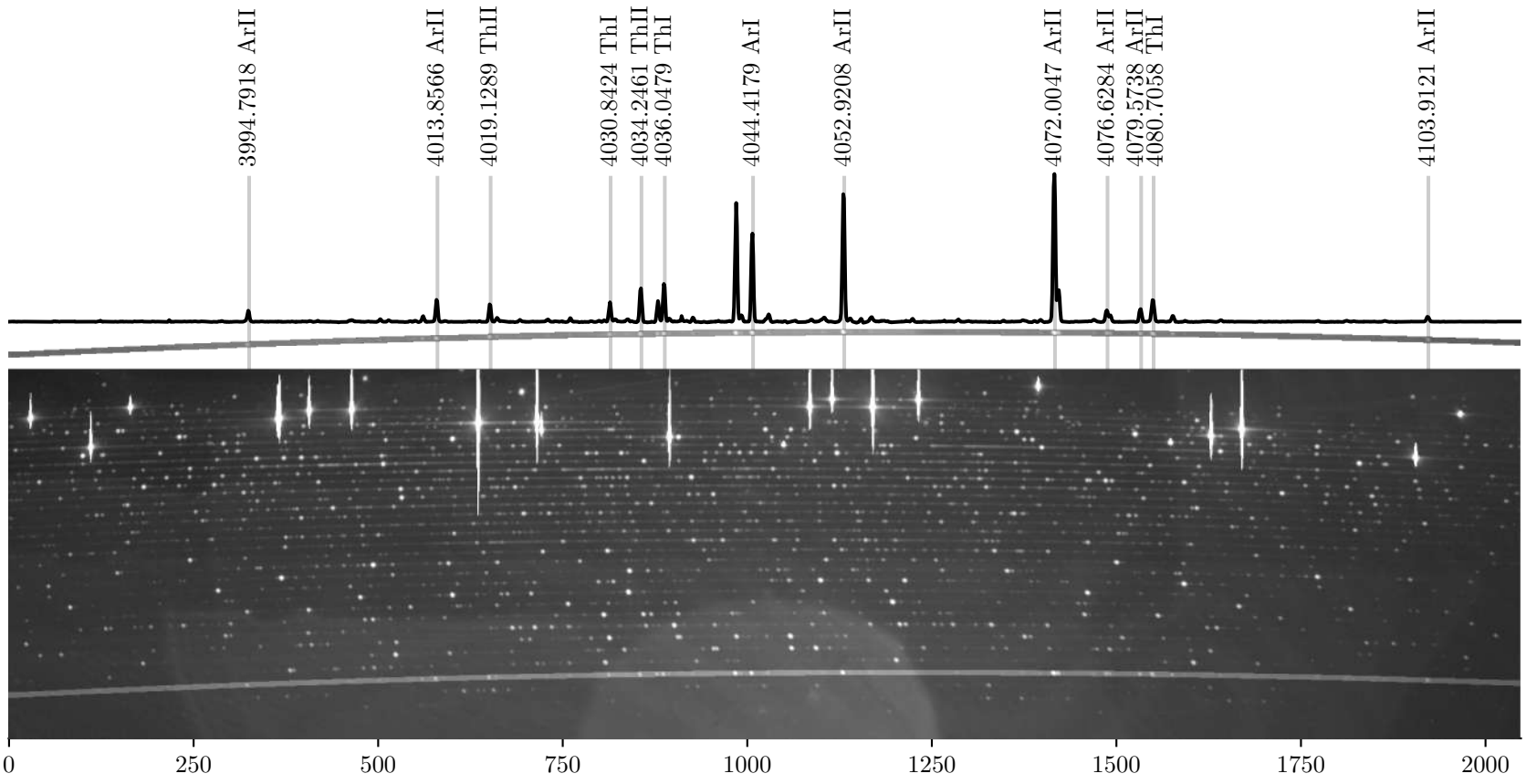


Approximate resolution:	16300
Dispersion:	0.068 Å pix^{-1}
Central wavelength:	3974 Å
Wavelength range:	3900 Å – 4039 Å

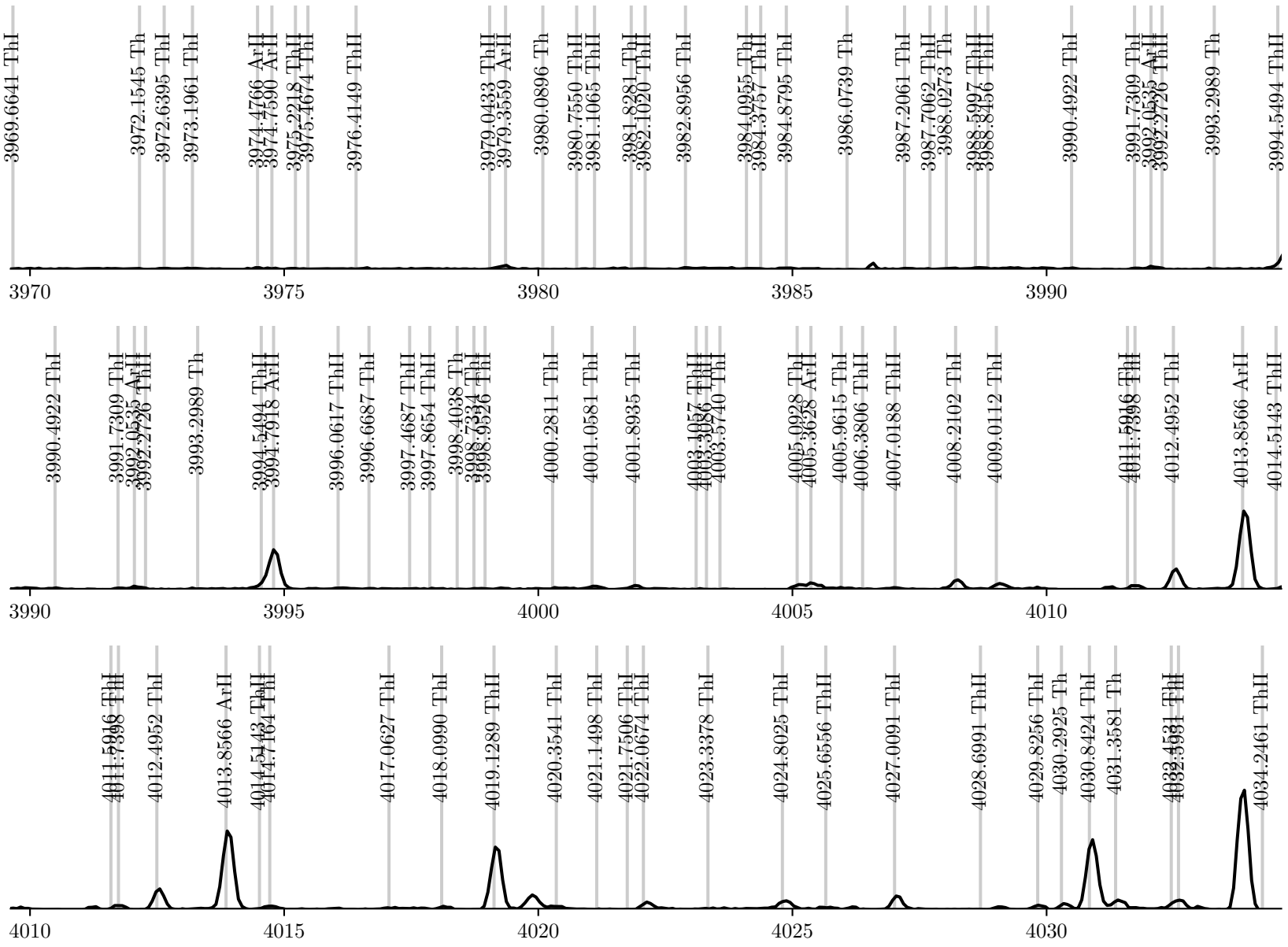

3900.5767 ThI
3900.8785 ThII
3901.1522 ThII
3901.6621 ThI
3902.1708 Th
3903.1024 ThI
3903.4812 ThI
3904.0828 ThII
3905.1865 ThII
3907.1603 ThI
3907.5432 ThI
3910.5210 ThI
3910.7734 ThI
3911.5760 ArII
3911.9091 ThI
3912.2810 ThII
3913.6450 ThI
3914.1623 Th
3914.7675 ArII
3915.2953 ThI
3915.8489 ThI
3916.4176 ThI
3917.2693 ThI
3918.0701 Th
3918.5115 Th
3919.0234 ThI
3919.4602 Th
3922.2191 ThII
3923.7995 ThI
3924.4034 Th
3900
3905
3910
3915
3920
3922.2191 ThII
3923.7995 ThI
3924.4034 Th
3925.0934 ThI
3925.7188 ArII
3926.0456 ArII
3926.8630 ThI
3927.1762 ThII
3927.8062 ThI
3928.6233 ArII
3929.2904 ThI
3929.6693 ThII
3930.3303 Th
3931.2566 ThII
3931.9941 Th
3932.2264 ThII
3932.5466 ArII
3932.9113 ThI
3933.2375 ThI
3933.6611 Th
3934.2742 Th
3935.2037 Th
3935.9388 ThII
3936.3532 Th
3937.0406 ThII
3937.9237 ThII
3938.1553 Th
3938.4227 ThII
3939.1503 Th
3939.3188 ThI
3940.3686 ThII
3940.8387 ThI
3942.0732 ThI
3942.3580 Th
3944.2534 ThI
3920
3925
3930
3935
3940
3940.3686 ThII
3940.8387 ThI
3942.0732 ThI
3942.3580 Th
3944.2534 ThI
3945.1352 Th
3945.5072 ThII
3945.8201 ThII
3946.1455 ThI
3946.3921 ThI
3947.1364 Th
3947.3046 ArI
3948.0305 ThI
3948.3789 ArI
3950.3951 ThI
3950.8050 ThI
3951.1085 ThII
3951.5149 ThII
3952.2355 Th
3952.7608 ThI
3953.4448 Th
3954.7267 Th
3955.1699 ThI
3955.8903 ThI
3956.6908 ThII
3958.2372 Th
3959.3000 ThI
3960.8796 Th
3961.5208 Th
3962.4196 ThI
3963.2199 ThII
3963.4688 ThII
3964.0302 ThI
3964.7257 Th
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3945
3950
3955
3960

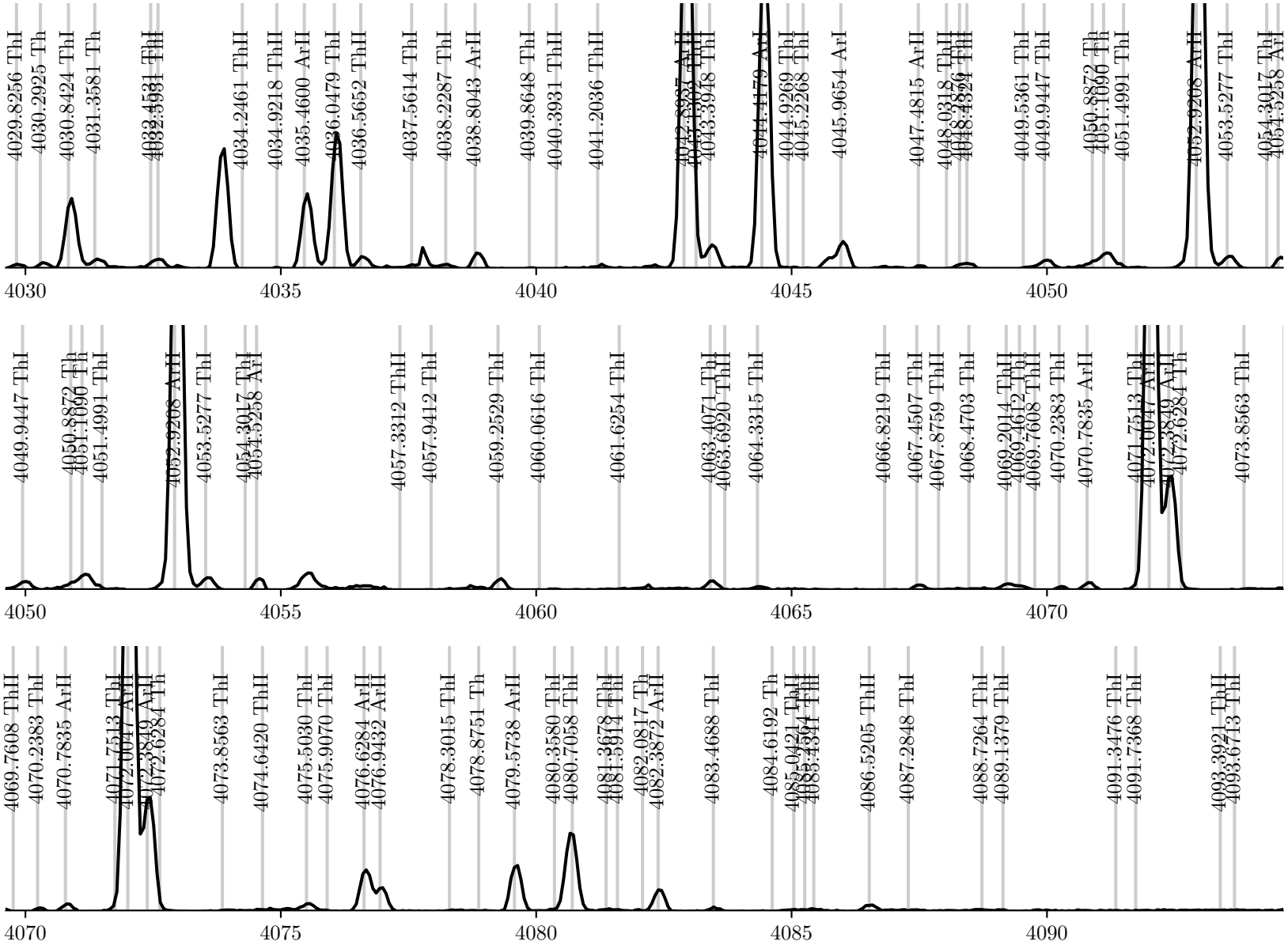

3960.8796 Th
3961.5208 Th
3962.4196 ThI
3964.0302 ThI
3964.7257 Th
3966.9647 Th
3969.0026 ThII
3969.6641 ThI
3972.1545 Th
3972.6395 ThI
3973.1961 ThI
3976.4149 ThII
3980.0896 Th
3980.7550 ThII
3981.1065 ThII
3982.8956 ThI
3984.8795 ThI
3960
3965
3970
3975
3980
3980.0896 Th
3980.7550 ThII
3981.1065 ThII
3982.8956 ThI
3984.8795 ThI
3986.0739 Th
3987.2061 ThI
3990.4922 ThI
3991.7309 ThI
3993.2989 Th
3996.0617 ThII
3996.6687 ThI
3997.4687 ThII
3997.8654 ThII
4000.2811 ThI
4001.0581 ThI
4001.8935 ThI
3980
3985
3990
3995
4000
4000.2811 ThI
4001.0581 ThI
4001.8935 ThI
4005.3628 ArII
4005.9615 ThI
4006.3806 ThII
4007.0188 ThII
4008.2102 ThI
4009.0112 ThI
4012.4952 ThI
4013.8566 ArII
4017.0627 ThI
4018.0990 ThI
4019.1289 ThII
4020.3541 ThI
4021.1498 ThI
4021.7506 ThI
4022.0674 ThI
4023.3378 ThI
4024.8025 ThI
4000
4005
4010
4015
4020

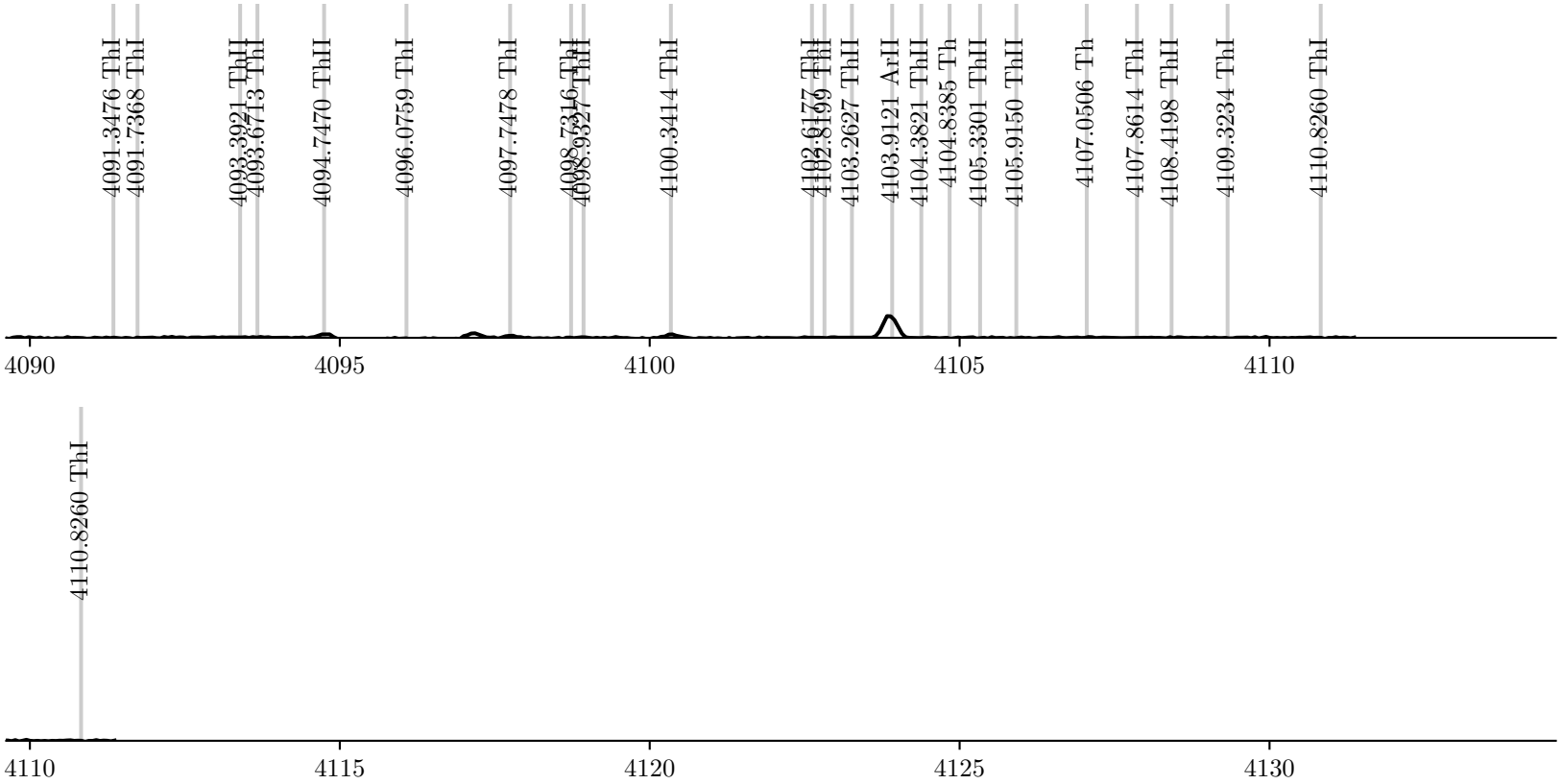
Order 56



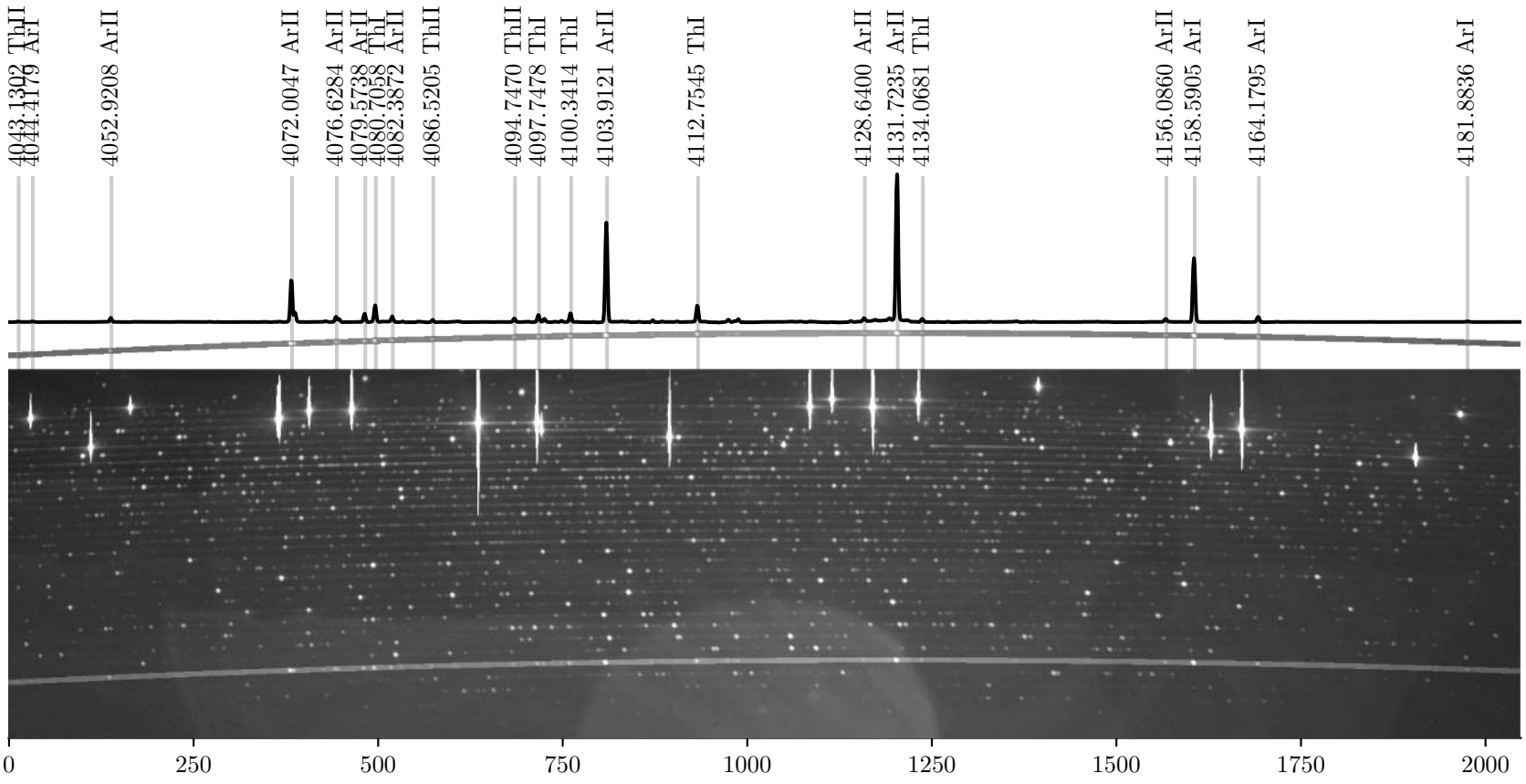
Approximate resolution:	15900
Dispersion:	0.069 Å pix^{-1}
Central wavelength:	4046 Å
Wavelength range:	3970 Å – 4111 Å


3969.6641 ThI
3972.1545 Th
3972.6395 ThI
3973.1961 ThI
3974.4766 ArII
3974.7590 ArII
3975.2218 ThII
3975.4674 ThI
3976.4149 ThII
3979.0433 ThII
3979.3559 ArII
3980.0896 Th
3980.7550 ThII
3981.1065 ThII
3981.8281 ThI
3982.1020 ThII
3982.8956 ThI
3984.0955 ThI
3984.3757 ThII
3984.8795 ThI
3986.0739 Th
3987.2061 ThI
3987.7062 ThII
3988.0273 Th
3990.4922 ThI
3991.7309 ThI
3992.0535 ArII
3992.2726 ThII
3993.2989 Th
3994.5494 ThII
3970
3975
3980
3985
3990
3990.4922 ThI
3991.7309 ThI
3992.0535 ArII
3992.2726 ThII
3993.2989 Th
3994.5494 ThII
3994.7918 ArII
3996.0617 ThII
3996.6687 ThI
3997.4687 ThII
3997.8654 ThII
3998.4038 Th
3998.7334 ThI
3998.9526 ThI
4000.2811 ThI
4001.0581 ThI
4001.8935 ThI
4003.1057 ThII
4003.3086 ThI
4003.5740 ThI
4005.0028 ThI
4005.3628 ArII
4005.9615 ThI
4006.3806 ThII
4007.0188 ThII
4008.2102 ThI
4009.0112 ThI
4011.7916 ThI
4011.7998 ThI
4012.4952 ThI
4013.8566 ArII
4014.5143 ThII
3990
3995
4000
4005
4010
4011.7916 ThI
4011.7998 ThI
4012.4952 ThI
4013.8566 ArII
4014.5143 ThII
4014.7164 ThII
4017.0627 ThI
4018.0990 ThI
4019.1289 ThII
4020.3541 ThI
4021.1498 ThI
4021.7506 ThI
4022.0674 ThI
4023.3378 ThI
4024.8025 ThI
4025.6556 ThII
4027.0091 ThI
4028.6991 ThII
4029.8256 ThI
4030.2925 Th
4030.8424 ThI
4031.3581 Th
4032.3531 ThI
4034.2461 ThII
4010
4015
4020
4025
4030

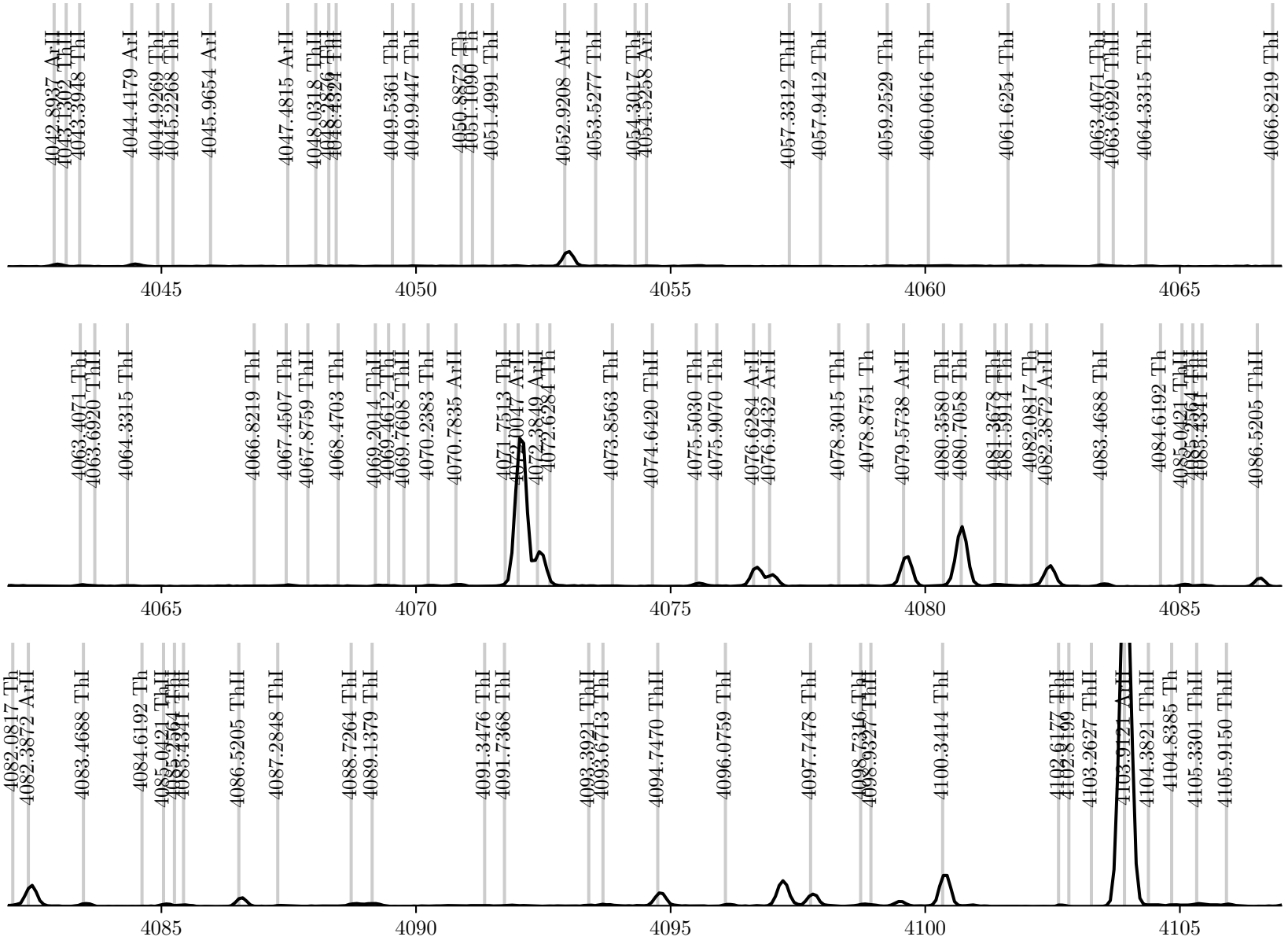

4029.8256 ThI
4030.2925 Th
4030.8424 ThI
4031.3581 Th
4034.2461 ThII
4034.9218 ThII
4035.4600 ArII
4036.0479 ThI
4036.5652 ThII
4037.5614 ThI
4038.2287 ThI
4038.8043 ArII
4039.8648 ThI
4040.3931 ThII
4041.2036 ThII
4045.9654 ArI
4047.4815 ArII
4049.5361 ThI
4049.9447 ThI
4051.4991 ThI
4053.5277 ThI
4030
4035
4040
4045
4050
4049.9447 ThI
4051.4991 ThI
4052.9208 ArII
4053.5277 ThI
4057.3312 ThII
4057.9412 ThI
4059.2529 ThI
4060.0616 ThI
4061.6254 ThI
4064.3315 ThI
4066.8219 ThI
4067.4507 ThI
4067.8759 ThII
4068.4703 ThI
4070.2383 ThI
4070.7835 ArII
4071.7513 ThI
4073.8563 ThI
4050
4055
4060
4065
4070
4069.7608 ThII
4070.2383 ThI
4070.7835 ArII
4073.8563 ThI
4074.6420 ThII
4075.5030 ThI
4075.9070 ThI
4076.6284 ArII
4076.9432 ArII
4078.3015 ThI
4078.8751 Th
4079.5738 ArII
4080.3580 ThI
4080.7058 ThI
4082.0817 Th
4082.3872 ArII
4083.4688 ThI
4084.6192 Th
4086.5205 ThII
4087.2848 ThI
4088.7264 ThI
4089.1379 ThI
4091.3476 ThI
4091.7368 ThI
4093.3921 ThII
4070
4075
4080
4085
4090

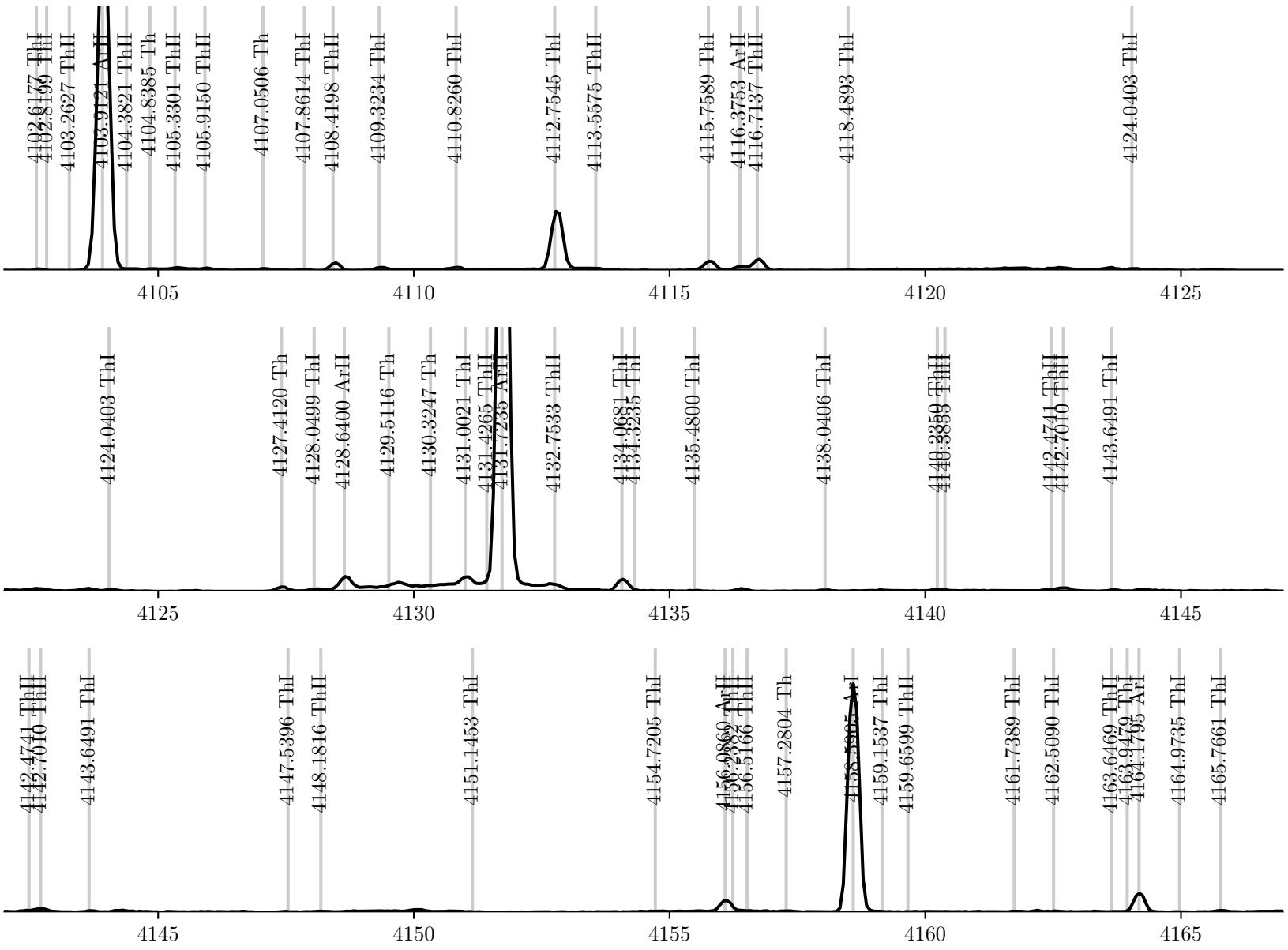

4091.3476 ThI
4091.7368 ThI
4093.3921 ThII
4093.6713 ThI
4094.7470 ThII
4096.0759 ThI
4097.7478 ThI
4098.7316 ThI
4098.9327 ThII
4100.3414 ThI
4102.6177 ThI
4102.8199 ThII
4103.2627 ThII
4103.9121 ArII
4104.3821 ThII
4104.8385 Th
4105.3301 ThII
4105.9150 ThII
4107.0506 Th
4107.8614 ThI
4108.4198 ThII
4109.3234 ThI
4110.8260 ThI
4090
4095
4100
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4110.8260 ThI
4110
4115
4120
4125
4130

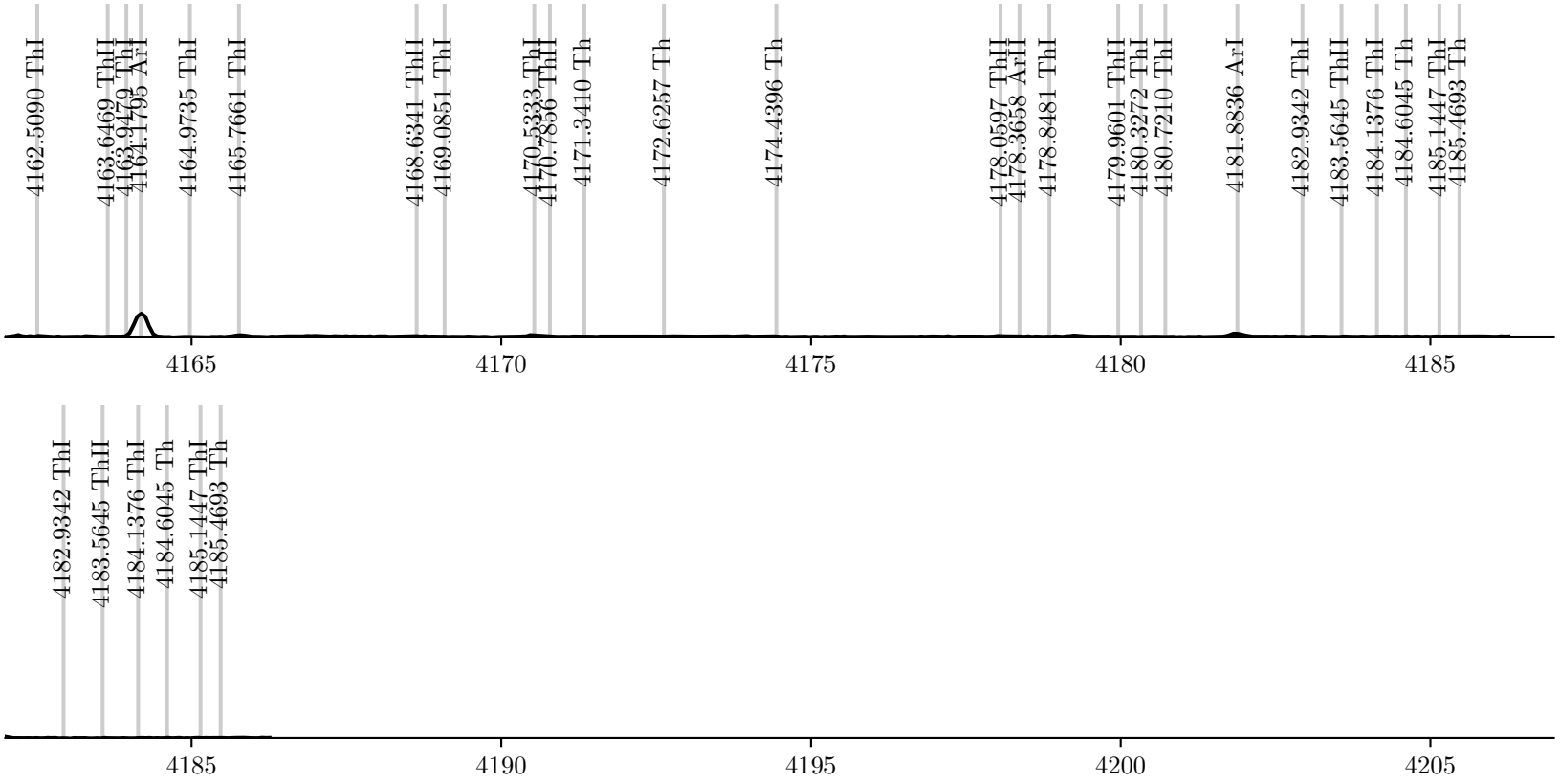
Order 55



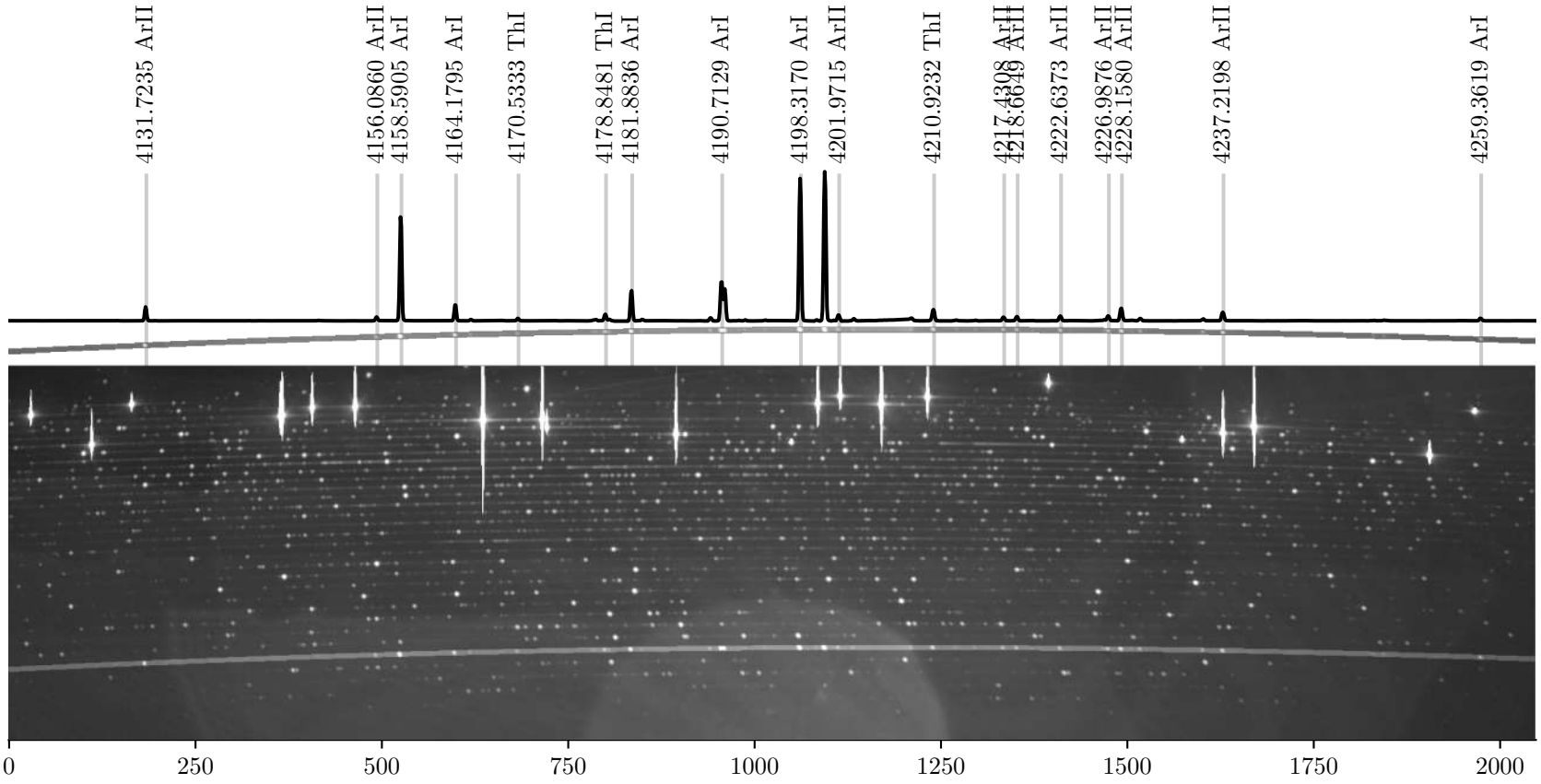
Approximate resolution:	16400
Dispersion:	0.070 Å pix^{-1}
Central wavelength:	4119 Å
Wavelength range:	4042 Å – 4186 Å


4042.8937 ArII
4043.1303 ThI
4043.3948 ThI
4044.4179 ArI
4044.9269 ThI
4045.2268 ThI
4045.9654 ArI
4047.4815 ArII
4048.0318 ThII
4048.4824 ThI
4049.5361 ThI
4049.9447 ThI
4050.8872 Th
4051.1090 Th
4051.4991 ThI
4052.9208 ArII
4053.5277 ThI
4054.3017 ThI
4054.3258 ArI
4057.3312 ThII
4057.9412 ThI
4059.2529 ThI
4060.0616 ThI
4061.6254 ThI
4063.4071 ThI
4063.6920 ThII
4064.3315 ThI
4066.8219 ThI
4045
4050
4055
4060
4065
4063.4071 ThI
4063.6920 ThII
4064.3315 ThI
4066.8219 ThI
4067.4507 ThI
4067.8759 ThII
4068.4703 ThI
4069.2014 ThII
4069.4612 Th
4069.7608 ThII
4070.2383 ThI
4070.7835 ArII
4071.7513 ThI
4072.3849 ArII
4072.6284 Th
4073.8563 ThI
4074.6420 ThII
4075.5030 ThI
4075.9070 ThI
4076.6284 ArII
4076.9432 ArII
4078.3015 ThI
4078.8751 Th
4079.5738 ArII
4080.3580 ThI
4080.7058 ThI
4081.3678 ThI
4081.5914 ThI
4082.0817 Th
4082.3872 ArII
4083.4688 ThI
4084.6192 Th
4085.0421 ThII
4085.4341 ThI
4086.5205 ThII
4065
4070
4075
4080
4085
4082.0817 Th
4082.3872 ArII
4083.4688 ThI
4084.6192 Th
4085.0421 ThII
4085.4341 ThI
4086.5205 ThII
4087.2848 ThI
4088.7264 ThI
4089.1379 ThI
4091.3476 ThI
4091.7368 ThI
4093.3921 ThII
4093.6713 ThI
4094.7470 ThII
4096.0759 ThI
4097.7478 ThI
4098.7316 Th
4098.9327 ThII
4100.3414 ThI
4102.6177 ThI
4102.8199 ThI
4103.2627 ThII
4103.9121 ArII
4104.3821 ThII
4104.8385 Th
4105.3301 ThII
4105.9150 ThII
4085
4090
4095
4100
4105

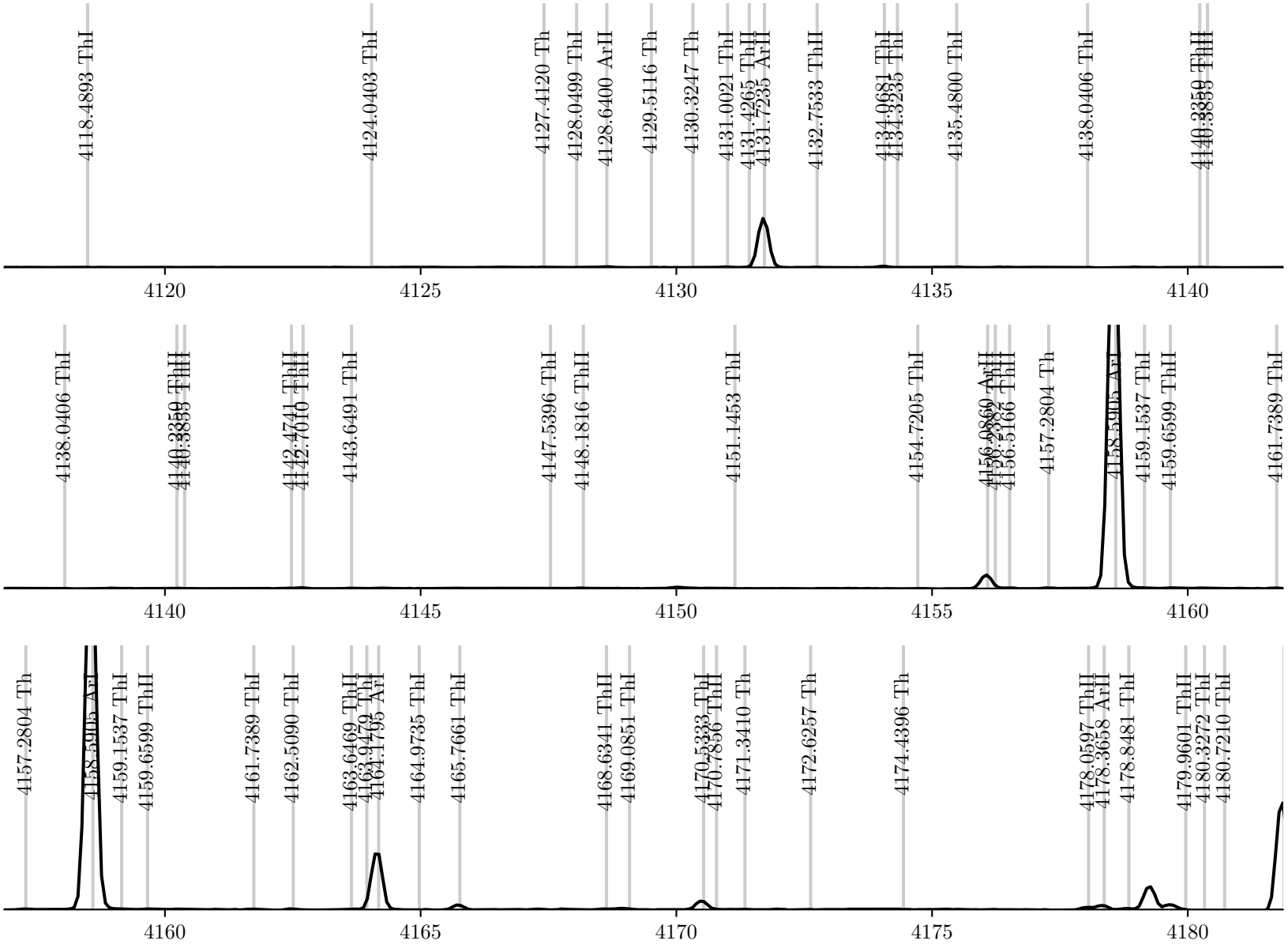

4102.6177 ThI
4102.8160 ThI
4103.2627 ThII
4103.9121 ArII
4104.3821 ThII
4104.8385 Th
4105.3301 ThII
4105.9150 ThII
4107.0506 Th
4107.8614 ThI
4108.4198 ThII
4109.3234 ThI
4110.8260 ThI
4112.7545 ThI
4113.5575 ThII
4115.7589 ThI
4116.3753 ArII
4116.7137 ThII
4118.4893 ThI
4124.0403 ThI
4105
4110
4115
4120
4125
4124.0403 ThI
4127.4120 Th
4128.0499 ThI
4128.6400 ArII
4129.5116 Th
4130.3247 Th
4131.0021 ThI
4131.4265 ThII
4131.7235 ArII
4132.7533 ThII
4134.0681 ThI
4135.4800 ThI
4138.0406 ThI
4142.4741 ThII
4143.6491 ThI
4125
4130
4135
4140
4145
4142.4741 ThII
4143.6491 ThI
4147.5396 ThI
4148.1816 ThII
4151.1453 ThI
4154.7205 ThI
4157.2804 Th
4159.1537 ThI
4159.6599 ThII
4161.7389 ThI
4162.5090 ThI
4163.6469 ThII
4164.9735 ThI
4165.7661 ThI
4145
4150
4155
4160
4165

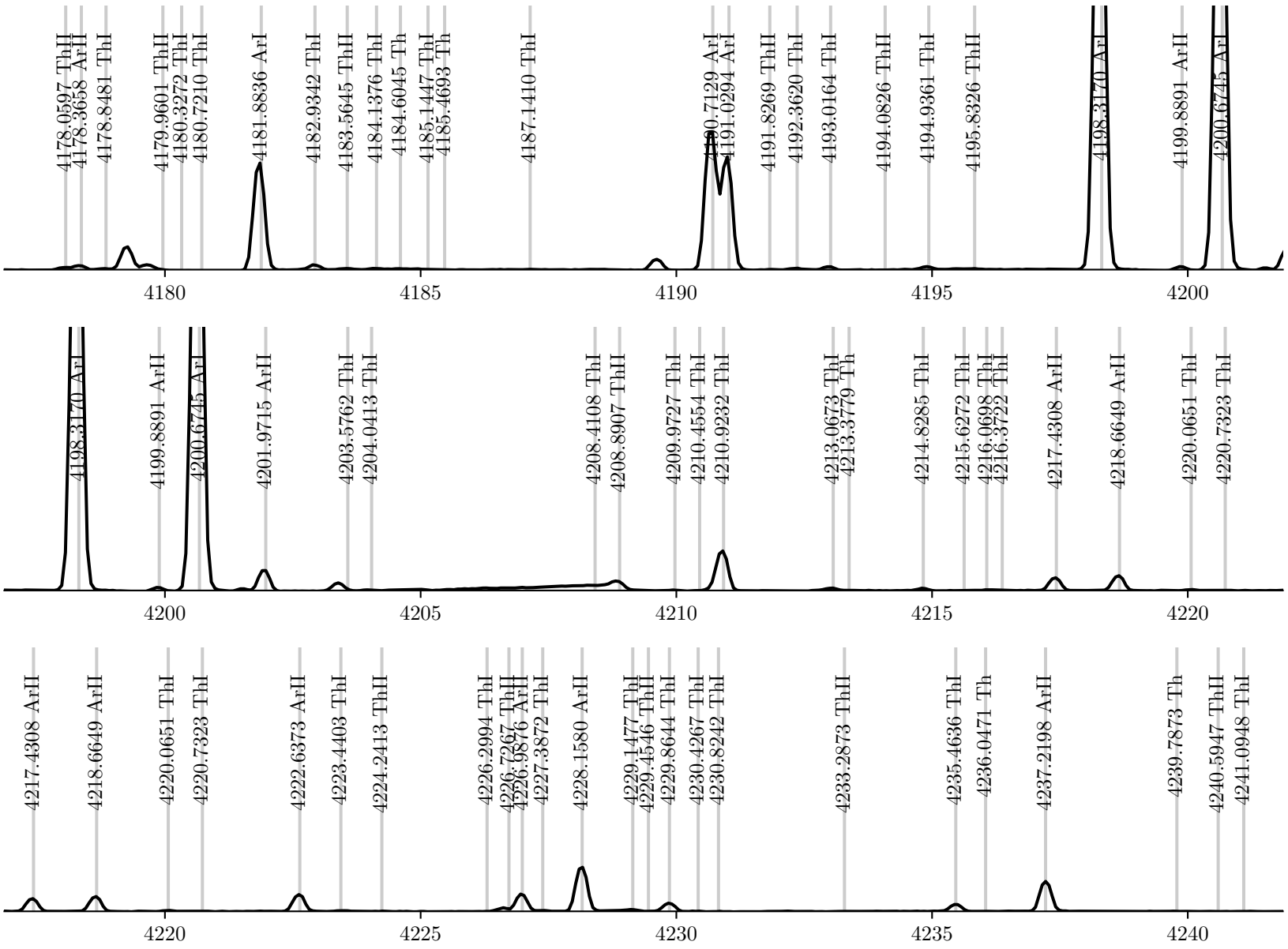

4162.5090 ThI
4163.6469 ThII
4163.9479 ThI
4164.1795 ArI
4164.9735 ThI
4165.7661 ThI
4168.6341 ThII
4169.0851 ThI
4170.5333 ThI
4170.7856 ThII
4171.3410 Th
4172.6257 Th
4174.4396 Th
4178.0597 ThII
4178.3658 ArII
4178.8481 ThI
4179.9601 ThII
4180.3272 ThI
4180.7210 ThI
4181.8836 ArI
4182.9342 ThI
4183.5645 ThII
4184.1376 ThI
4184.6045 Th
4185.1447 ThI
4185.4693 Th
4165
4170
4175
4180
4185
4182.9342 ThI
4183.5645 ThII
4184.1376 ThI
4184.6045 Th
4185.1447 ThI
4185.4693 Th
4185
4190
4195
4200
4205

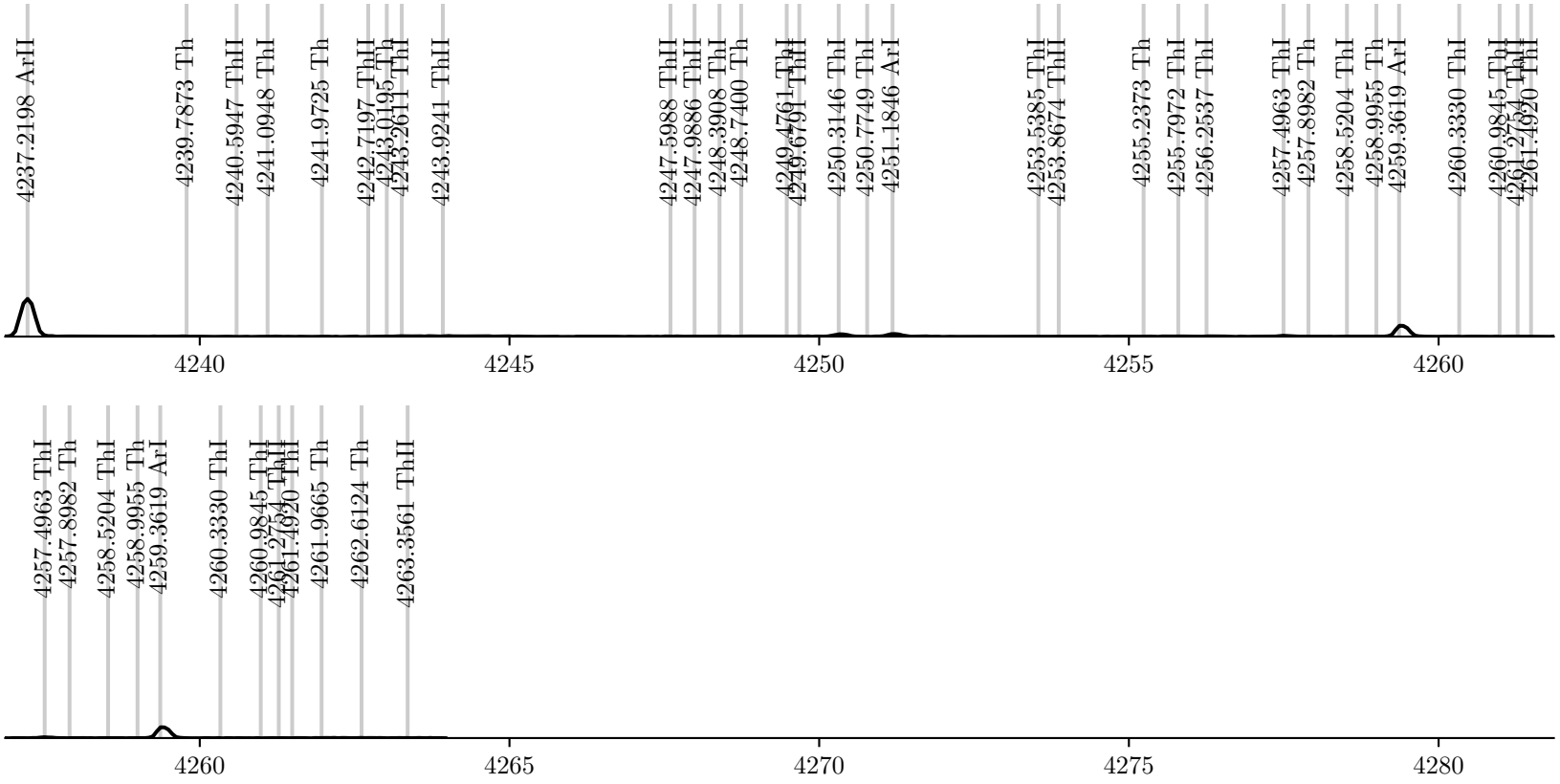
Order 54



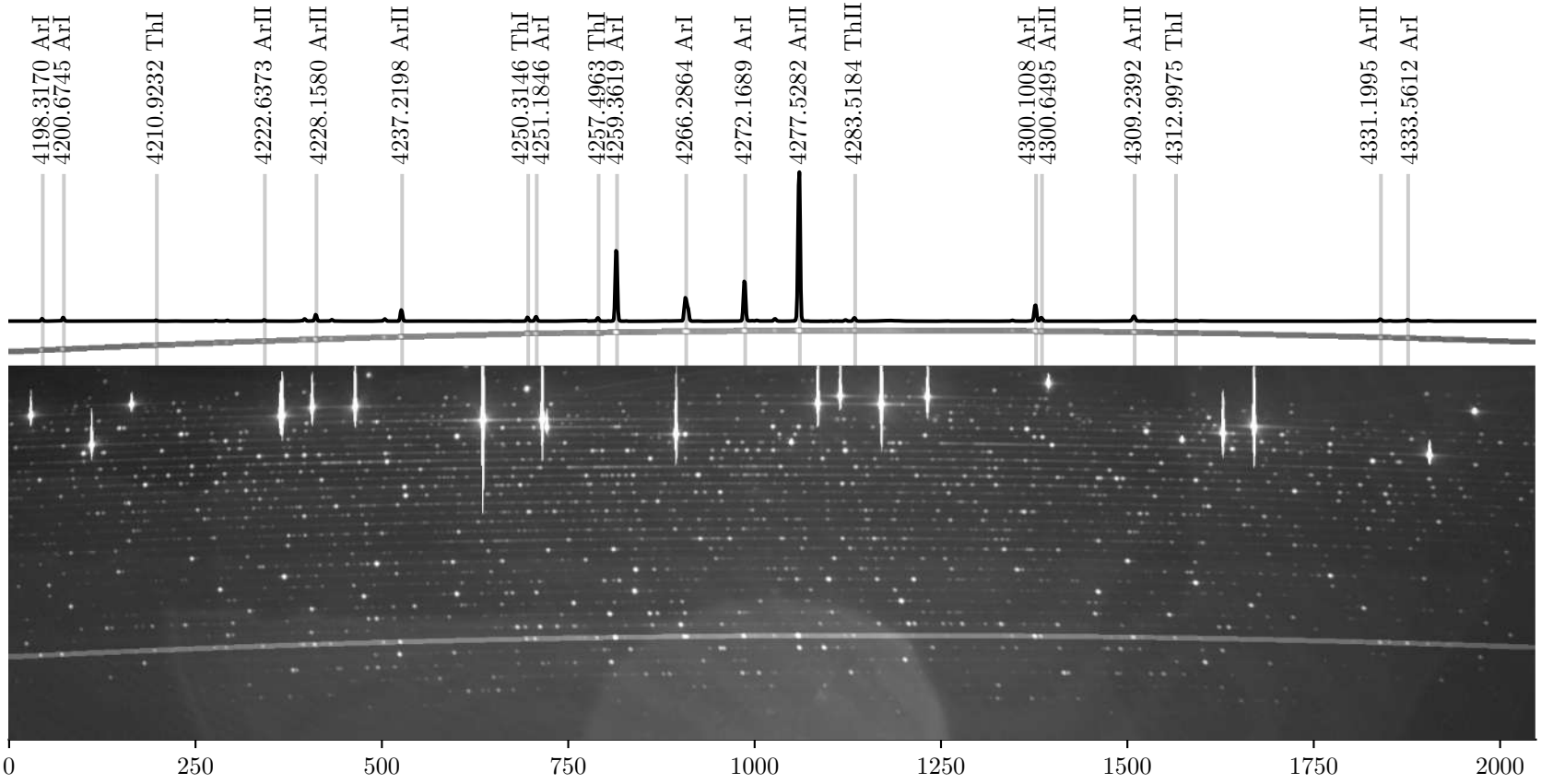
Approximate resolution:	16200
Dispersion:	0.072 Å pix^{-1}
Central wavelength:	4196 Å
Wavelength range:	4117 Å – 4264 Å


4118.4893 ThI
4124.0403 ThI
4127.4120 Th
4128.0499 ThI
4128.6400 ArII
4129.5116 Th
4130.3247 Th
4131.0021 ThI
4131.4265 ThII
4131.7235 ArII
4132.7533 ThII
4134.0681 ThI
4134.3235 ThI
4135.4800 ThII
4138.0406 ThI
4120
4125
4130
4135
4140
4138.0406 ThI
4140.3359 ThII
4142.4741 ThII
4142.7010 ThII
4143.6491 ThI
4147.5396 ThI
4148.1816 ThII
4151.1453 ThI
4154.7205 ThI
4156.0860 ArII
4156.2382 ThII
4156.5166 ThII
4157.2804 Th
4158.5905 ArI
4159.1537 ThI
4159.6599 ThII
4161.7389 ThI
4140
4145
4150
4155
4160
4157.2804 Th
4158.5905 ArI
4159.1537 ThI
4159.6599 ThII
4161.7389 ThI
4162.5090 ThI
4163.6469 ThII
4163.9479 ThI
4164.1795 ArI
4164.9735 ThI
4165.7661 ThI
4168.6341 ThII
4169.0851 ThI
4170.5333 ThI
4170.7856 ThI
4171.3410 Th
4172.6257 Th
4174.4396 Th
4178.0597 ThII
4178.3658 ArII
4178.8481 ThI
4179.9601 ThII
4180.3272 ThI
4180.7210 ThI
4160
4165
4170
4175
4180

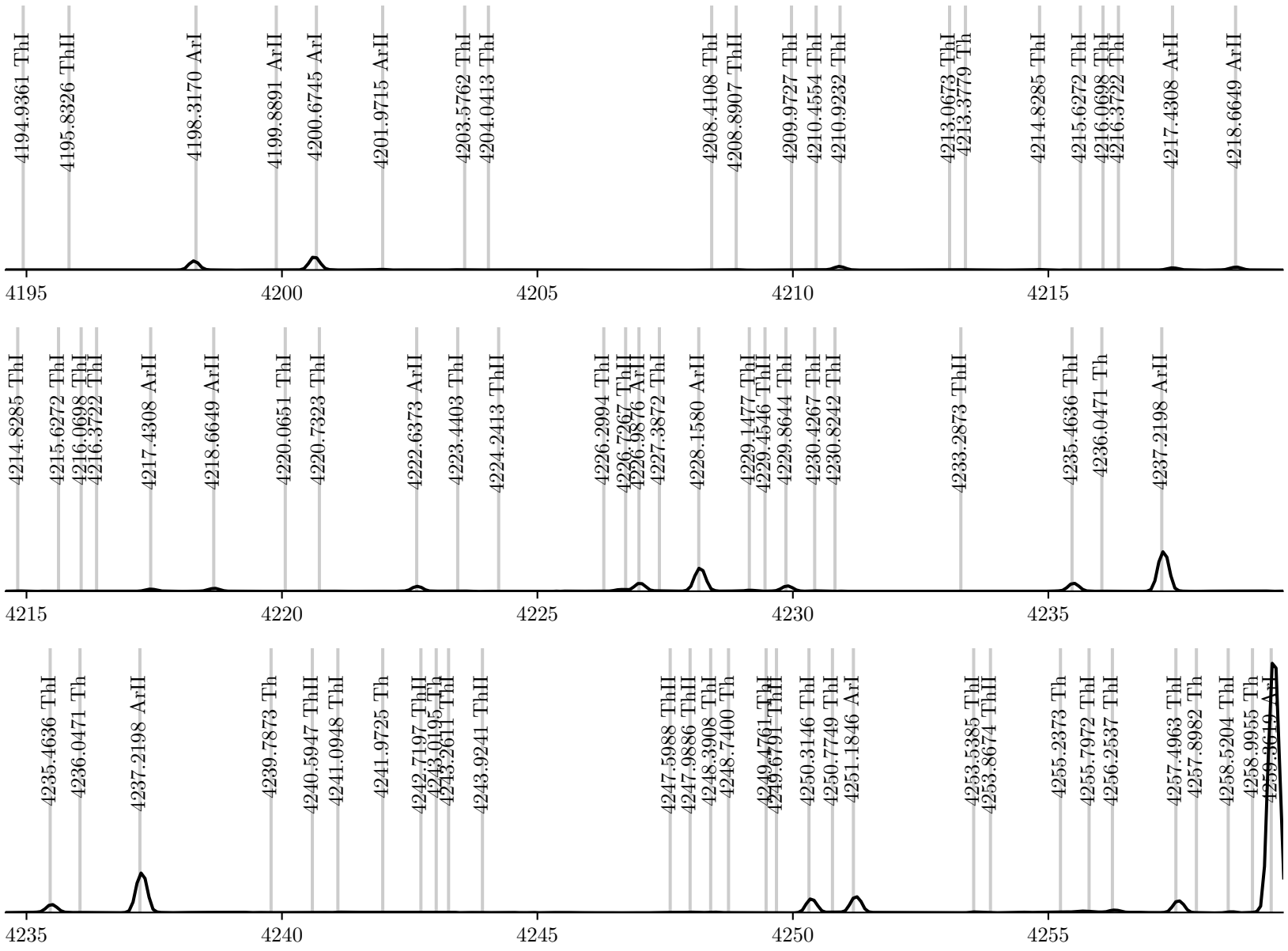

4178.0597 ThII
4178.3658 ArII
4178.8481 ThI
4179.9601 ThII
4180.3272 ThI
4180.7210 ThI
4181.8836 ArI
4182.9342 ThI
4183.5645 ThII
4184.1376 ThI
4184.6045 Th
4185.1447 ThI
4185.4693 Th
4187.1410 ThI
4190.7129 ArI
4191.0294 ArI
4191.8269 ThII
4192.3620 ThI
4193.0164 ThI
4194.0826 ThII
4194.9361 ThI
4195.8326 ThII
4198.3170 ArI
4199.8891 ArII
4200.6745 ArI
4180
4185
4190
4195
4200
4198.3170 ArI
4199.8891 ArII
4200.6745 ArI
4201.9715 ArII
4203.5762 ThI
4204.0413 ThI
4208.4108 ThI
4208.8907 ThII
4209.9727 ThI
4210.4554 ThI
4210.9232 ThI
4213.0673 ThI
4213.3779 Th
4214.8285 ThI
4215.6272 ThI
4216.0698 ThI
4216.3722 ThI
4217.4308 ArII
4218.6649 ArII
4220.0651 ThI
4220.7323 ThI
4200
4205
4210
4215
4220
4217.4308 ArII
4218.6649 ArII
4220.0651 ThI
4220.7323 ThI
4222.6373 ArII
4223.4403 ThI
4224.2413 ThII
4226.2994 ThI
4226.7267 ThII
4226.9876 ArII
4227.3872 ThI
4228.1580 ArII
4229.1477 ThI
4229.4546 ThII
4229.8644 ThI
4230.4267 ThI
4230.8242 ThI
4233.2873 ThII
4235.4636 ThI
4236.0471 Th
4237.2198 ArII
4239.7873 Th
4240.5947 ThII
4241.0948 ThI
4220
4225
4230
4235
4240

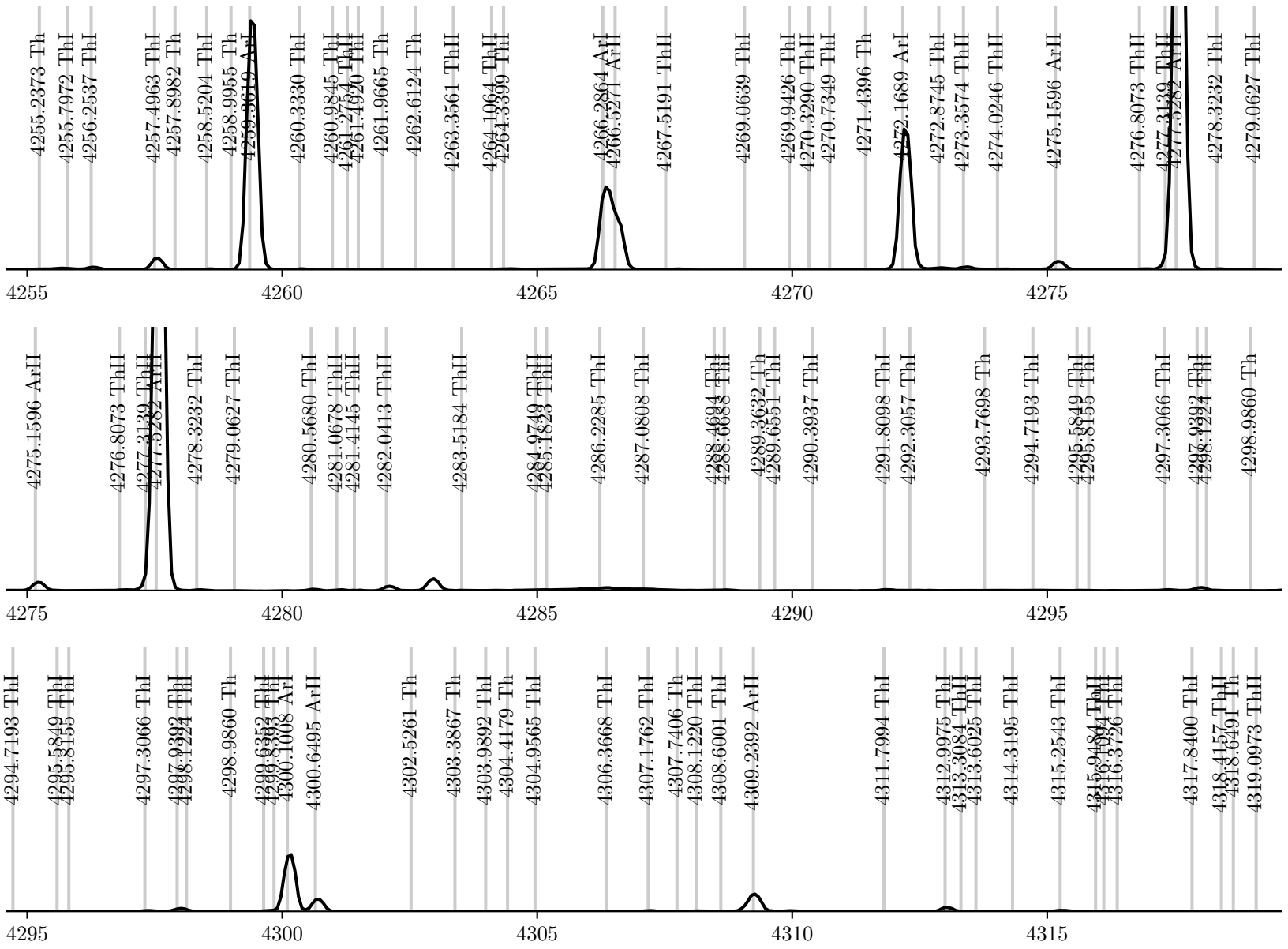

4237.2198 ArII
4239.7873 Th
4240.5947 ThII
4241.0948 ThI
4241.9725 Th
4242.7197 ThII
4243.0195 Th
4243.2611 ThI
4243.9241 ThII
4247.5988 ThII
4247.9886 ThII
4248.3908 ThI
4248.7400 Th
4249.4761 ThI
4249.6791 ThII
4250.3146 ThI
4250.7749 ThI
4251.1846 ArI
4253.5385 ThI
4253.8674 ThII
4255.2373 Th
4255.7972 ThI
4256.2537 ThI
4257.4963 ThI
4257.8982 Th
4258.5204 ThI
4258.9955 Th
4259.3619 ArI
4260.3330 ThI
4260.9845 ThI
4261.2754 ThII
4261.4920 ThI
4240
4245
4250
4255
4260
4257.4963 ThI
4257.8982 Th
4258.5204 ThI
4258.9955 Th
4259.3619 ArI
4260.3330 ThI
4260.9845 ThI
4261.4920 ThI
4261.9665 Th
4262.6124 Th
4263.3561 ThII
4260
4265
4270
4275
4280

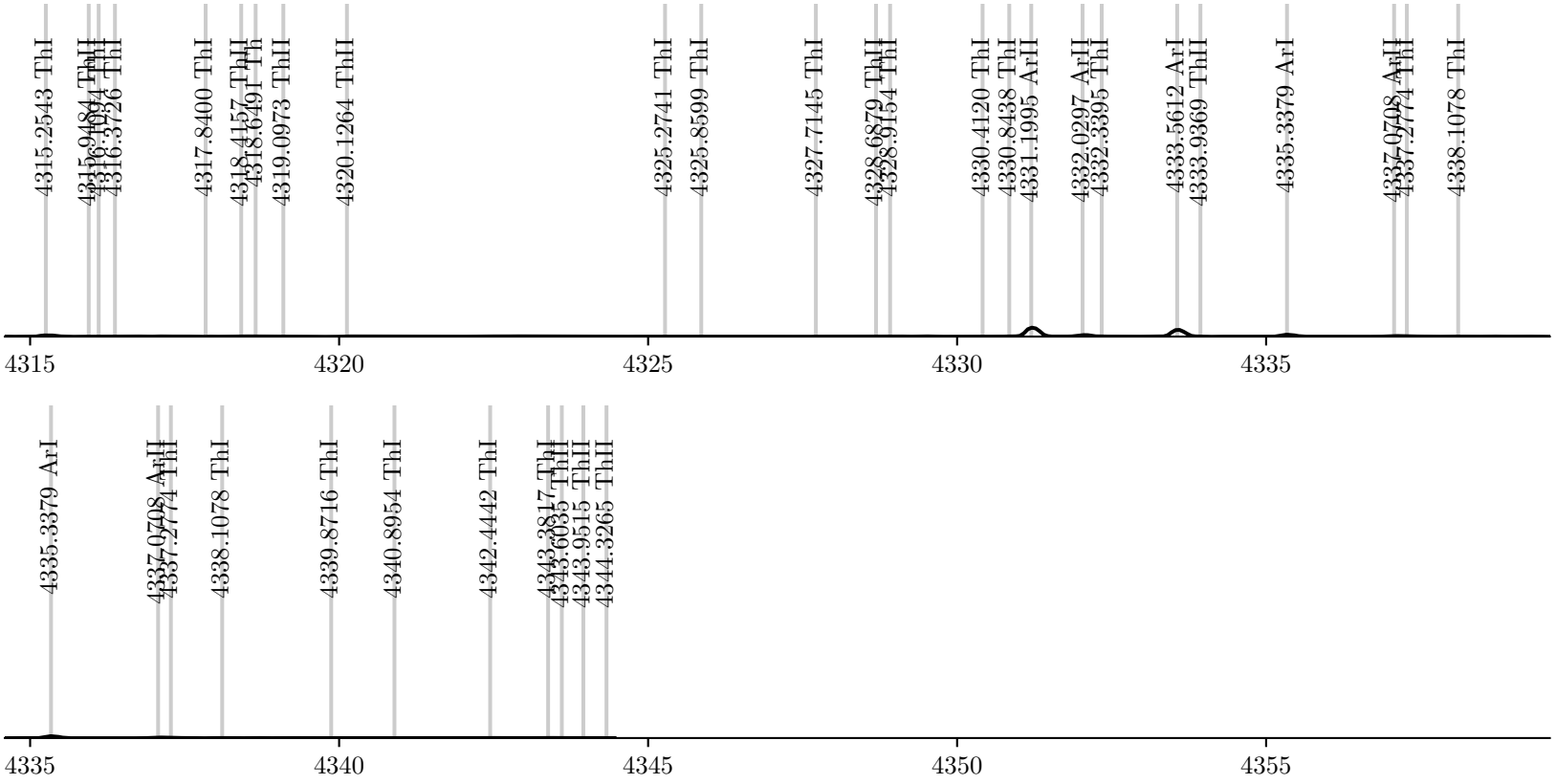
Order 53



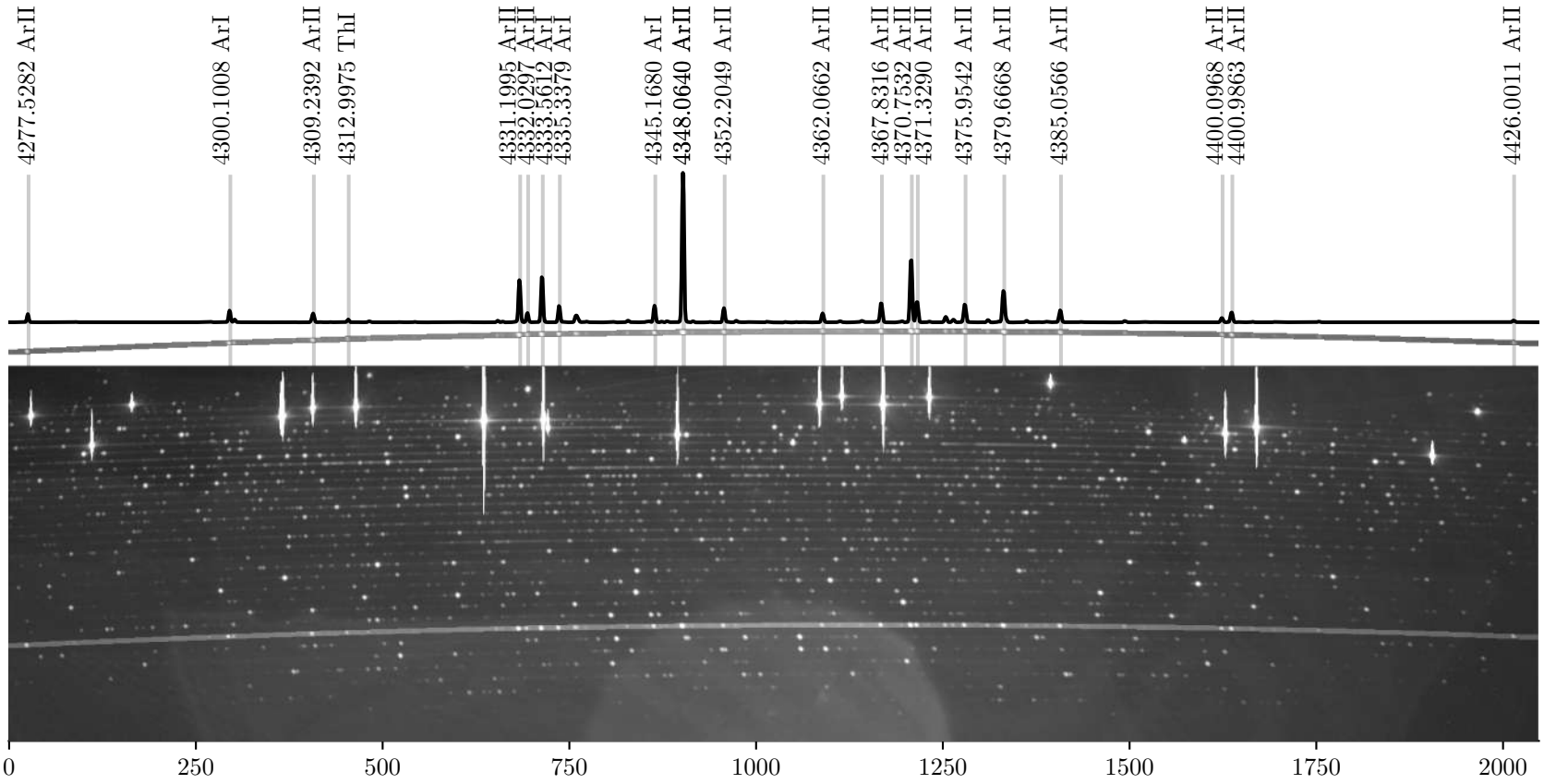
Approximate resolution:	17000
Dispersion:	0.073 Å pix^{-1}
Central wavelength:	4275 Å
Wavelength range:	4195 Å – 4344 Å


4194.9361 ThI
4195.8326 ThII
4198.3170 ArI
4199.8891 ArII
4200.6745 ArI
4201.9715 ArII
4203.5762 ThI
4204.0413 ThI
4208.4108 ThI
4208.8907 ThII
4209.9727 ThI
4210.4554 ThI
4210.9232 ThI
4213.0673 ThI
4213.3779 Th
4214.8285 ThI
4215.6272 ThI
4216.0698 ThI
4216.3722 ThI
4217.4308 ArII
4218.6649 ArII
4195
4200
4205
4210
4215
4214.8285 ThI
4215.6272 ThI
4216.0698 ThI
4216.3722 ThI
4217.4308 ArII
4218.6649 ArII
4220.0651 ThI
4220.7323 ThI
4222.6373 ArII
4223.4403 ThI
4224.2413 ThII
4226.2994 ThI
4226.7367 ThII
4226.9876 ArII
4227.3872 ThI
4228.1580 ArII
4229.1477 ThI
4229.4546 ThII
4229.8644 ThI
4230.4267 ThI
4230.8242 ThI
4233.2873 ThII
4235.4636 ThI
4236.0471 Th
4237.2198 ArII
4215
4220
4225
4230
4235
4235.4636 ThI
4236.0471 Th
4237.2198 ArII
4239.7873 Th
4240.5947 ThII
4241.0948 ThI
4241.9725 Th
4242.7197 ThII
4243.0195 Th
4243.2611 ThI
4243.9241 ThII
4247.5988 ThII
4247.9886 ThII
4248.3908 ThI
4248.7400 Th
4249.4761 ThI
4250.3146 ThI
4250.7749 ThI
4251.1846 ArI
4253.5385 ThI
4253.8674 ThII
4255.2373 Th
4255.7972 ThI
4256.2537 ThI
4257.4963 ThI
4257.8982 Th
4258.5204 ThI
4258.9955 Th
4259.3619 ArI
4235
4240
4245
4250
4255

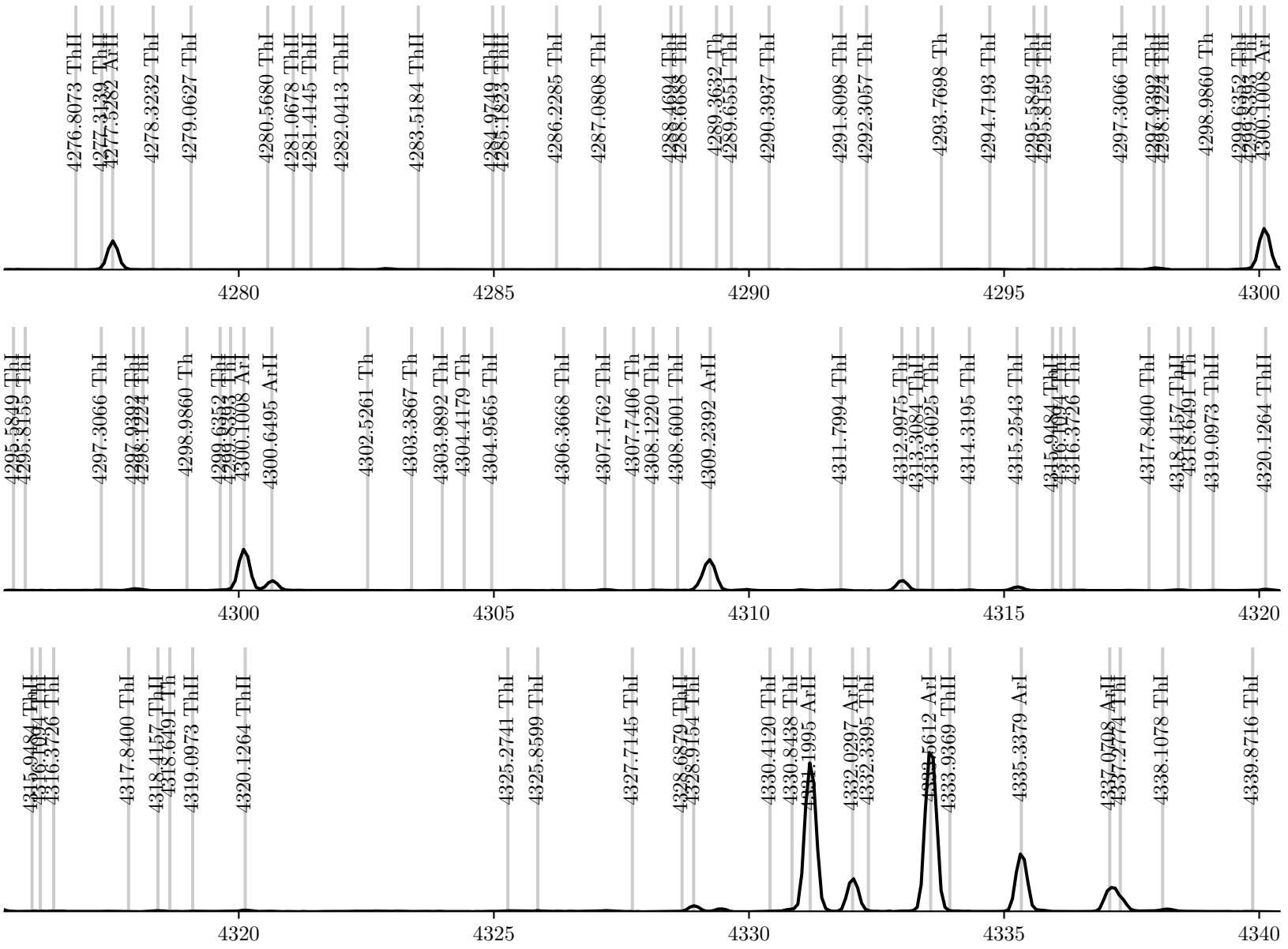

4255.2373 Th
4255.7972 ThI
4256.2537 ThI
4257.4963 ThI
4257.8982 Th
4258.5204 ThI
4258.9955 Th
4260.3330 ThI
4260.9845 ThI
4261.9665 Th
4262.6124 Th
4263.3561 ThII
4266.2864 ArI
4266.5271 ArII
4267.5191 ThII
4269.0639 ThI
4269.9426 ThI
4270.3290 ThII
4270.7349 ThI
4271.4396 Th
4272.1689 ArI
4272.8745 ThI
4273.3574 ThII
4274.0246 ThII
4275.1596 ArII
4276.8073 ThII
4277.3139 ThII
4278.3232 ThI
4279.0627 ThI
4255
4260
4265
4270
4275
4275.1596 ArII
4276.8073 ThII
4277.3139 ThII
4277.3282 ArII
4278.3232 ThI
4279.0627 ThI
4280.5680 ThI
4281.0678 ThII
4281.4145 ThII
4282.0413 ThII
4283.5184 ThII
4286.2285 ThI
4287.0808 ThI
4289.3632 Th
4289.6551 ThI
4290.3937 ThI
4291.8098 ThI
4292.3057 ThI
4293.7698 Th
4294.7193 ThI
4297.3066 ThI
4298.9860 Th
4275
4280
4285
4290
4295
4294.7193 ThI
4295.5849 ThI
4297.3066 ThI
4298.9860 Th
4300.6495 ArII
4300.1008 ArI
4302.5261 Th
4303.3867 Th
4303.9892 ThI
4304.4179 Th
4304.9565 ThI
4306.3668 ThI
4307.1762 ThI
4307.7406 Th
4308.1220 ThI
4308.6001 ThI
4309.2392 ArII
4311.7994 ThI
4312.9975 ThI
4313.3084 ThII
4313.6025 ThI
4314.3195 ThI
4315.2543 ThI
4317.8400 ThI
4318.4157 ThII
4319.0973 ThII
4295
4300
4305
4310
4315

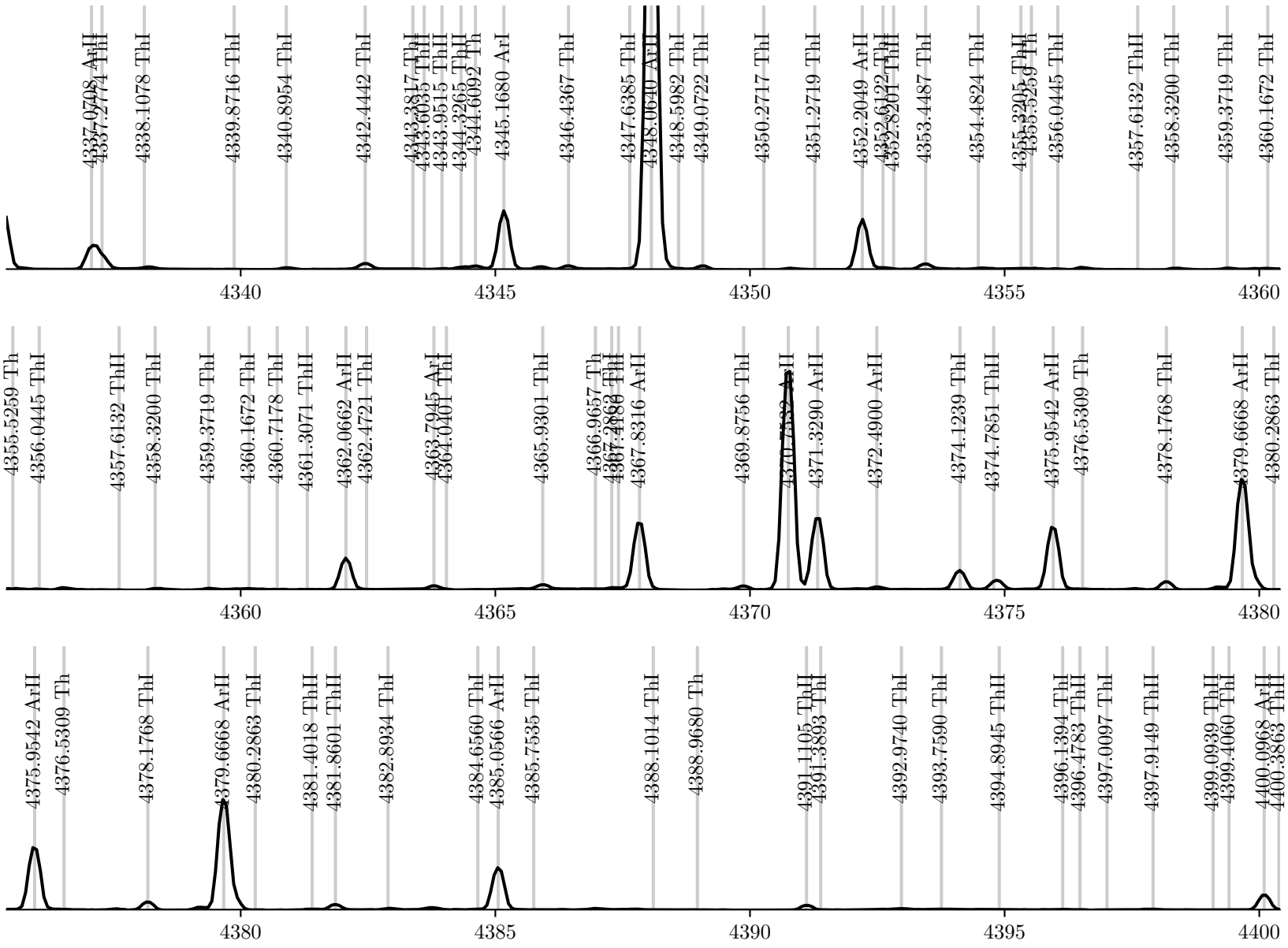

4315.2543 ThI
4315.9484 ThII
4316.1034 ThII
4316.3726 ThI
4317.8400 ThI
4318.4157 ThII
4318.6491 ThI
4319.0973 ThII
4320.1264 ThII
4325.2741 ThI
4325.8599 ThI
4327.7145 ThI
4328.6879 ThII
4328.9154 ThI
4330.4120 ThI
4330.8438 ThI
4331.1995 ArII
4332.0297 ArII
4332.3395 ThI
4333.5612 ArI
4333.9369 ThII
4335.3379 ArI
4337.0708 ArII
4337.2774 ThI
4338.1078 ThI
4315
4320
4325
4330
4335
4335.3379 ArI
4337.0708 ArII
4337.2774 ThI
4338.1078 ThI
4339.8716 ThI
4340.8954 ThI
4342.4442 ThI
4343.3817 ThI
4343.6035 ThII
4343.9515 ThII
4344.3265 ThII
4335
4340
4345
4350
4355

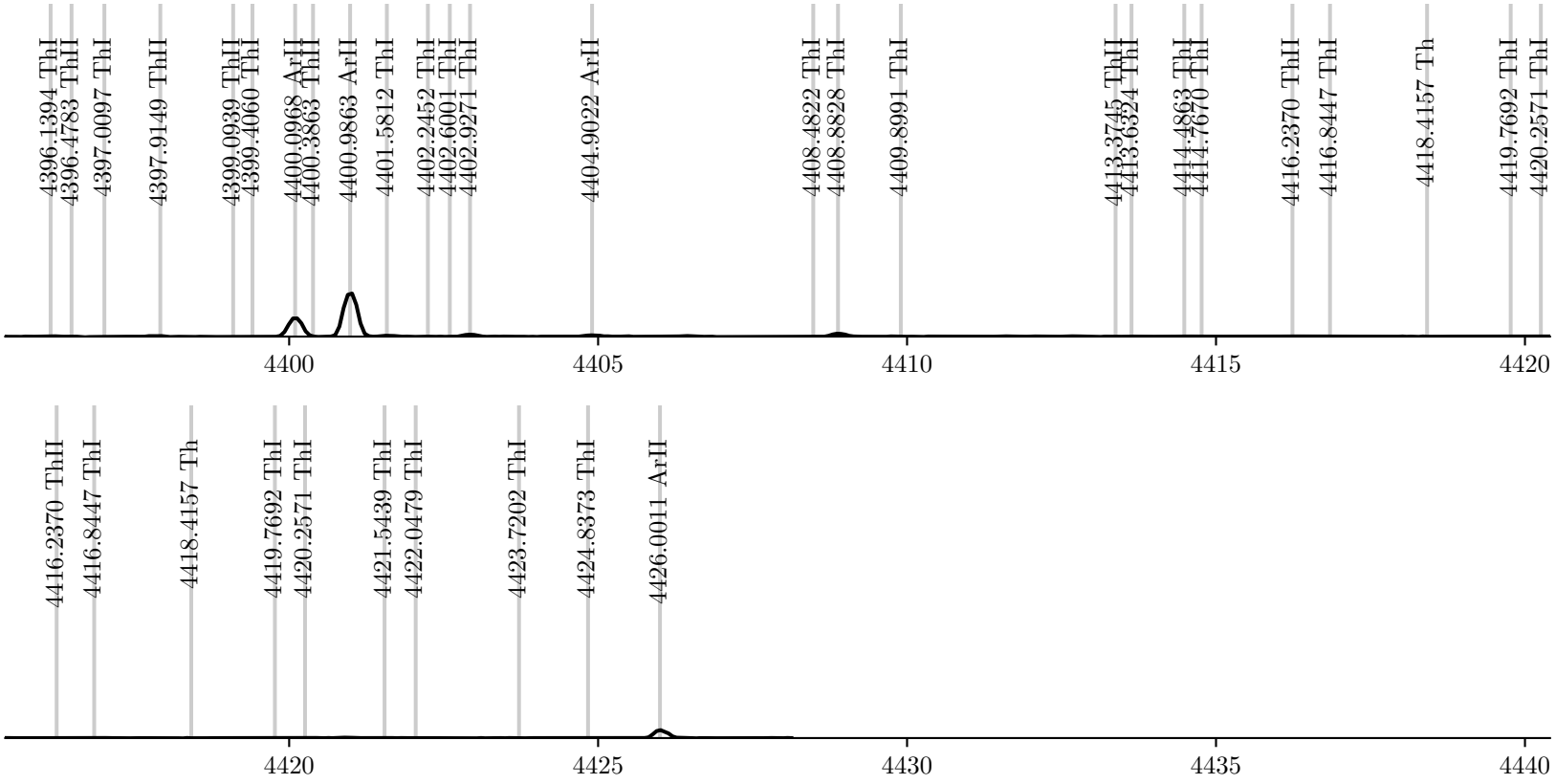
Order 52



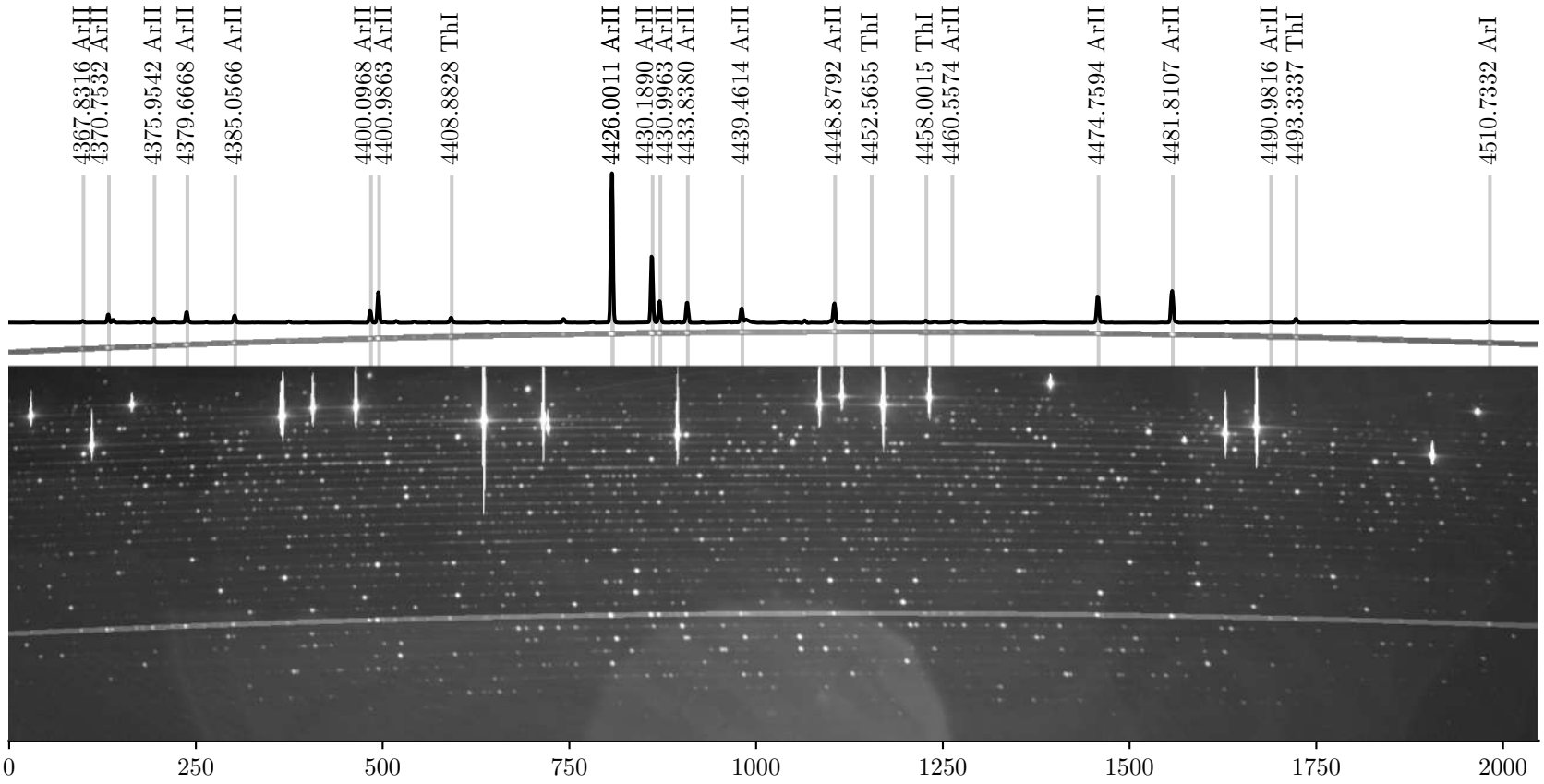
Approximate resolution:	16500
Dispersion:	0.075 Å pix^{-1}
Central wavelength:	4357 Å
Wavelength range:	4275 Å – 4428 Å


4276.8073 ThII
4277.3139 ThII
4277.5282 ArII
4278.3232 ThI
4279.0627 ThI
4280.5680 ThI
4281.0678 ThII
4281.4145 ThII
4282.0413 ThII
4283.5184 ThII
4284.9749 ThII
4285.1823 ThII
4286.2285 ThI
4287.0808 ThI
4288.4604 ThII
4288.6088 ThII
4289.3632 Th
4289.6551 ThI
4290.3937 ThI
4291.8098 ThI
4292.3057 ThI
4293.7698 Th
4294.7193 ThI
4295.5849 ThII
4295.8155 ThII
4297.3066 ThI
4297.9392 ThII
4298.1224 ThII
4298.9860 Th
4299.6352 ThI
4300.1008 ArI
4280
4285
4290
4295
4300
4295.5849 ThII
4295.8155 ThII
4297.3066 ThI
4297.9392 ThII
4298.1224 ThII
4298.9860 Th
4299.6352 ThI
4300.1008 ArI
4300.6495 ArII
4302.5261 Th
4303.3867 Th
4303.9892 ThI
4304.4179 Th
4304.9565 ThI
4306.3668 ThI
4307.1762 ThI
4307.7406 Th
4308.1220 ThI
4308.6001 ThI
4309.2392 ArII
4311.7994 ThI
4312.9975 ThI
4313.3084 ThII
4313.6025 ThI
4314.3195 ThI
4315.2543 ThI
4315.9484 ThII
4316.1694 ThII
4316.3726 ThI
4317.8400 ThI
4318.4157 ThII
4318.6491 Th
4319.0973 ThII
4320.1264 ThII
4300
4305
4310
4315
4320
4315.9484 ThII
4316.1694 ThII
4316.3726 ThI
4317.8400 ThI
4318.4157 ThII
4318.6491 Th
4319.0973 ThII
4320.1264 ThII
4325.2741 ThI
4325.8599 ThI
4327.7145 ThI
4328.6879 ThII
4328.9154 ThI
4330.4120 ThI
4330.8438 ThI
4331.1995 ArII
4332.0297 ArII
4332.3395 ThI
4333.5612 ArI
4333.9369 ThII
4335.3379 ArI
4337.0708 ArII
4337.2774 ThI
4338.1078 ThI
4339.8716 ThI
4320
4325
4330
4335
4340

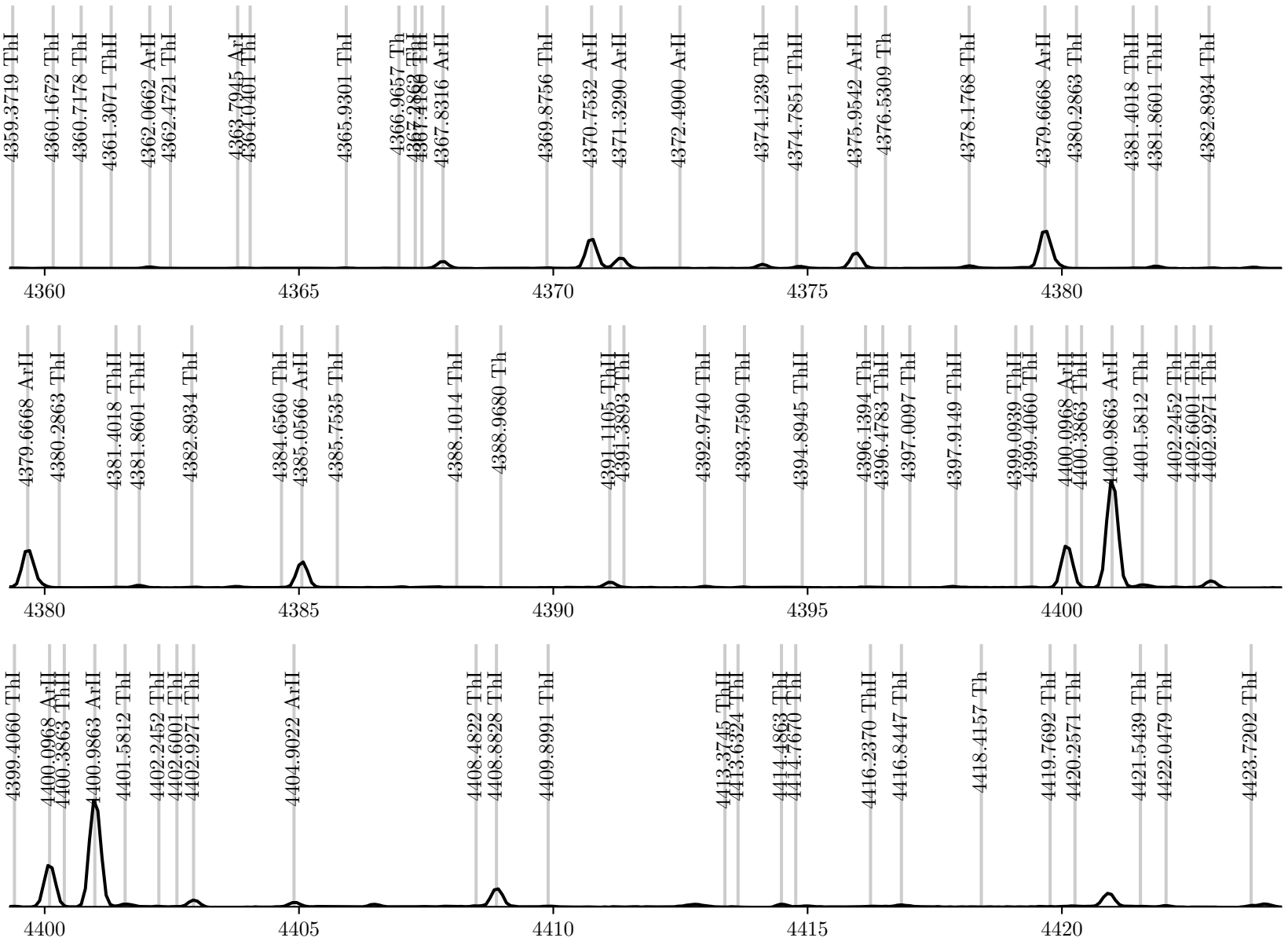

4337.0708 ArII
4337.2774 ThI
4338.1078 ThI
4339.8716 ThI
4340.8954 ThI
4342.4442 ThI
4343.3817 ThI
4343.6035 ThII
4343.9515 ThII
4344.3265 ThII
4344.6092 Th
4345.1680 ArI
4346.4367 ThI
4347.6385 ThI
4348.0640 ArII
4348.5982 ThI
4349.0722 ThI
4350.2717 ThI
4351.2719 ThI
4352.2049 ArII
4352.6122 ThI
4352.8201 ThII
4353.4487 ThI
4354.4824 ThI
4355.3205 ThII
4355.5259 Th
4356.0445 ThI
4357.6132 ThII
4358.3200 ThI
4359.3719 ThI
4360.1672 ThI
4340
4345
4350
4355
4360
4355.5259 Th
4356.0445 ThI
4357.6132 ThII
4358.3200 ThI
4359.3719 ThI
4360.1672 ThI
4360.7178 ThI
4361.3071 ThII
4362.0662 ArII
4362.4721 ThI
4363.7945 ArI
4364.0401 ThI
4365.9301 ThI
4366.9657 Th
4367.2862 ThI
4367.8316 ArII
4369.8756 ThI
4370.7532 ArII
4371.3290 ArII
4372.4900 ArII
4374.1239 ThI
4374.7851 ThII
4375.9542 ArII
4376.5309 Th
4378.1768 ThI
4379.6668 ArII
4380.2863 ThI
4360
4365
4370
4375
4380
4375.9542 ArII
4376.5309 Th
4378.1768 ThI
4379.6668 ArII
4380.2863 ThI
4381.4018 ThII
4381.8601 ThII
4382.8934 ThI
4384.6560 ThI
4385.0566 ArII
4385.7535 ThI
4388.1014 ThI
4388.9680 Th
4391.1105 ThI
4391.3893 ThI
4392.9740 ThI
4393.7590 ThI
4394.8945 ThII
4396.1394 ThI
4396.4783 ThII
4397.0097 ThI
4397.9149 ThII
4399.0939 ThII
4399.4060 ThI
4400.0968 ArII
4400.3863 ThII
4380
4385
4390
4395
4400

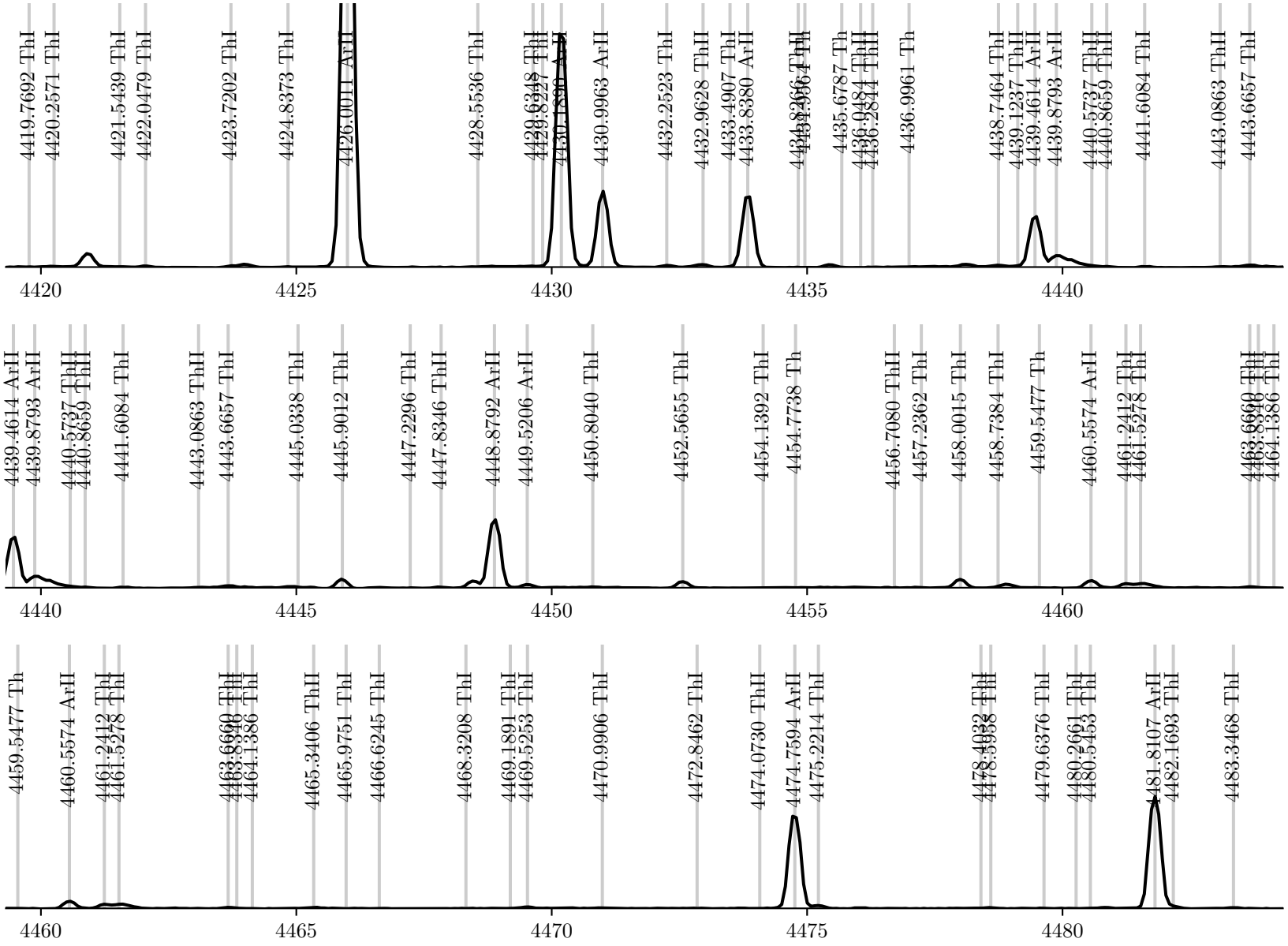

4396.1394 ThI
4396.4783 ThII
4397.0097 ThI
4397.9149 ThII
4399.0939 ThII
4399.4060 ThI
4400.0968 ArII
4400.3863 ThII
4400.9863 ArII
4401.5812 ThI
4402.2452 ThI
4402.6001 ThI
4402.9271 ThI
4404.9022 ArII
4408.4822 ThI
4408.8828 ThI
4409.8991 ThI
4413.3745 ThII
4413.6324 ThI
4414.4863 ThI
4414.7670 ThI
4416.2370 ThII
4416.8447 ThI
4418.4157 Th
4419.7692 ThI
4420.2571 ThI
4400
4405
4410
4415
4420
4416.2370 ThII
4416.8447 ThI
4418.4157 Th
4419.7692 ThI
4420.2571 ThI
4421.5439 ThI
4422.0479 ThI
4423.7202 ThI
4424.8373 ThI
4426.0011 ArII
4420
4425
4430
4435
4440

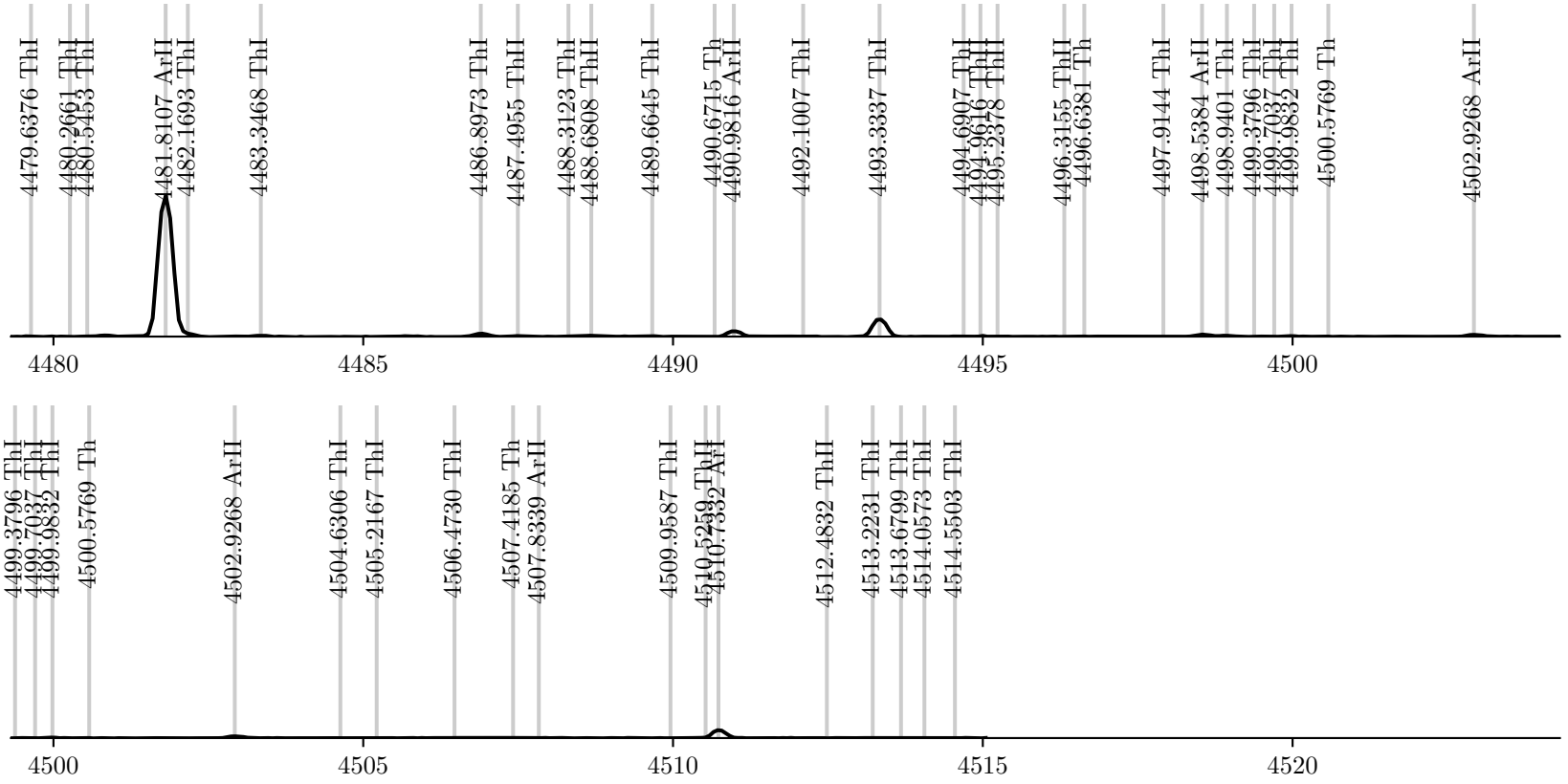
Order 51



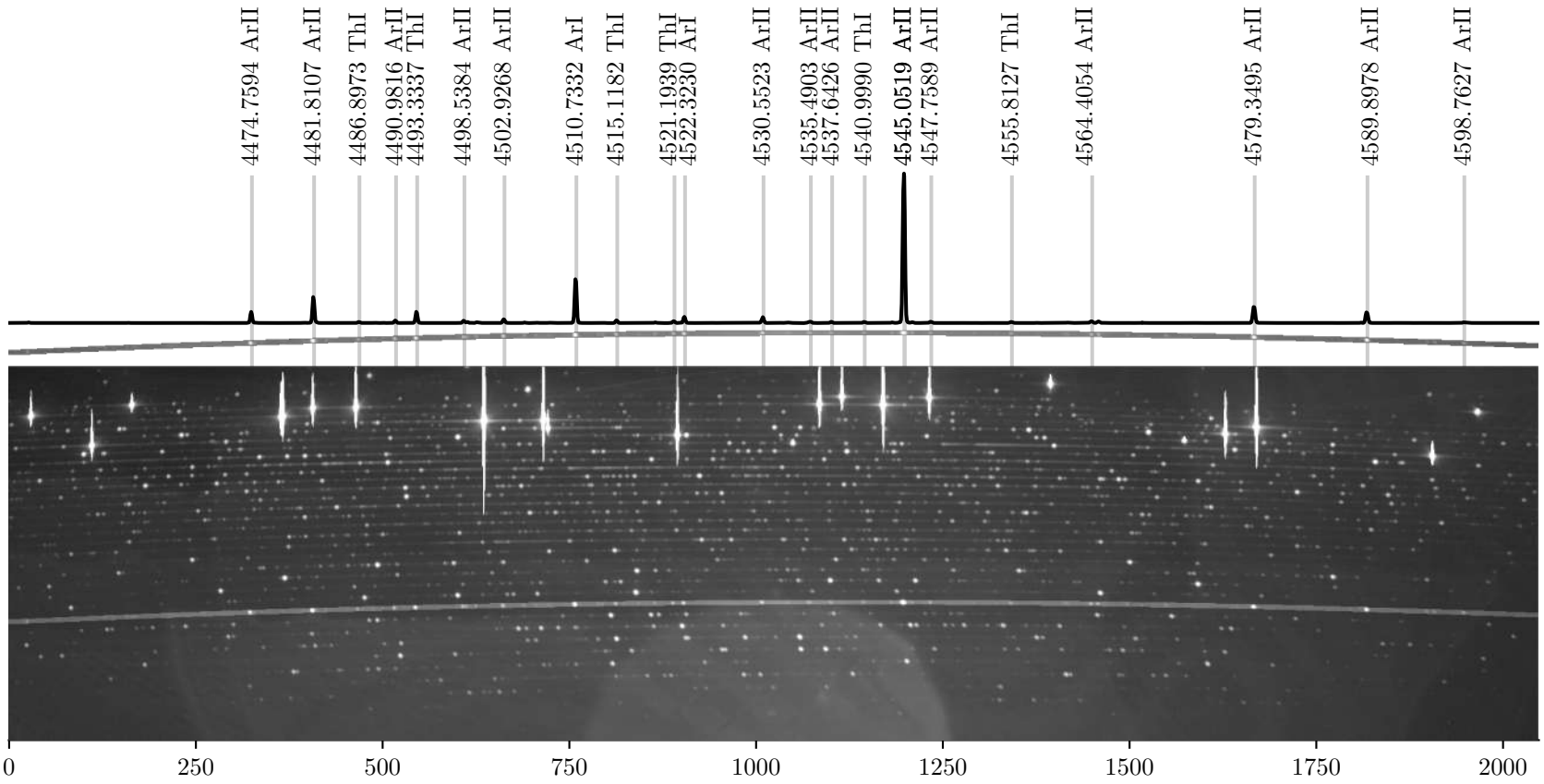
Approximate resolution:	16700
Dispersion:	0.076 Å pix^{-1}
Central wavelength:	4443 Å
Wavelength range:	4359 Å – 4515 Å


4359.3719 ThI
4360.1672 ThI
4360.7178 ThI
4361.3071 ThII
4362.0662 ArII
4362.4721 ThI
4363.7945 ArI
4364.0401 ThI
4365.9301 ThI
4366.9657 Th
4367.2862 ThI
4367.4180 ThI
4367.8316 ArII
4369.8756 ThI
4370.7532 ArII
4371.3290 ArII
4372.4900 ArII
4374.1239 ThI
4374.7851 ThII
4375.9542 ArII
4376.5309 Th
4378.1768 ThI
4379.6668 ArII
4380.2863 ThI
4381.4018 ThII
4381.8601 ThII
4382.8934 ThI
4360
4365
4370
4375
4380
4379.6668 ArII
4380.2863 ThI
4381.4018 ThII
4381.8601 ThII
4382.8934 ThI
4384.6560 ThI
4385.0566 ArII
4385.7535 ThI
4388.1014 ThI
4388.9680 Th
4391.1105 ThII
4391.3803 ThI
4392.9740 ThI
4393.7590 ThI
4394.8945 ThII
4396.1394 ThI
4396.4783 ThII
4397.0097 ThI
4397.9149 ThII
4399.0939 ThII
4399.4060 ThI
4400.0968 ArII
4400.3863 ThII
4400.9863 ArII
4401.5812 ThI
4402.2452 ThI
4402.6001 ThI
4402.9271 ThI
4380
4385
4390
4395
4400
4399.4060 ThI
4400.0968 ArII
4400.3863 ThII
4400.9863 ArII
4401.5812 ThI
4402.2452 ThI
4402.6001 ThI
4402.9271 ThI
4404.9022 ArII
4408.4822 ThI
4408.8828 ThI
4409.8991 ThI
4413.3745 ThII
4413.6324 ThI
4414.4863 ThI
4414.7670 ThI
4416.2370 ThII
4416.8447 ThI
4418.4157 Th
4419.7692 ThI
4420.2571 ThI
4421.5439 ThI
4422.0479 ThI
4423.7202 ThI
4400
4405
4410
4415
4420

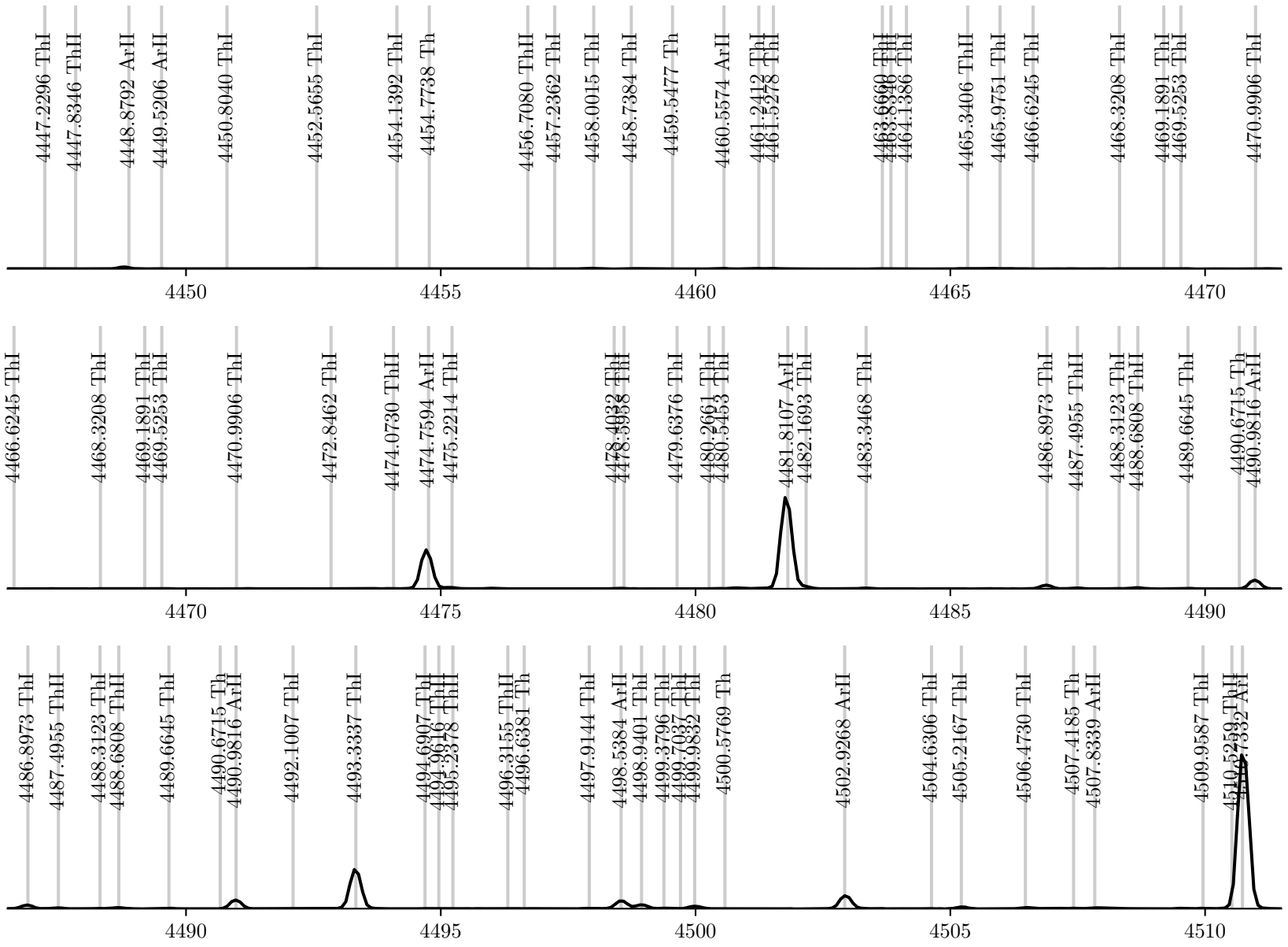

4419.7692 ThI
4420.2571 ThI
4421.5439 ThI
4422.0479 ThI
4423.7202 ThI
4424.8373 ThI
4426.0011 ArII
4428.5536 ThI
4430.9963 ArII
4432.2523 ThI
4432.9628 ThII
4433.4907 ThI
4433.8380 ArII
4435.6787 Th
4438.7464 ThI
4439.1237 ThII
4439.4614 ArII
4439.8793 ArII
4440.5737 ThII
4440.8659 ThII
4441.6084 ThI
4443.0863 ThII
4443.6657 ThI
4420
4425
4430
4435
4440
4439.4614 ArII
4439.8793 ArII
4440.5737 ThII
4440.8659 ThII
4441.6084 ThI
4443.0863 ThII
4443.6657 ThI
4445.0338 ThI
4445.9012 ThI
4447.2296 ThI
4447.8346 ThII
4448.8792 ArII
4449.5206 ArII
4450.8040 ThI
4452.5655 ThI
4454.1392 ThI
4454.7738 Th
4456.7080 ThII
4457.2362 ThI
4458.0015 ThI
4458.7384 ThI
4459.5477 Th
4460.5574 ArII
4461.2412 ThI
4461.5278 ThI
4463.6660 ThI
4463.8340 ThI
4464.1386 ThI
4440
4445
4450
4455
4460
4459.5477 Th
4460.5574 ArII
4461.2412 ThI
4461.5278 ThI
4463.6660 ThI
4463.8340 ThI
4464.1386 ThI
4465.3406 ThII
4465.9751 ThI
4466.6245 ThI
4468.3208 ThI
4469.1891 ThI
4469.5253 ThI
4470.9906 ThI
4472.8462 ThI
4474.0730 ThII
4474.7594 ArII
4475.2214 ThI
4478.4032 ThI
4478.5938 ThI
4479.6376 ThI
4480.2661 ThI
4480.5453 ThI
4481.8107 ArII
4482.1693 ThI
4483.3468 ThI
4460
4465
4470
4475
4480

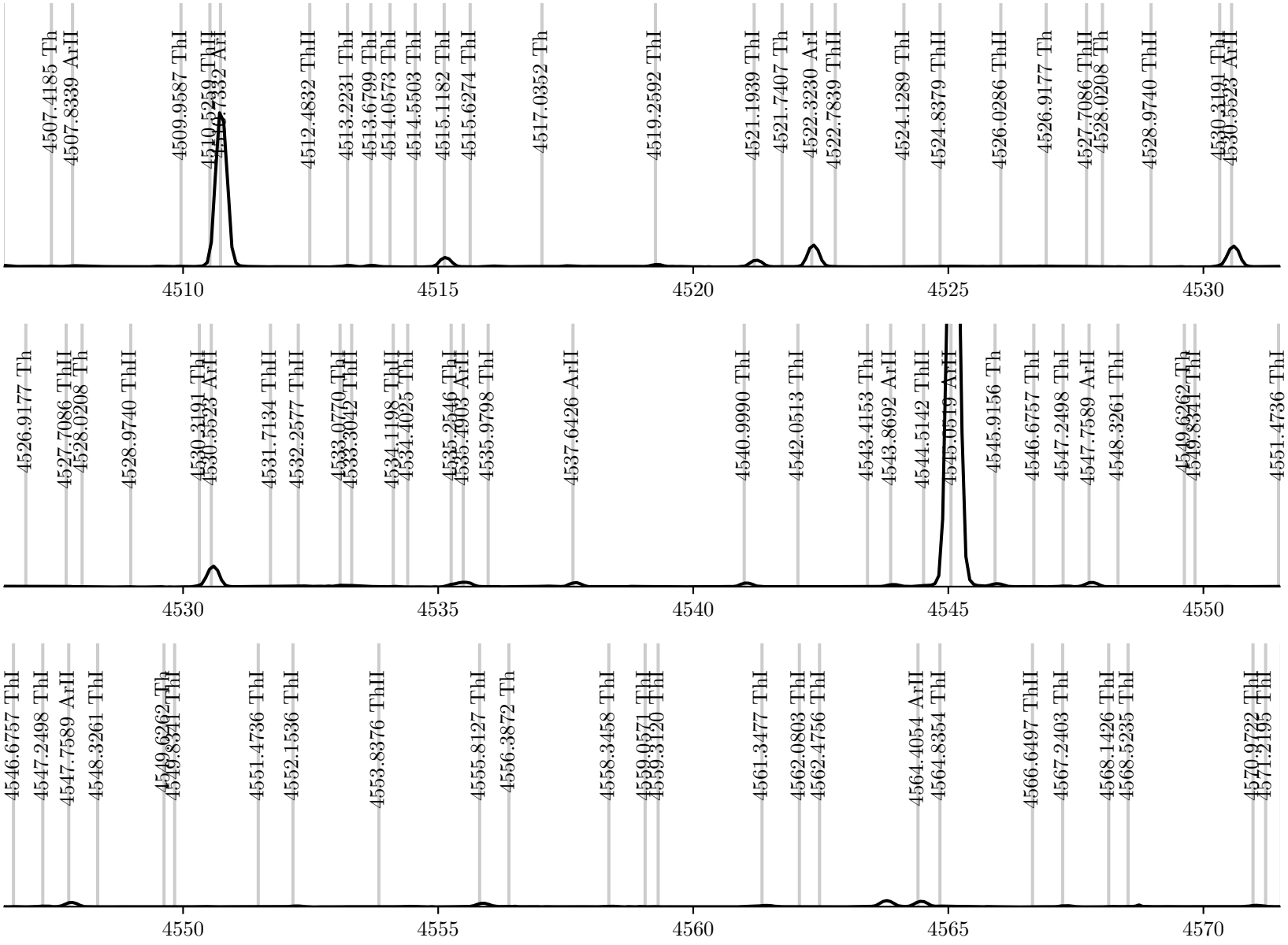

4479.6376 ThI
4480.2661 ThI
4480.5453 ThI
4481.8107 ArII
4482.1693 ThI
4483.3468 ThI
4486.8973 ThI
4487.4955 ThII
4488.3123 ThI
4488.6808 ThII
4489.6645 ThI
4490.6715 Th
4490.9816 ArII
4492.1007 ThI
4493.3337 ThI
4494.6907 ThI
4495.2378 ThII
4496.3155 ThII
4496.6381 Th
4497.9144 ThI
4498.5384 ArII
4498.9401 ThI
4499.3796 ThI
4499.7037 ThI
4499.9832 ThI
4500.5769 Th
4502.9268 ArII
4480
4485
4490
4495
4500
4499.3796 ThI
4499.7037 ThI
4499.9832 ThI
4500.5769 Th
4502.9268 ArII
4504.6306 ThI
4505.2167 ThI
4506.4730 ThI
4507.4185 Th
4507.8339 ArII
4509.9587 ThI
4510.5259 ThII
4510.7332 ArII
4512.4832 ThII
4513.2231 ThI
4513.6799 ThI
4514.0573 ThI
4514.5503 ThI
4500
4505
4510
4515
4520

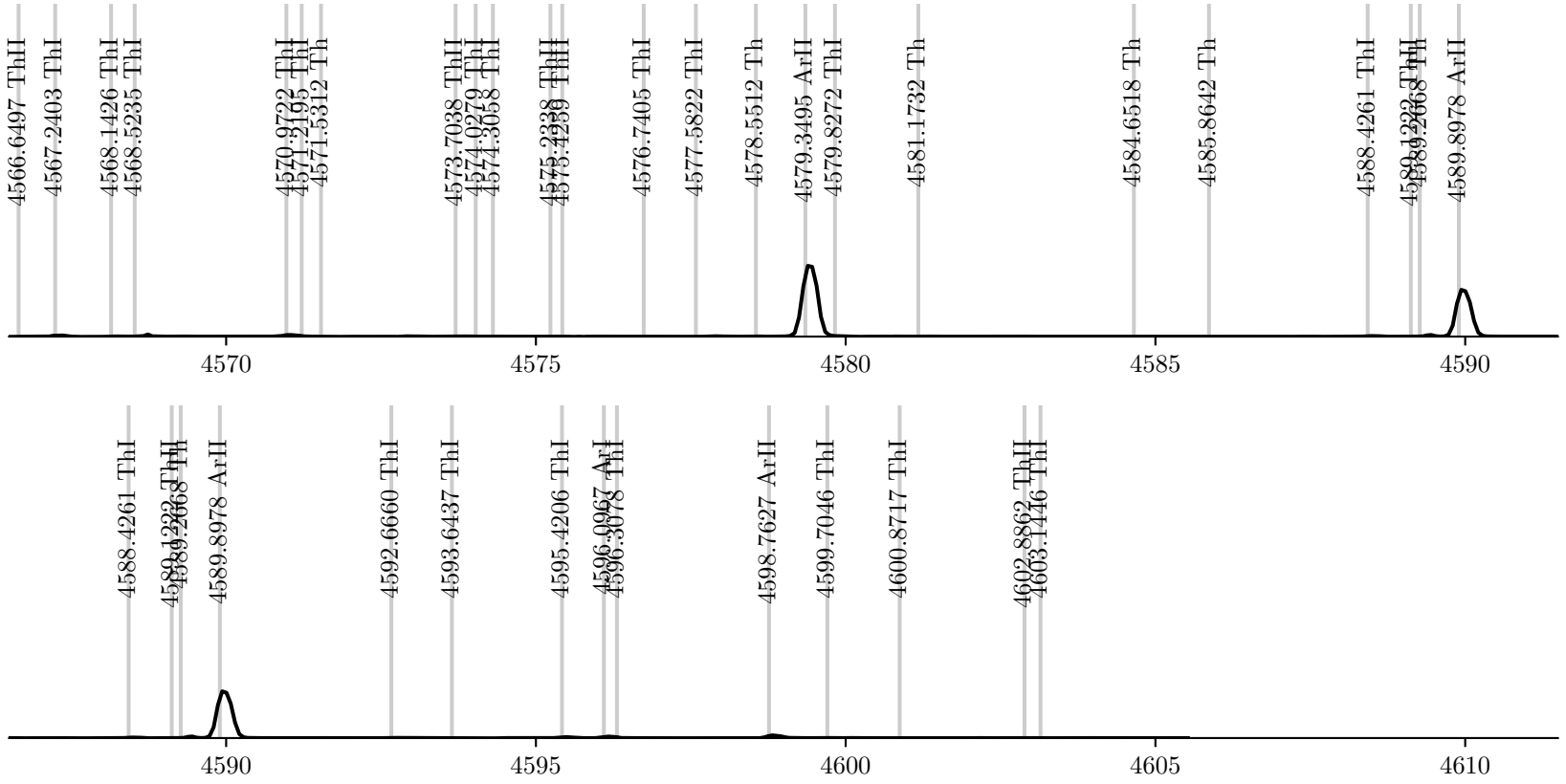
Order 50



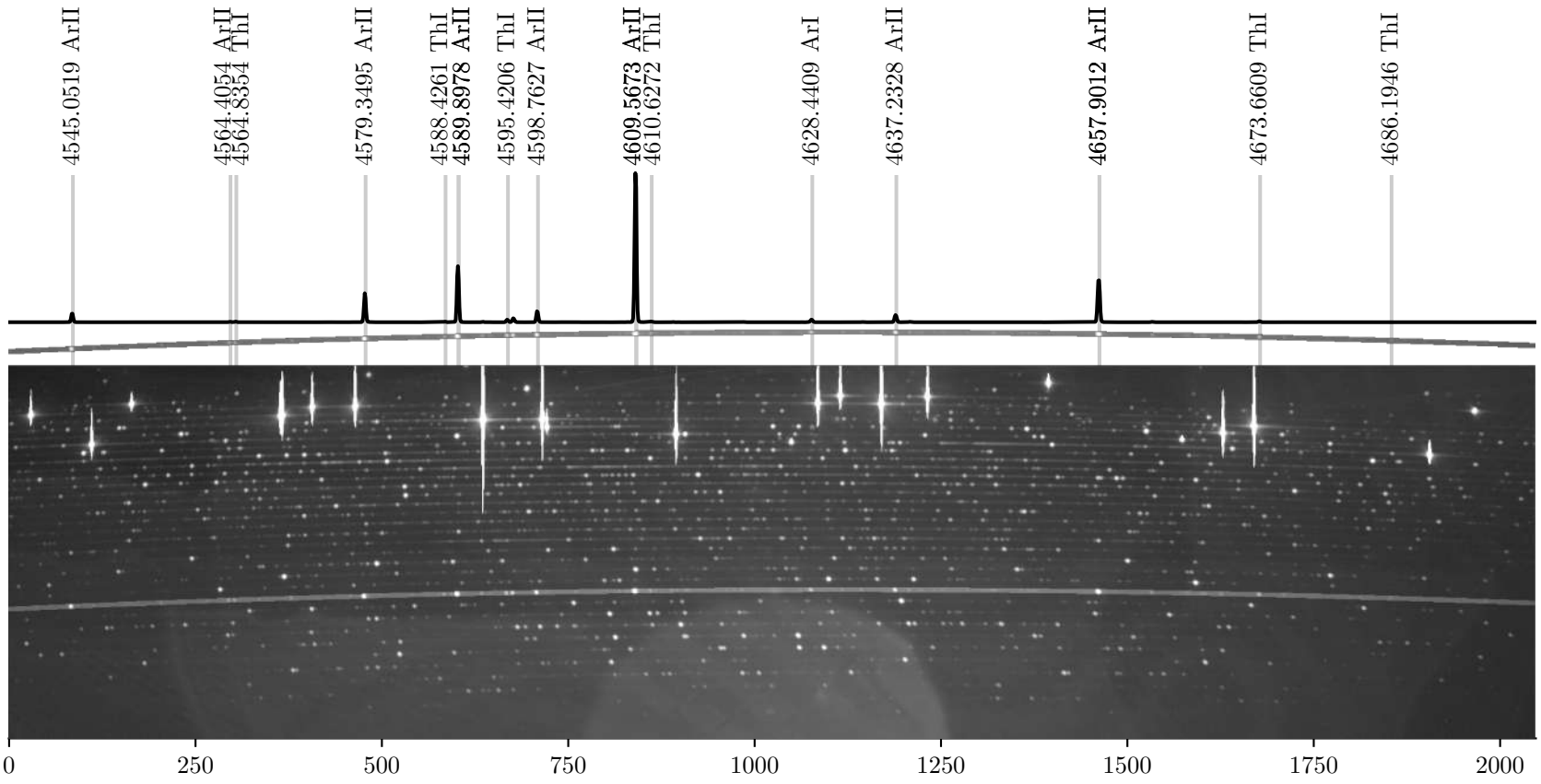
Approximate resolution:	16800
Dispersion:	0.078 Å pix^{-1}
Central wavelength:	4532 Å
Wavelength range:	4446 Å – 4606 Å


4447.2296 ThI
4447.8346 ThII
4448.8792 ArII
4449.5206 ArII
4450.8040 ThI
4452.5655 ThI
4454.1392 ThI
4454.7738 Th
4456.7080 ThII
4457.2362 ThI
4458.0015 ThI
4458.7384 ThI
4459.5477 Th
4460.5574 ArII
4465.3406 ThII
4465.9751 ThI
4466.6245 ThI
4468.3208 ThI
4469.1891 ThI
4469.5253 ThI
4470.9906 ThI
4450
4455
4460
4465
4470
4466.6245 ThI
4468.3208 ThI
4469.1891 ThI
4469.5253 ThI
4470.9906 ThI
4472.8462 ThI
4474.0730 ThII
4474.7594 ArII
4475.2214 ThI
4479.6376 ThI
4481.8107 ArII
4482.1693 ThI
4483.3468 ThI
4486.8973 ThI
4487.4955 ThII
4488.3123 ThI
4488.6808 ThII
4489.6645 ThI
4490.6715 Th
4490.9816 ArII
4470
4475
4480
4485
4490
4486.8973 ThI
4487.4955 ThII
4488.3123 ThI
4488.6808 ThII
4489.6645 ThI
4490.6715 Th
4490.9816 ArII
4492.1007 ThI
4493.3337 ThI
4496.3155 ThII
4496.6381 Th
4497.9144 ThI
4498.5384 ArII
4498.9401 ThI
4499.3796 ThI
4500.5769 Th
4502.9268 ArII
4504.6306 ThI
4505.2167 ThI
4506.4730 ThI
4507.4185 Th
4507.8339 ArII
4509.9587 ThI
4510.5259 ThII
4511.7332 ArII
4490
4495
4500
4505
4510

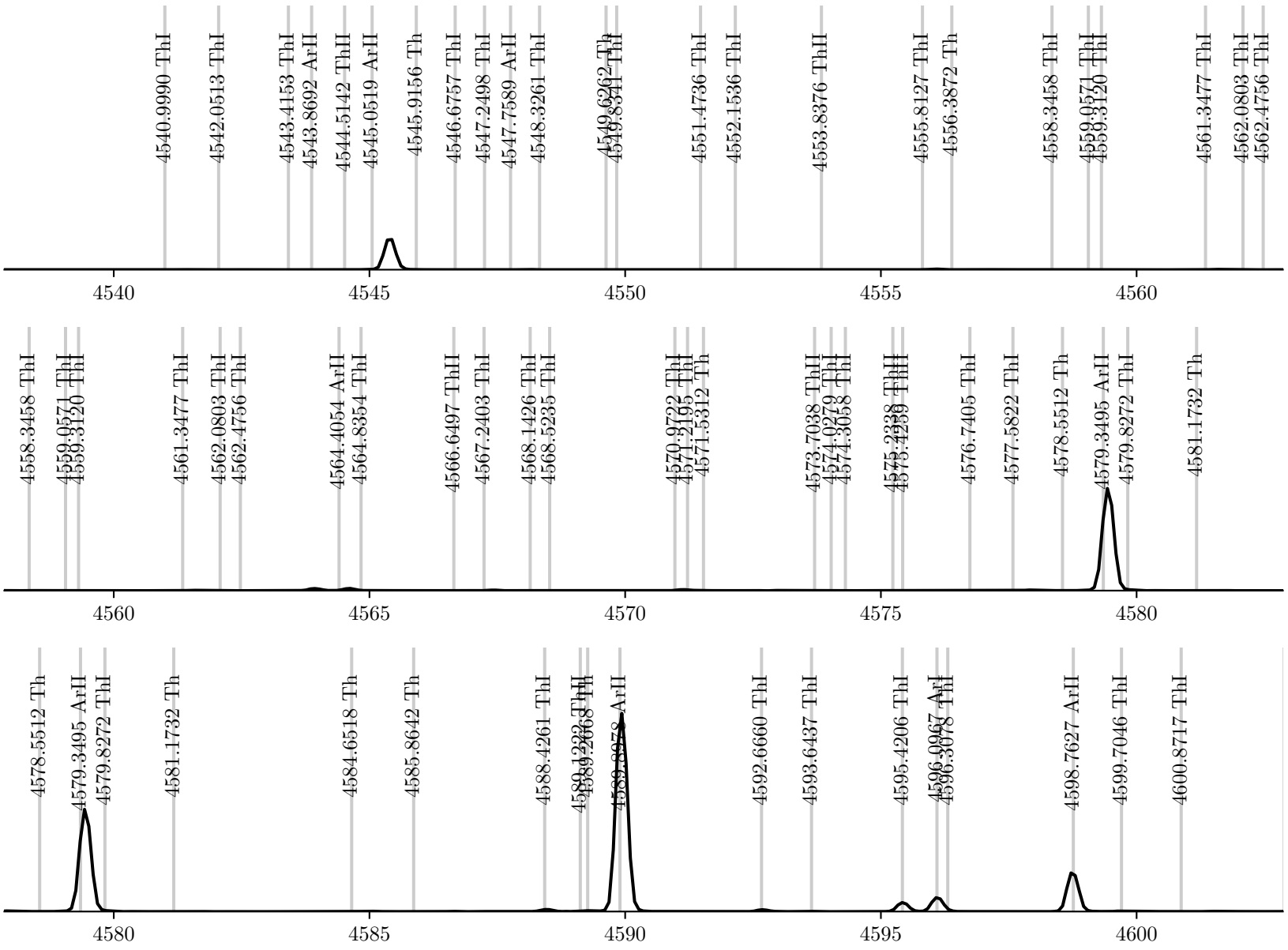

4507.4185 Th
4507.8339 ArII
4509.9587 ThI
4510.5259 ThI
4510.7332 ArI
4512.4832 ThII
4513.2231 ThI
4513.6799 ThI
4514.0573 ThI
4514.5503 ThI
4515.1182 ThI
4515.6274 ThI
4517.0352 Th
4519.2592 ThI
4521.1939 ThI
4521.7407 Th
4522.3230 ArI
4522.7839 ThII
4524.1289 ThI
4524.8379 ThII
4526.0286 ThII
4526.9177 Th
4527.7086 ThII
4528.0208 Th
4528.9740 ThII
4530.3191 ThI
4530.5523 ArII
4510
4515
4520
4525
4530
4526.9177 Th
4527.7086 ThII
4528.0208 Th
4528.9740 ThII
4530.3191 ThI
4530.5523 ArII
4531.7134 ThII
4532.2577 ThII
4533.0770 ThI
4533.3042 ThII
4534.1198 ThII
4534.4025 ThI
4535.2546 ThI
4535.4903 ArII
4535.9798 ThI
4537.6426 ArII
4540.9990 ThI
4542.0513 ThI
4543.4153 ThI
4543.8692 ArII
4544.5142 ThII
4545.0519 ArII
4545.9156 Th
4546.6757 ThI
4547.2498 ThI
4547.7589 ArII
4548.3261 ThI
4549.6262 Th
4549.8341 ThI
4551.4736 ThI
4530
4535
4540
4545
4550
4546.6757 ThI
4547.2498 ThI
4547.7589 ArII
4548.3261 ThI
4549.6262 Th
4549.8341 ThI
4551.4736 ThI
4552.1536 ThI
4553.8376 ThII
4555.8127 ThI
4556.3872 Th
4558.3458 ThI
4559.0571 ThI
4559.3120 ThI
4561.3477 ThI
4562.0803 ThI
4562.4756 ThI
4564.4054 ArII
4564.8354 ThI
4566.6497 ThII
4567.2403 ThI
4568.1426 ThI
4568.5235 ThI
4570.3722 ThI
4571.2195 ThI
4550
4555
4560
4565
4570

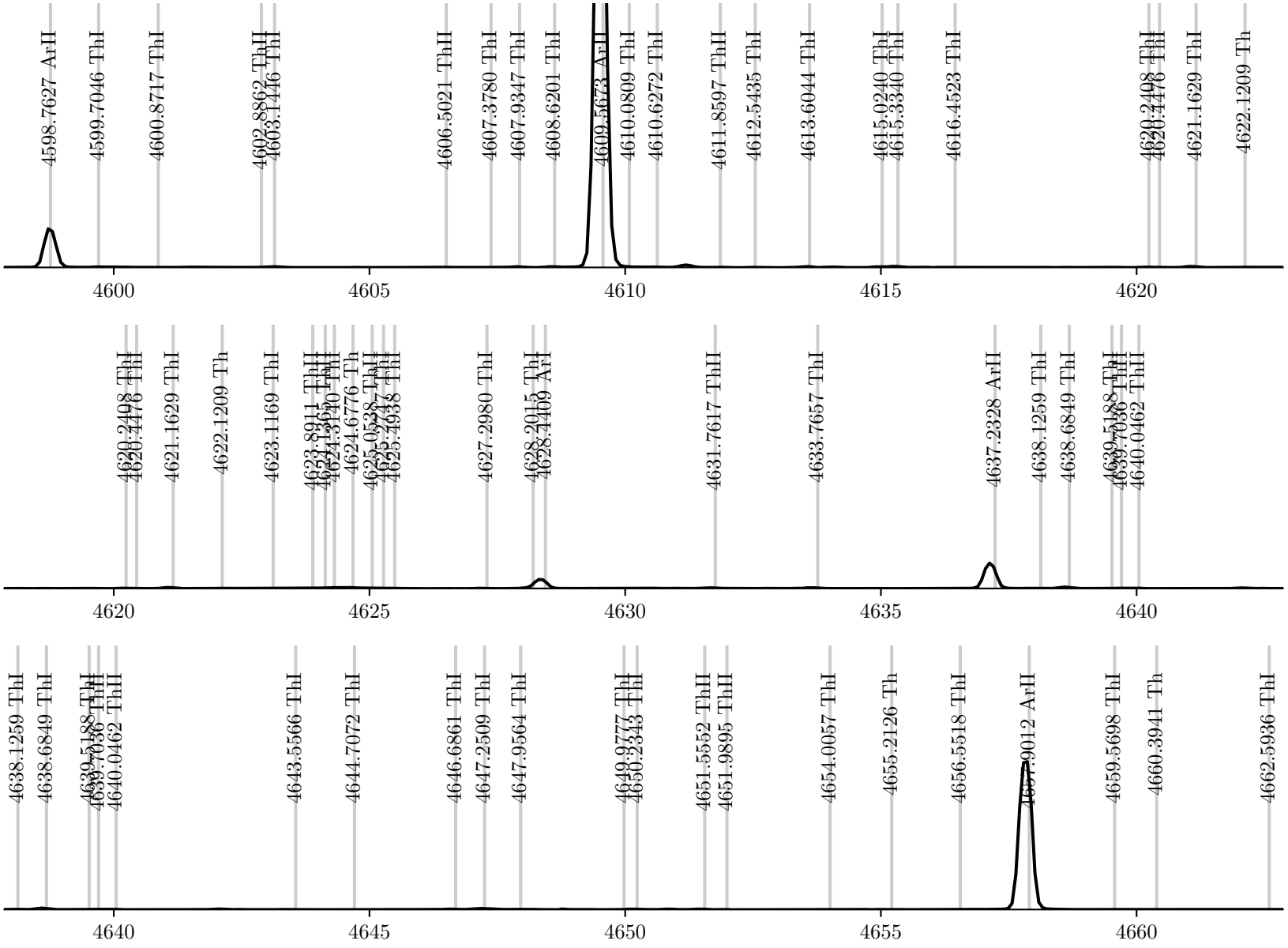

4566.6497 ThII
4567.2403 ThI
4568.1426 ThI
4568.5235 ThI
4570.9722 ThI
4571.2195 ThI
4571.5312 Th
4573.7038 ThII
4574.0279 ThI
4574.3058 ThI
4576.7405 ThI
4577.5822 ThI
4578.5512 Th
4579.3495 ArII
4579.8272 ThI
4581.1732 Th
4584.6518 Th
4585.8642 Th
4588.4261 ThI
4589.8978 ArII
4570
4575
4580
4585
4590
4588.4261 ThI
4589.8978 ArII
4592.6660 ThI
4593.6437 ThI
4595.4206 ThI
4598.7627 ArII
4599.7046 ThI
4600.8717 ThI
4602.8862 ThII
4603.1446 ThI
4590
4595
4600
4605
4610

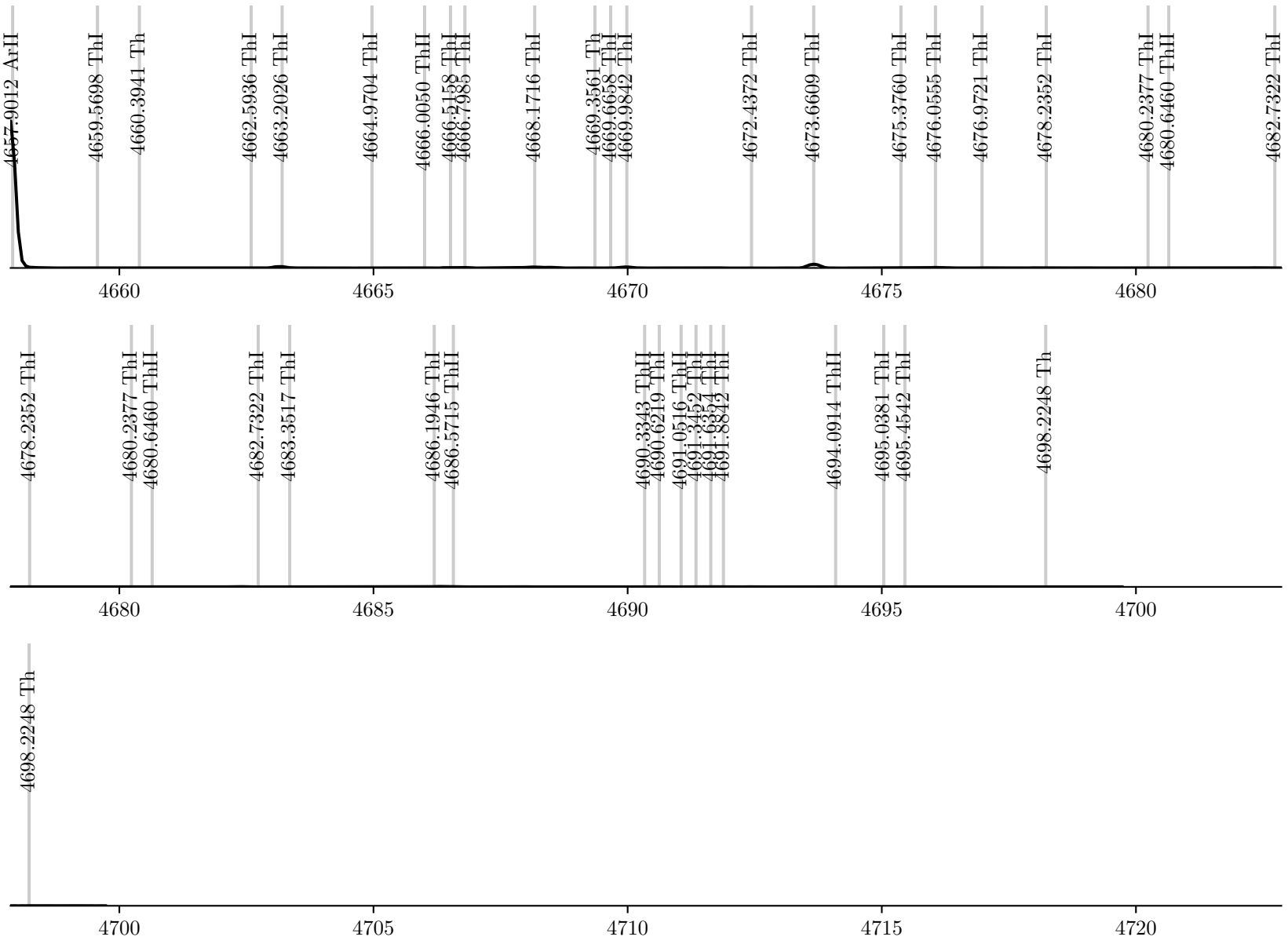
Order 49



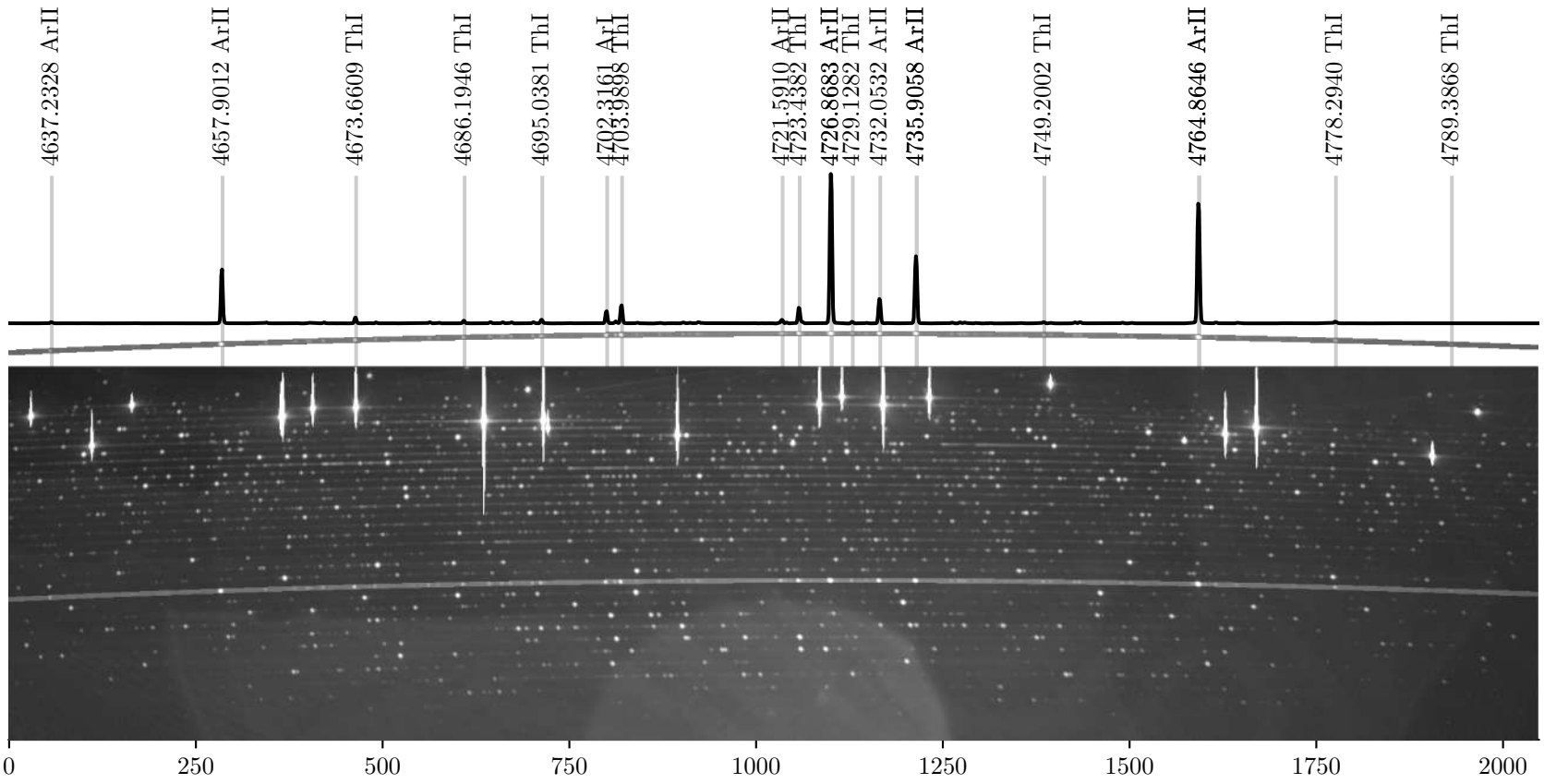
Approximate resolution:	17400
Dispersion:	0.079 Å pix^{-1}
Central wavelength:	4624 Å
Wavelength range:	4538 Å – 4700 Å


4540.9990 ThI
4542.0513 ThI
4543.4153 ThI
4543.8692 ArII
4544.5142 ThII
4545.0519 ArII
4545.9156 Th
4546.6757 ThI
4547.2498 ThI
4547.7589 ArII
4548.3261 ThI
4549.6262 Th
4549.8341 ThI
4551.4736 ThI
4552.1536 ThI
4553.8376 ThII
4555.8127 ThI
4556.3872 Th
4558.3458 ThI
4559.0571 ThI
4559.3120 ThI
4561.3477 ThI
4562.0803 ThI
4562.4756 ThI
4540
4545
4550
4555
4560
4558.3458 ThI
4559.0571 ThI
4559.3120 ThI
4561.3477 ThI
4562.0803 ThI
4562.4756 ThI
4564.4054 ArII
4564.8354 ThI
4566.6497 ThII
4567.2403 ThI
4568.1426 ThI
4568.5235 ThI
4570.9722 ThI
4571.2195 ThI
4571.5312 Th
4573.7038 ThII
4574.0279 ThI
4574.3058 ThI
4576.7405 ThI
4577.5822 ThI
4578.5512 Th
4579.3495 ArII
4579.8272 ThI
4581.1732 Th
4560
4565
4570
4575
4580
4578.5512 Th
4579.3495 ArII
4579.8272 ThI
4581.1732 Th
4584.6518 Th
4585.8642 Th
4588.4261 ThI
4589.8978 ArII
4592.6660 ThI
4593.6437 ThI
4595.4206 ThI
4598.7627 ArII
4599.7046 ThI
4600.8717 ThI
4580
4585
4590
4595
4600

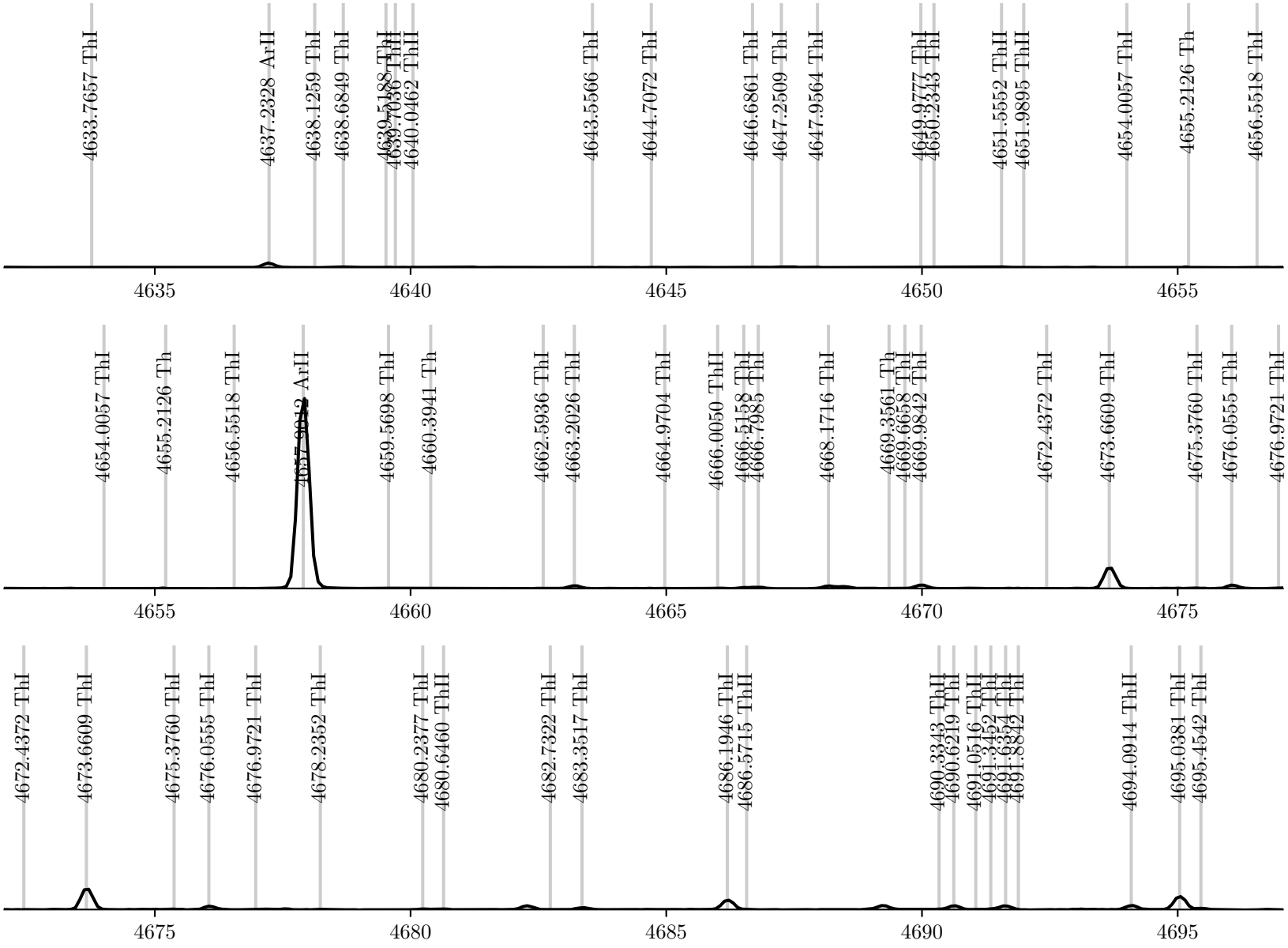

4598.7627 ArII
4599.7046 ThI
4600.8717 ThI
4602.8862 ThII
4603.1446 ThI
4606.5021 ThII
4607.3780 ThI
4607.9347 ThI
4608.6201 ThI
4609.5673 ArII
4610.0809 ThI
4610.6272 ThI
4611.8597 ThII
4612.5435 ThI
4613.6044 ThI
4615.0240 ThI
4615.3340 ThI
4616.4523 ThI
4620.2408 ThI
4620.4478 ThI
4621.1629 ThI
4622.1209 Th
4600
4605
4610
4615
4620
4620.2408 ThI
4620.4478 ThI
4621.1629 ThI
4622.1209 Th
4623.1169 ThI
4627.2980 ThI
4628.2015 ThI
4628.4409 ArI
4631.7617 ThII
4633.7657 ThI
4637.2328 ArII
4638.1259 ThI
4638.6849 ThI
4640.0462 ThII
4620
4625
4630
4635
4640
4638.1259 ThI
4638.6849 ThI
4640.0462 ThII
4643.5566 ThI
4644.7072 ThI
4646.6861 ThI
4647.2509 ThI
4647.9564 ThI
4651.5552 ThII
4651.9895 ThII
4654.0057 ThI
4655.2126 Th
4656.5518 ThI
4659.5698 ThI
4660.3941 Th
4662.5936 ThI
4640
4645
4650
4655
4660

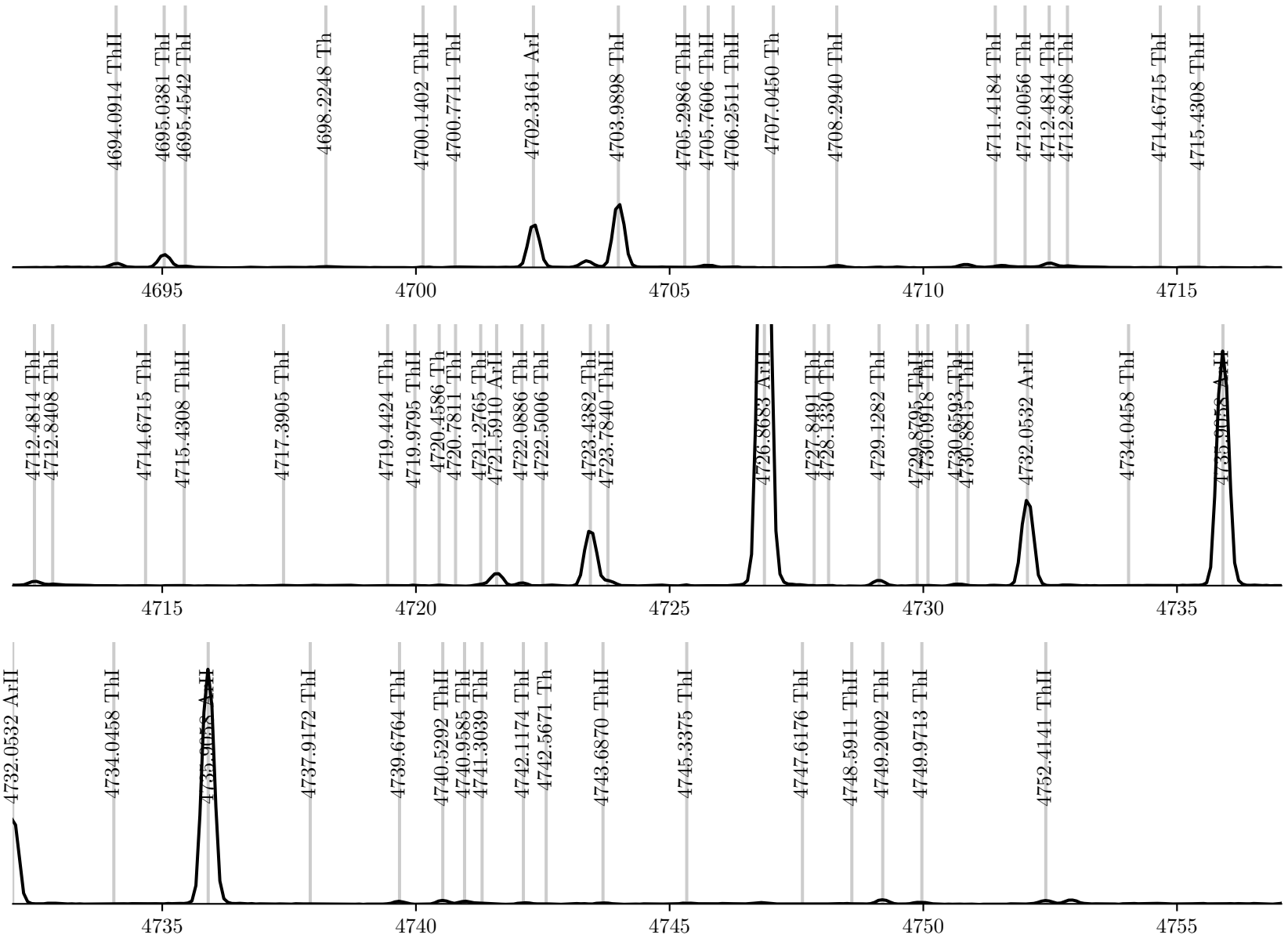

4657.9012 ArII
4659.5698 ThI
4660.3941 Th
4662.5936 ThI
4663.2026 ThI
4664.9704 ThI
4666.0050 ThII
4668.1716 ThI
4669.3561 Th
4669.6658 ThI
4669.9842 ThI
4672.4372 ThI
4673.6609 ThI
4675.3760 ThI
4676.0555 ThI
4676.9721 ThI
4678.2352 ThI
4680.2377 ThI
4680.6460 ThII
4682.7322 ThI
4660
4665
4670
4675
4680
4678.2352 ThI
4680.2377 ThI
4680.6460 ThII
4682.7322 ThI
4683.3517 ThI
4686.1946 ThI
4686.5715 ThII
4690.3343 ThII
4690.6219 ThI
4691.0516 ThII
4691.3452 ThI
4691.6354 ThI
4691.8842 ThI
4694.0914 ThII
4695.0381 ThI
4695.4542 ThI
4698.2248 Th
4680
4685
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4695
4700
4698.2248 Th
4700
4705
4710
4715
4720

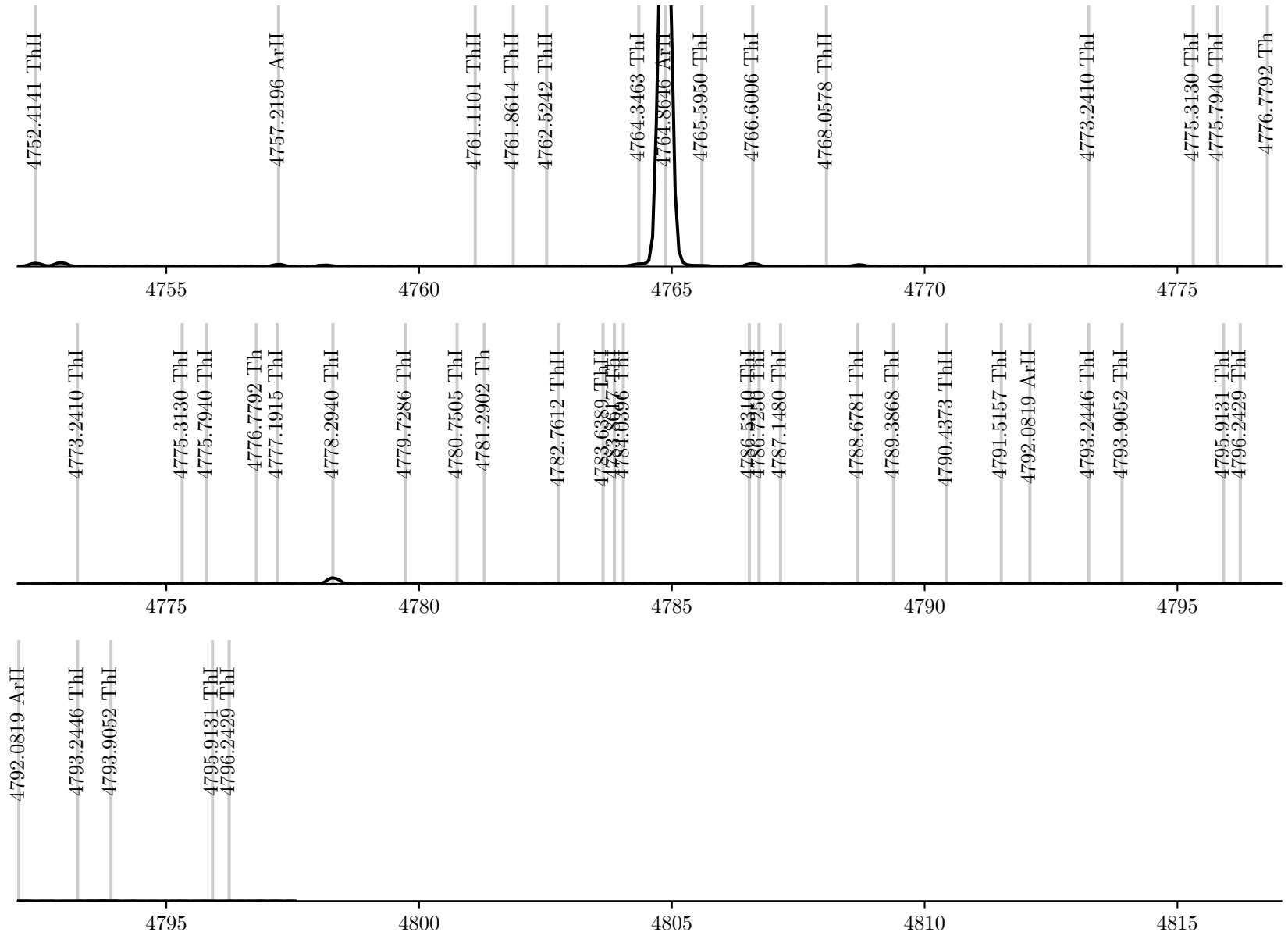
Order 48



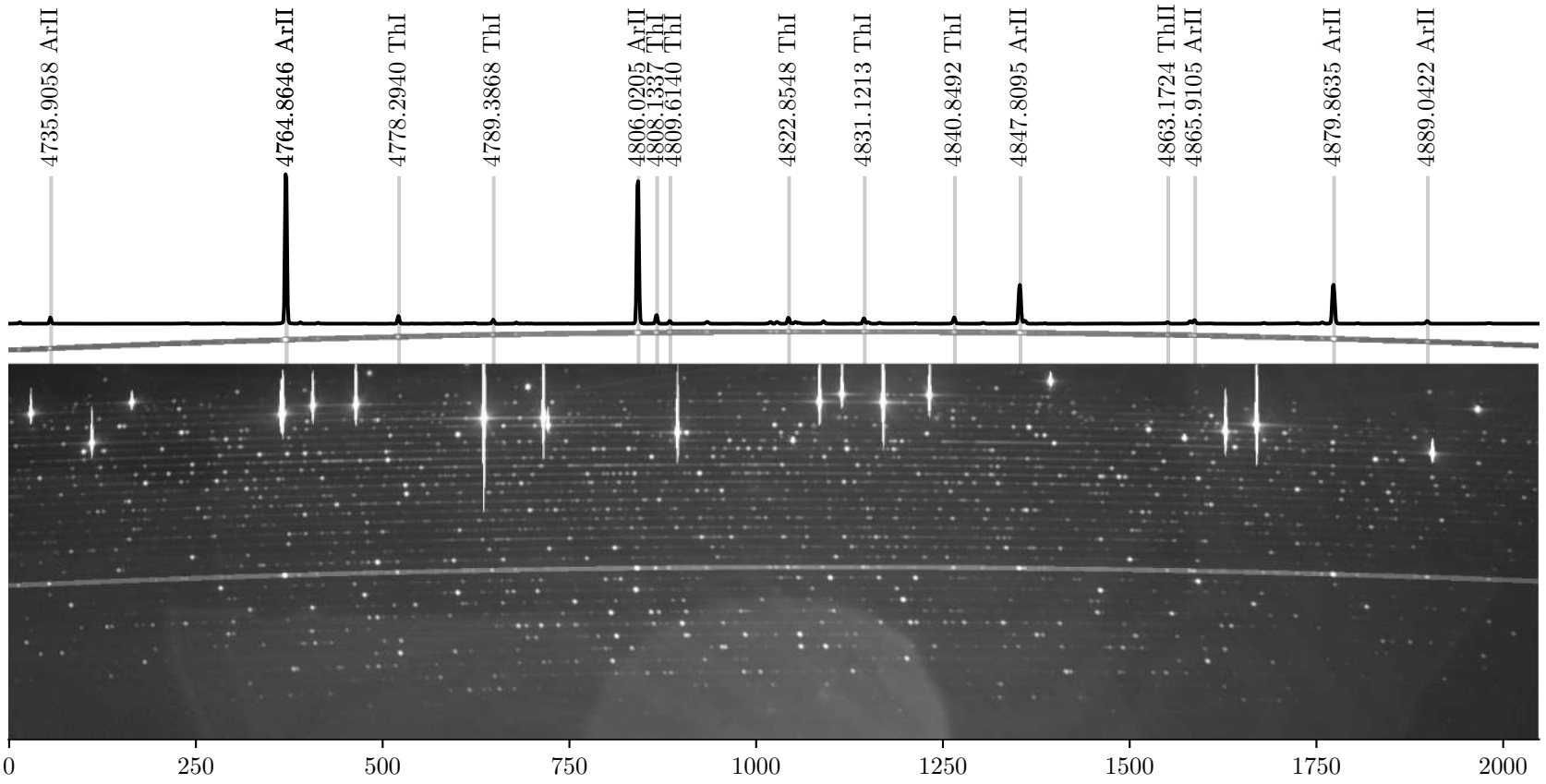
Approximate resolution:	17100
Dispersion:	0.081 Å pix^{-1}
Central wavelength:	4721 Å
Wavelength range:	4632 Å – 4798 Å


4633.7657 ThI
4637.2328 ArII
4638.1259 ThI
4638.6849 ThI
4639.5188 ThI
4639.7036 ThI
4640.0462 ThII
4643.5566 ThI
4644.7072 ThI
4646.6861 ThI
4647.2509 ThI
4647.9564 ThI
4649.9777 ThI
4650.2343 ThI
4651.5552 ThII
4651.9895 ThII
4654.0057 ThI
4655.2126 Th
4656.5518 ThI
4635
4640
4645
4650
4655
4654.0057 ThI
4655.2126 Th
4656.5518 ThI
4657.9012 ArII
4659.5698 ThI
4660.3941 Th
4662.5936 ThI
4663.2026 ThI
4664.9704 ThI
4666.0050 ThII
4666.5158 ThI
4666.7985 ThI
4668.1716 ThI
4669.3561 Th
4669.6658 ThI
4669.9842 ThI
4672.4372 ThI
4673.6609 ThI
4675.3760 ThI
4676.0555 ThI
4676.9721 ThI
4655
4660
4665
4670
4675
4672.4372 ThI
4673.6609 ThI
4675.3760 ThI
4676.0555 ThI
4676.9721 ThI
4678.2352 ThI
4680.2377 ThI
4680.6460 ThII
4682.7322 ThI
4683.3517 ThI
4686.1946 ThI
4686.5715 ThII
4690.3343 ThI
4690.6219 ThI
4691.0516 ThII
4691.3452 Th
4691.6354 Th
4691.8842 ThI
4694.0914 ThII
4695.0381 ThI
4695.4542 ThI
4675
4680
4685
4690
4695

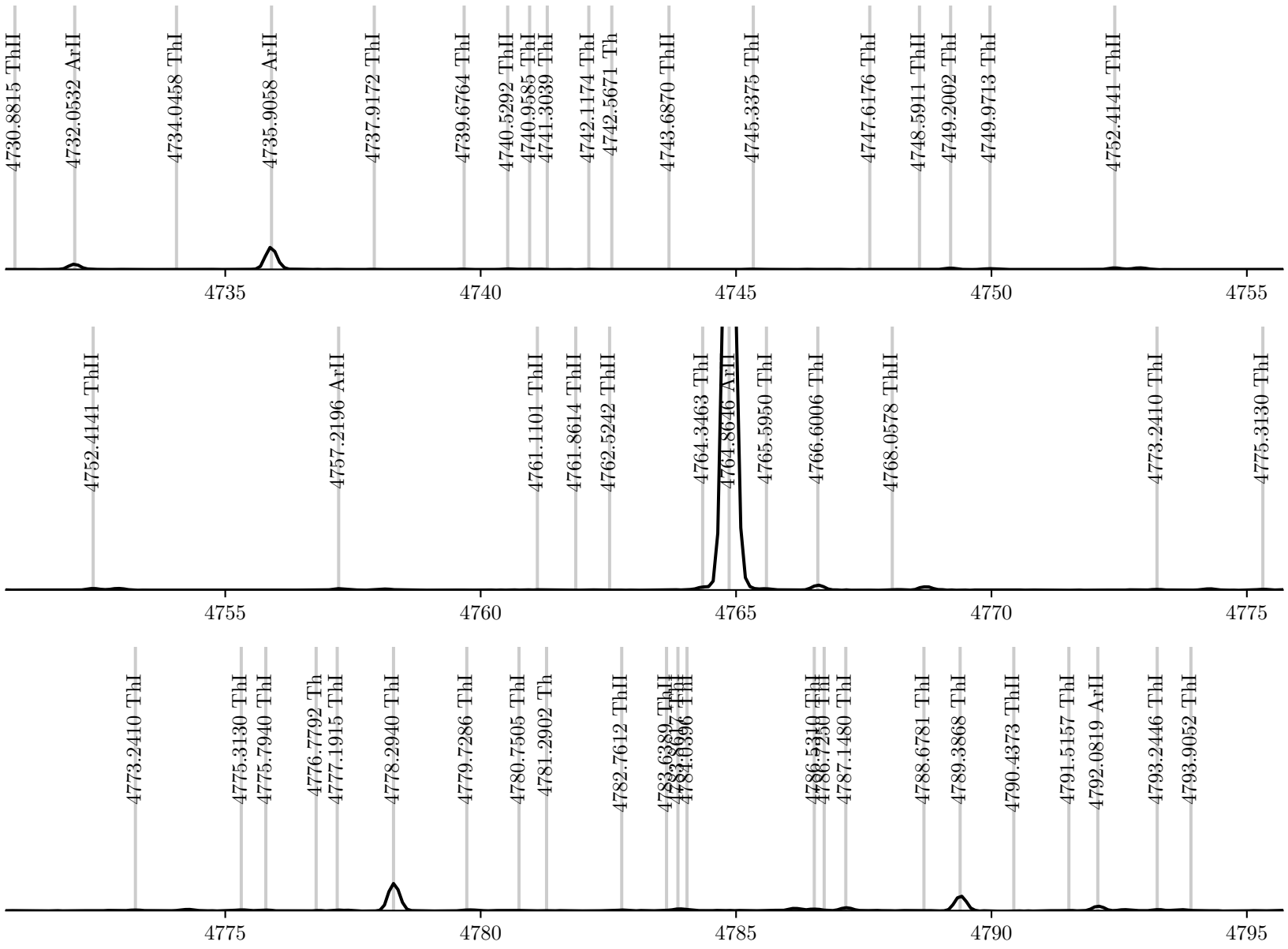

4694.0914 ThII
4695.0381 ThI
4695.4542 ThI
4698.2248 Th
4700.1402 ThII
4700.7711 ThI
4702.3161 ArI
4703.9898 ThI
4705.2986 ThII
4705.7606 ThII
4706.2511 ThII
4707.0450 Th
4708.2940 ThI
4711.4184 ThI
4712.0056 ThI
4712.4814 ThI
4712.8408 ThI
4714.6715 ThI
4715.4308 ThII
4695
4700
4705
4710
4715
4712.4814 ThI
4712.8408 ThI
4714.6715 ThI
4715.4308 ThII
4717.3905 ThI
4719.4424 ThI
4719.9795 ThII
4720.4586 Th
4720.7811 ThI
4721.2765 ThI
4721.5910 ArII
4722.0886 ThI
4722.5006 ThI
4723.4382 ThI
4723.7840 ThII
4726.8683 ArII
4727.8491 ThI
4728.1330 ThI
4729.1282 ThI
4729.8795 ThII
4730.0918 ThI
4730.6593 ThI
4730.8815 ThII
4732.0532 ArII
4734.0458 ThI
4735.9053 ArII
4715
4720
4725
4730
4735
4732.0532 ArII
4734.0458 ThI
4735.9053 ArII
4737.9172 ThI
4739.6764 ThI
4740.5292 ThII
4740.9585 ThI
4741.3039 ThI
4742.1174 ThI
4742.5671 Th
4743.6870 ThII
4745.3375 ThI
4747.6176 ThI
4748.5911 ThII
4749.2002 ThI
4749.9713 ThI
4752.4141 ThII
4735
4740
4745
4750
4755

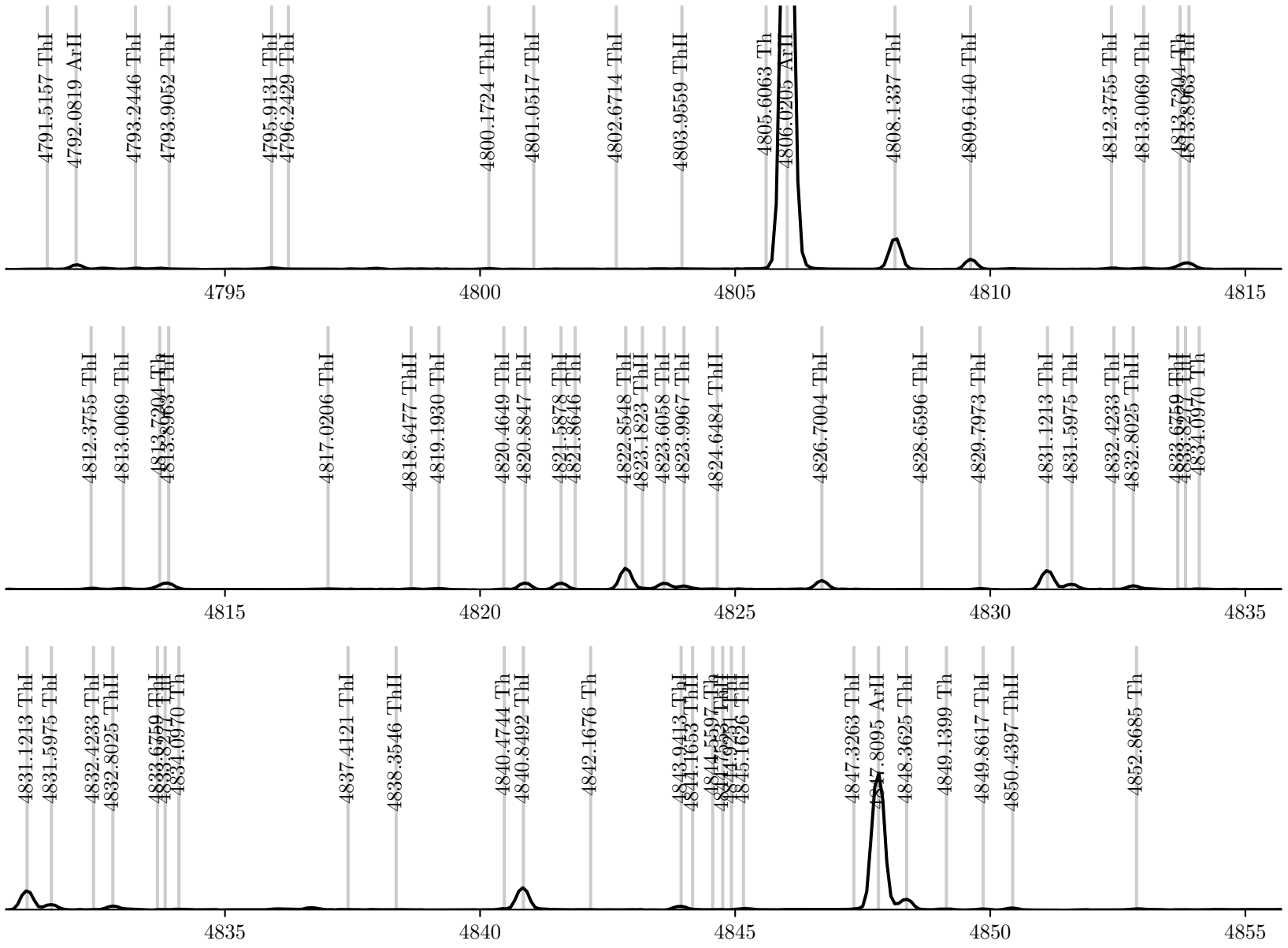

4752.4141 ThII
4757.2196 ArII
4761.1101 ThII
4761.8614 ThII
4762.5242 ThII
4764.3463 ThI
4764.8646 ArII
4765.5950 ThI
4766.6006 ThI
4768.0578 ThII
4773.2410 ThI
4775.3130 ThI
4775.7940 ThI
4776.7792 Th
4755
4760
4765
4770
4775
4773.2410 ThI
4775.3130 ThI
4775.7940 ThI
4776.7792 Th
4777.1915 ThI
4778.2940 ThI
4779.7286 ThI
4780.7505 ThI
4781.2902 Th
4782.7612 ThII
4783.6389 ThII
4784.0396 ThI
4786.5310 ThI
4786.7250 ThI
4787.1480 ThI
4788.6781 ThI
4789.3868 ThI
4790.4373 ThII
4791.5157 ThI
4792.0819 ArII
4793.2446 ThI
4793.9052 ThI
4795.9131 ThI
4796.2429 ThI
4775
4780
4785
4790
4795
4792.0819 ArII
4793.2446 ThI
4793.9052 ThI
4795.9131 ThI
4796.2429 ThI
4795
4800
4805
4810
4815

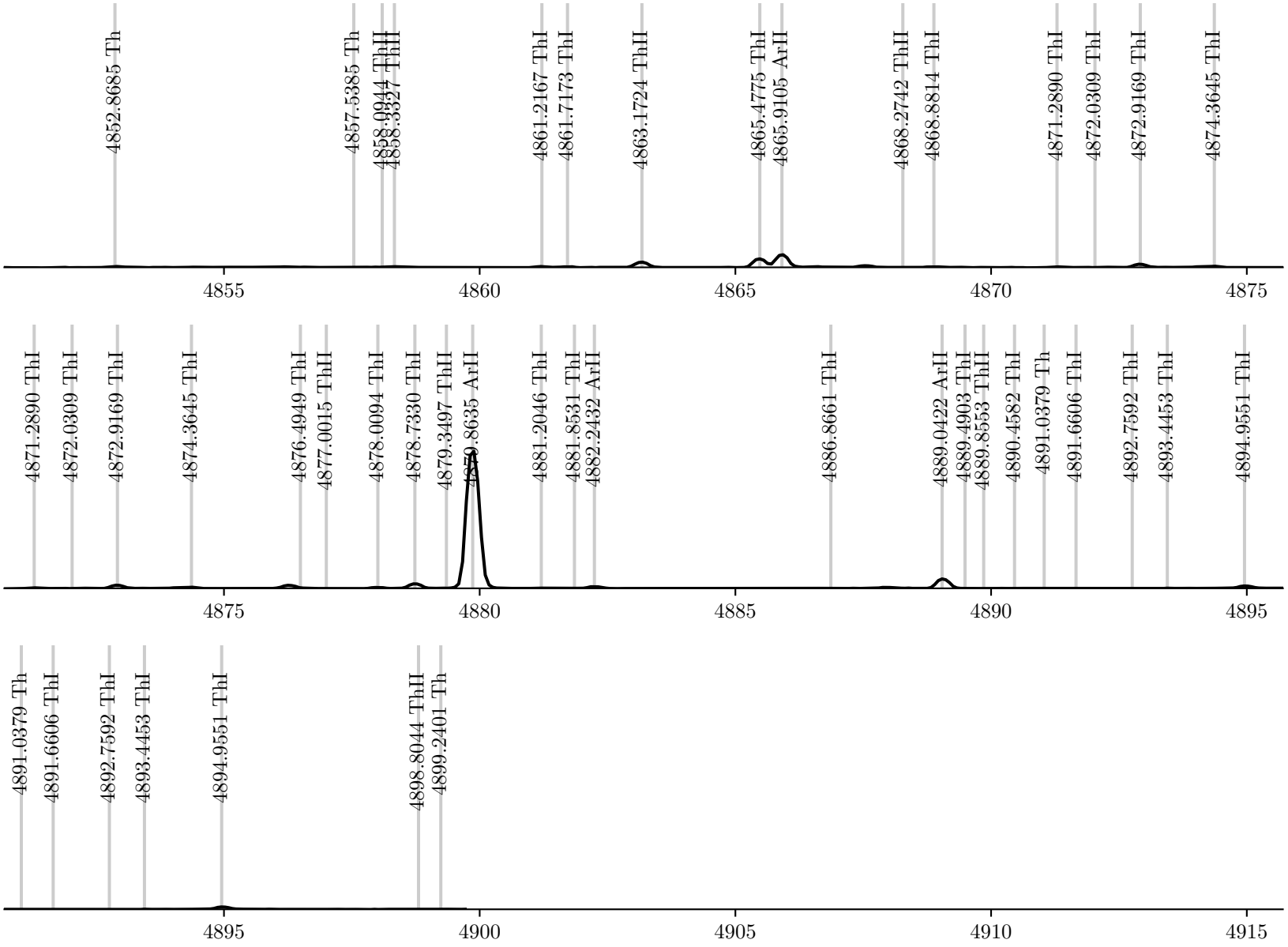
Order 47



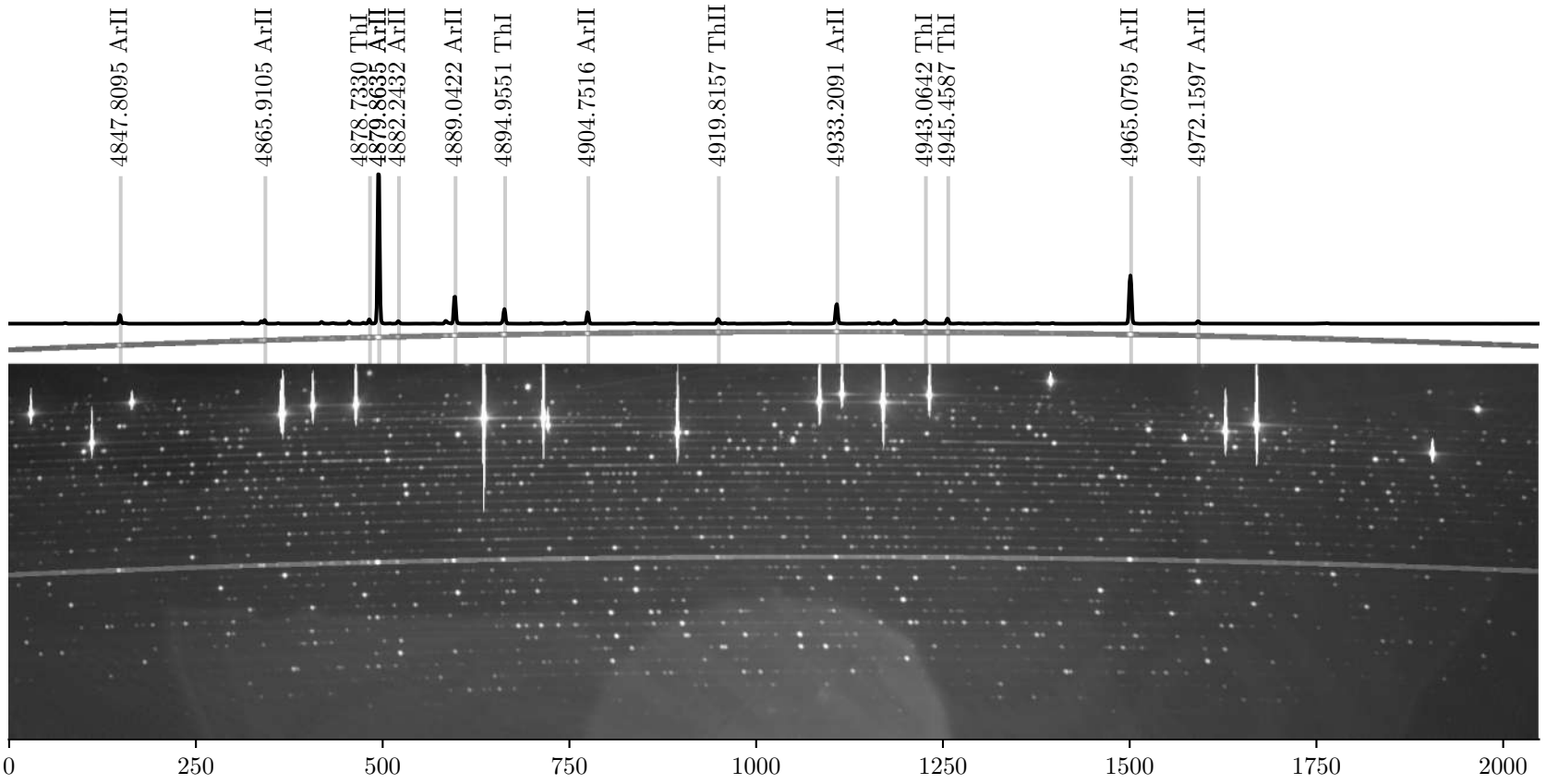
Approximate resolution:	17100
Dispersion:	0.083 Å pix^{-1}
Central wavelength:	4821 Å
Wavelength range:	4731 Å – 4900 Å


4730.8815 ThII
4732.0532 ArII
4734.0458 ThI
4735.9058 ArII
4737.9172 ThI
4739.6764 ThI
4740.5292 ThII
4740.9585 ThI
4741.3039 ThI
4742.1174 ThI
4742.5671 Th
4743.6870 ThII
4745.3375 ThI
4747.6176 ThI
4748.5911 ThII
4749.2002 ThI
4749.9713 ThI
4752.4141 ThII
4735
4740
4745
4750
4755
4752.4141 ThII
4757.2196 ArII
4761.1101 ThII
4761.8614 ThII
4762.5242 ThII
4764.3463 ThI
4764.8646 ArII
4765.5950 ThI
4766.6006 ThI
4768.0578 ThII
4773.2410 ThI
4775.3130 ThI
4755
4760
4765
4770
4775
4773.2410 ThI
4775.3130 ThI
4775.7940 ThI
4776.7792 Th
4777.1915 ThI
4778.2940 ThI
4779.7286 ThI
4780.7505 ThI
4781.2902 Th
4782.7612 ThII
4787.1480 ThI
4788.6781 ThI
4789.3868 ThI
4790.4373 ThII
4791.5157 ThI
4792.0819 ArII
4793.2446 ThI
4793.9052 ThI
4775
4780
4785
4790
4795

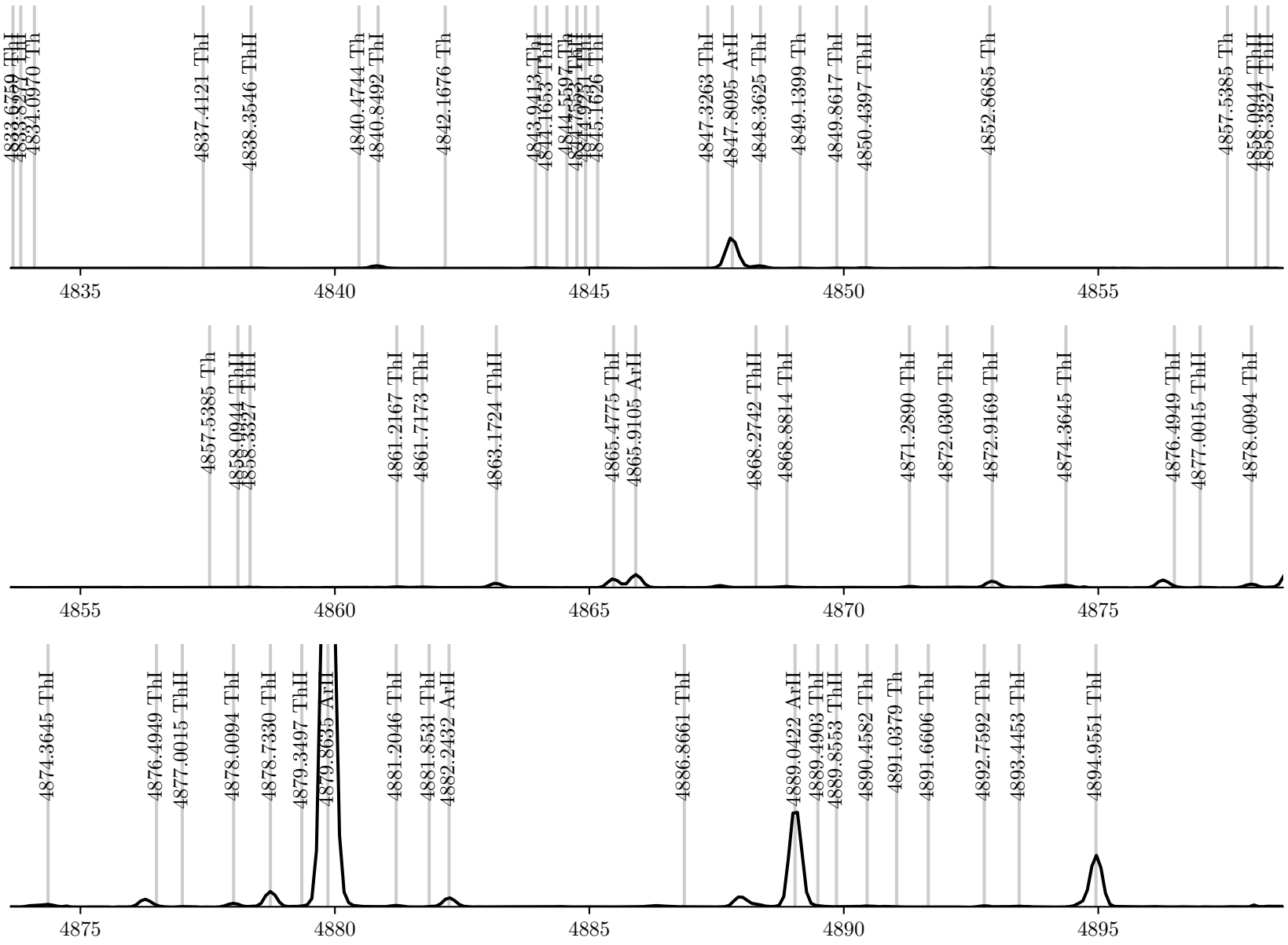

4791.5157 ThI
4792.0819 ArII
4793.2446 ThI
4793.9052 ThI
4795.9131 ThI
4796.2429 ThI
4800.1724 ThII
4801.0517 ThI
4802.6714 ThI
4803.9559 ThII
4805.6063 Th
4806.0205 ArII
4808.1337 ThI
4809.6140 ThI
4812.3755 ThI
4813.0069 ThI
4813.7204 Th
4813.8963 ThI
4795
4800
4805
4810
4815
4812.3755 ThI
4813.0069 ThI
4813.7204 Th
4813.8963 ThI
4817.0206 ThI
4818.6477 ThII
4819.1930 ThI
4820.4649 ThI
4820.8847 ThI
4821.5878 ThI
4821.8646 ThI
4822.8548 ThI
4823.1823 ThII
4823.6058 ThI
4823.9967 ThI
4824.6484 ThII
4826.7004 ThI
4828.6596 ThI
4829.7973 ThI
4831.1213 ThI
4831.5975 ThI
4832.4233 ThI
4832.8025 ThII
4833.6759 ThI
4834.0970 Th
4815
4820
4825
4830
4835
4831.1213 ThI
4831.5975 ThI
4832.4233 ThI
4832.8025 ThII
4833.6759 ThI
4834.0970 Th
4837.4121 ThI
4838.3546 ThII
4840.4744 Th
4840.8492 ThI
4842.1676 Th
4843.9413 ThI
4844.1633 ThII
4844.5597 Th
4845.1626 ThI
4847.3263 ThI
4847.8095 ArII
4848.3625 ThI
4849.1399 Th
4849.8617 ThI
4850.4397 ThII
4852.8685 Th
4835
4840
4845
4850
4855

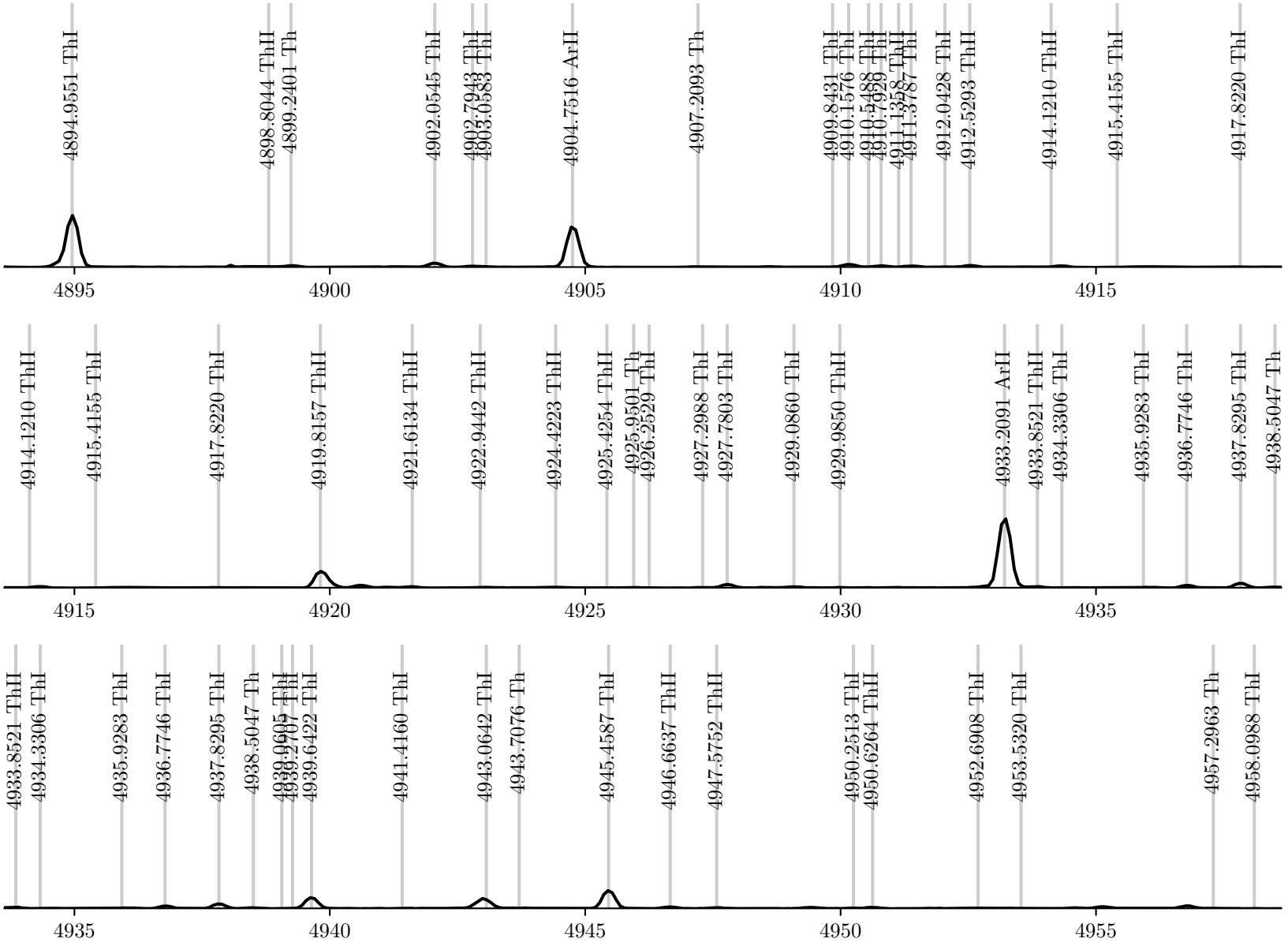

4852.8685 Th
4857.5385 Th
4858.0944 ThII
4858.3327 ThII
4861.2167 ThI
4861.7173 ThI
4863.1724 ThII
4865.4775 ThI
4865.9105 ArII
4868.2742 ThII
4868.8814 ThI
4871.2890 ThI
4872.0309 ThI
4872.9169 ThI
4874.3645 ThI
4855
4860
4865
4870
4875
4871.2890 ThI
4872.0309 ThI
4872.9169 ThI
4874.3645 ThI
4876.4949 ThI
4877.0015 ThII
4878.0094 ThI
4878.7330 ThI
4879.3497 ThII
4879.8635 ArII
4881.2046 ThI
4881.8531 ThI
4882.2432 ArII
4886.8661 ThI
4889.0422 ArII
4889.4903 ThI
4889.8553 ThII
4890.4582 ThI
4891.0379 Th
4891.6606 ThI
4892.7592 ThI
4893.4453 ThI
4894.9551 ThI
4875
4880
4885
4890
4895
4891.0379 Th
4891.6606 ThI
4892.7592 ThI
4893.4453 ThI
4894.9551 ThI
4898.8044 ThII
4899.2401 Th
4895
4900
4905
4910
4915

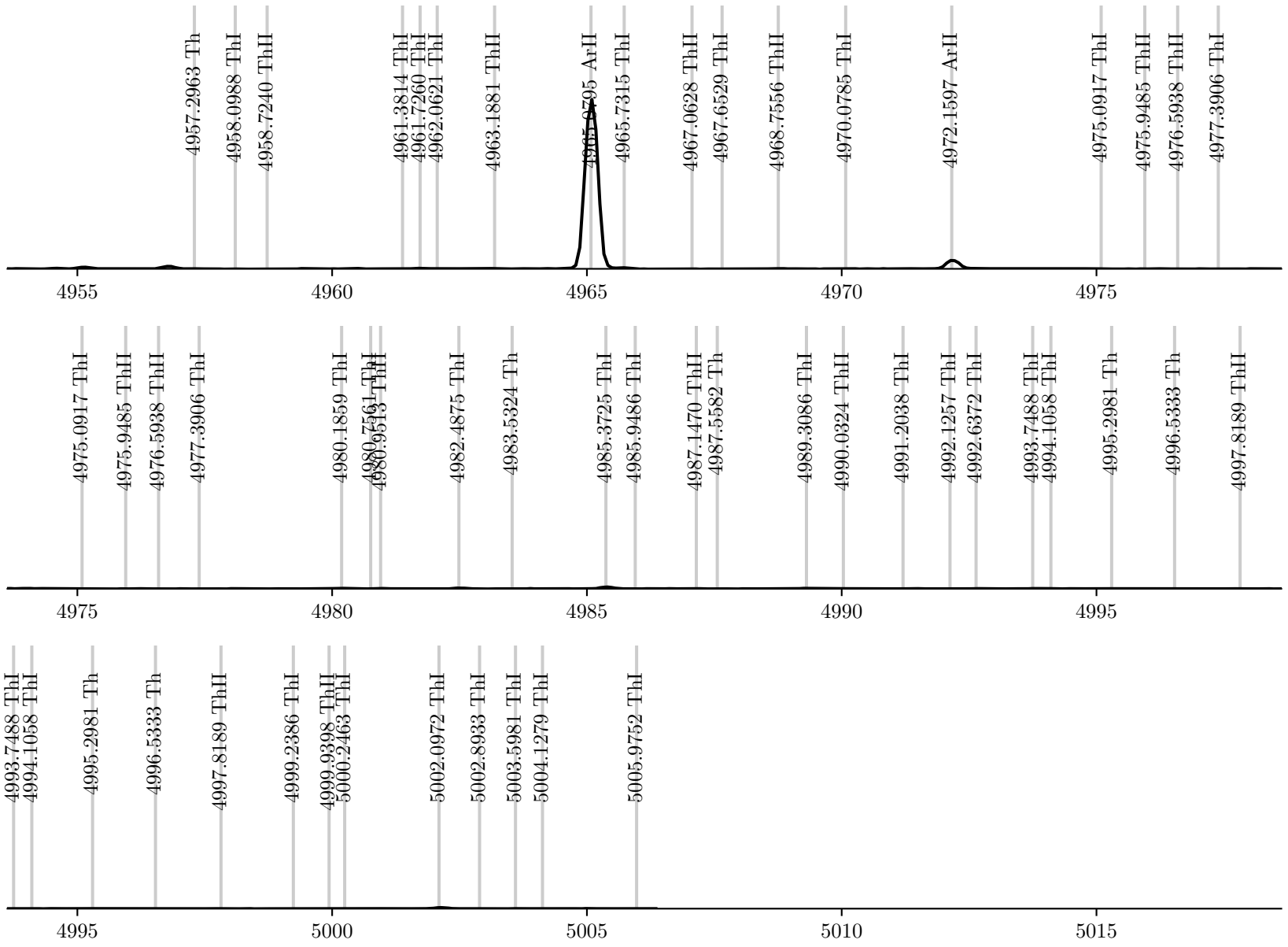
Order 46



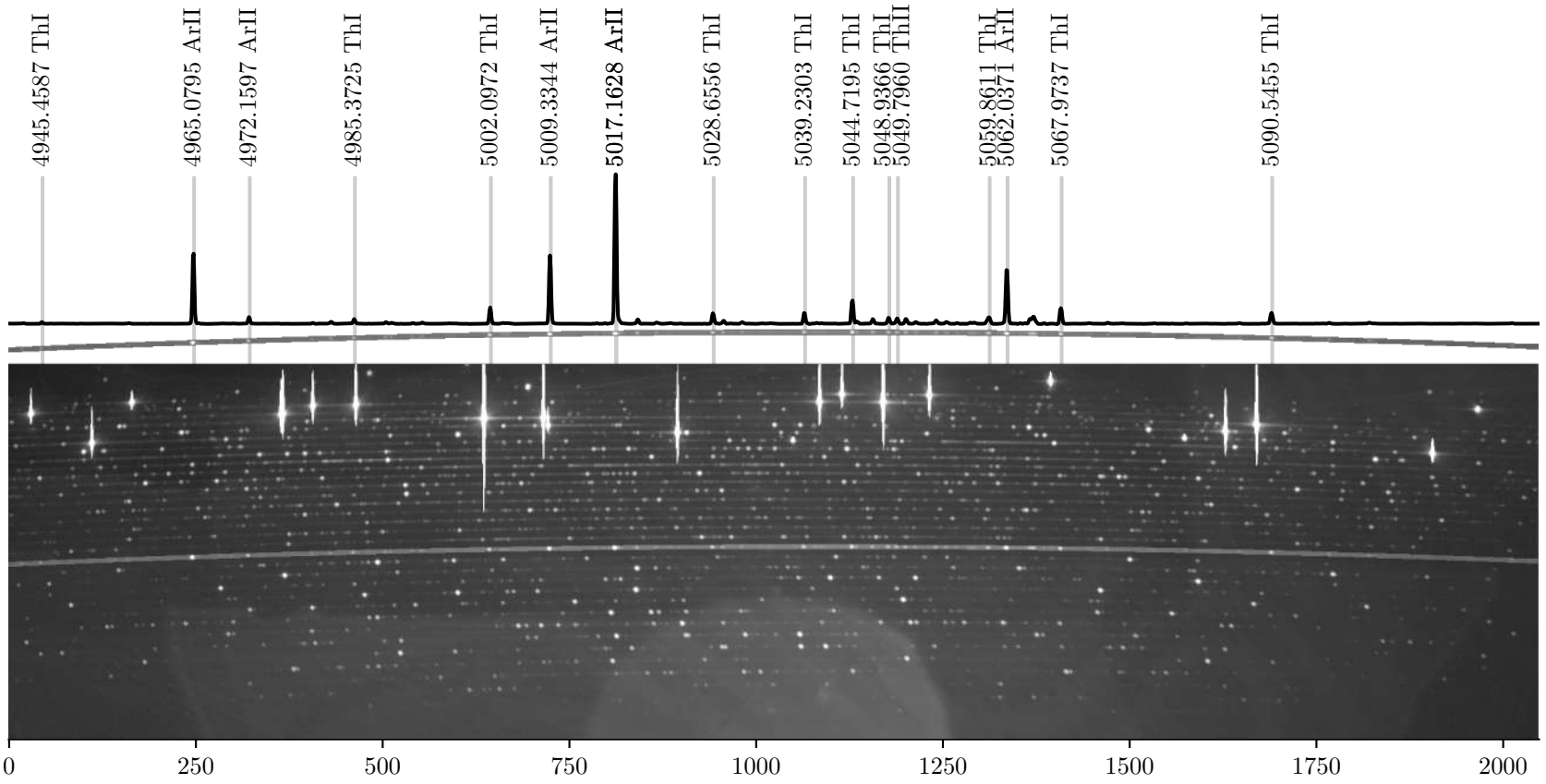
Approximate resolution:	17000
Dispersion:	0.084 Å pix^{-1}
Central wavelength:	4926 Å
Wavelength range:	4834 Å – 5006 Å


4833.6759 ThI
4834.0370 Th
4837.4121 ThI
4838.3546 ThII
4840.4744 Th
4840.8492 ThI
4842.1676 Th
4843.9413 ThI
4844.1653 ThI
4847.3263 ThI
4847.8095 ArII
4848.3625 ThI
4849.1399 Th
4849.8617 ThI
4850.4397 ThII
4852.8685 Th
4857.5385 Th
4858.0944 ThI
4858.3327 ThII
4835
4840
4845
4850
4855
4857.5385 Th
4858.0944 ThII
4858.3327 ThII
4861.2167 ThI
4861.7173 ThI
4863.1724 ThII
4865.4775 ThI
4865.9105 ArII
4868.2742 ThII
4868.8814 ThI
4871.2890 ThI
4872.0309 ThI
4872.9169 ThI
4874.3645 ThI
4876.4949 ThI
4877.0015 ThII
4878.0094 ThI
4855
4860
4865
4870
4875
4874.3645 ThI
4876.4949 ThI
4877.0015 ThII
4878.0094 ThI
4878.7330 ThI
4879.3497 ThII
4879.8635 ArII
4881.2046 ThI
4881.8531 ThI
4882.2432 ArII
4886.8661 ThI
4889.0422 ArII
4889.4903 ThI
4889.8553 ThII
4890.4582 ThI
4891.0379 Th
4891.6606 ThI
4892.7592 ThI
4893.4453 ThI
4894.9551 ThI
4875
4880
4885
4890
4895

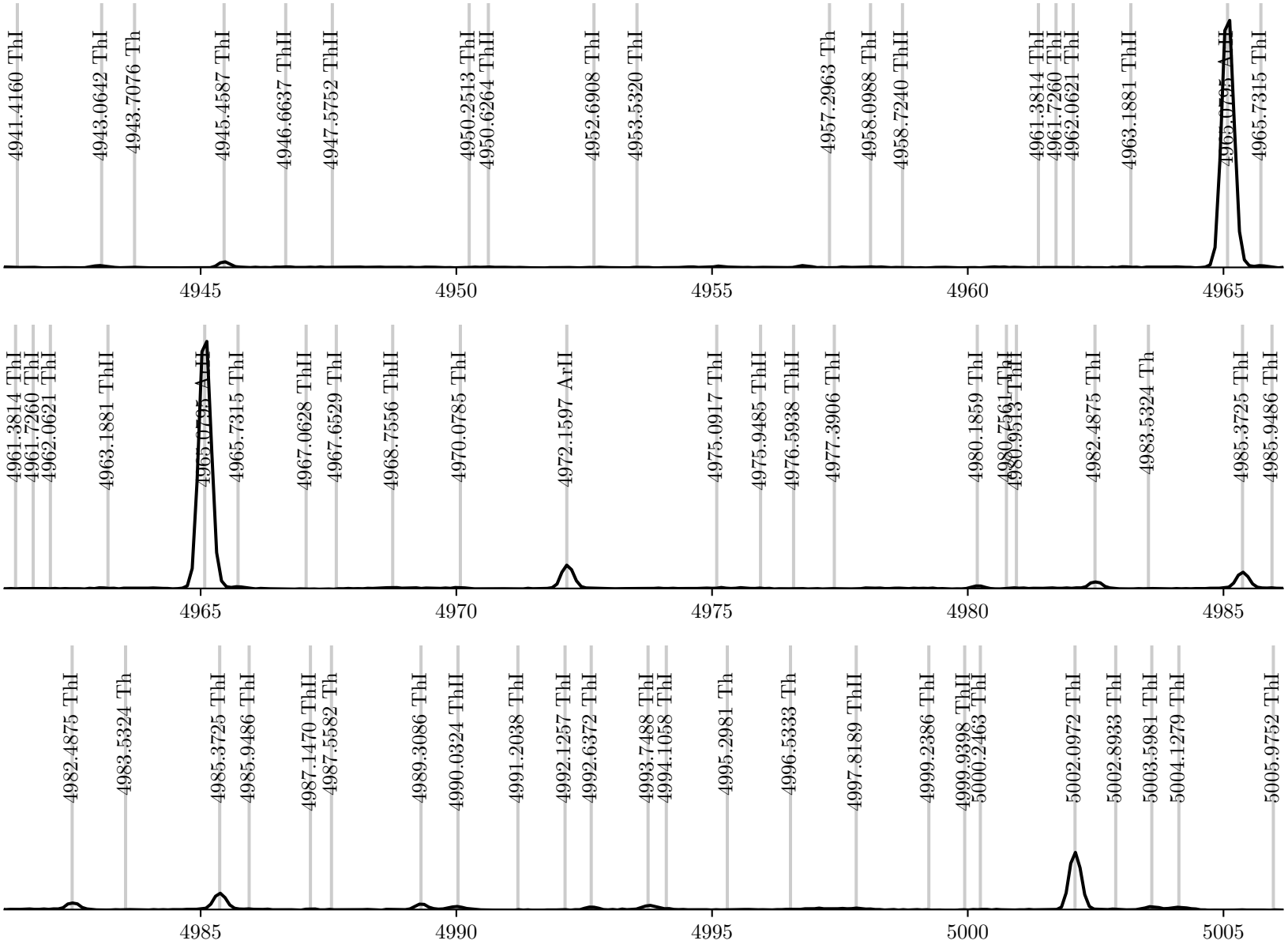

4894.9551 ThI
4898.8044 ThII
4899.2401 Th
4902.0545 ThI
4902.7943 ThI
4903.0583 ThI
4904.7516 ArII
4907.2093 Th
4909.8431 ThI
4910.1576 ThI
4910.5488 ThI
4910.7929 ThI
4911.1358 ThI
4911.3787 ThI
4912.0428 ThI
4912.5293 ThII
4914.1210 ThII
4915.4155 ThI
4917.8220 ThI
4895
4900
4905
4910
4915
4914.1210 ThII
4915.4155 ThI
4917.8220 ThI
4919.8157 ThII
4921.6134 ThII
4922.9442 ThII
4924.4223 ThII
4925.4254 ThII
4925.9501 Th
4926.2529 ThI
4927.2988 ThI
4927.7803 ThI
4929.0860 ThI
4929.9850 ThII
4933.2091 ArII
4933.8521 ThII
4934.3306 ThI
4935.9283 ThI
4936.7746 ThI
4937.8295 ThI
4938.5047 Th
4915
4920
4925
4930
4935
4933.8521 ThII
4934.3306 ThI
4935.9283 ThI
4936.7746 ThI
4937.8295 ThI
4938.5047 Th
4939.0605 ThI
4939.2707 ThI
4939.6422 ThI
4941.4160 ThI
4943.0642 ThI
4943.7076 Th
4945.4587 ThI
4946.6637 ThII
4947.5752 ThII
4950.2513 ThI
4950.6264 ThII
4952.6908 ThI
4953.5320 ThI
4957.2963 Th
4958.0988 ThI
4935
4940
4945
4950
4955

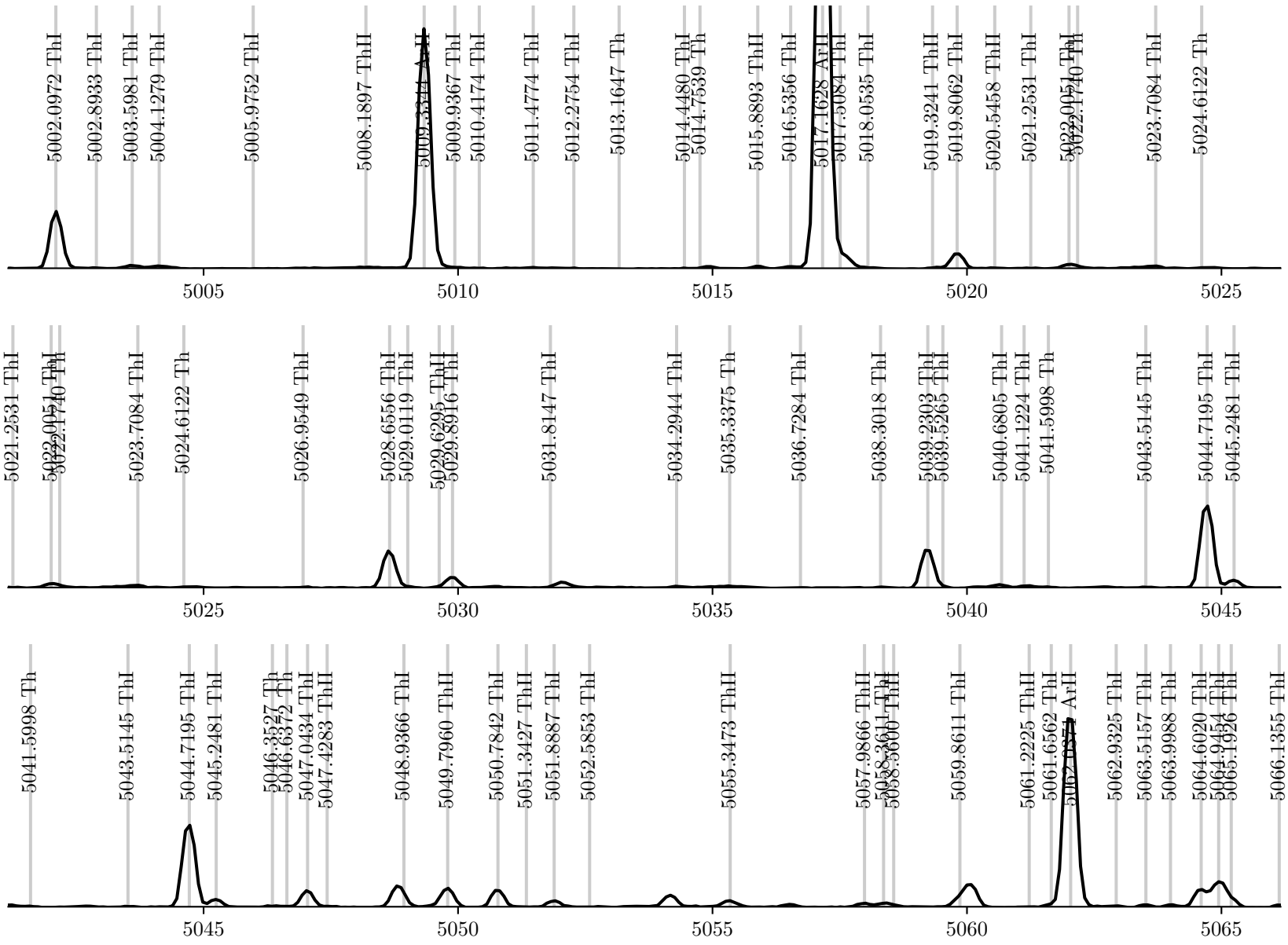

4957.2963 Th
4958.0988 ThI
4958.7240 ThII
4961.3814 ThI
4961.7260 ThI
4962.0621 ThI
4963.1881 ThII
4965.0795 ArII
4965.7315 ThI
4967.0628 ThII
4967.6529 ThI
4968.7556 ThII
4970.0785 ThI
4972.1597 ArII
4975.0917 ThI
4975.9485 ThII
4976.5938 ThII
4977.3906 ThI
4955
4960
4965
4970
4975
4975.0917 ThI
4975.9485 ThII
4976.5938 ThII
4977.3906 ThI
4980.1859 ThI
4980.7561 ThI
4980.9513 ThII
4982.4875 ThI
4983.5324 Th
4985.3725 ThI
4985.9486 ThI
4987.1470 ThII
4987.5582 Th
4989.3086 ThI
4990.0324 ThII
4991.2038 ThI
4992.1257 ThI
4992.6372 ThI
4993.7488 ThI
4994.1058 ThI
4995.2981 Th
4996.5333 Th
4997.8189 ThII
4975
4980
4985
4990
4995
4993.7488 ThI
4994.1058 ThI
4995.2981 Th
4996.5333 Th
4997.8189 ThII
4999.2386 ThI
4999.9398 ThII
5000.2463 ThI
5002.0972 ThI
5002.8933 ThI
5003.5981 ThI
5004.1279 ThI
5005.9752 ThI
4995
5000
5005
5010
5015

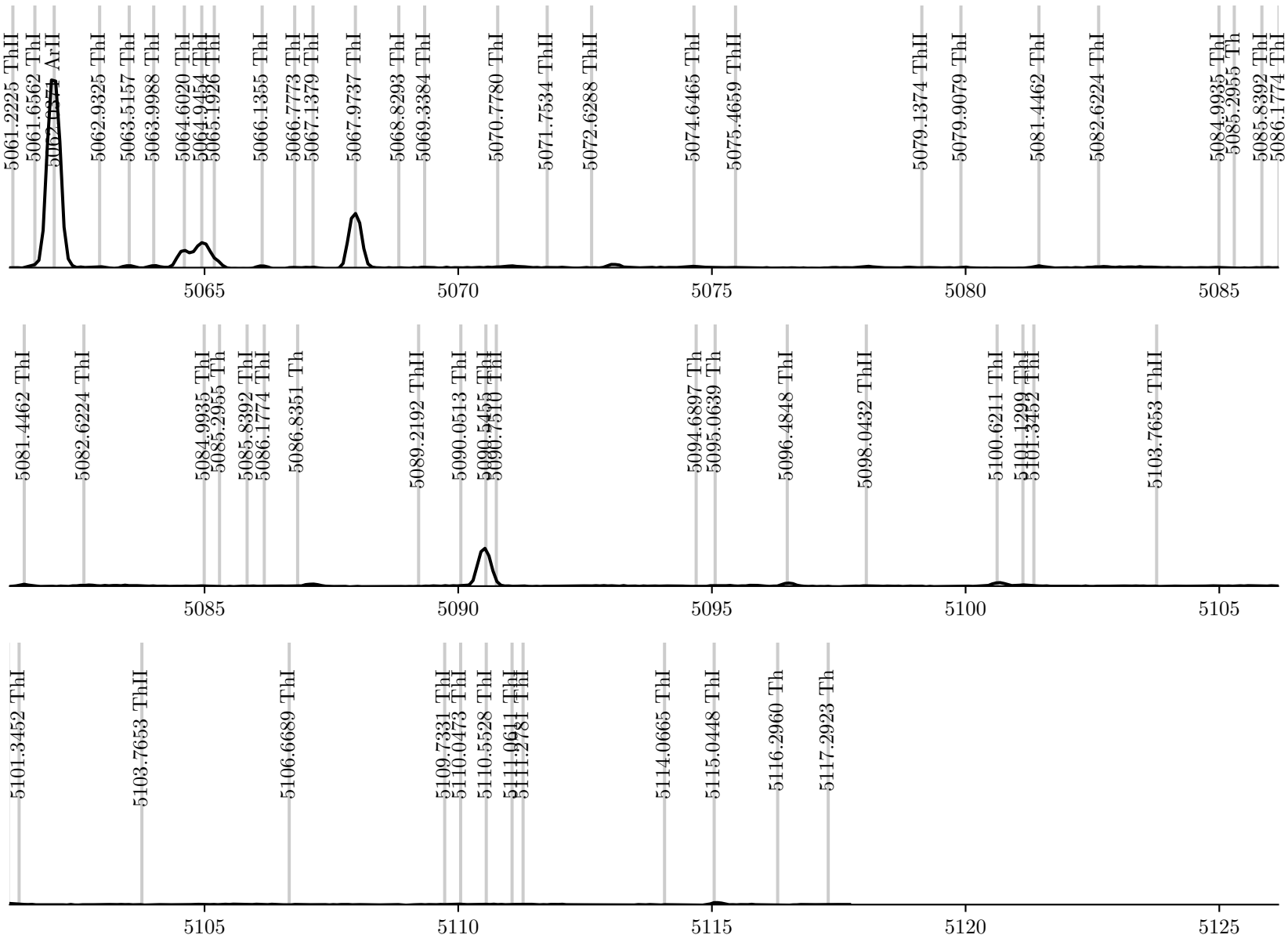
Order 45



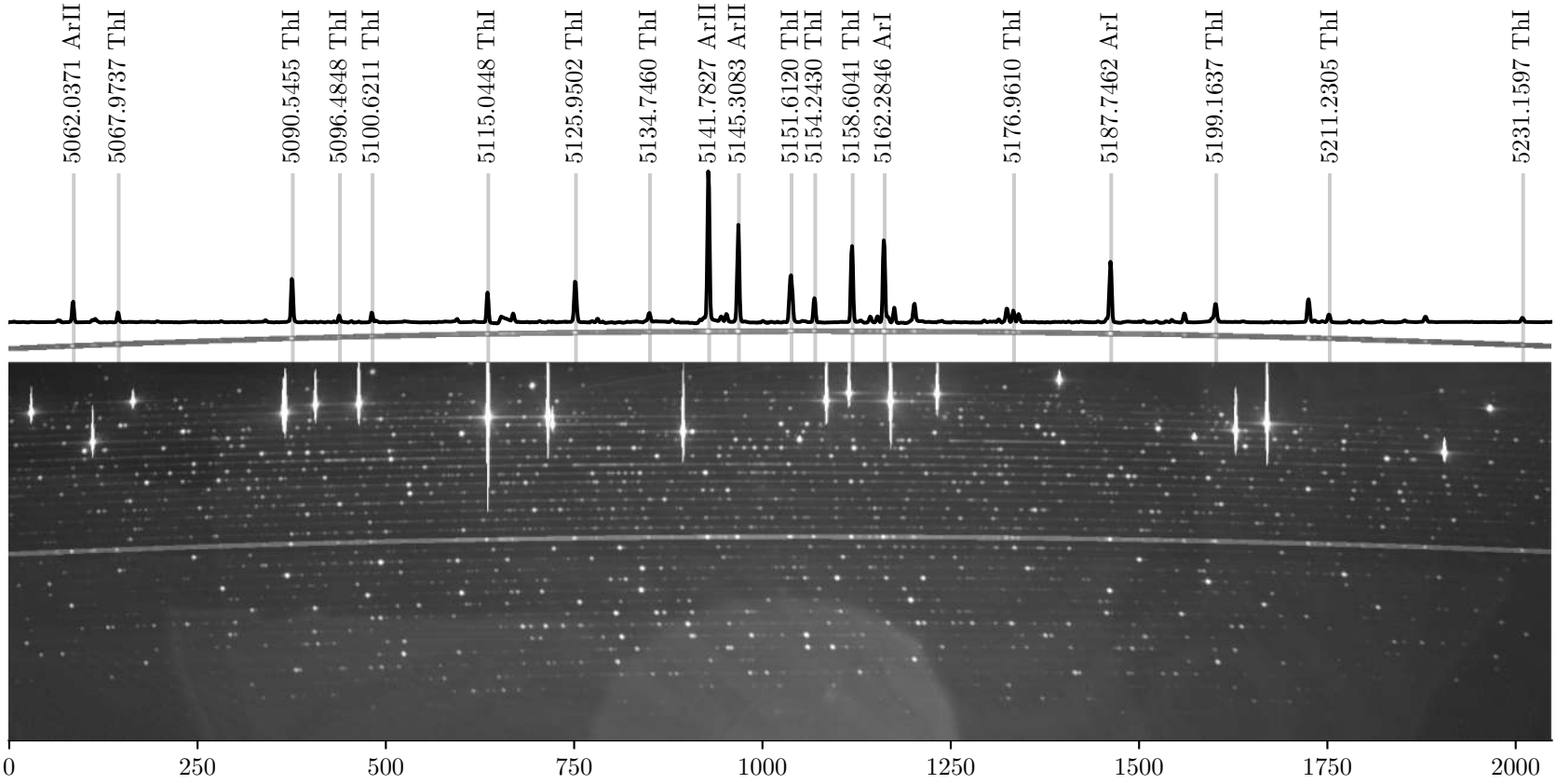
Approximate resolution:	17200
Dispersion:	0.086 Å pix^{-1}
Central wavelength:	5036 Å
Wavelength range:	4941 Å – 5118 Å


4941.4160 ThI
4943.0642 ThI
4943.7076 Th
4945.4587 ThI
4946.6637 ThII
4947.5752 ThII
4950.2513 ThI
4950.6264 ThII
4952.6908 ThI
4953.5320 ThI
4957.2963 Th
4958.0988 ThI
4958.7240 ThII
4961.3814 ThI
4961.7260 ThI
4962.0621 ThI
4963.1881 ThII
4965.0795 ArII
4965.7315 ThI
4945
4950
4955
4960
4965
4961.3814 ThI
4961.7260 ThI
4962.0621 ThI
4963.1881 ThII
4965.0795 ArII
4965.7315 ThI
4967.0628 ThII
4967.6529 ThI
4968.7556 ThII
4970.0785 ThI
4972.1597 ArII
4975.0917 ThI
4975.9485 ThII
4976.5938 ThII
4977.3906 ThI
4980.1859 ThI
4980.7561 ThI
4980.9513 ThII
4982.4875 ThI
4983.5324 Th
4985.3725 ThI
4985.9486 ThI
4965
4970
4975
4980
4985
4982.4875 ThI
4983.5324 Th
4985.3725 ThI
4985.9486 ThI
4987.1470 ThII
4987.5582 Th
4989.3086 ThI
4990.0324 ThII
4991.2038 ThI
4992.1257 ThI
4992.6372 ThI
4993.7488 ThI
4994.1058 ThI
4995.2981 Th
4996.5333 Th
4997.8189 ThII
4999.2386 ThI
4999.9398 ThII
5000.2463 ThI
5002.0972 ThI
5002.8933 ThI
5003.5981 ThI
5004.1279 ThI
5005.9752 ThI
4985
4990
4995
5000
5005

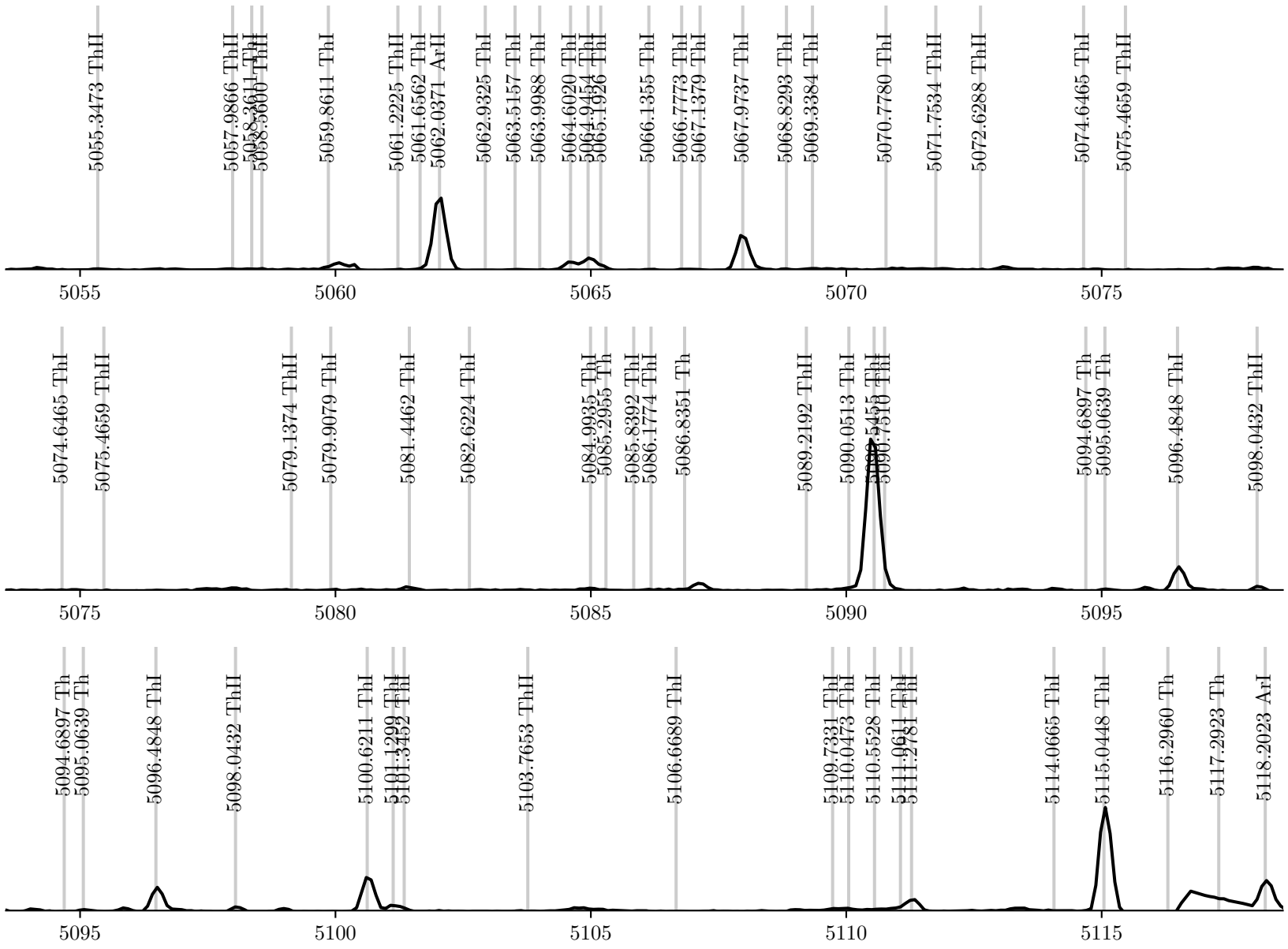

5002.0972 ThI
5002.8933 ThI
5003.5981 ThI
5004.1279 ThI
5005.9752 ThI
5008.1897 ThII
5009.3341 ArII
5009.9367 ThI
5010.4174 ThI
5011.4774 ThI
5012.2754 ThI
5013.1647 Th
5014.4480 ThI
5014.7539 Th
5015.8893 ThII
5016.5356 ThI
5017.1628 ArII
5017.5084 ThI
5018.0535 ThI
5019.3241 ThII
5019.8062 ThI
5020.5458 ThII
5021.2531 ThI
5023.7084 ThI
5024.6122 Th
5005
5010
5015
5020
5025
5021.2531 ThI
5023.7084 ThI
5024.6122 Th
5026.9549 ThI
5028.6556 ThI
5029.0119 ThI
5031.8147 ThI
5034.2944 ThI
5035.3375 Th
5036.7284 ThI
5038.3018 ThI
5039.2303 ThI
5039.5265 ThI
5040.6805 ThI
5041.1224 ThI
5041.5998 Th
5043.5145 ThI
5044.7195 ThI
5045.2481 ThI
5025
5030
5035
5040
5045
5041.5998 Th
5043.5145 ThI
5044.7195 ThI
5045.2481 ThI
5047.0434 ThI
5047.4283 ThII
5048.9366 ThI
5049.7960 ThII
5050.7842 ThI
5051.3427 ThII
5051.8887 ThI
5052.5853 ThI
5055.3473 ThII
5059.8611 ThI
5061.2225 ThII
5061.6562 ThI
5062.1877 ArII
5062.9325 ThI
5063.5157 ThI
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5064.6020 ThI
5045
5050
5055
5060
5065

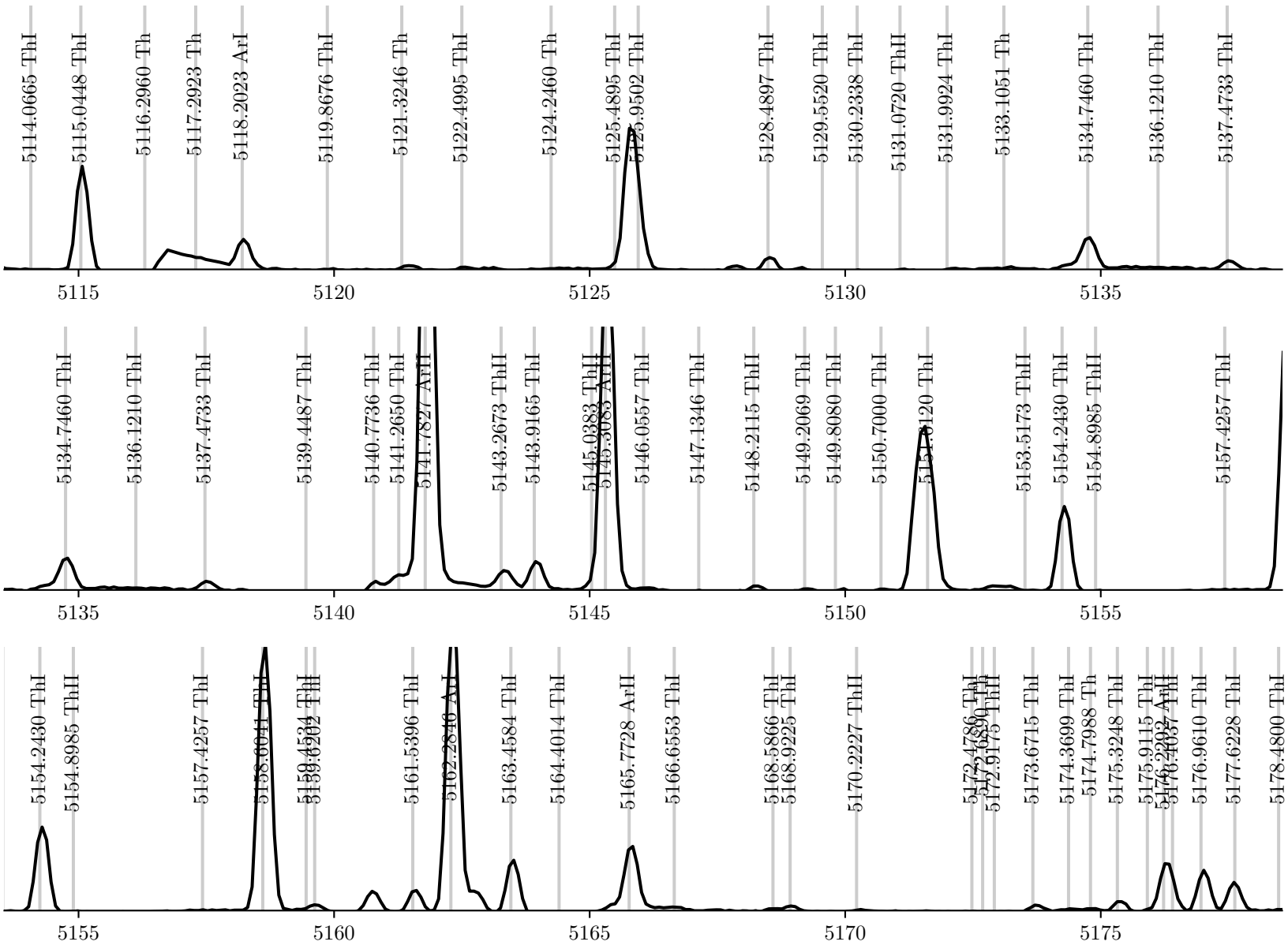

5061.2225 ThII
5061.6562 ThI
5062.0879 ArII
5062.9325 ThI
5063.5157 ThI
5063.9988 ThI
5064.6020 ThI
5064.9434 ThI
5065.1926 ThI
5066.1355 ThI
5066.7773 ThI
5067.1379 ThI
5067.9737 ThI
5068.8293 ThI
5069.3384 ThI
5070.7780 ThI
5071.7534 ThII
5072.6288 ThII
5074.6465 ThI
5075.4659 ThII
5079.1374 ThII
5079.9079 ThI
5081.4462 ThI
5082.6224 ThI
5084.9935 ThI
5085.2955 Th
5085.8392 ThI
5086.1774 ThI
5065
5070
5075
5080
5085
5081.4462 ThI
5082.6224 ThI
5084.9935 ThI
5085.2955 Th
5085.8392 ThI
5086.1774 ThI
5086.8351 Th
5089.2192 ThII
5090.0513 ThI
5090.5455 ThI
5090.7510 ThI
5094.6897 Th
5095.0639 Th
5096.4848 ThI
5098.0432 ThII
5100.6211 ThI
5101.1299 ThI
5101.3452 ThI
5103.7653 ThII
5085
5090
5095
5100
5105
5101.3452 ThI
5103.7653 ThII
5106.6689 ThI
5109.7331 ThI
5110.0473 ThI
5110.5528 ThI
5111.0611 ThI
5111.2781 ThI
5114.0665 ThI
5115.0448 ThI
5116.2960 Th
5117.2923 Th
5105
5110
5115
5120
5125

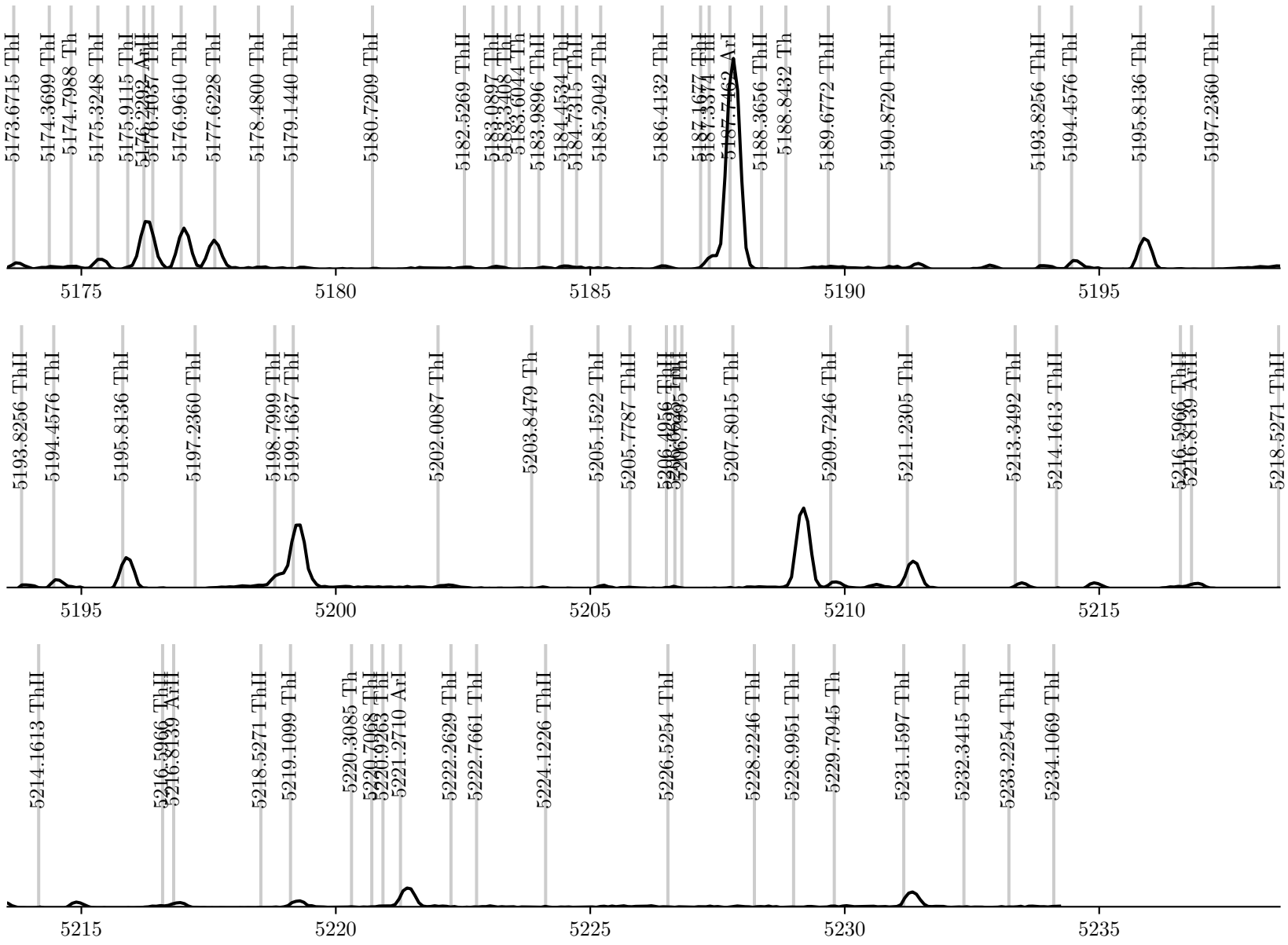
Order 44



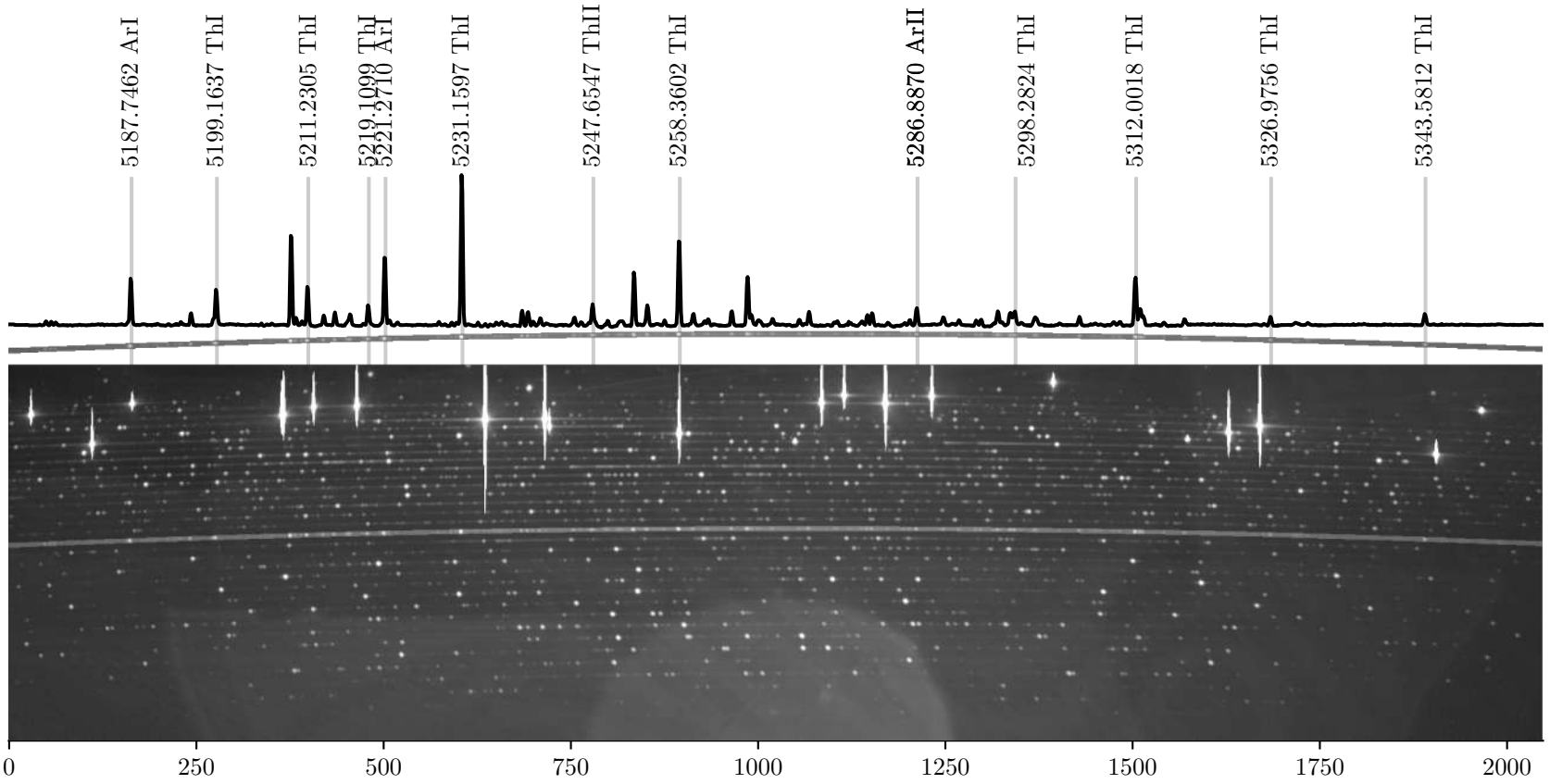
Approximate resolution:	17300
Dispersion:	0.088 Å pix^{-1}
Central wavelength:	5150 Å
Wavelength range:	5054 Å – 5234 Å


5055.3473 ThII
5057.9866 ThII
5059.8611 ThI
5061.2225 ThII
5061.6562 ThI
5062.0371 ArII
5062.9325 ThI
5063.5157 ThI
5063.9988 ThI
5064.6020 ThI
5066.1355 ThI
5066.7773 ThI
5067.1379 ThI
5067.9737 ThI
5068.8293 ThI
5069.3384 ThI
5070.7780 ThI
5071.7534 ThII
5072.6288 ThII
5074.6465 ThI
5075.4659 ThII
5055
5060
5065
5070
5075
5074.6465 ThI
5075.4659 ThII
5079.1374 ThII
5079.9079 ThI
5081.4462 ThI
5082.6224 ThI
5084.9935 ThI
5085.2955 Th
5085.8392 ThI
5086.1774 ThI
5086.8351 Th
5089.2192 ThII
5090.0513 ThI
5094.6897 Th
5095.0639 Th
5096.4848 ThI
5098.0432 ThII
5075
5080
5085
5090
5095
5094.6897 Th
5095.0639 Th
5096.4848 ThI
5098.0432 ThII
5100.6211 ThI
5103.7653 ThII
5106.6689 ThI
5109.7331 ThI
5110.0473 ThI
5110.5528 ThI
5114.0665 ThI
5115.0448 ThI
5116.2960 Th
5117.2923 Th
5118.2023 ArI
5095
5100
5105
5110
5115

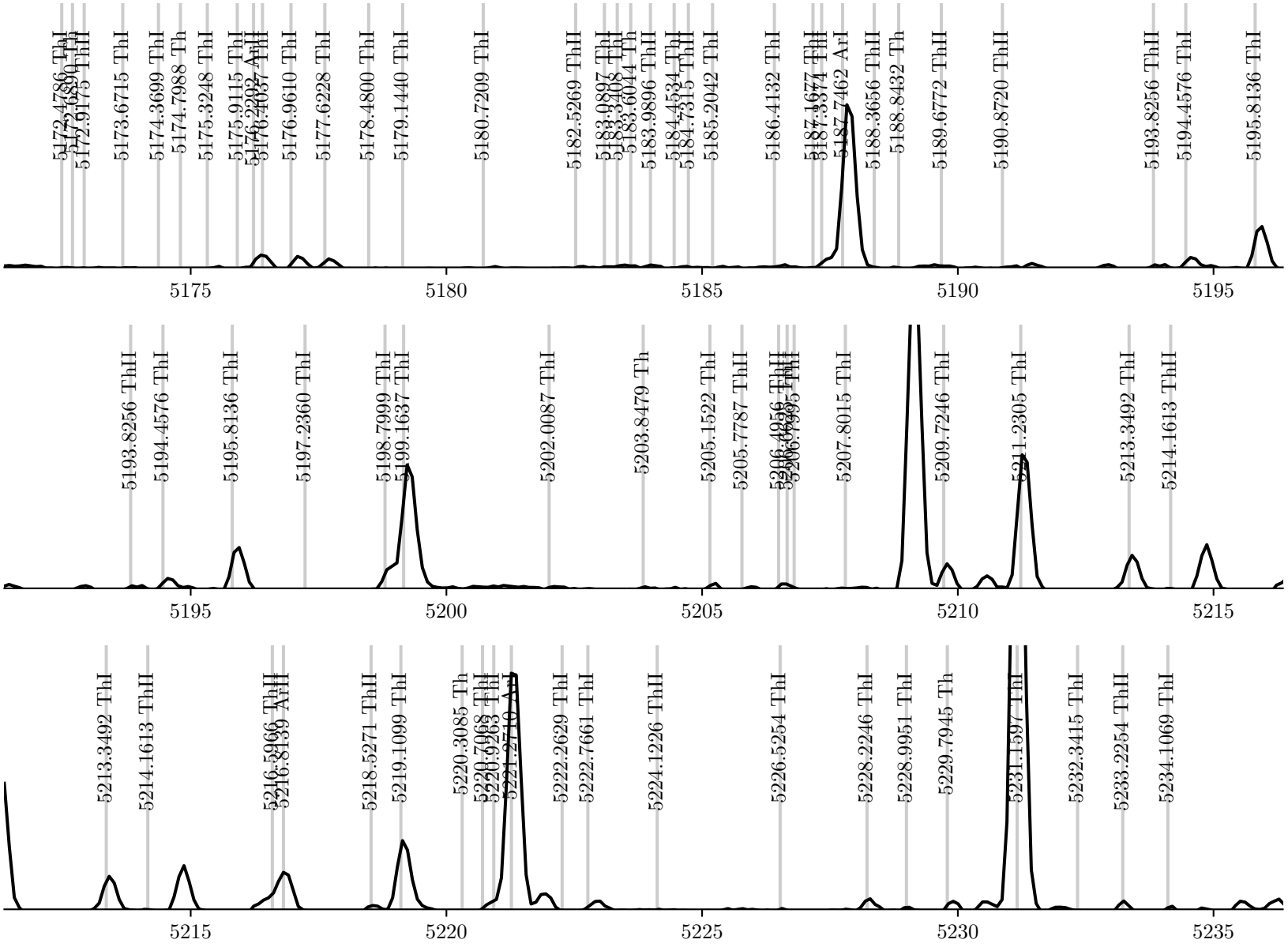

5114.0665 ThI
5115.0448 ThI
5116.2960 Th
5117.2923 Th
5118.2023 ArI
5119.8676 ThI
5121.3246 Th
5122.4995 ThI
5124.2460 Th
5125.4895 ThI
5125.9502 ThI
5128.4897 ThI
5129.5520 ThI
5130.2338 ThI
5131.0720 ThII
5131.9924 ThI
5133.1051 Th
5134.7460 ThI
5136.1210 ThI
5137.4733 ThI
5115
5120
5125
5130
5135
5134.7460 ThI
5136.1210 ThI
5137.4733 ThI
5139.4487 ThI
5140.7736 ThI
5141.2650 ThI
5141.7827 ArII
5143.2673 ThII
5143.9165 ThI
5145.0383 ThII
5145.3083 ArII
5146.0557 ThI
5147.1346 ThI
5148.2115 ThII
5149.2069 ThI
5149.8080 ThI
5150.7000 ThI
5151.8120 ThI
5153.5173 ThII
5154.2430 ThI
5154.8985 ThII
5157.4257 ThI
5135
5140
5145
5150
5155
5154.2430 ThI
5154.8985 ThII
5157.4257 ThI
5158.6041 ThI
5159.4524 ThI
5159.6202 ThII
5161.5396 ThI
5162.2846 ArI
5163.4584 ThI
5164.4014 ThI
5165.7728 ArII
5166.6553 ThI
5168.5866 ThI
5168.9225 ThI
5170.2227 ThII
5172.4786 ThI
5172.9175 ThII
5173.6715 ThI
5174.3699 ThI
5174.7988 Th
5175.3248 ThI
5175.9115 ThI
5176.2292 ArII
5176.4037 ArII
5176.9610 ThI
5177.6228 ThI
5178.4800 ThI
5155
5160
5165
5170
5175

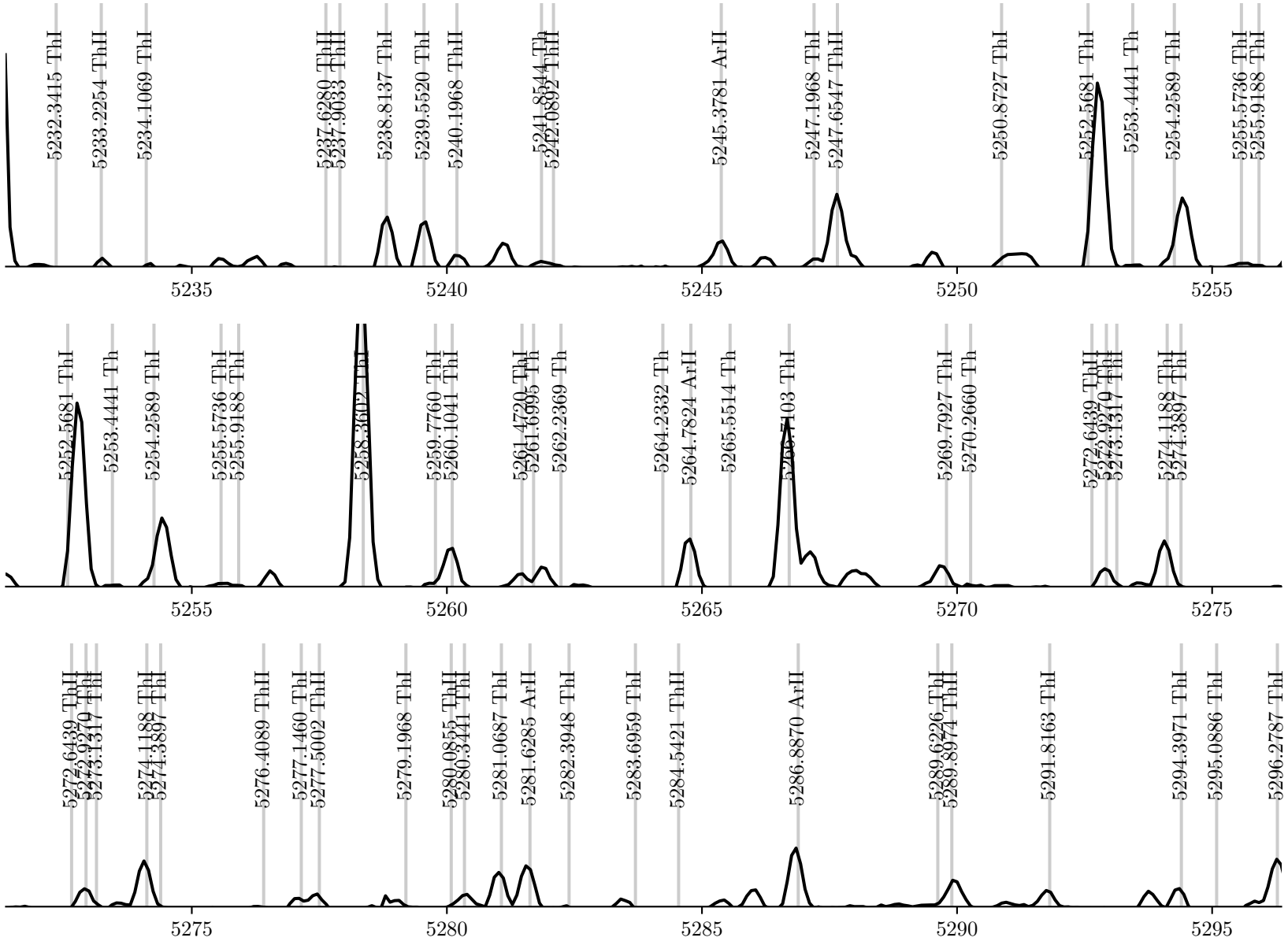

5173.6715 ThI
5174.3699 ThI
5174.7988 Th
5175.3248 ThI
5175.9115 ThI
5176.4037 ThII
5176.9610 ThI
5177.6228 ThI
5178.4800 ThI
5179.1440 ThI
5180.7209 ThI
5182.5269 ThII
5183.0897 ThI
5183.6044 Th
5183.9896 ThII
5184.4534 ThI
5184.7315 ThII
5185.2042 ThI
5186.4132 ThI
5187.1677 ThI
5187.7462 ArI
5188.3656 ThII
5188.8432 Th
5189.6772 ThII
5190.8720 ThII
5193.8256 ThII
5194.4576 ThI
5195.8136 ThI
5197.2360 ThI
5175
5180
5185
5190
5195
5193.8256 ThII
5194.4576 ThI
5195.8136 ThI
5197.2360 ThI
5198.7999 ThI
5199.1637 ThI
5202.0087 ThI
5203.8479 Th
5205.1522 ThI
5205.7787 ThII
5207.8015 ThI
5209.7246 ThI
5211.2305 ThI
5213.3492 ThI
5214.1613 ThII
5216.5966 ThI
5216.8139 ArII
5218.5271 ThII
5195
5200
5205
5210
5215
5214.1613 ThII
5216.5966 ThI
5216.8139 ArII
5218.5271 ThII
5219.1099 ThI
5220.3085 Th
5221.2710 ArI
5222.2629 ThI
5222.7661 ThI
5224.1226 ThII
5226.5254 ThI
5228.2246 ThI
5228.9951 ThI
5229.7945 Th
5231.1597 ThI
5232.3415 ThI
5233.2254 ThII
5234.1069 ThI
5215
5220
5225
5230
5235

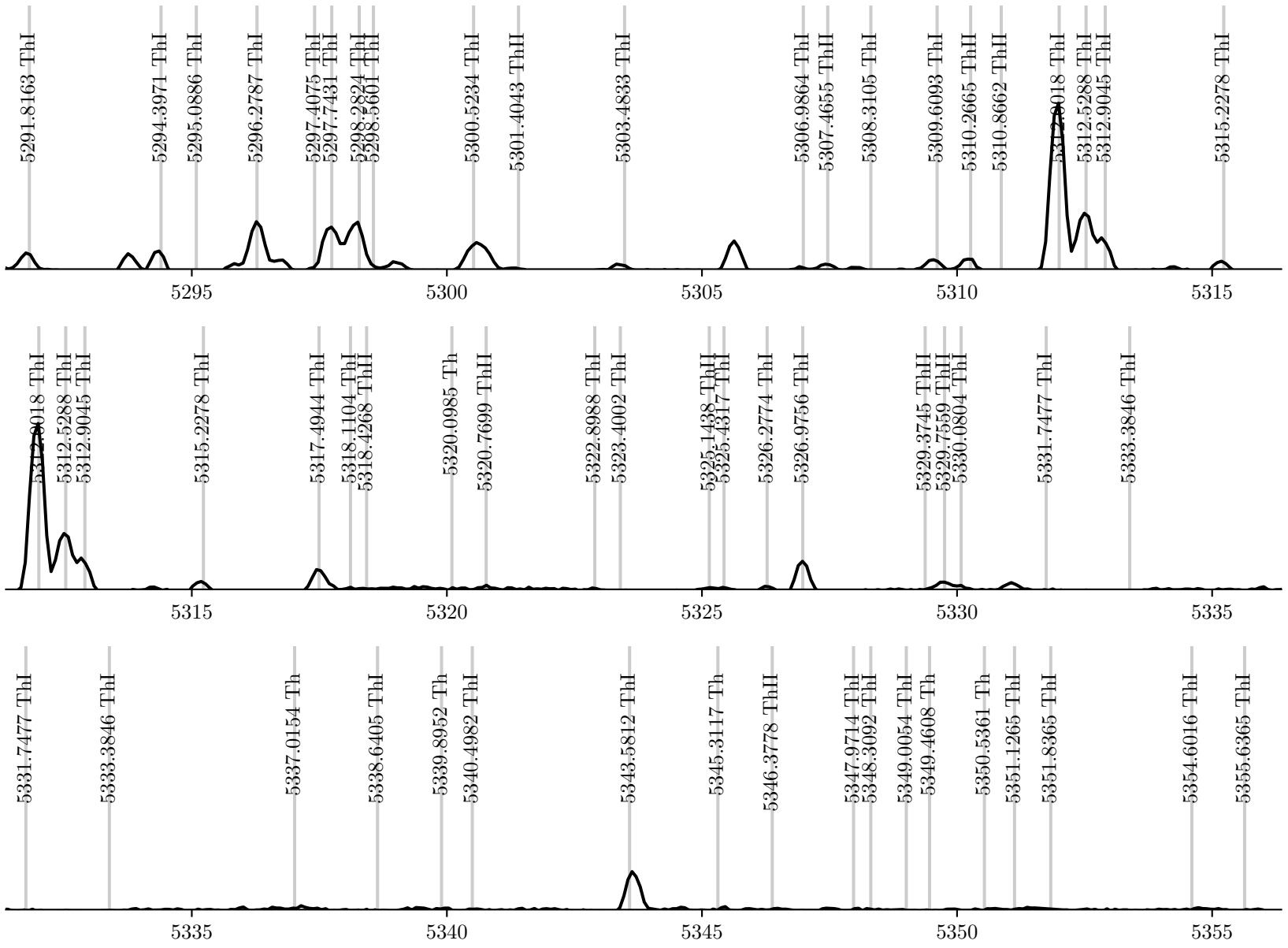
Order 43



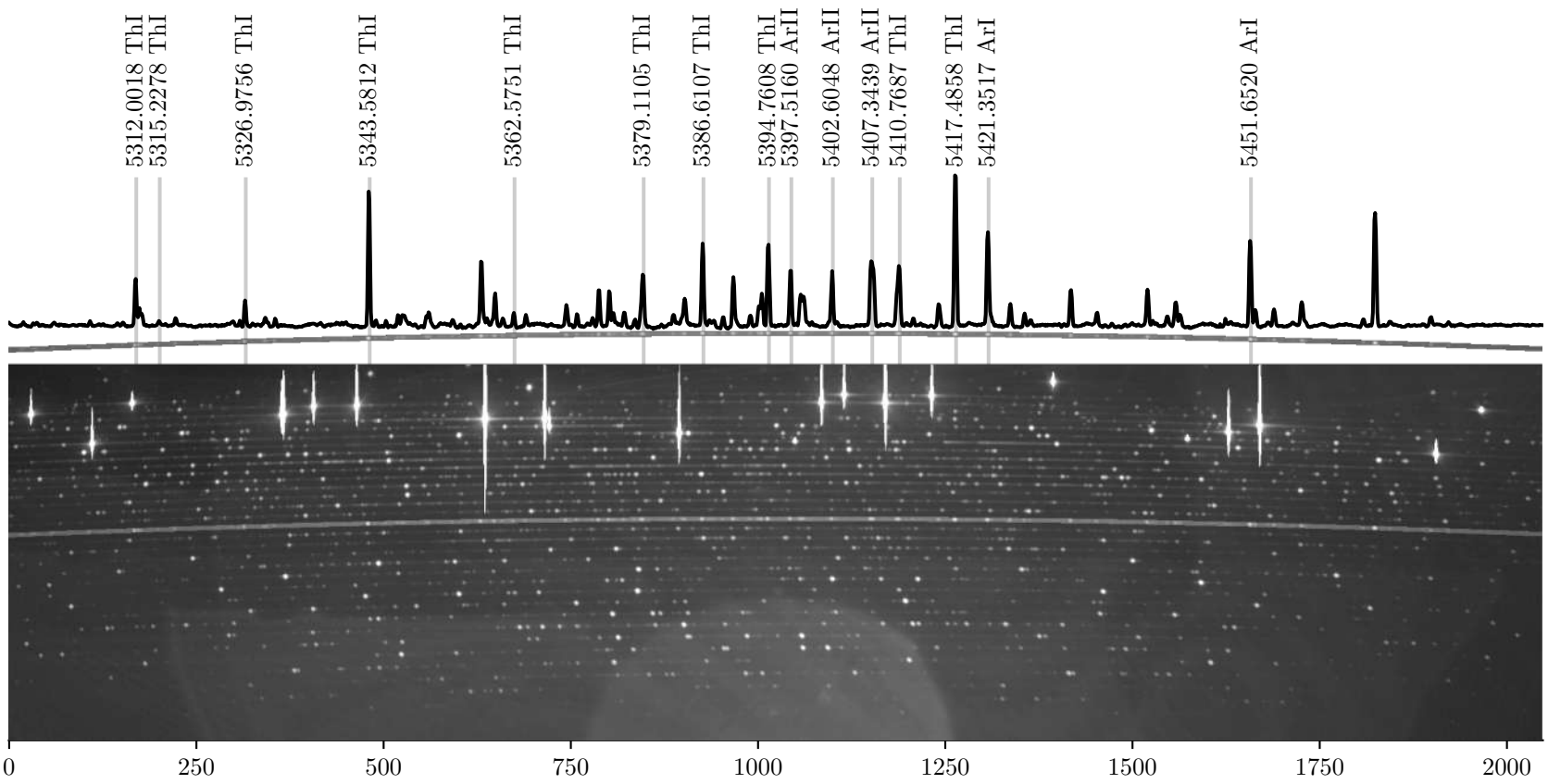
Approximate resolution:	17400
Dispersion:	0.090 Å pix^{-1}
Central wavelength:	5270 Å
Wavelength range:	5171 Å – 5356 Å


5172.4786 ThI
5172.5175 ThII
5173.6715 ThI
5174.3699 ThI
5174.7988 Th
5175.3248 ThI
5175.9115 ThI
5176.4037 ArII
5176.9610 ThI
5177.6228 ThI
5178.4800 ThI
5179.1440 ThI
5180.7209 ThI
5182.5269 ThII
5183.0897 ThI
5183.3408 ThI
5183.6044 Th
5183.9896 ThII
5184.4534 ThI
5184.7315 ThII
5185.2042 ThI
5186.4132 ThI
5187.1677 ThI
5187.3871 ThI
5187.7462 ArI
5188.3656 ThII
5188.8432 Th
5189.6772 ThII
5190.8720 ThII
5193.8256 ThII
5194.4576 ThI
5195.8136 ThI
5175
5180
5185
5190
5195
5193.8256 ThII
5194.4576 ThI
5195.8136 ThI
5197.2360 ThI
5198.7999 ThI
5199.1637 ThI
5202.0087 ThI
5203.8479 Th
5205.1522 ThI
5205.7787 ThII
5207.8015 ThI
5209.7246 ThI
5211.2305 ThI
5213.3492 ThI
5214.1613 ThII
5195
5200
5205
5210
5215
5213.3492 ThI
5214.1613 ThII
5216.5966 ThII
5216.8139 ArII
5218.5271 ThII
5219.1099 ThI
5220.3085 Th
5220.7068 ThI
5220.9263 ThI
5221.2710 ArI
5222.2629 ThI
5222.7661 ThI
5224.1226 ThII
5226.5254 ThI
5228.2246 ThI
5228.9951 ThI
5229.7945 Th
5231.1597 ThI
5232.3415 ThI
5233.2254 ThII
5234.1069 ThI
5215
5220
5225
5230
5235

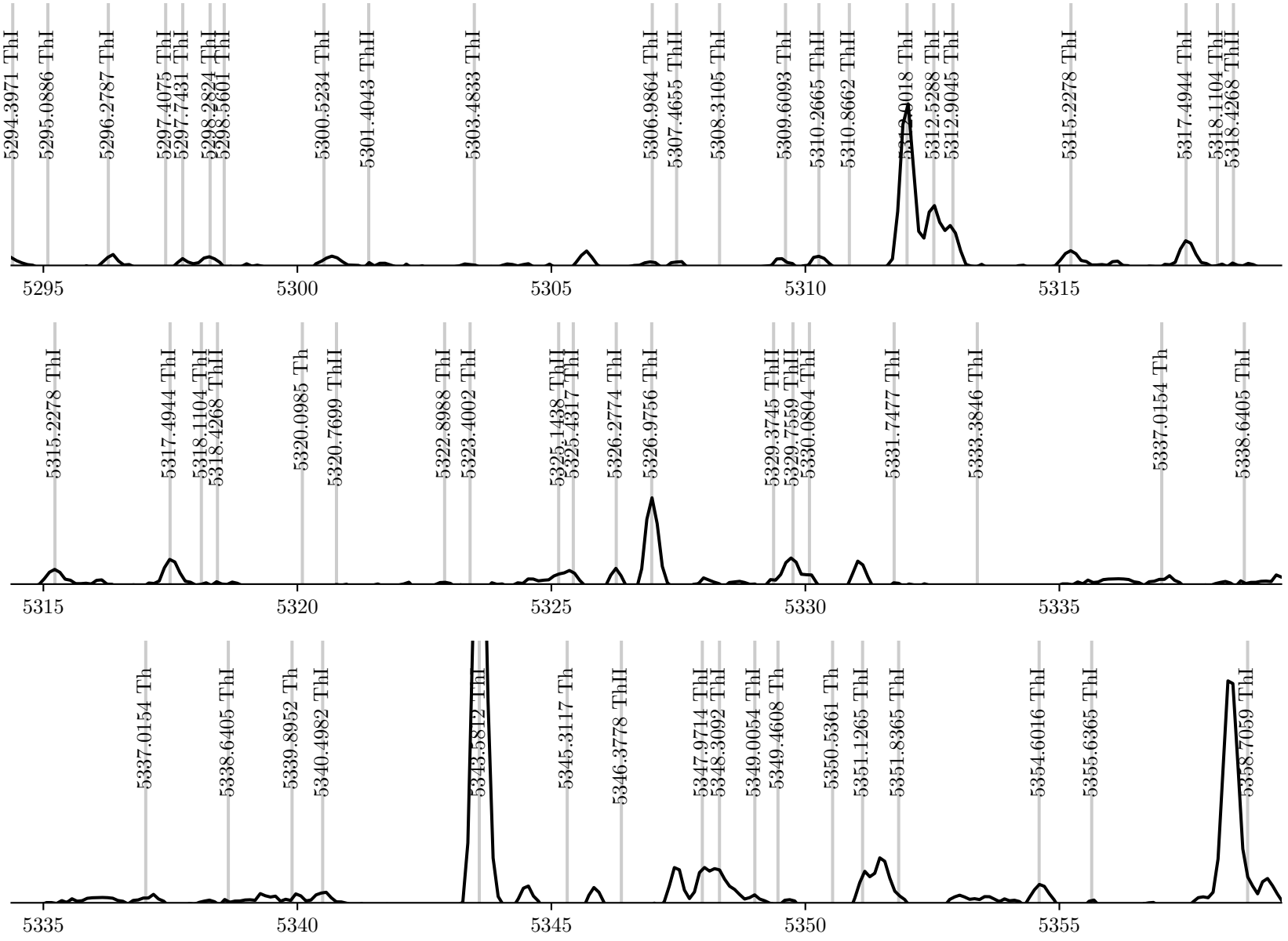

5232.3415 ThI
5233.2254 ThII
5234.1069 ThI
5237.6280 ThII
5237.9033 ThII
5238.8137 ThI
5239.5520 ThI
5240.1968 ThII
5241.8544 Th
5242.0892 ThII
5245.3781 ArII
5247.1968 ThI
5247.6547 ThII
5250.8727 ThI
5252.5681 ThI
5253.4441 Th
5254.2589 ThI
5255.5736 ThI
5255.9188 ThI
5235
5240
5245
5250
5255
5252.5681 ThI
5253.4441 Th
5254.2589 ThI
5255.5736 ThI
5255.9188 ThI
5258.3602 ThI
5259.7760 ThI
5260.1041 ThI
5261.4720 ThI
5261.6995 Th
5262.2369 Th
5264.2332 Th
5264.7824 ArII
5265.5514 Th
5266.7103 ThI
5269.7927 ThI
5270.2660 Th
5272.6439 ThII
5273.1317 ThI
5274.1188 ThI
5274.3897 ThI
5255
5260
5265
5270
5275
5272.6439 ThII
5273.0320 Th
5273.1317 ThI
5274.1188 ThI
5274.3897 ThI
5276.4089 ThII
5277.1460 ThI
5277.5002 ThII
5279.1968 ThI
5280.0855 ThII
5280.3441 ThI
5281.0687 ThI
5281.6285 ArII
5282.3948 ThI
5283.6959 ThI
5284.5421 ThII
5286.8870 ArII
5289.6226 ThI
5289.8974 ThII
5291.8163 ThI
5294.3971 ThI
5295.0886 ThI
5296.2787 ThI
5275
5280
5285
5290
5295

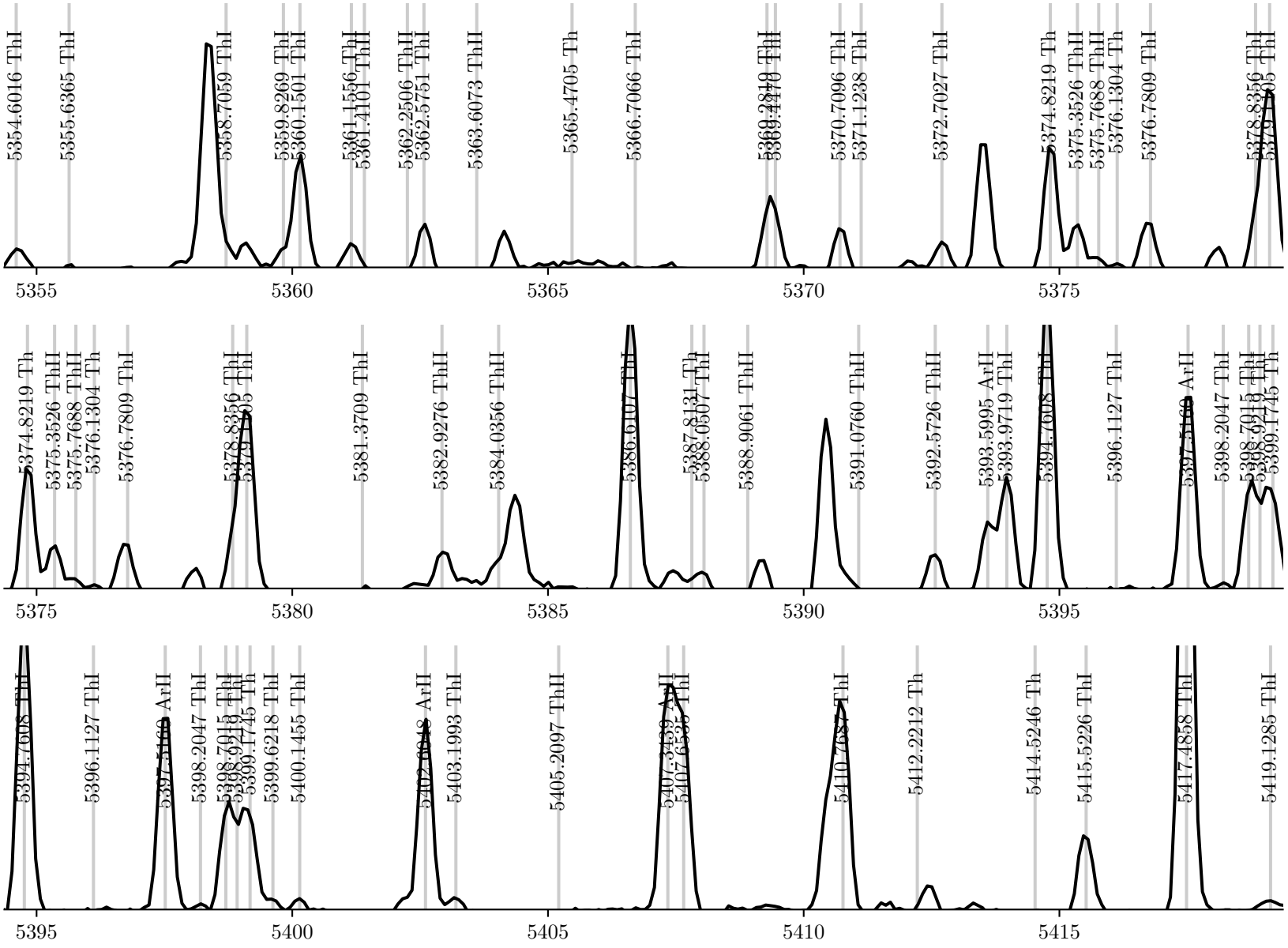

5291.8163 ThI
5294.3971 ThI
5295.0886 ThI
5296.2787 ThI
5297.4075 ThI
5297.7431 ThI
5298.2824 ThI
5298.3601 ThI
5300.5234 ThI
5301.4043 ThII
5303.4833 ThI
5306.9864 ThI
5307.4655 ThII
5308.3105 ThI
5309.6093 ThI
5310.2665 ThII
5310.8662 ThII
5312.5288 ThI
5312.9045 ThI
5315.2278 ThI
5295
5300
5305
5310
5315
5312.5288 ThI
5312.9045 ThI
5315.2278 ThI
5317.4944 ThI
5318.1104 ThI
5318.4268 ThII
5320.0985 Th
5320.7699 ThII
5322.8988 ThI
5323.4002 ThI
5325.1438 ThII
5325.4317 ThI
5326.2774 ThI
5326.9756 ThI
5329.3745 ThII
5329.7559 ThII
5330.0804 ThI
5331.7477 ThI
5333.3846 ThI
5315
5320
5325
5330
5335
5331.7477 ThI
5333.3846 ThI
5337.0154 Th
5338.6405 ThI
5339.8952 Th
5340.4982 ThI
5343.5812 ThI
5345.3117 Th
5346.3778 ThII
5347.9714 ThI
5348.3092 ThI
5349.0054 ThI
5349.4608 Th
5350.5361 Th
5351.1265 ThI
5351.8365 ThI
5354.6016 ThI
5355.6365 ThI
5335
5340
5345
5350
5355

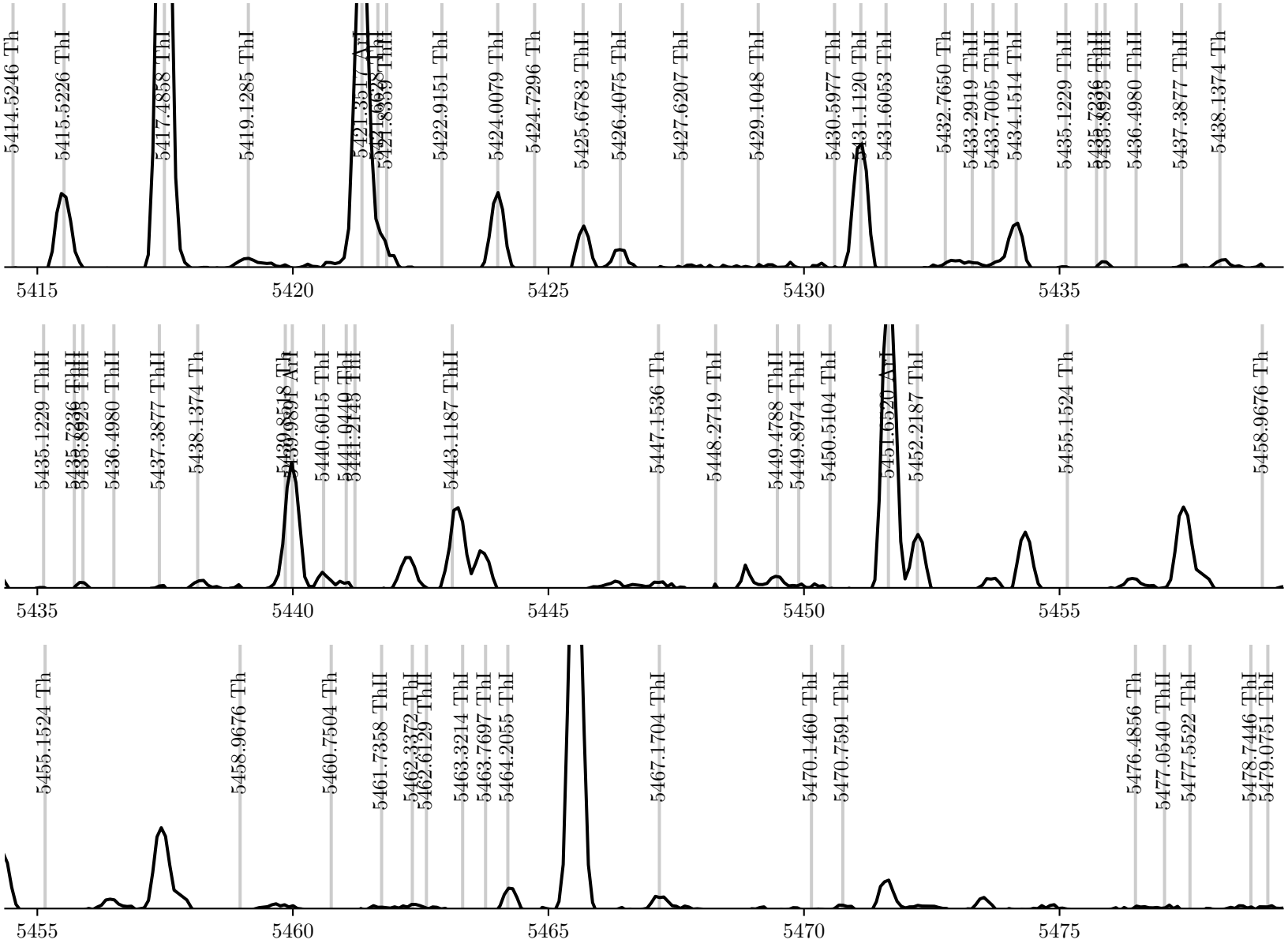
Order 42



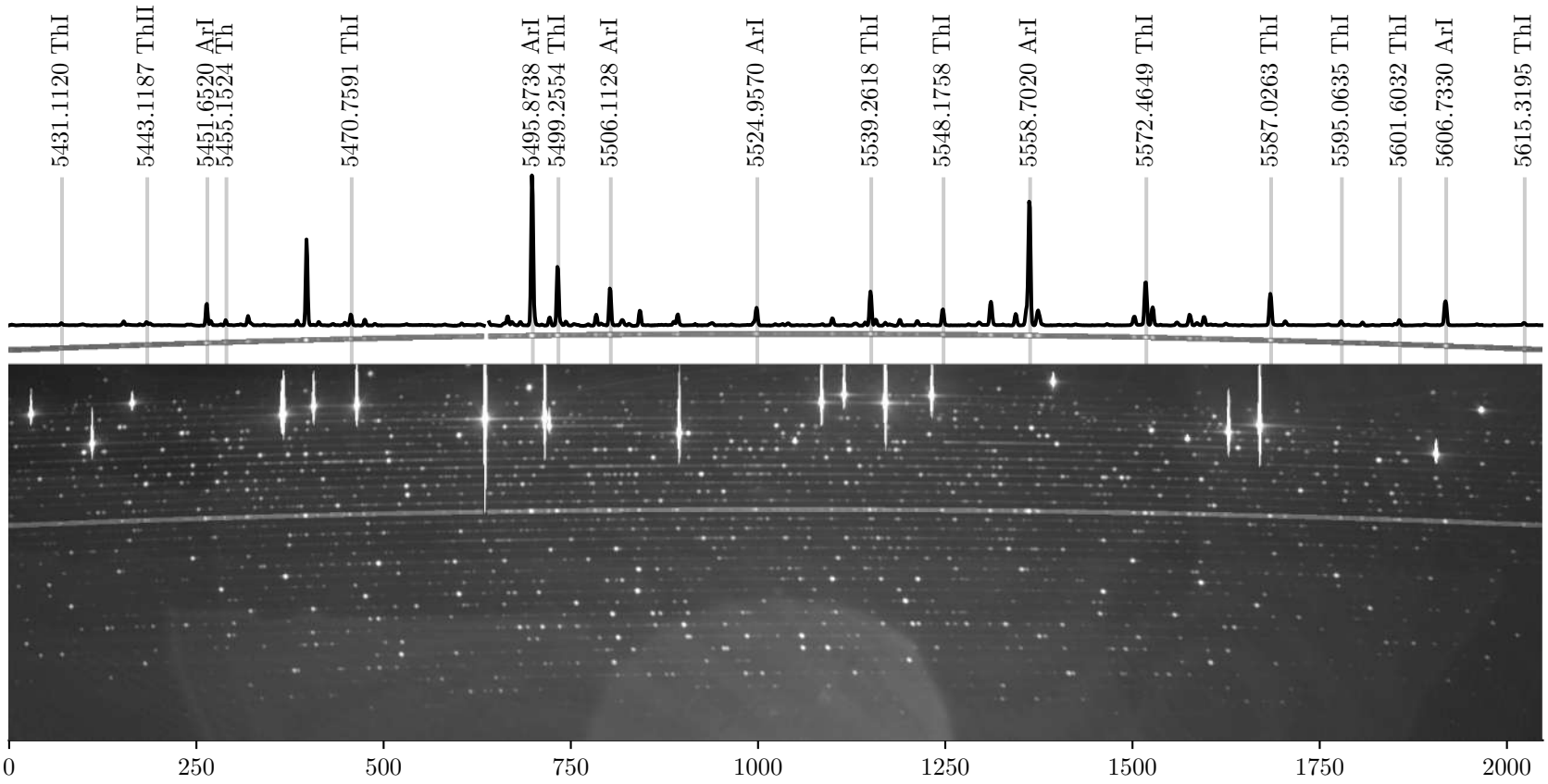
Approximate resolution:	17300
Dispersion:	0.092 Å pix^{-1}
Central wavelength:	5396 Å
Wavelength range:	5294 Å – 5484 Å


5294.3971 ThI
5295.0886 ThI
5296.2787 ThI
5297.4075 ThI
5297.7431 ThI
5298.2824 ThI
5298.5601 ThI
5300.5234 ThI
5301.4043 ThII
5303.4833 ThI
5306.9864 ThI
5307.4655 ThII
5308.3105 ThI
5309.6093 ThI
5310.2665 ThII
5310.8662 ThII
5312.5288 ThI
5312.9045 ThI
5315.2278 ThI
5317.4944 ThI
5318.1104 ThI
5318.4268 ThII
5295
5300
5305
5310
5315
5315.2278 ThI
5317.4944 ThI
5318.1104 ThI
5318.4268 ThII
5320.0985 Th
5320.7699 ThII
5322.8988 ThI
5323.4002 ThI
5325.1438 ThII
5325.4317 ThI
5326.2774 ThI
5326.9756 ThI
5329.3745 ThII
5329.7559 ThII
5330.0804 ThI
5331.7477 ThI
5333.3846 ThI
5337.0154 Th
5338.6405 ThI
5315
5320
5325
5330
5335
5337.0154 Th
5338.6405 ThI
5339.8952 Th
5340.4982 ThI
5343.5812 ThI
5345.3117 Th
5346.3778 ThII
5347.9714 ThI
5348.3092 ThI
5349.0054 ThI
5349.4608 Th
5350.5361 Th
5351.1265 ThI
5351.8365 ThI
5354.6016 ThI
5355.6365 ThI
5358.7059 ThI
5335
5340
5345
5350
5355

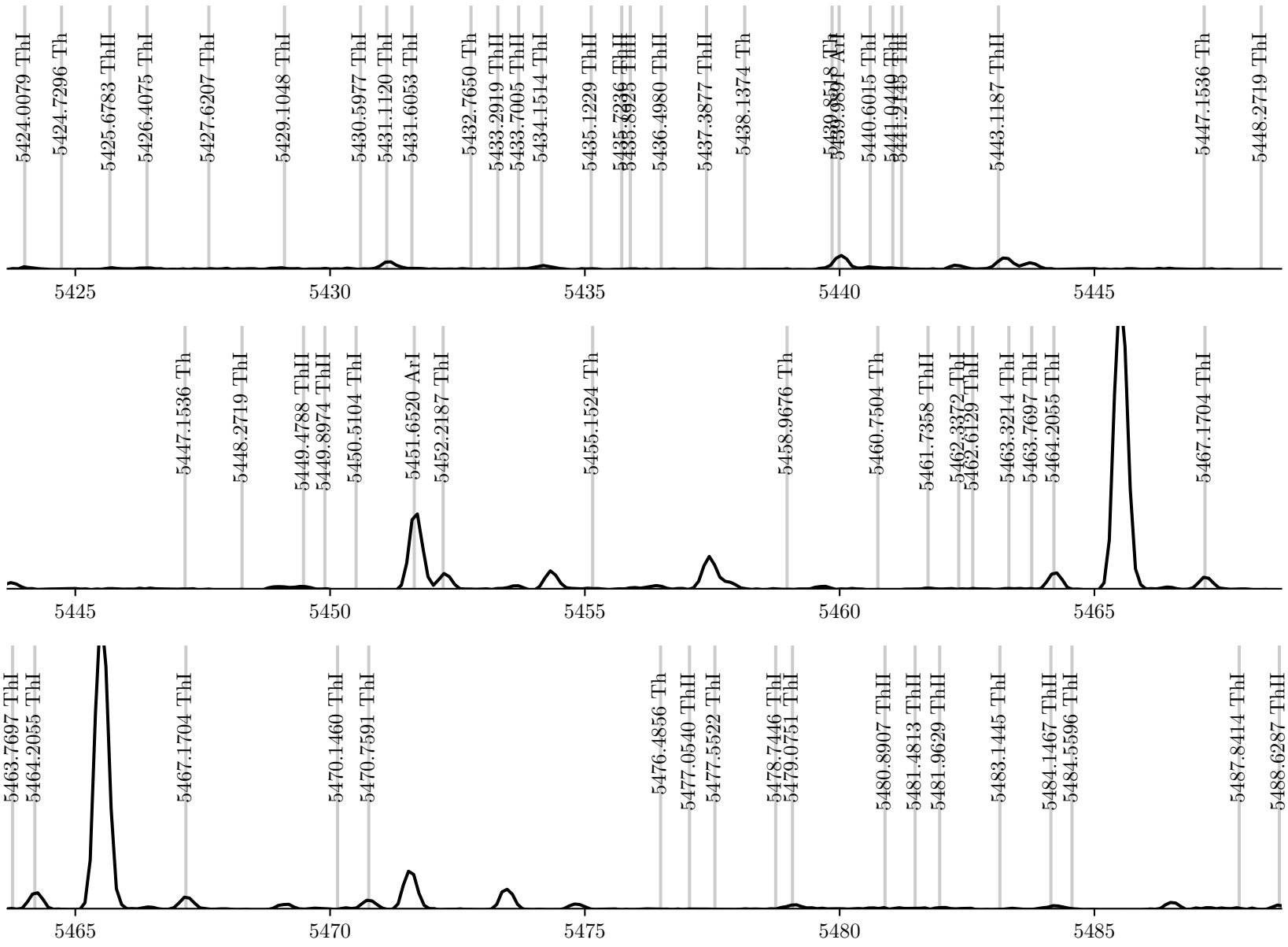

5354.6016 ThI
5355.6365 ThI
5358.7059 ThI
5359.8269 ThI
5360.1501 ThI
5361.1556 ThI
5361.4101 ThII
5362.2506 ThII
5362.5751 ThI
5363.6073 ThII
5365.4705 Th
5366.7066 ThI
5370.7096 ThI
5371.1238 ThI
5372.7027 ThI
5374.8219 Th
5375.3526 ThII
5375.7688 ThII
5376.1304 Th
5376.7809 ThI
5355
5360
5365
5370
5375
5374.8219 Th
5375.3526 ThII
5375.7688 ThII
5376.1304 Th
5376.7809 ThI
5378.8356 ThI
5379.1305 ThI
5381.3709 ThI
5382.9276 ThII
5384.0356 ThII
5386.6107 ThI
5387.8131 Th
5388.0507 ThI
5388.9061 ThII
5391.0760 ThII
5392.5726 ThII
5393.5995 ArII
5393.9719 ThI
5394.7608 ThI
5396.1127 ThI
5397.5169 ArII
5398.2047 ThI
5398.7015 ThI
5399.1745 Th
5399.7745 Th
5375
5380
5385
5390
5395
5394.7608 ThI
5396.1127 ThI
5397.5169 ArII
5398.2047 ThI
5398.7015 ThI
5399.1745 Th
5399.6218 ThI
5400.1455 ThI
5402.6048 ArII
5403.1993 ThI
5405.2097 ThII
5407.3439 ArII
5407.6535 ThI
5410.7687 ThI
5412.2212 Th
5414.5246 Th
5415.5226 ThI
5417.4858 ThI
5419.1285 ThI
5395
5400
5405
5410
5415

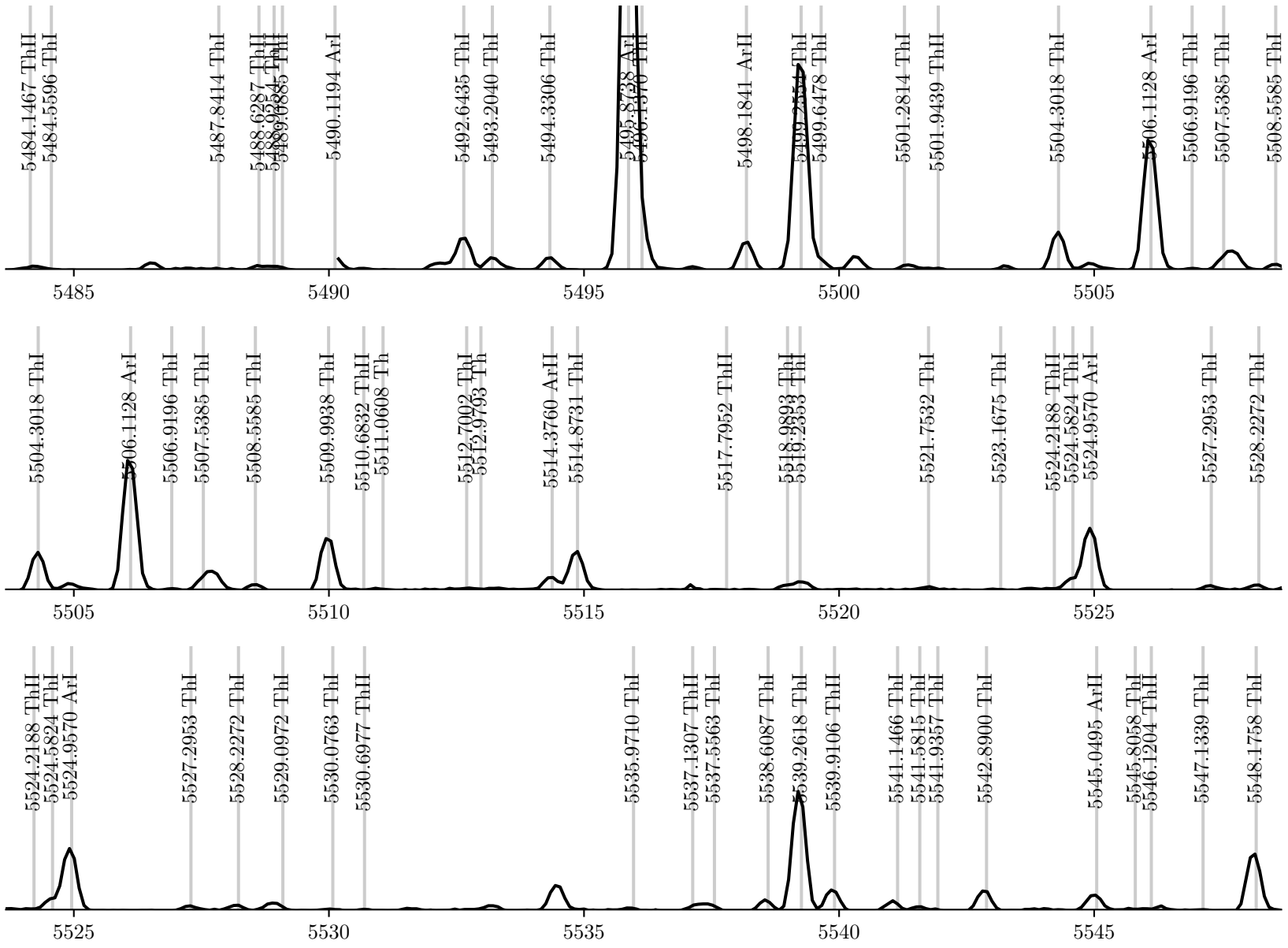

5414.5246 Th
5415.5226 ThI
5417.4858 ThI
5419.1285 ThI
5422.9151 ThI
5424.0079 ThI
5424.7296 Th
5425.6783 ThII
5426.4075 ThI
5427.6207 ThI
5429.1048 ThI
5430.5977 ThI
5431.1120 ThI
5431.6053 ThI
5432.7650 Th
5433.2919 ThII
5433.7005 ThII
5434.1514 ThI
5435.1229 ThII
5436.4980 ThII
5437.3877 ThII
5438.1374 Th
5415
5420
5425
5430
5435
5435.1229 ThII
5436.4980 ThII
5437.3877 ThII
5438.1374 Th
5440.6015 ThI
5443.1187 ThII
5447.1536 Th
5448.2719 ThI
5449.4788 ThII
5449.8974 ThII
5450.5104 ThI
5452.2187 ThI
5455.1524 Th
5458.9676 Th
5435
5440
5445
5450
5455
5455.1524 Th
5458.9676 Th
5460.7504 Th
5461.7358 ThII
5463.3214 ThI
5463.7697 ThI
5464.2055 ThI
5467.1704 ThI
5470.1460 ThI
5470.7591 ThI
5476.4856 Th
5477.0540 ThII
5477.5522 ThI
5478.7446 ThI
5479.0751 ThI
5455
5460
5465
5470
5475

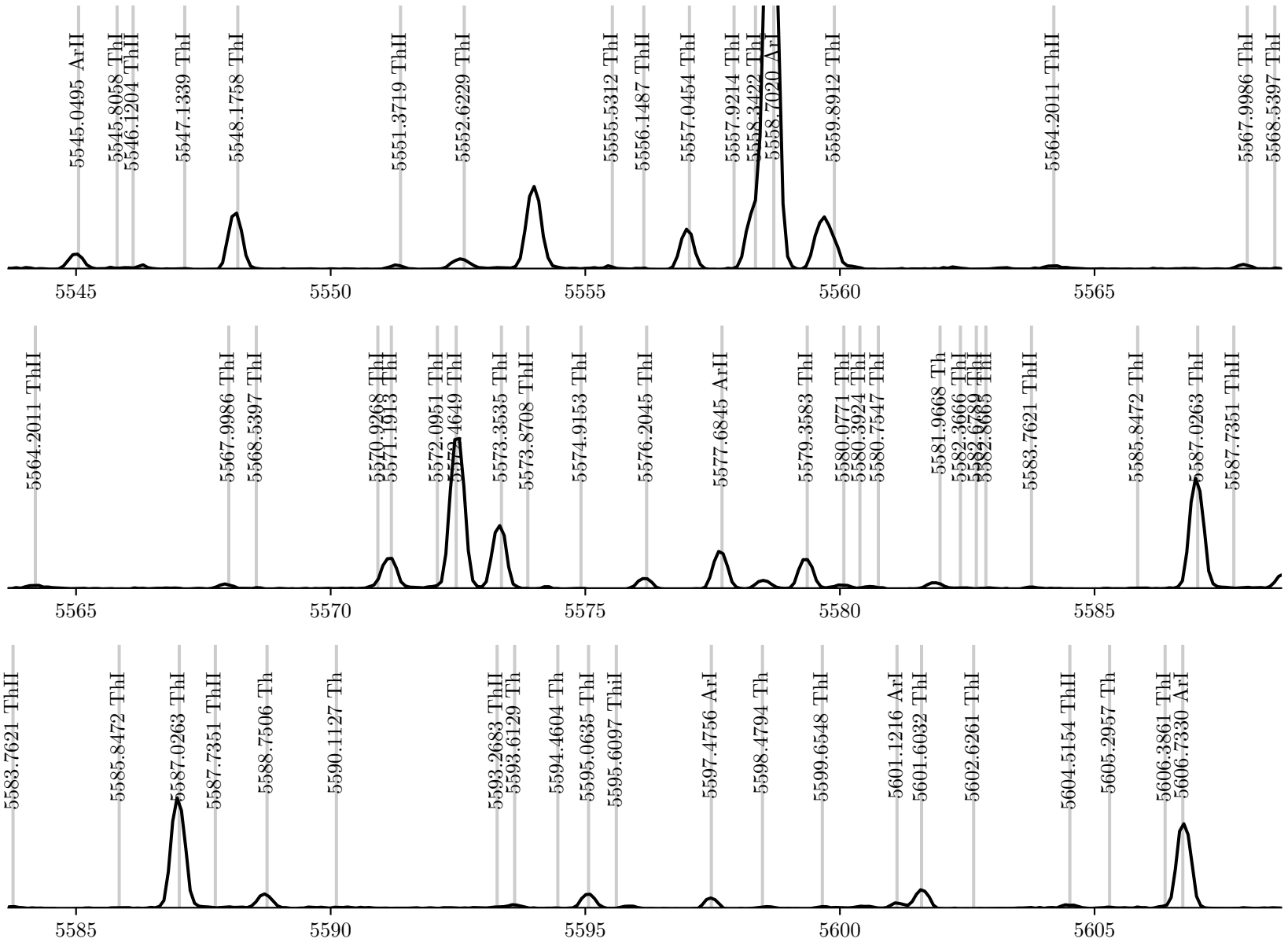
Order 41



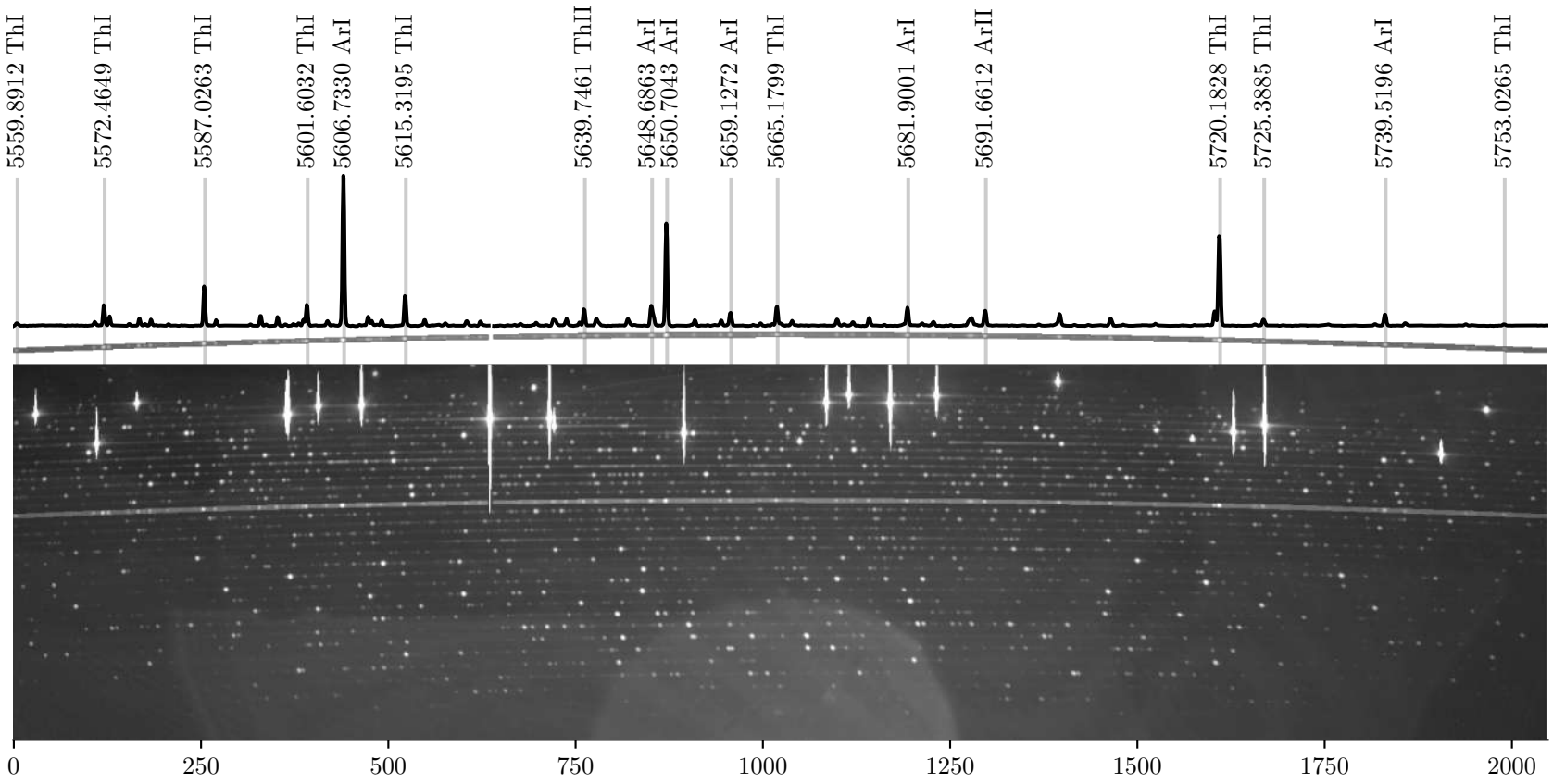
Approximate resolution:	17300
Dispersion:	0.095 Å pix^{-1}
Central wavelength:	5527 Å
Wavelength range:	5424 Å – 5617 Å


5424.0079 ThI
5424.7296 Th
5425.6783 ThII
5426.4075 ThI
5427.6207 ThI
5429.1048 ThI
5430.5977 ThI
5431.1120 ThI
5431.6053 ThI
5432.7650 Th
5433.2919 ThII
5433.7005 ThII
5434.1514 ThI
5435.1229 ThII
5436.4980 ThII
5437.3877 ThII
5438.1374 Th
5440.6015 ThI
5443.1187 ThII
5447.1536 Th
5448.2719 ThI
5425
5430
5435
5440
5445
5447.1536 Th
5448.2719 ThI
5449.4788 ThII
5449.8974 ThII
5450.5104 ThI
5451.6520 ArI
5452.2187 ThI
5455.1524 Th
5458.9676 Th
5460.7504 Th
5461.7358 ThII
5463.3214 ThI
5463.7697 ThI
5464.2055 ThI
5467.1704 ThI
5445
5450
5455
5460
5465
5463.7697 ThI
5464.2055 ThI
5467.1704 ThI
5470.1460 ThI
5470.7591 ThI
5476.4856 Th
5477.0540 ThII
5477.5522 ThI
5478.7446 ThI
5479.0751 ThI
5480.8907 ThII
5481.4813 ThII
5481.9629 ThII
5483.1445 ThI
5484.1467 ThII
5484.5596 ThI
5487.8414 ThI
5488.6287 ThII
5465
5470
5475
5480
5485

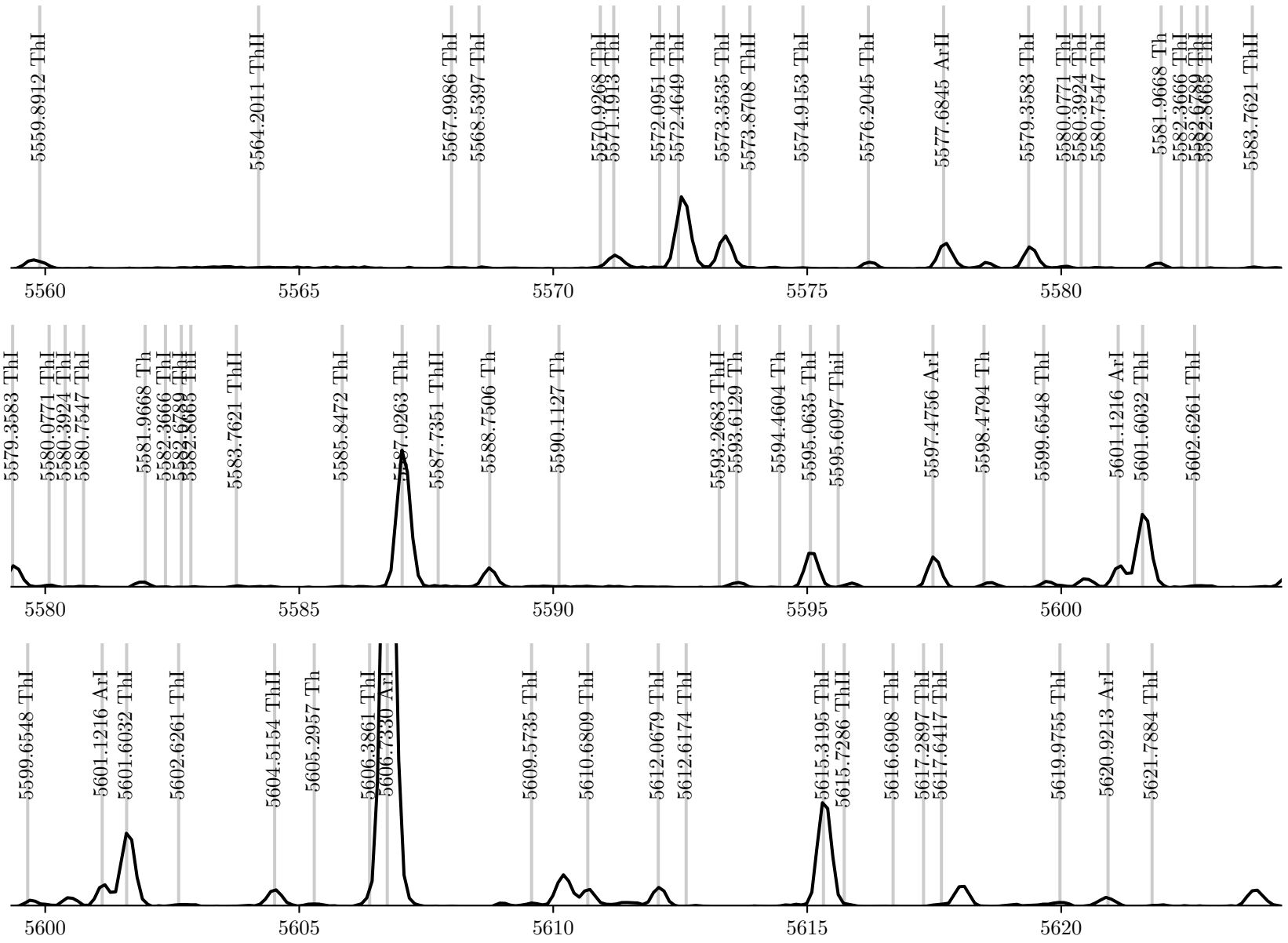

5484.1467 ThII
5484.5596 ThI
5487.8414 ThI
5488.6287 ThII
5490.1194 ArI
5492.6435 ThI
5493.2040 ThI
5494.3306 ThI
5495.8738 ArI
5498.1841 ArII
5499.6478 ThI
5501.2814 ThI
5501.9439 ThII
5504.3018 ThI
5506.1128 ArI
5506.9196 ThI
5507.5385 ThI
5508.5585 ThI
5485
5490
5495
5500
5505
5504.3018 ThI
5506.1128 ArI
5506.9196 ThI
5507.5385 ThI
5508.5585 ThI
5509.9938 ThI
5510.6832 ThII
5511.0608 Th
5512.7002 ThI
5514.3760 ArII
5514.8731 ThI
5517.7952 ThII
5521.7532 ThI
5523.1675 ThI
5524.2188 ThII
5524.5824 ThI
5524.9570 ArI
5527.2953 ThI
5528.2272 ThI
5505
5510
5515
5520
5525
5524.2188 ThII
5524.5824 ThI
5524.9570 ArI
5527.2953 ThI
5528.2272 ThI
5529.0972 ThI
5530.0763 ThI
5530.6977 ThII
5535.9710 ThI
5537.1307 ThII
5537.5563 ThI
5538.6087 ThI
5539.2618 ThI
5539.9106 ThII
5541.1466 ThI
5541.5815 ThI
5541.9357 ThI
5542.8900 ThI
5545.0495 ArII
5545.8058 ThI
5546.1204 ThII
5547.1339 ThI
5548.1758 ThI
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5545

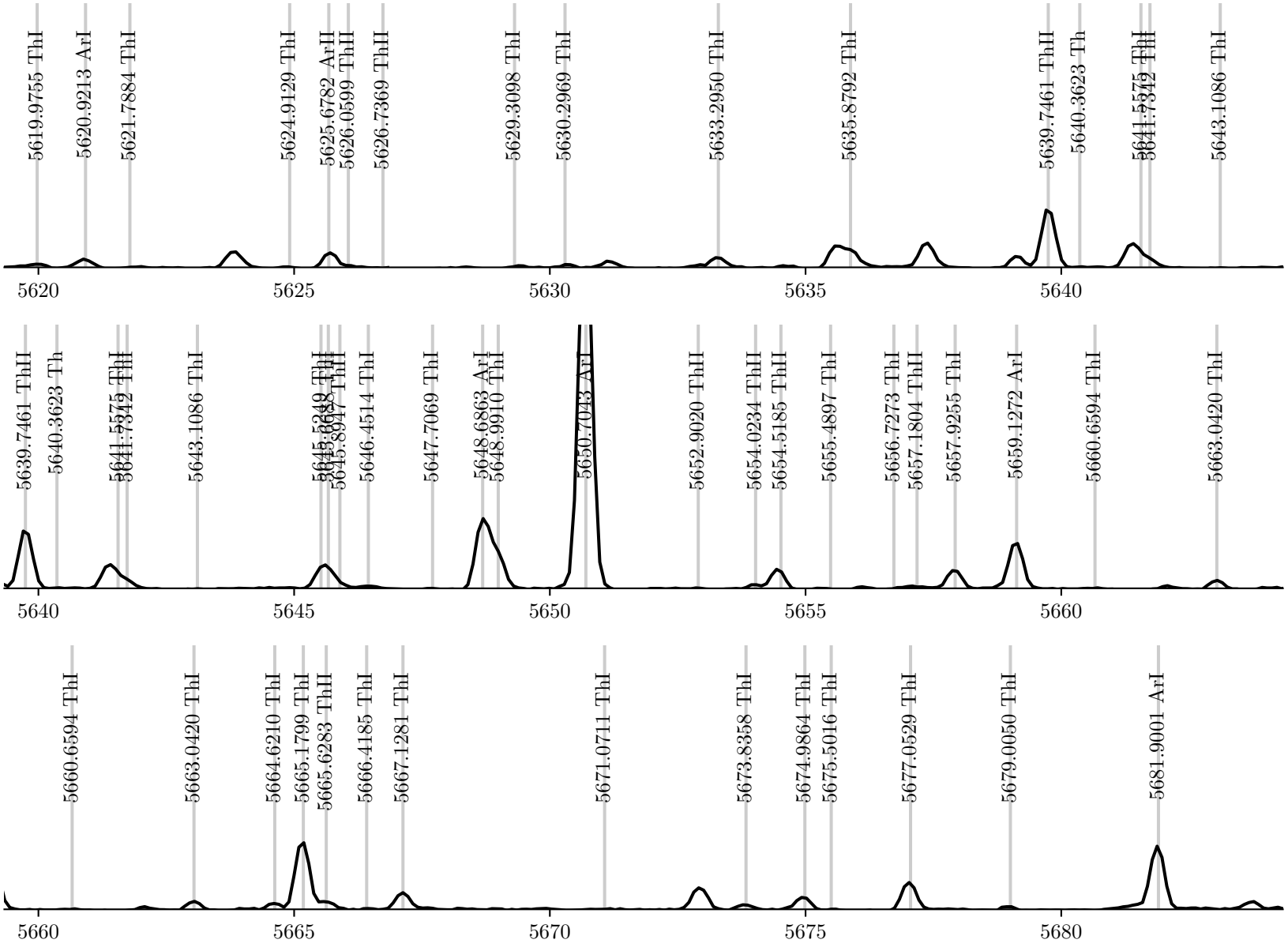

5545.0495 ArII
5545.8058 ThI
5546.1204 ThII
5547.1339 ThI
5548.1758 ThI
5551.3719 ThII
5552.6229 ThI
5555.5312 ThI
5556.1487 ThII
5557.0454 ThI
5557.9214 ThI
5558.3422 ThI
5558.7020 ArI
5559.8912 ThI
5564.2011 ThII
5567.9986 ThI
5568.5397 ThI
5545
5550
5555
5560
5565
5564.2011 ThII
5567.9986 ThI
5568.5397 ThI
5570.9268 ThI
5571.1913 ThI
5572.0951 ThI
5573.3535 ThI
5573.8708 ThII
5574.9153 ThI
5576.2045 ThI
5577.6845 ArII
5579.3583 ThI
5580.0771 ThI
5580.3924 ThI
5580.7547 ThI
5581.9668 Th
5583.7621 ThII
5585.8472 ThI
5587.0263 ThI
5587.7351 ThII
5565
5570
5575
5580
5585
5583.7621 ThII
5585.8472 ThI
5587.0263 ThI
5587.7351 ThII
5588.7506 Th
5590.1127 Th
5593.2683 ThII
5593.6129 Th
5594.4604 Th
5595.0635 ThI
5595.6097 ThiI
5597.4756 ArI
5598.4794 Th
5599.6548 ThI
5601.1216 ArI
5601.6032 ThI
5602.6261 ThI
5604.5154 ThII
5605.2957 Th
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5595
5600
5605

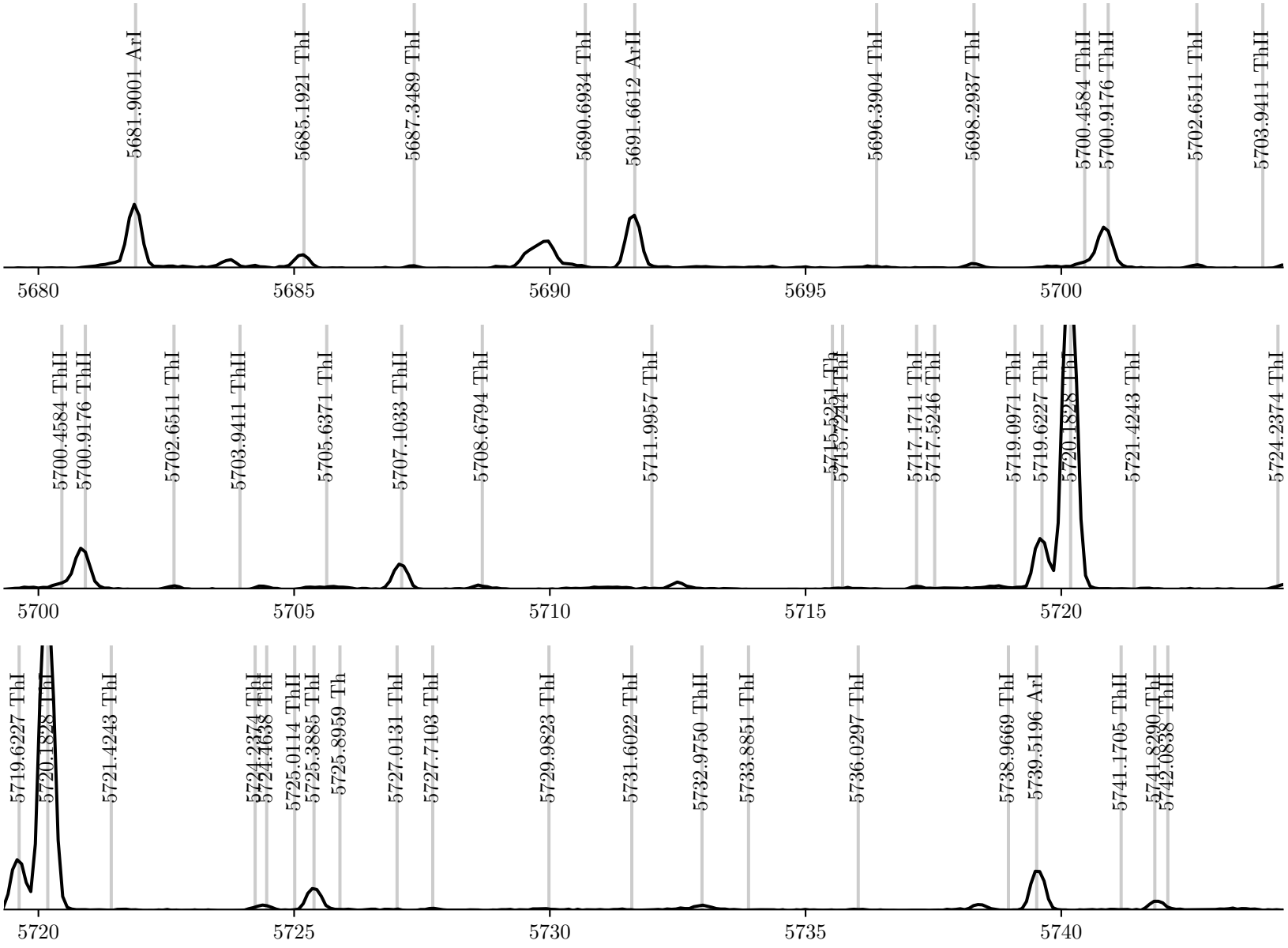
Order 40



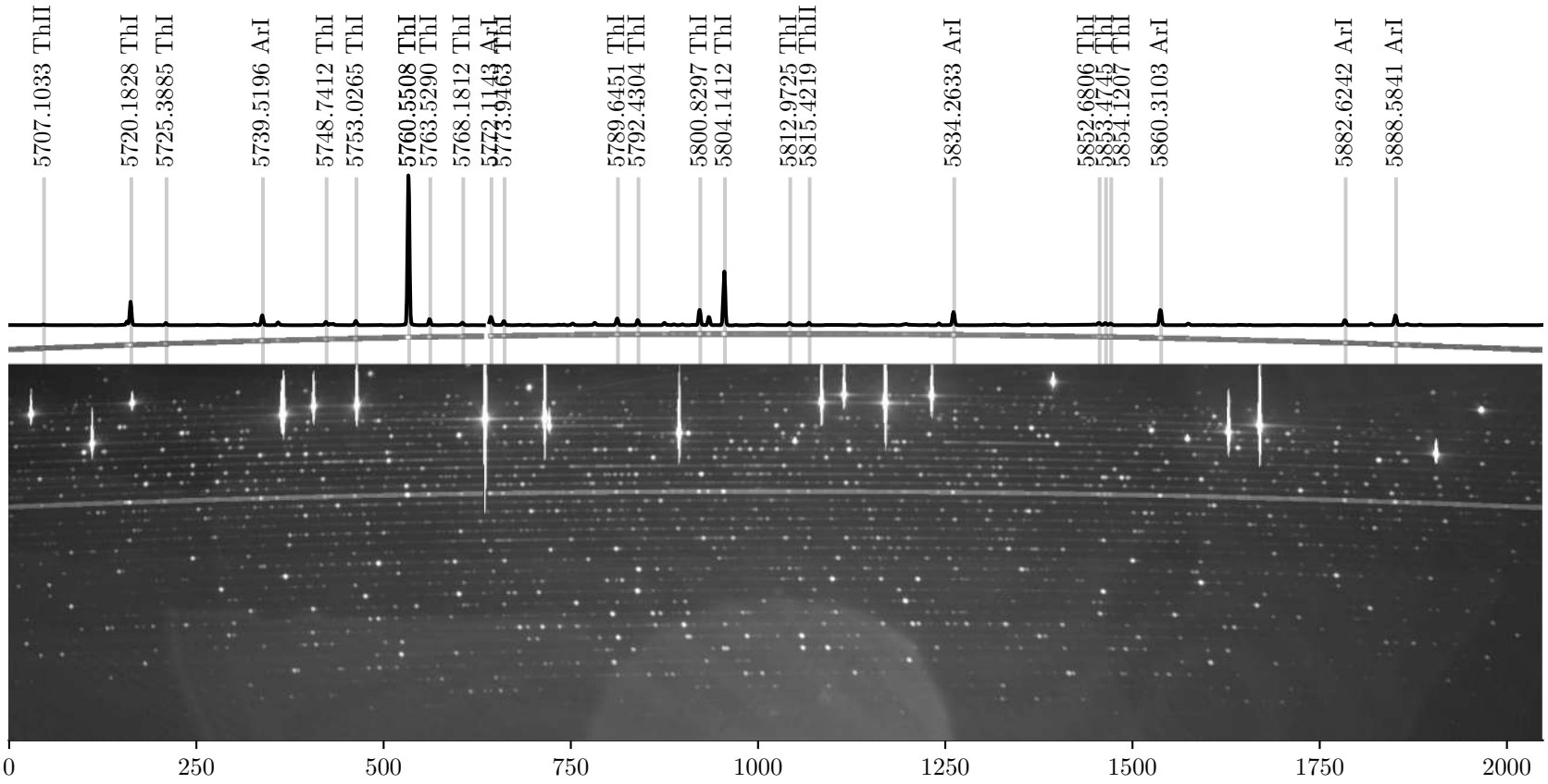
Approximate resolution:	17400
Dispersion:	0.097 Å pix^{-1}
Central wavelength:	5666 Å
Wavelength range:	5559 Å – 5758 Å


5559.8912 ThI
5564.2011 ThII
5567.9986 ThI
5568.5397 ThI
5570.9268 ThI
5571.1013 ThI
5572.0951 ThI
5572.4649 ThI
5573.3535 ThI
5573.8708 ThII
5574.9153 ThI
5576.2045 ThI
5577.6845 ArII
5579.3583 ThI
5580.0771 ThI
5580.3924 ThI
5580.7547 ThI
5581.9668 Th
5582.3666 ThI
5583.7621 ThII
5585.8472 ThI
5587.0263 ThI
5587.7351 ThII
5588.7506 Th
5590.1127 Th
5593.2683 ThII
5593.6129 Th
5594.4604 Th
5595.0635 ThI
5595.6097 ThiI
5597.4756 ArI
5598.4794 Th
5599.6548 ThI
5601.1216 ArI
5601.6032 ThI
5602.6261 ThI
5604.5154 ThII
5605.2957 Th
5606.3861 ThI
5606.7330 ArI
5609.5735 ThI
5610.6809 ThI
5612.0679 ThI
5612.6174 ThI
5615.3195 ThI
5615.7286 ThII
5616.6908 ThI
5617.2897 ThI
5617.6417 ThI
5619.9755 ThI
5620.9213 ArI
5621.7884 ThI
5560
5565
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5575
5580
5580
5585
5590
5595
5600
5600
5605
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5615
5620

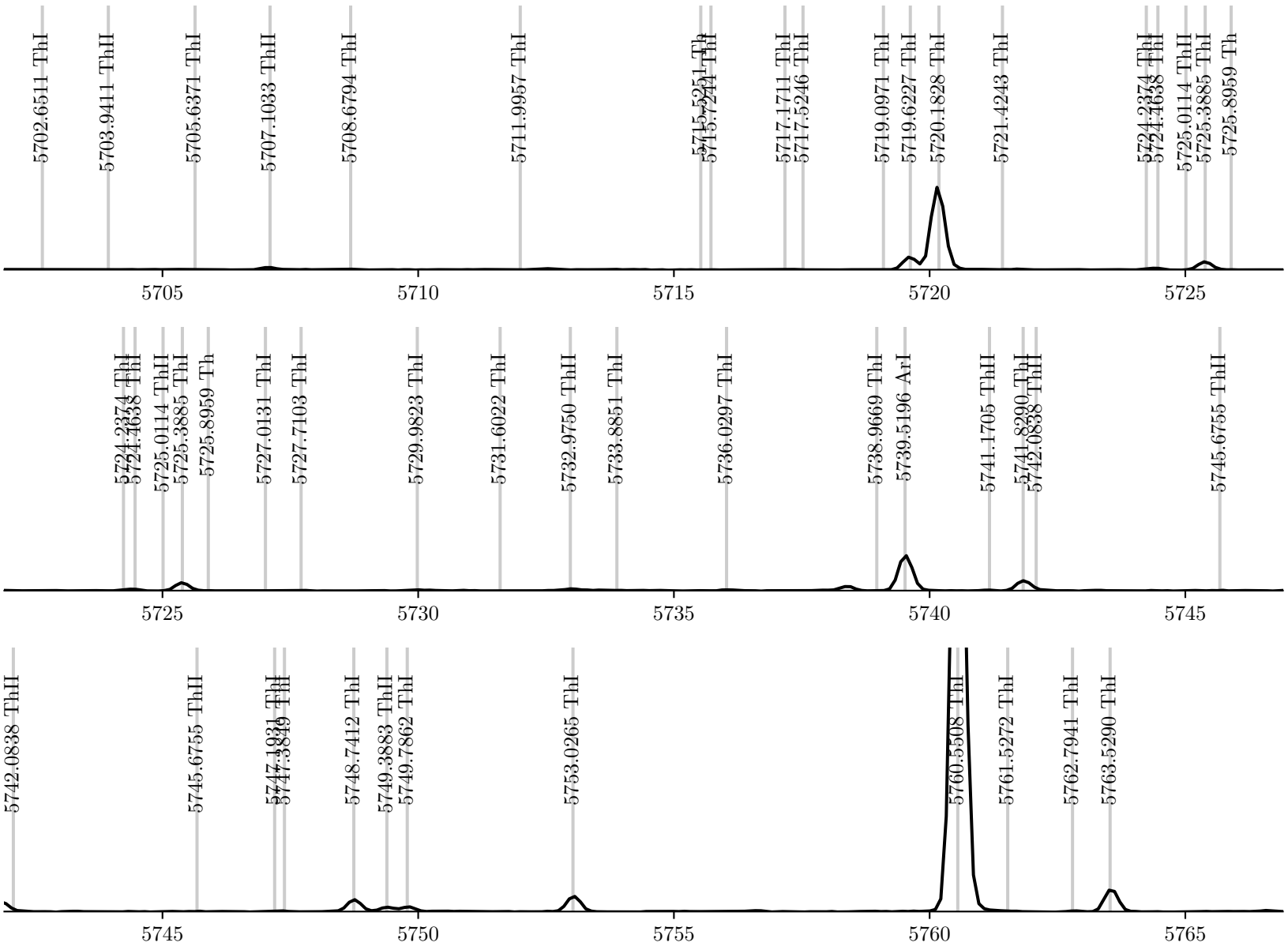

5619.9755 ThI
5620.9213 ArI
5621.7884 ThI
5624.9129 ThI
5625.6782 ArII
5626.0599 ThII
5626.7369 ThII
5629.3098 ThI
5630.2969 ThI
5633.2950 ThI
5635.8792 ThI
5639.7461 ThII
5640.3623 Th
5643.1086 ThI
5620
5625
5630
5635
5640
5639.7461 ThII
5640.3623 Th
5643.1086 ThI
5646.4514 ThI
5647.7069 ThI
5648.6863 ArI
5648.9910 ThI
5650.7043 ArI
5652.9020 ThII
5654.0234 ThII
5654.5185 ThII
5655.4897 ThI
5656.7273 ThI
5657.1804 ThII
5657.9255 ThI
5659.1272 ArI
5660.6594 ThI
5663.0420 ThI
5640
5645
5650
5655
5660
5660.6594 ThI
5663.0420 ThI
5664.6210 ThI
5665.1799 ThI
5665.6283 ThII
5666.4185 ThI
5667.1281 ThI
5671.0711 ThI
5673.8358 ThI
5674.9864 ThI
5675.5016 ThI
5677.0529 ThI
5679.0050 ThI
5681.9001 ArI
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5670
5675
5680

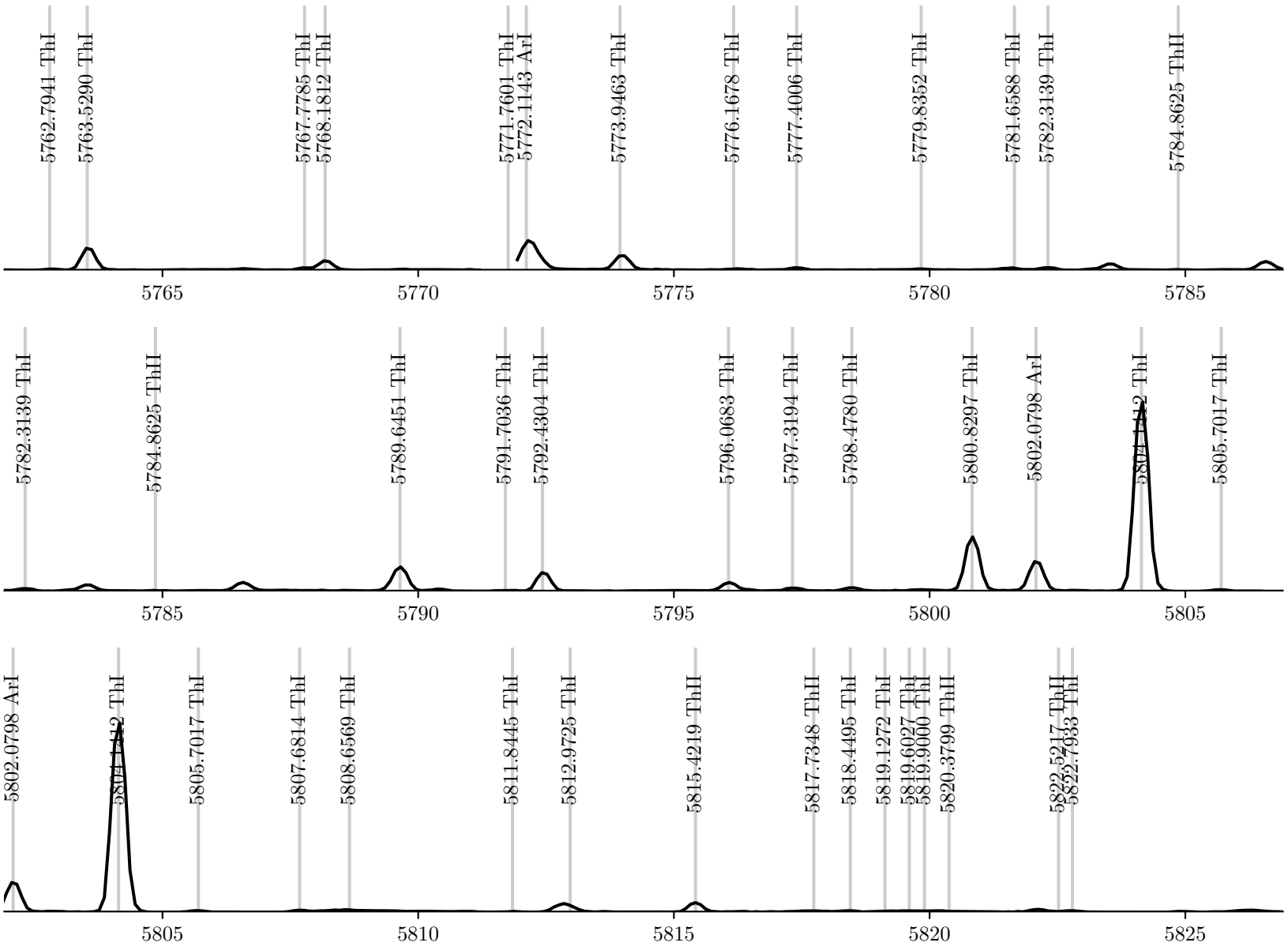

5681.9001 ArI
5685.1921 ThI
5687.3489 ThI
5690.6934 ThI
5691.6612 ArII
5696.3904 ThI
5698.2937 ThI
5700.4584 ThII
5700.9176 ThII
5702.6511 ThI
5703.9411 ThII
5680
5685
5690
5695
5700
5700.4584 ThII
5700.9176 ThII
5702.6511 ThI
5703.9411 ThII
5705.6371 ThI
5707.1033 ThII
5708.6794 ThI
5711.9957 ThI
5715.5251 Th
5715.7241 ThI
5717.1711 ThI
5717.5246 ThI
5719.0971 ThI
5719.6227 ThI
5720.1828 ThI
5721.4243 ThI
5724.2374 ThI
5700
5705
5710
5715
5720
5719.6227 ThI
5720.1828 ThI
5721.4243 ThI
5724.2374 ThI
5724.4638 ThI
5725.0114 ThII
5725.3885 ThI
5725.8959 Th
5727.0131 ThI
5727.7103 ThI
5729.9823 ThI
5731.6022 ThI
5732.9750 ThII
5733.8851 ThI
5736.0297 ThI
5738.9669 ThI
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5742.0838 ThII
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5725
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5735
5740

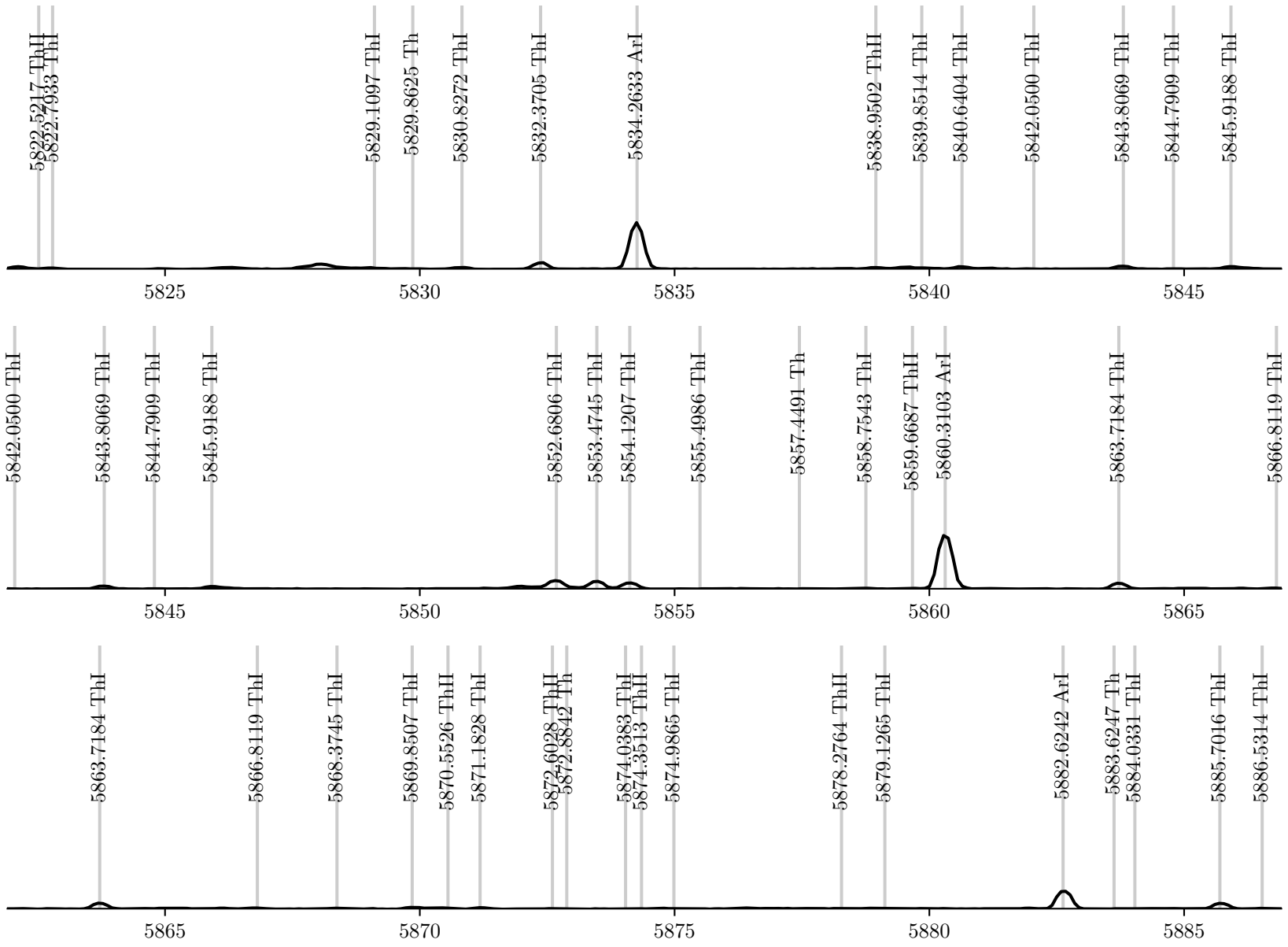
Order 39

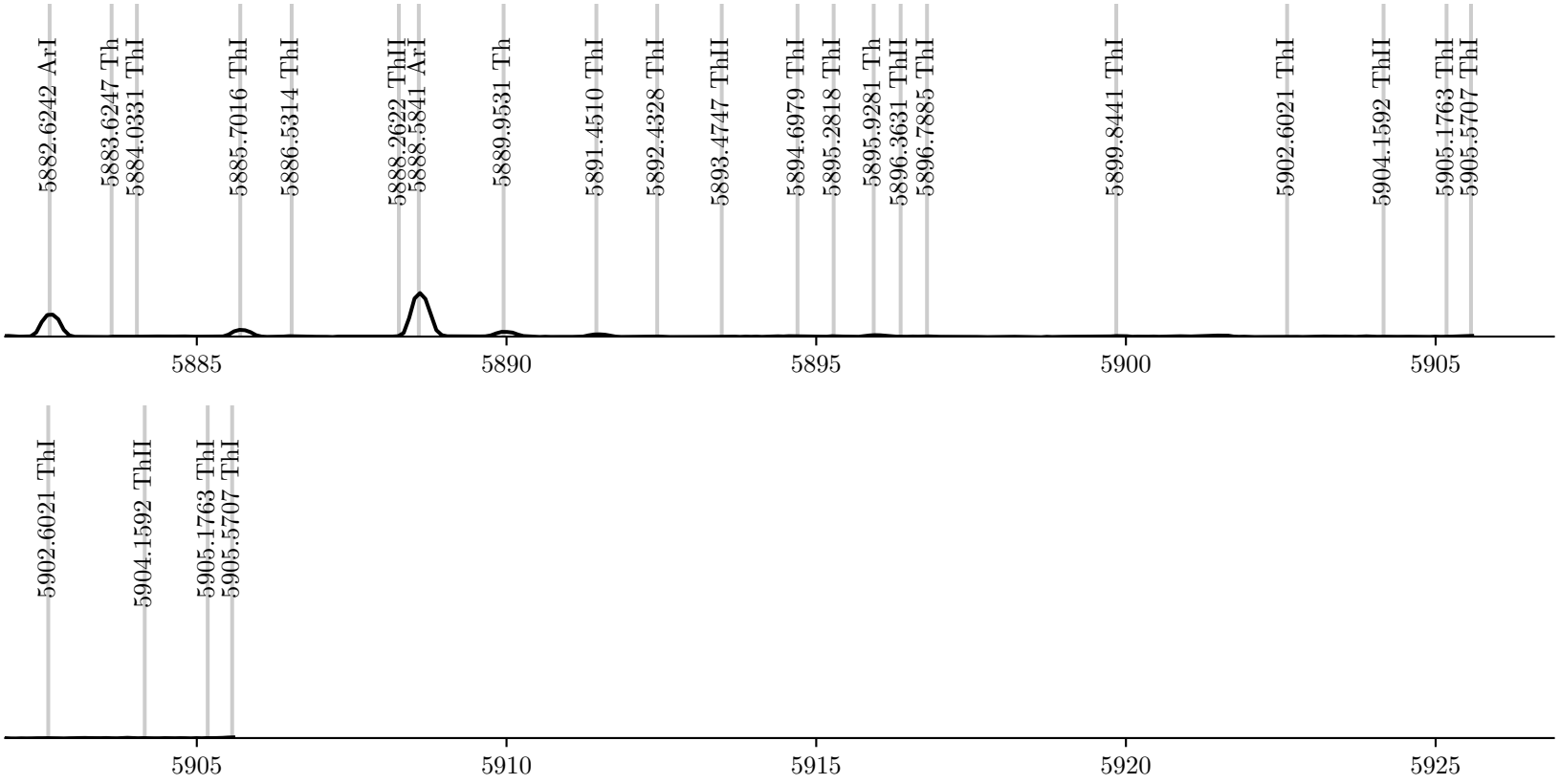


Approximate resolution:	17400
Dispersion:	0.100 Å pix^{-1}
Central wavelength:	5811 Å
Wavelength range:	5702 Å – 5906 Å

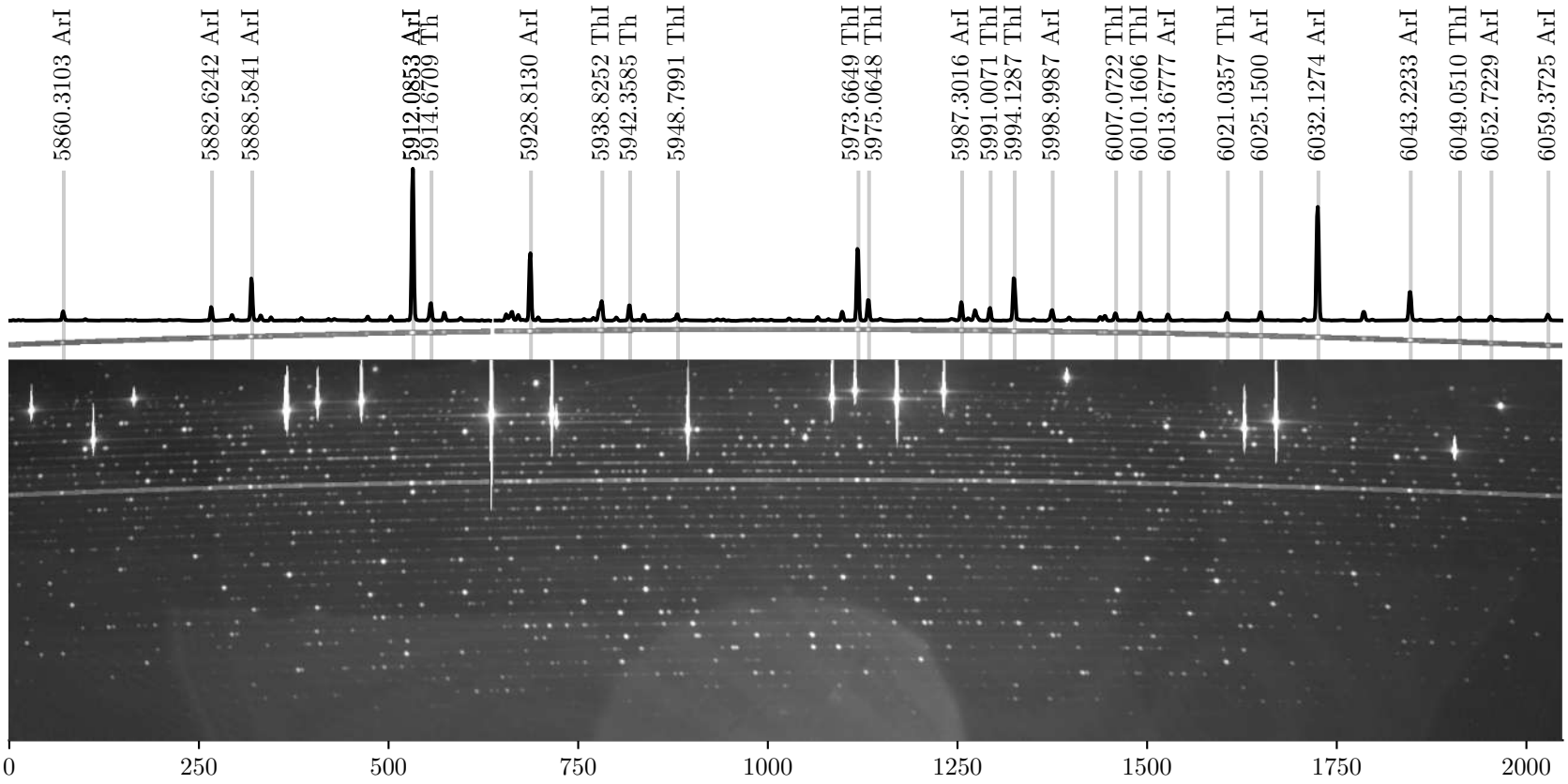

5702.6511 ThI
5703.9411 ThII
5705.6371 ThI
5707.1033 ThII
5708.6794 ThI
5711.9957 ThI
5715.5251 Th
5715.7241 ThI
5717.1711 ThI
5717.5246 ThI
5719.0971 ThI
5719.6227 ThI
5720.1828 ThI
5721.4243 ThI
5724.2374 ThI
5724.4638 ThI
5725.0114 ThII
5725.3885 ThI
5725.8959 Th
5705
5710
5715
5720
5725
5724.2374 ThI
5724.4638 ThI
5725.0114 ThII
5725.3885 ThI
5725.8959 Th
5727.0131 ThI
5727.7103 ThI
5729.9823 ThI
5731.6022 ThI
5732.9750 ThII
5733.8851 ThI
5736.0297 ThI
5738.9669 ThI
5739.5196 ArI
5741.1705 ThII
5741.8290 ThI
5742.0838 ThII
5745.6755 ThII
5725
5730
5735
5740
5745
5742.0838 ThII
5745.6755 ThII
5747.1931 ThI
5747.3849 ThI
5748.7412 ThI
5749.3883 ThII
5749.7862 ThI
5753.0265 ThI
5760.5508 ThI
5761.5272 ThI
5762.7941 ThI
5763.5290 ThI
5745
5750
5755
5760
5765


5762.7941 ThI
5763.5290 ThI
5767.7785 ThI
5768.1812 ThI
5771.7601 ThI
5772.1143 ArI
5773.9463 ThI
5776.1678 ThI
5777.4006 ThI
5779.8352 ThI
5781.6588 ThI
5782.3139 ThI
5784.8625 ThII
5765
5770
5775
5780
5785
5782.3139 ThI
5784.8625 ThII
5789.6451 ThI
5791.7036 ThI
5792.4304 ThI
5796.0683 ThI
5797.3194 ThI
5798.4780 ThI
5800.8297 ThI
5802.0798 ArI
5804.1412 ThI
5805.7017 ThI
5785
5790
5795
5800
5805
5802.0798 ArI
5804.1412 ThI
5805.7017 ThI
5807.6814 ThI
5808.6569 ThI
5811.8445 ThI
5812.9725 ThI
5815.4219 ThII
5817.7348 ThII
5818.4495 ThI
5819.1272 ThI
5819.6027 ThI
5819.9000 ThI
5820.3799 ThII
5822.5217 ThII
5822.7933 ThI
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5815
5820
5825

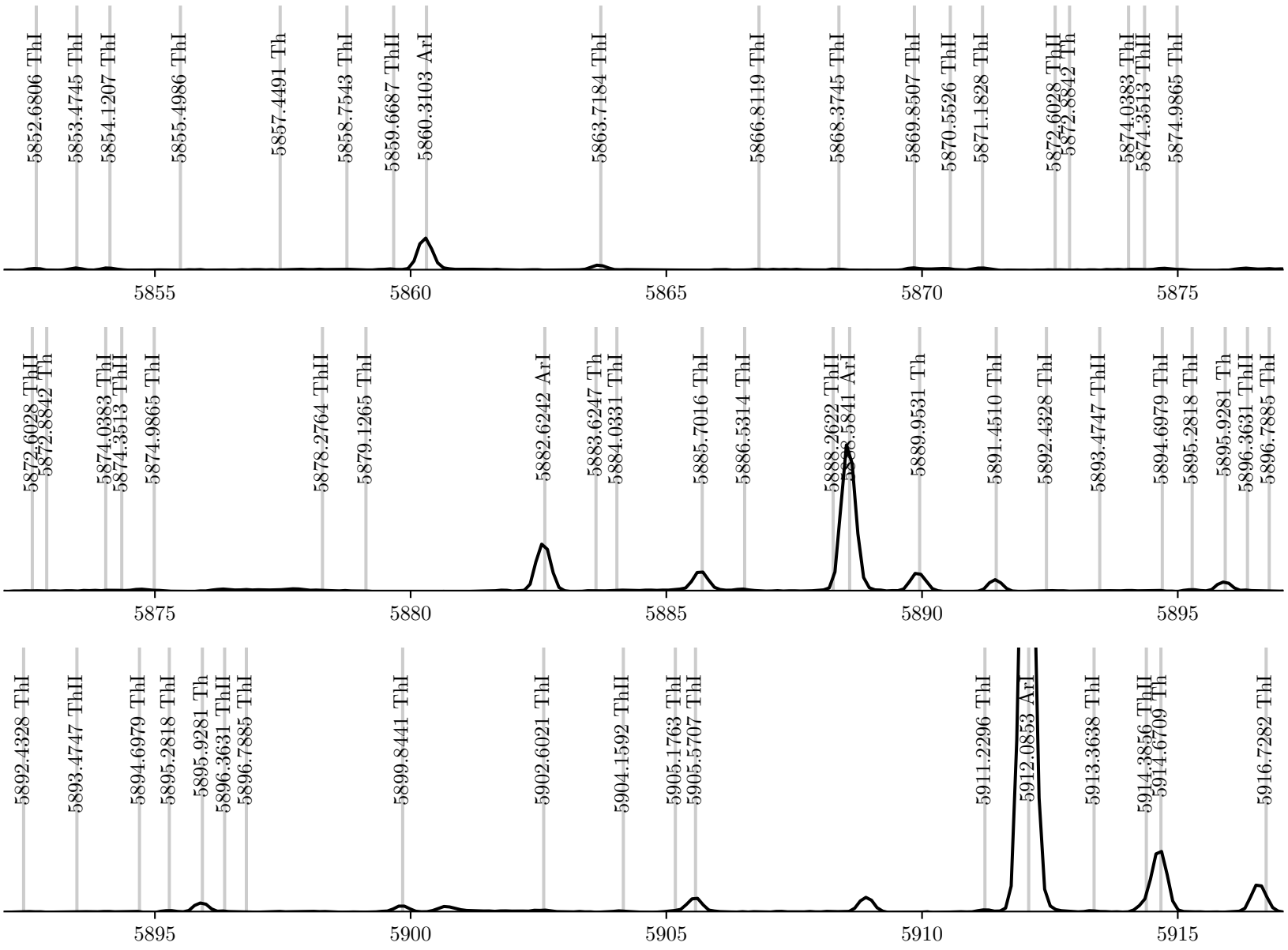

5822.5217 ThII
5822.7933 ThI
5829.1097 ThI
5829.8625 Th
5830.8272 ThI
5832.3705 ThI
5834.2633 ArI
5838.9502 ThII
5839.8514 ThI
5840.6404 ThI
5842.0500 ThI
5843.8069 ThI
5844.7909 ThI
5845.9188 ThI
5825
5830
5835
5840
5845
5842.0500 ThI
5843.8069 ThI
5844.7909 ThI
5845.9188 ThI
5852.6806 ThI
5853.4745 ThI
5854.1207 ThI
5855.4986 ThI
5857.4491 Th
5858.7543 ThI
5859.6687 ThII
5860.3103 ArI
5863.7184 ThI
5866.8119 ThI
5845
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5863.7184 ThI
5866.8119 ThI
5868.3745 ThI
5869.8507 ThI
5870.5526 ThII
5871.1828 ThI
5872.6028 ThII
5872.8842 Th
5874.0383 ThI
5874.3513 ThII
5874.9865 ThI
5878.2764 ThII
5879.1265 ThI
5882.6242 ArI
5883.6247 Th
5884.0331 ThI
5885.7016 ThI
5886.5314 ThI
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5880
5885

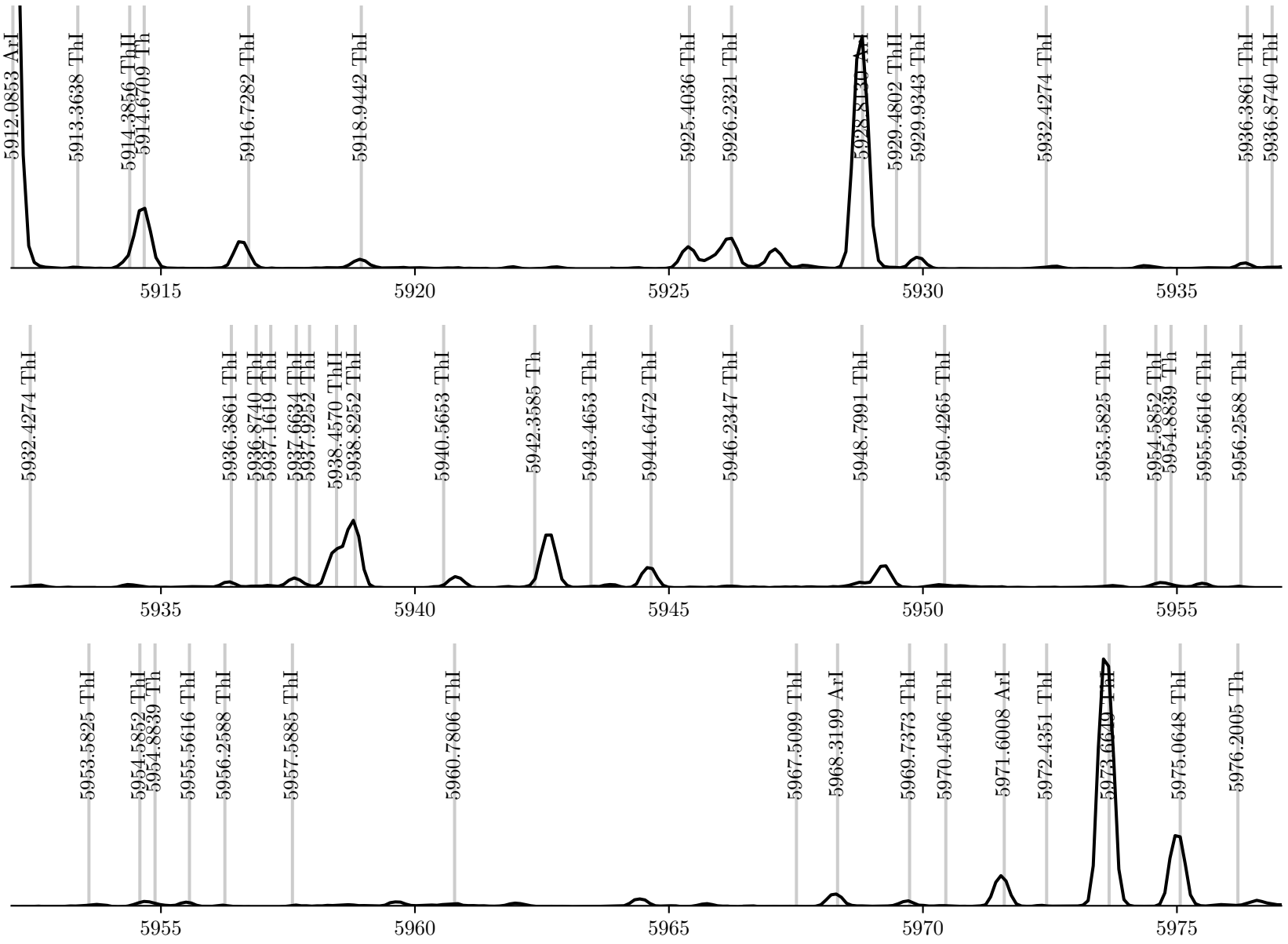

5882.6242 ArI
5883.6247 Th
5884.0331 ThI
5885.7016 ThI
5886.5314 ThI
5888.2622 ThII
5888.5841 ArI
5889.9531 Th
5891.4510 ThI
5892.4328 ThI
5893.4747 ThII
5894.6979 ThI
5895.2818 ThI
5895.9281 Th
5896.3631 ThII
5896.7885 ThI
5899.8441 ThI
5902.6021 ThI
5904.1592 ThII
5905.1763 ThI
5905.5707 ThI
5885
5890
5895
5900
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5902.6021 ThI
5904.1592 ThII
5905.1763 ThI
5905.5707 ThI
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5920
5925

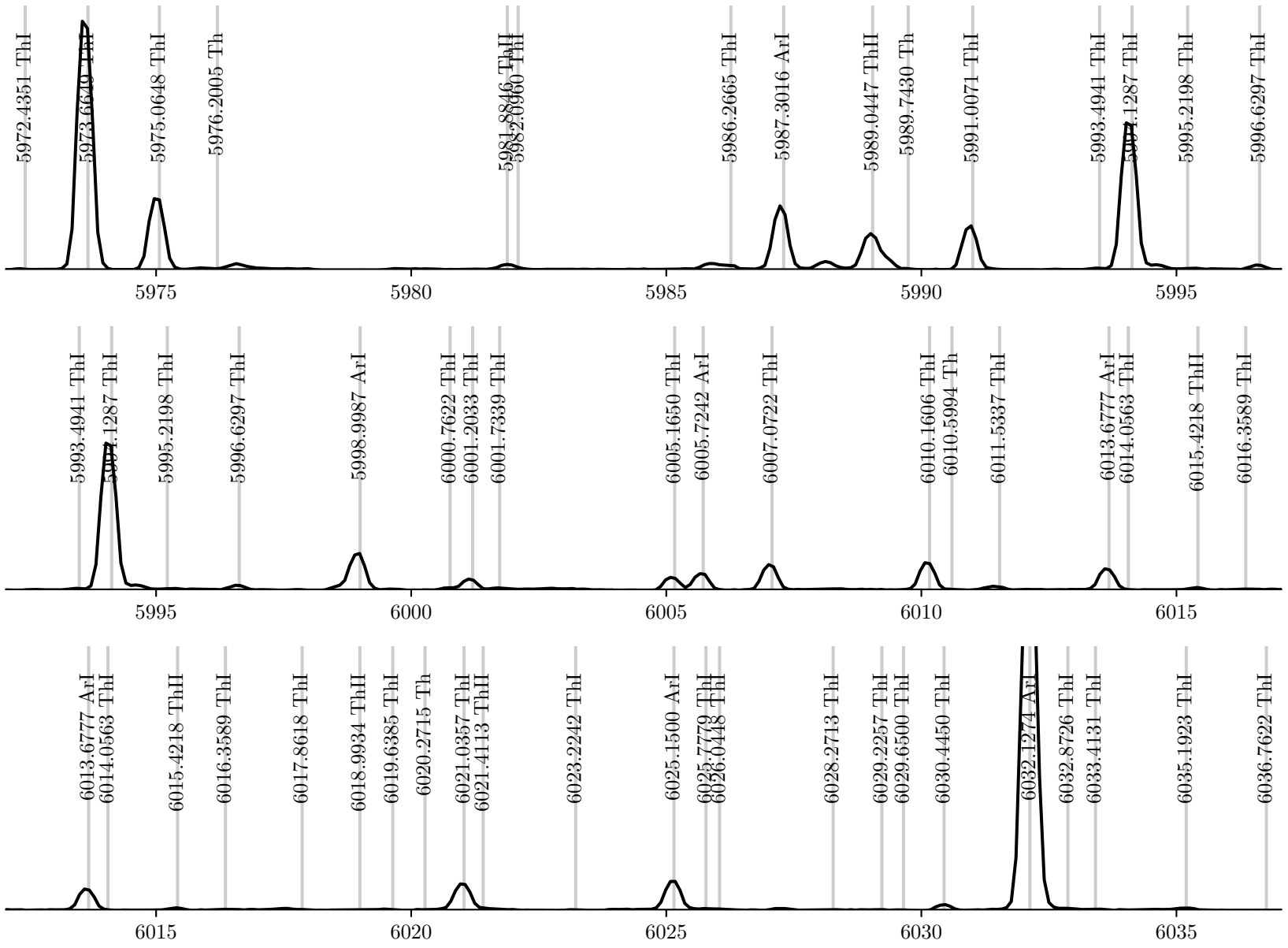
Order 38

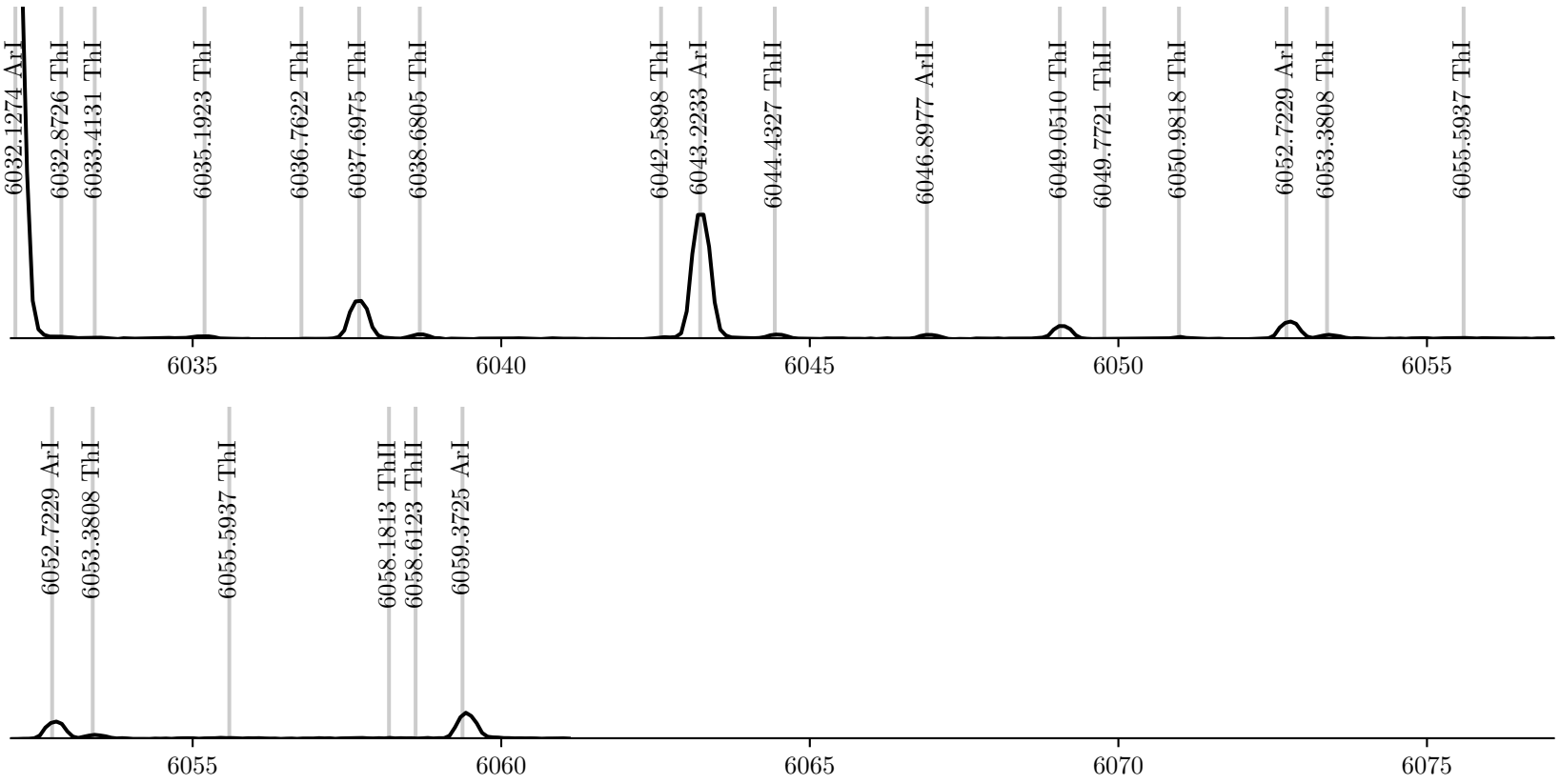


Approximate resolution:	17100
Dispersion:	0.102 Å pix^{-1}
Central wavelength:	5964 Å
Wavelength range:	5852 Å – 6061 Å

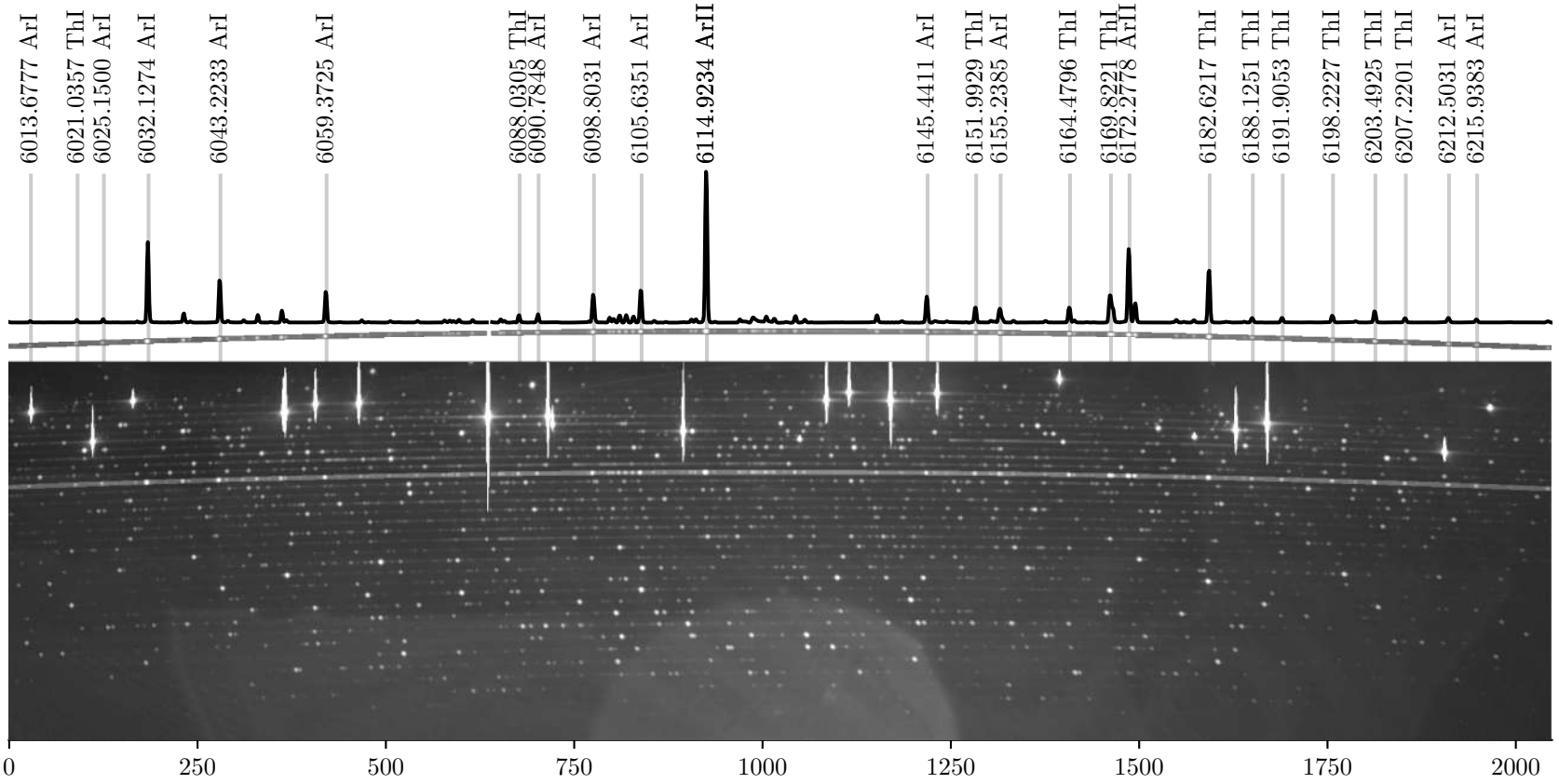

5852.6806 ThI
5853.4745 ThI
5854.1207 ThI
5855.4986 ThI
5857.4491 Th
5858.7543 ThI
5859.6687 ThII
5860.3103 ArI
5863.7184 ThI
5866.8119 ThI
5868.3745 ThI
5869.8507 ThI
5870.5526 ThII
5871.1828 ThI
5872.6028 ThII
5872.8842 Th
5874.0383 ThI
5874.3513 ThII
5874.9865 ThI
5855
5860
5865
5870
5875
5872.6028 ThII
5872.8842 Th
5874.0383 ThI
5874.3513 ThII
5874.9865 ThI
5878.2764 ThII
5879.1265 ThI
5882.6242 ArI
5883.6247 Th
5884.0331 ThI
5885.7016 ThI
5886.5314 ThI
5888.2622 ThII
5888.5841 ArI
5889.9531 Th
5891.4510 ThI
5892.4328 ThI
5893.4747 ThII
5894.6979 ThI
5895.2818 ThI
5895.9281 Th
5896.3631 ThII
5896.7885 ThI
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5880
5885
5890
5895
5892.4328 ThI
5893.4747 ThII
5894.6979 ThI
5895.2818 ThI
5895.9281 Th
5896.3631 ThII
5896.7885 ThI
5899.8441 ThI
5902.6021 ThI
5904.1592 ThII
5905.1763 ThI
5905.5707 ThI
5911.2296 ThI
5912.0853 ArI
5913.3638 ThI
5914.3856 ThII
5914.6709 Th
5916.7282 ThI
5895
5900
5905
5910
5915


5912.0853 ArI
5913.3638 ThI
5914.3856 ThII
5914.6709 Th
5916.7282 ThI
5918.9442 ThI
5925.4036 ThI
5926.2321 ThI
5928.8130 ArI
5929.4802 ThII
5929.9343 ThI
5932.4274 ThI
5936.3861 ThI
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5937.1619 ThI
5937.6634 ThI
5937.9252 ThI
5938.4570 ThII
5938.8252 ThI
5940.5653 ThI
5942.3585 Th
5943.4653 ThI
5944.6472 ThI
5946.2347 ThI
5948.7991 ThI
5950.4265 ThI
5953.5825 ThI
5954.5852 ThI
5954.8839 Th
5955.5616 ThI
5956.2588 ThI
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5953.5825 ThI
5954.5852 ThI
5954.8839 Th
5955.5616 ThI
5956.2588 ThI
5957.5885 ThI
5960.7806 ThI
5967.5099 ThI
5968.3199 ArI
5969.7373 ThI
5970.4506 ThI
5971.6008 ArI
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5973.6649 ThII
5975.0648 ThI
5976.2005 Th
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5965
5970
5975

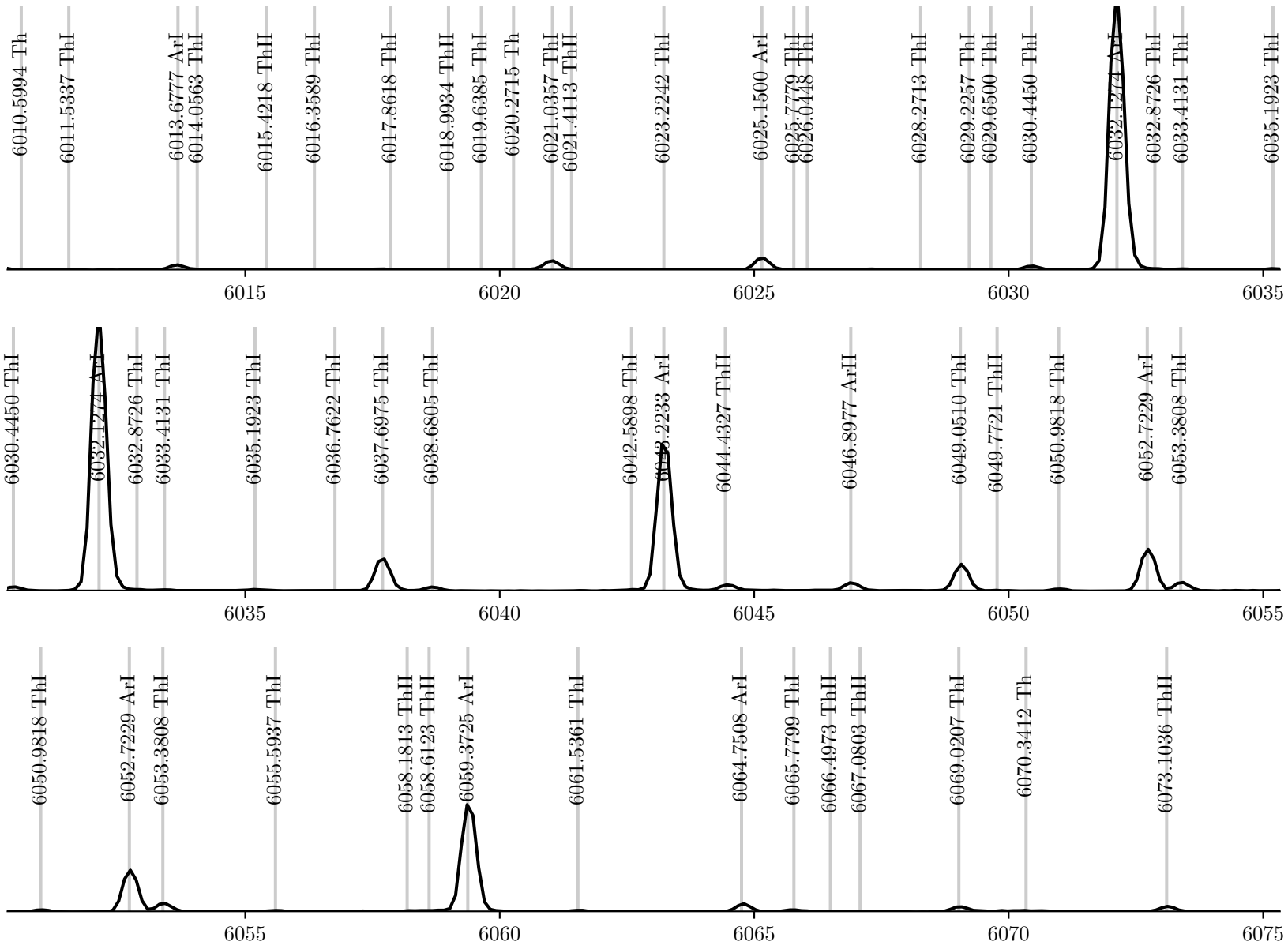

5972.4351 ThI
5973.6619 ThI
5975.0648 ThI
5976.2005 Th
5981.8846 ThII
5982.0960 ThI
5986.2665 ThI
5987.3016 ArI
5989.0447 ThII
5989.7430 Th
5991.0071 ThI
5993.4941 ThI
5994.1287 ThI
5995.2198 ThI
5996.6297 ThI
5975
5980
5985
5990
5995
5993.4941 ThI
5994.1287 ThI
5995.2198 ThI
5996.6297 ThI
5998.9987 ArI
6000.7622 ThI
6001.2033 ThI
6001.7339 ThI
6005.1650 ThI
6005.7242 ArI
6007.0722 ThI
6010.1606 ThI
6010.5994 Th
6011.5337 ThI
6013.6777 ArI
6014.0563 ThI
6015.4218 ThII
6016.3589 ThI
5995
6000
6005
6010
6015
6013.6777 ArI
6014.0563 ThI
6015.4218 ThII
6016.3589 ThI
6017.8618 ThI
6018.9934 ThII
6019.6385 ThI
6020.2715 Th
6021.0357 ThI
6021.4113 ThII
6023.2242 ThI
6025.1500 ArI
6025.7779 ThI
6026.0448 ThI
6028.2713 ThI
6029.2257 ThI
6029.6500 ThI
6030.4450 ThI
6032.1274 ArI
6032.8726 ThI
6033.4131 ThI
6035.1923 ThI
6036.7622 ThI
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6035

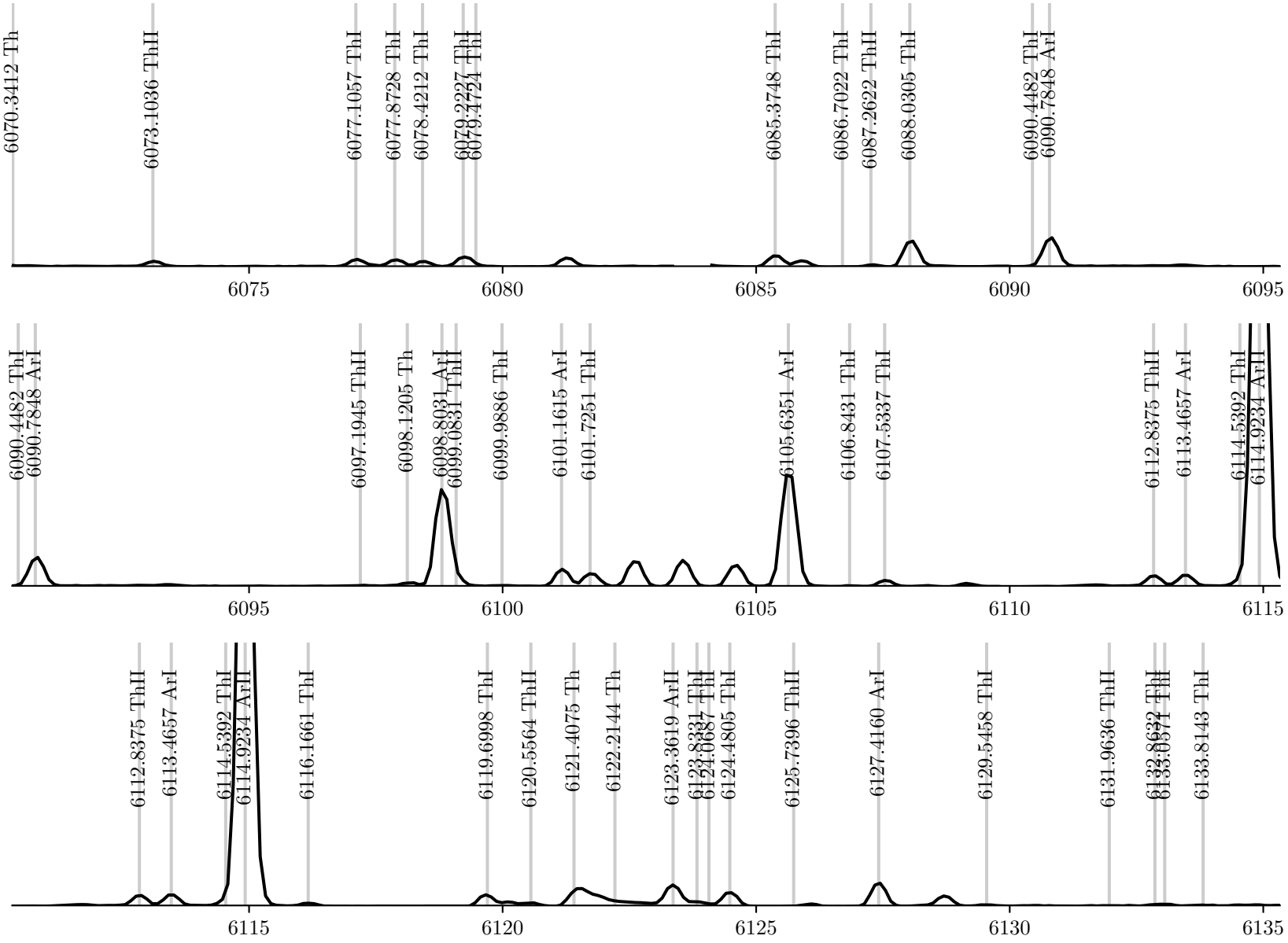

6032.1274 ArI
6032.8726 ThI
6033.4131 ThI
6035.1923 ThI
6036.7622 ThI
6037.6975 ThI
6038.6805 ThI
6042.5898 ThI
6043.2233 ArI
6044.4327 ThII
6046.8977 ArII
6049.0510 ThI
6049.7721 ThII
6050.9818 ThI
6052.7229 ArI
6053.3808 ThI
6055.5937 ThI
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6055
6052.7229 ArI
6053.3808 ThI
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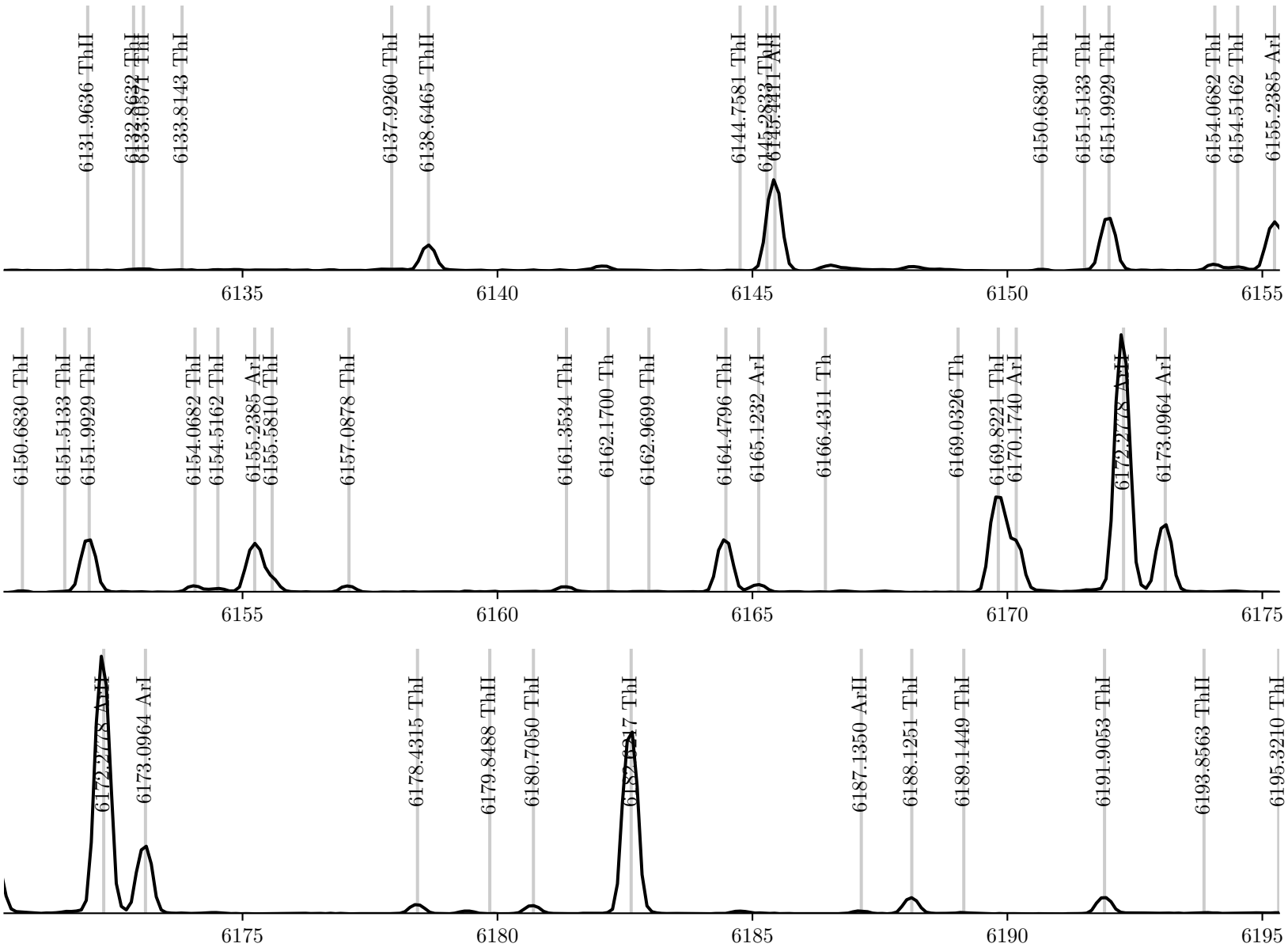
Order 37

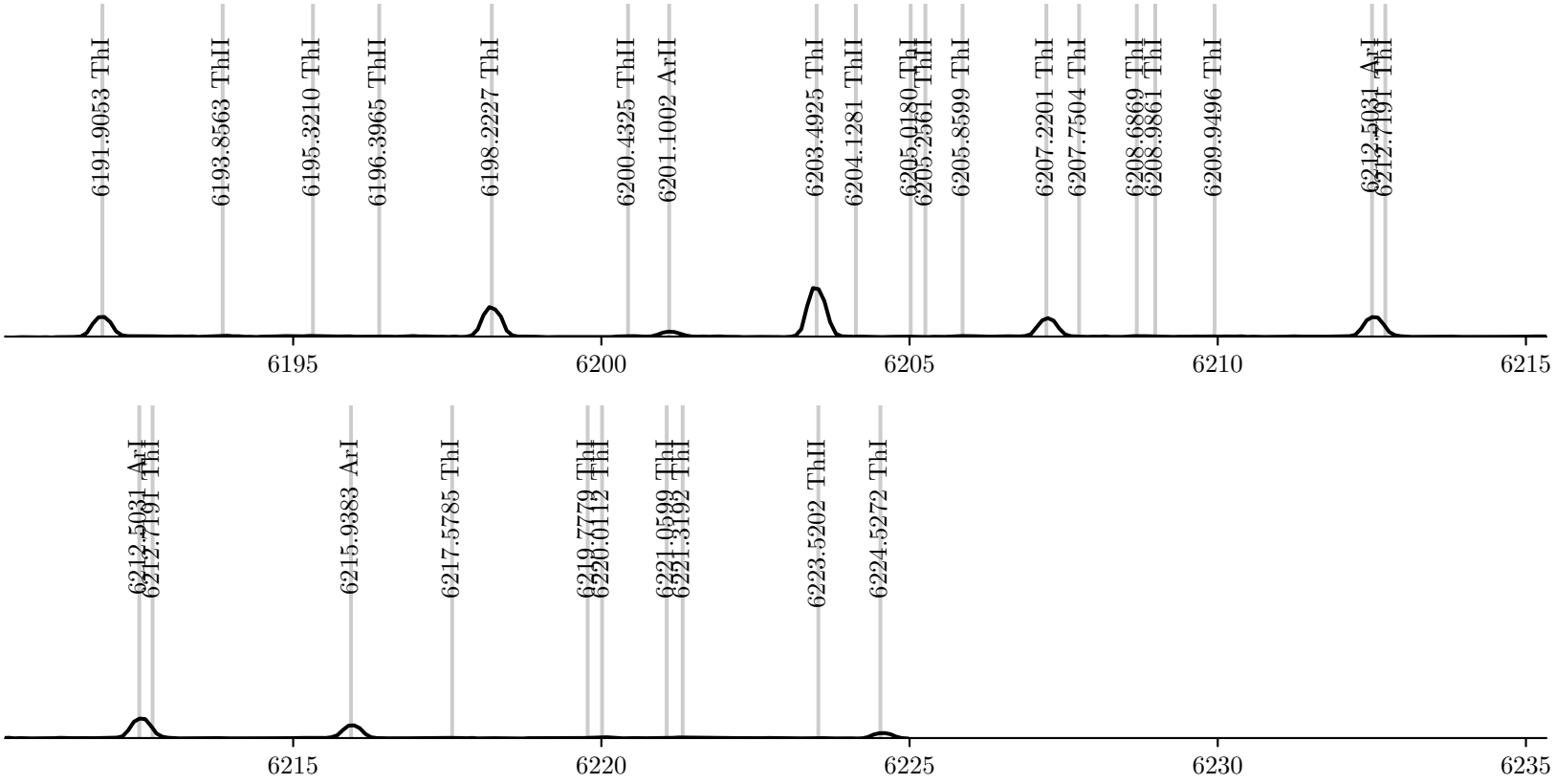


Approximate resolution:	17400
Dispersion:	0.105 Å pix^{-1}
Central wavelength:	6125 Å
Wavelength range:	6010 Å – 6225 Å

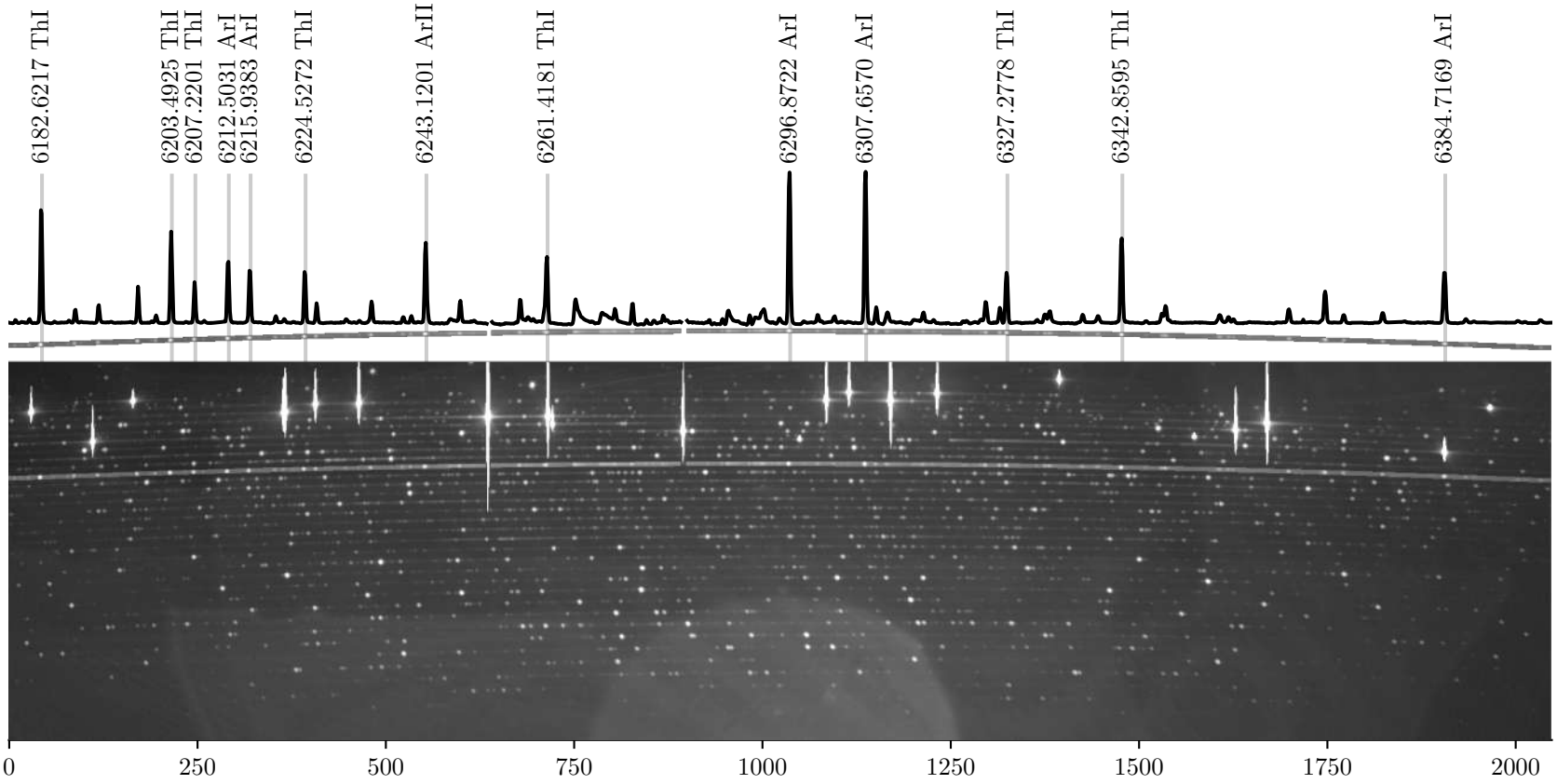

6010.5994 Th
6011.5337 ThI
6013.6777 ArI
6014.0563 ThI
6015.4218 ThII
6016.3589 ThI
6017.8618 ThI
6018.9934 ThII
6019.6385 ThI
6020.2715 Th
6021.0357 ThI
6021.4113 ThII
6023.2242 ThI
6025.1500 ArI
6025.7779 ThI
6026.0448 ThI
6028.2713 ThI
6029.2257 ThI
6029.6500 ThI
6030.4450 ThI
6032.1274 ArI
6032.8726 ThI
6033.4131 ThI
6035.1923 ThI
6015
6020
6025
6030
6035
6030.4450 ThI
6032.1274 ArI
6032.8726 ThI
6033.4131 ThI
6035.1923 ThI
6036.7622 ThI
6037.6975 ThI
6038.6805 ThI
6042.5898 ThI
6043.2233 ArI
6044.4327 ThII
6046.8977 ArII
6049.0510 ThI
6049.7721 ThII
6050.9818 ThI
6052.7229 ArI
6053.3808 ThI
6035
6040
6045
6050
6055
6050.9818 ThI
6052.7229 ArI
6053.3808 ThI
6055.5937 ThI
6058.1813 ThII
6058.6123 ThII
6059.3725 ArI
6061.5361 ThI
6064.7508 ArI
6065.7799 ThI
6066.4973 ThII
6067.0803 ThII
6069.0207 ThI
6070.3412 Th
6073.1036 ThII
6055
6060
6065
6070
6075


6070.3412 Th
6073.1036 ThII
6077.1057 ThI
6077.8728 ThI
6078.4212 ThI
6085.3748 ThI
6086.7022 ThI
6087.2622 ThII
6088.0305 ThI
6090.4482 ThI
6090.7848 ArI
6075
6080
6085
6090
6095
6090.4482 ThI
6090.7848 ArI
6097.1945 ThII
6098.1205 Th
6098.8031 ArI
6099.0831 ThII
6099.9886 ThI
6101.1615 ArI
6101.7251 ThI
6105.6351 ArI
6106.8431 ThI
6107.5337 ThI
6112.8375 ThII
6113.4657 ArI
6114.5392 ThI
6114.9234 ArII
6095
6100
6105
6110
6115
6112.8375 ThII
6113.4657 ArI
6114.5392 ThI
6114.9234 ArII
6116.1661 ThI
6119.6998 ThI
6120.5564 ThII
6121.4075 Th
6122.2144 Th
6123.3619 ArII
6123.8331 ThI
6124.0687 ThI
6124.4805 ThI
6125.7396 ThII
6127.4160 ArI
6129.5458 ThI
6131.9636 ThII
6133.8143 ThI
6115
6120
6125
6130
6135

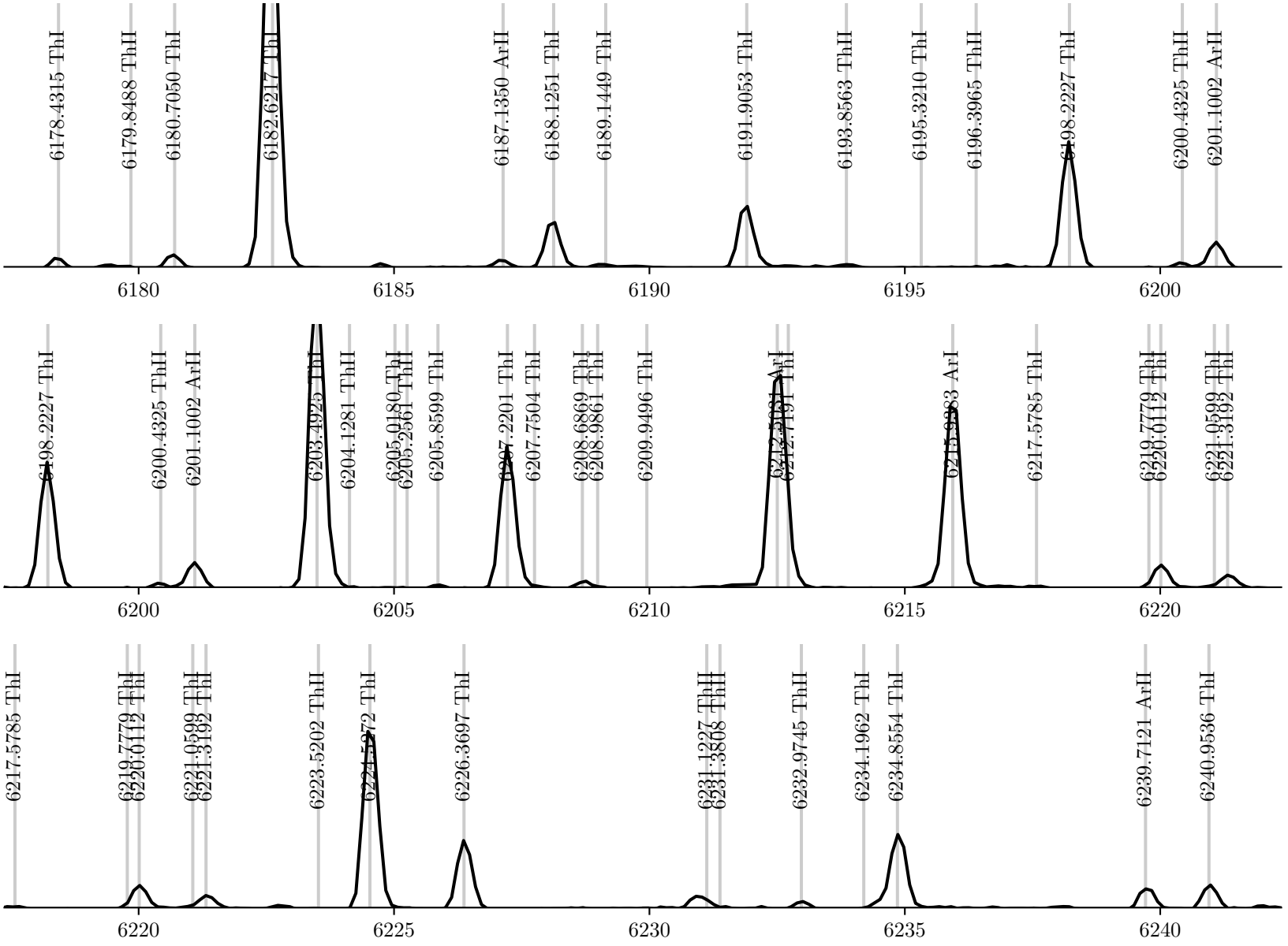

6131.9636 ThII
6133.8143 ThI
6137.9260 ThI
6138.6465 ThII
6144.7581 ThI
6150.6830 ThI
6151.5133 ThI
6151.9929 ThI
6154.0682 ThI
6154.5162 ThI
6155.2385 ArI
6135
6140
6145
6150
6155
6150.6830 ThI
6151.5133 ThI
6151.9929 ThI
6154.0682 ThI
6154.5162 ThI
6155.2385 ArI
6155.5810 ThI
6157.0878 ThI
6161.3534 ThI
6162.1700 Th
6162.9699 ThI
6164.4796 ThI
6165.1232 ArI
6166.4311 Th
6169.0326 Th
6169.8221 ThI
6170.1740 ArI
6172.2778 ArII
6173.0964 ArI
6155
6160
6165
6170
6175
6172.2778 ArII
6173.0964 ArI
6178.4315 ThI
6179.8488 ThII
6180.7050 ThI
6187.1350 ArII
6188.1251 ThI
6189.1449 ThI
6191.9053 ThI
6193.8563 ThII
6195.3210 ThI
6175
6180
6185
6190
6195

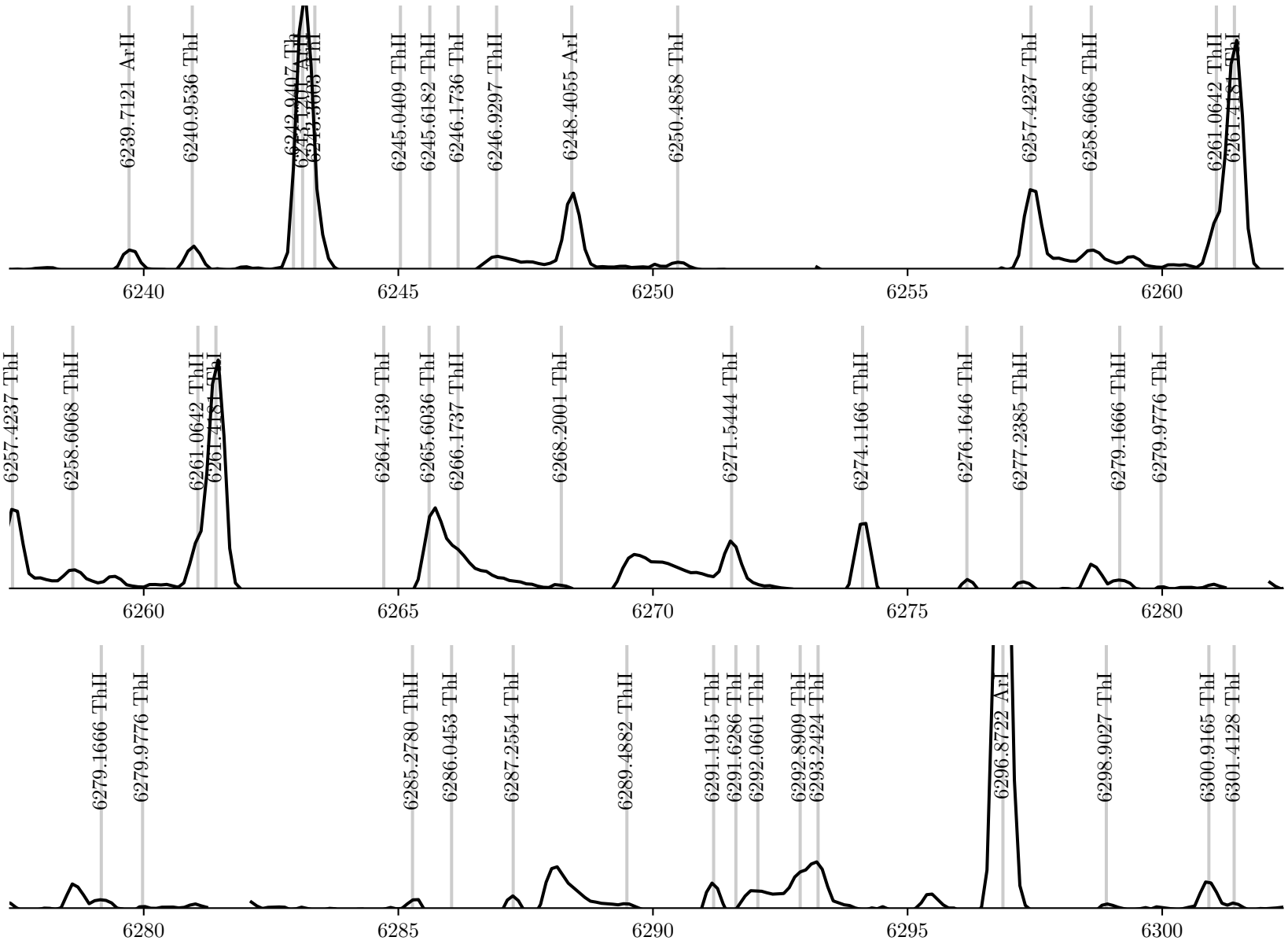

6191.9053 ThI
6193.8563 ThII
6195.3210 ThI
6196.3965 ThII
6198.2227 ThI
6200.4325 ThII
6201.1002 ArII
6203.4925 ThI
6204.1281 ThII
6205.0180 ThI
6205.2561 ThII
6205.8599 ThI
6207.2201 ThI
6207.7504 ThI
6208.6869 ThI
6208.9861 ThI
6209.9496 ThI
6212.5031 ArI
6212.7191 ThI
6195
6200
6205
6210
6215
6212.5031 ArI
6212.7191 ThI
6215.9383 ArI
6217.5785 ThI
6219.7779 ThI
6220.0112 ThI
6221.0599 ThI
6221.3192 ThI
6223.5202 ThII
6224.5272 ThI
6215
6220
6225
6230
6235

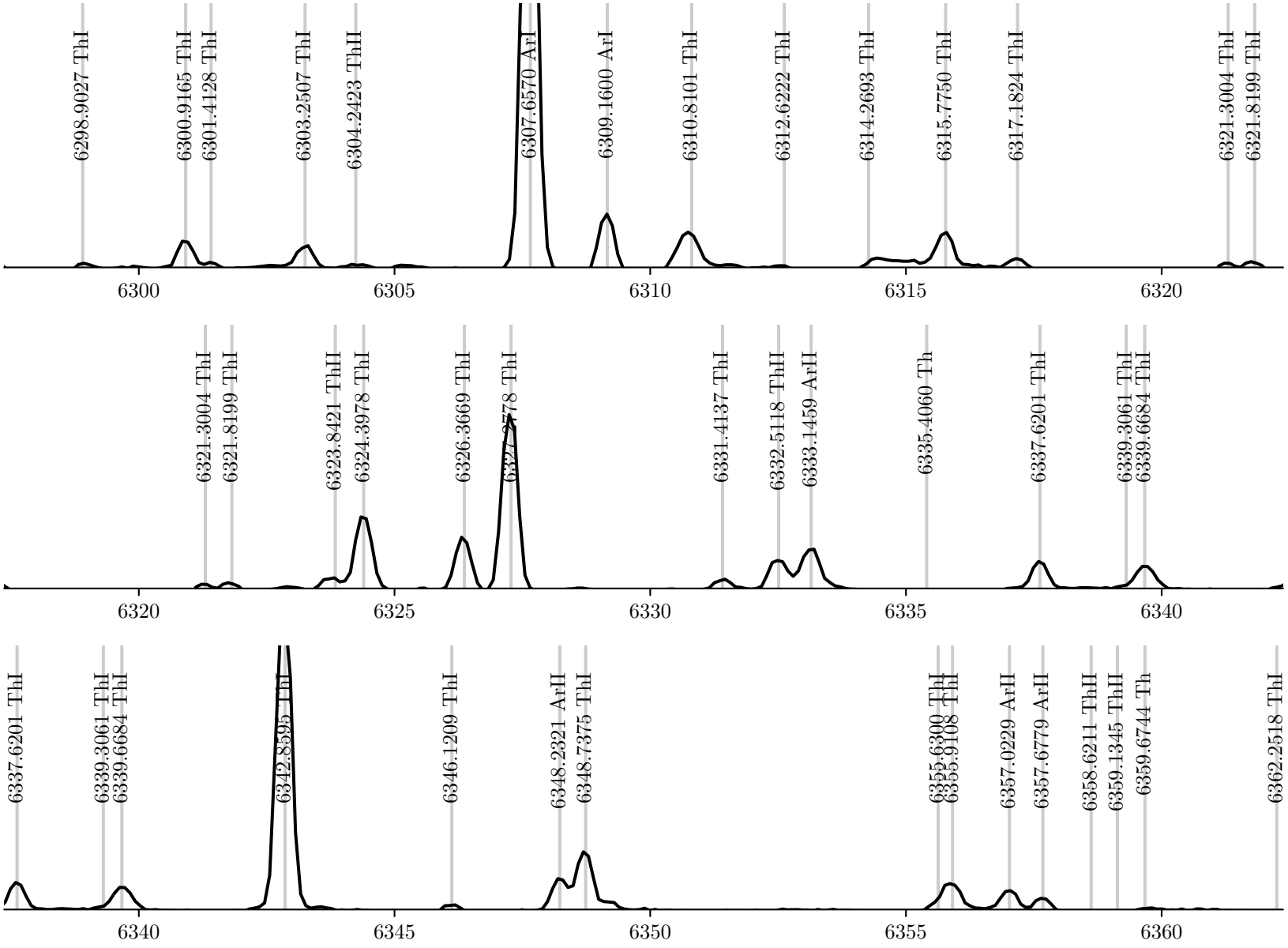
Order 36

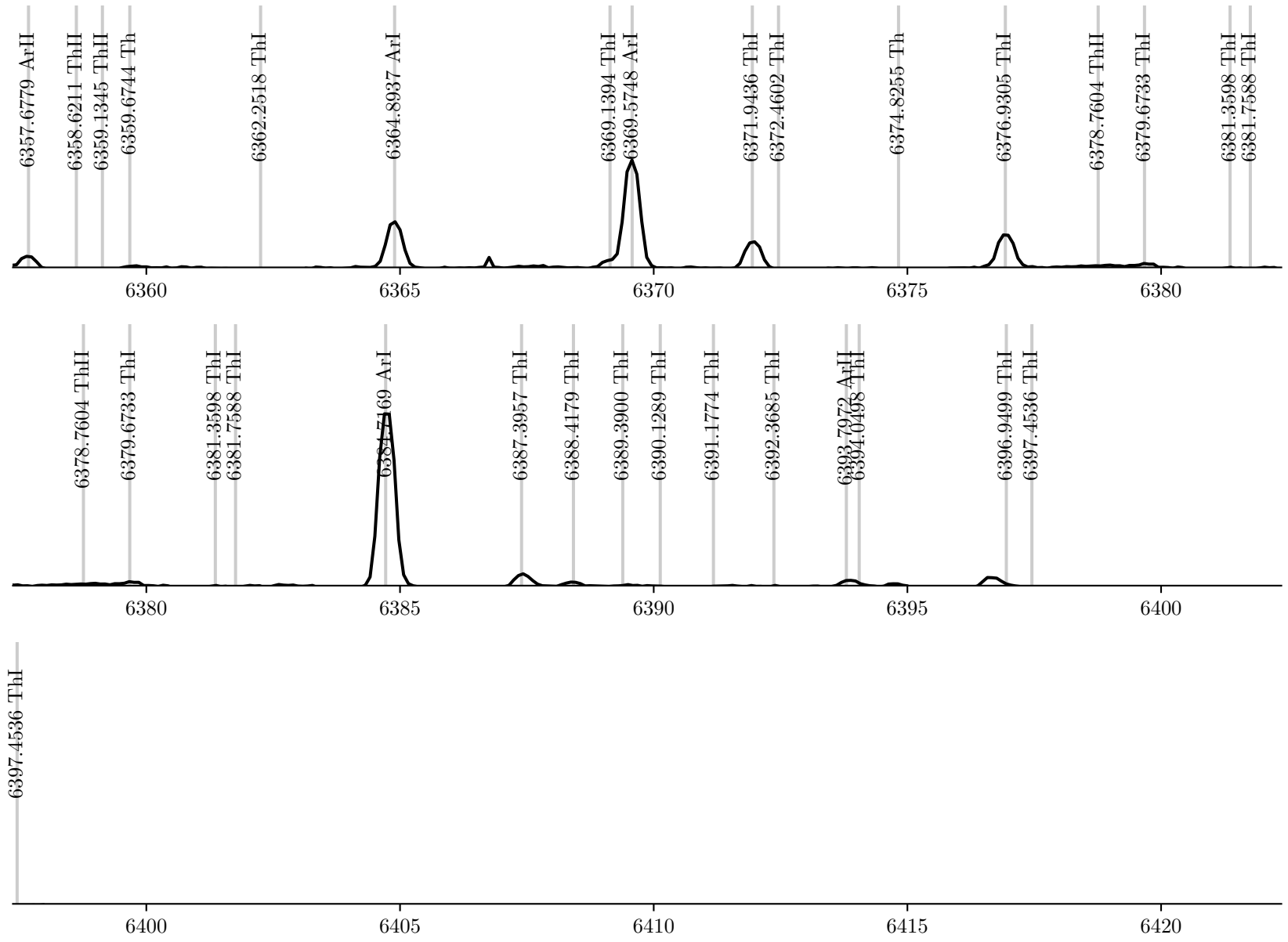


Approximate resolution:	17400
Dispersion:	0.108 Å pix^{-1}
Central wavelength:	6295 Å
Wavelength range:	6177 Å – 6398 Å

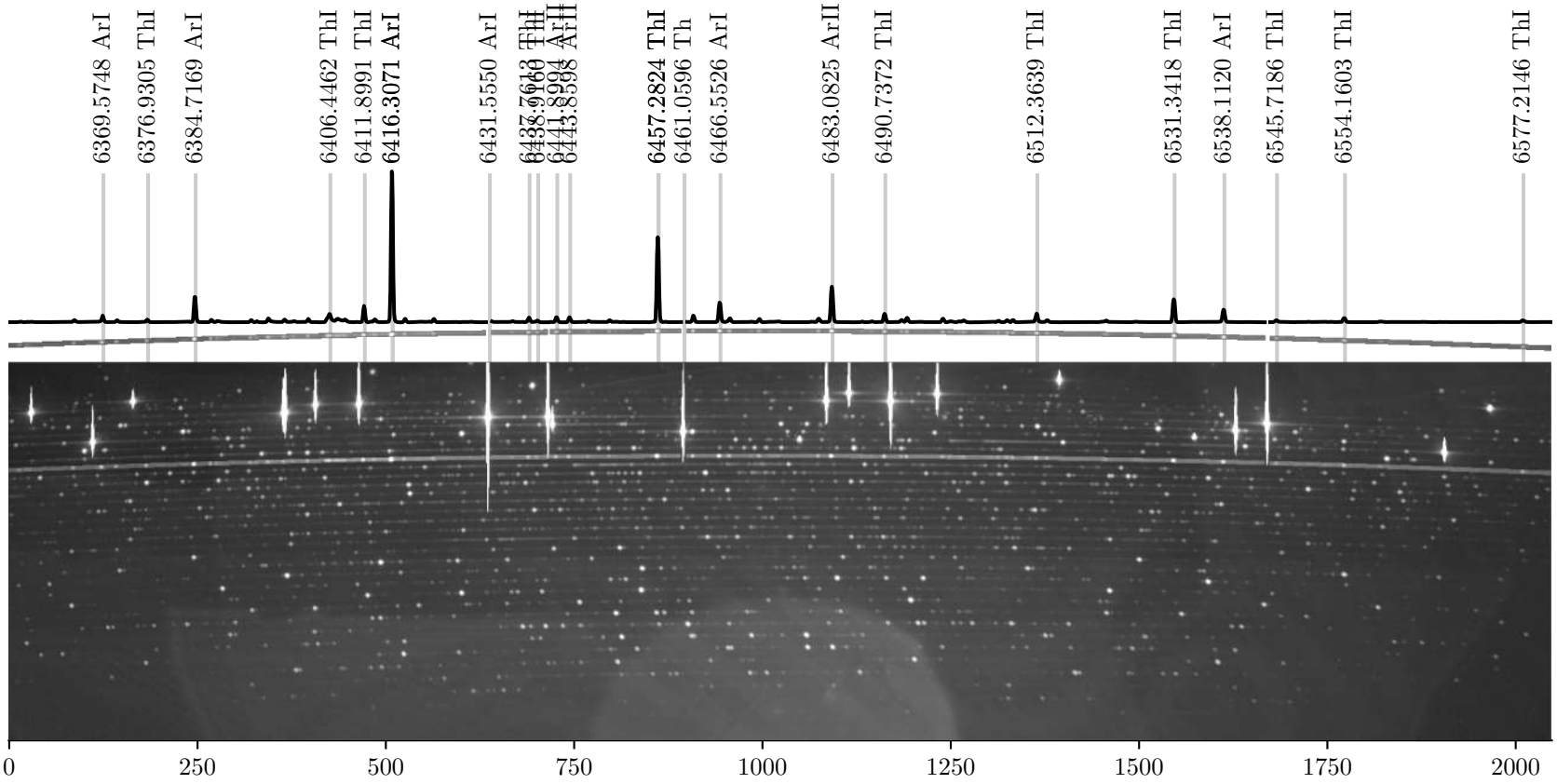

6178.4315 ThI
6179.8488 ThII
6180.7050 ThI
6182.6217 ThI
6187.1350 ArII
6188.1251 ThI
6189.1449 ThI
6191.9053 ThI
6193.8563 ThII
6195.3210 ThI
6196.3965 ThII
6198.2227 ThI
6200.4325 ThII
6201.1002 ArII
6180
6185
6190
6195
6200
6198.2227 ThI
6200.4325 ThII
6201.1002 ArII
6203.4925 ThI
6204.1281 ThII
6205.0180 ThI
6205.2561 ThII
6205.8599 ThI
6207.2201 ThI
6207.7504 ThI
6208.6869 ThI
6208.9861 ThI
6209.9496 ThI
6212.5081 ArI
6212.7161 ThI
6215.9283 ArI
6217.5785 ThI
6219.7779 ThI
6220.0112 ThI
6221.0599 ThI
6221.3192 ThI
6200
6205
6210
6215
6220
6217.5785 ThI
6219.7779 ThI
6220.0112 ThI
6221.0599 ThI
6221.3192 ThI
6223.5202 ThII
6224.5272 ThI
6226.3697 ThI
6231.1227 ThII
6231.3808 ThII
6232.9745 ThII
6234.1962 ThI
6234.8554 ThI
6239.7121 ArII
6240.9536 ThI
6220
6225
6230
6235
6240


6239.7121 ArII
6240.9536 ThI
6242.9407 Th
6243.1201 ArII
6243.3068 ThI
6245.0409 ThII
6245.6182 ThII
6246.1736 ThI
6246.9297 ThII
6248.4055 ArI
6250.4858 ThI
6257.4237 ThI
6258.6068 ThII
6261.0642 ThII
6261.4181 ThI
6240
6245
6250
6255
6260
6257.4237 ThI
6258.6068 ThII
6261.0642 ThII
6261.4181 ThI
6264.7139 ThI
6265.6036 ThI
6266.1737 ThII
6268.2001 ThI
6271.5444 ThI
6274.1166 ThII
6276.1646 ThI
6277.2385 ThII
6279.1666 ThII
6279.9776 ThI
6260
6265
6270
6275
6280
6279.1666 ThII
6279.9776 ThI
6285.2780 ThII
6286.0453 ThI
6287.2554 ThI
6289.4882 ThII
6291.1915 ThI
6291.6286 ThI
6292.0601 ThI
6292.8909 ThI
6293.2424 ThI
6296.8722 ArI
6298.9027 ThI
6300.9165 ThI
6301.4128 ThI
6280
6285
6290
6295
6300

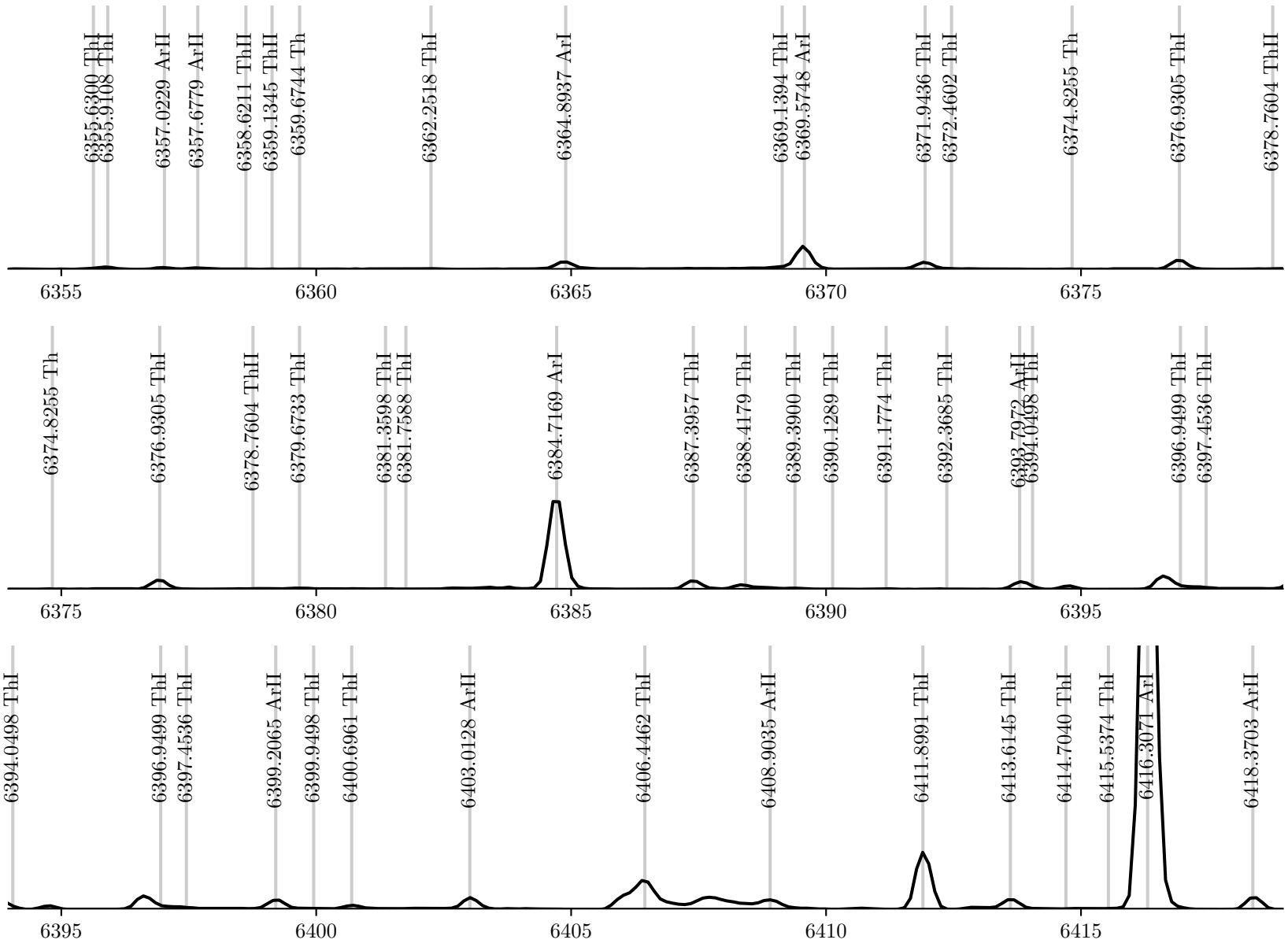

6298.9027 ThI
6300.9165 ThI
6301.4128 ThI
6303.2507 ThI
6304.2423 ThII
6307.6570 ArI
6309.1600 ArI
6310.8101 ThI
6312.6222 ThI
6314.2693 ThI
6315.7750 ThI
6317.1824 ThI
6321.3004 ThI
6321.8199 ThI
6300
6305
6310
6315
6320
6321.3004 ThI
6321.8199 ThI
6323.8421 ThII
6324.3978 ThI
6326.3669 ThI
6327.2778 ThI
6331.4137 ThI
6332.5118 ThII
6333.1459 ArII
6335.4060 Th
6337.6201 ThI
6339.3061 ThI
6339.6684 ThI
6320
6325
6330
6335
6340
6337.6201 ThI
6339.3061 ThI
6339.6684 ThI
6342.8595 ThI
6346.1209 ThI
6348.2321 ArII
6348.7375 ThI
6355.6300 ThI
6355.9108 ThI
6357.0229 ArII
6357.6779 ArII
6358.6211 ThII
6359.1345 ThII
6359.6744 Th
6362.2518 ThI
6340
6345
6350
6355
6360

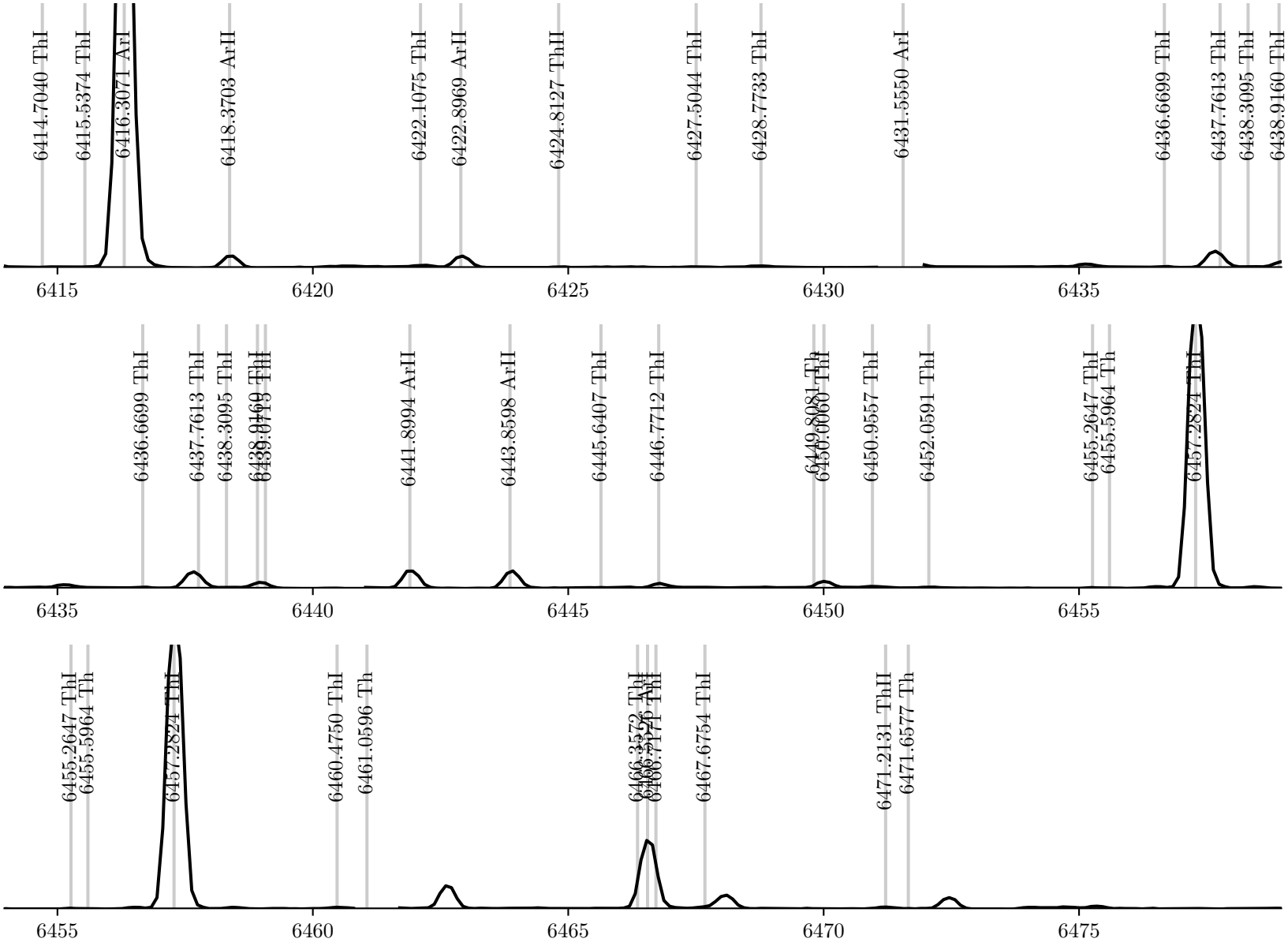

6357.6779 ArII
6358.6211 ThII
6359.1345 ThII
6359.6744 Th
6362.2518 ThI
6364.8937 ArI
6369.1394 ThI
6369.5748 ArI
6371.9436 ThI
6372.4602 ThI
6374.8255 Th
6376.9305 ThI
6378.7604 ThII
6379.6733 ThI
6381.3598 ThI
6381.7588 ThI
6360
6365
6370
6375
6380
6378.7604 ThII
6379.6733 ThI
6381.3598 ThI
6381.7588 ThI
6384.7169 ArI
6387.3957 ThI
6388.4179 ThI
6389.3900 ThI
6390.1289 ThI
6391.1774 ThI
6392.3685 ThI
6393.7972 ArII
6394.0498 ThI
6396.9499 ThI
6397.4536 ThI
6380
6385
6390
6395
6400
6397.4536 ThI
6400
6405
6410
6415
6420

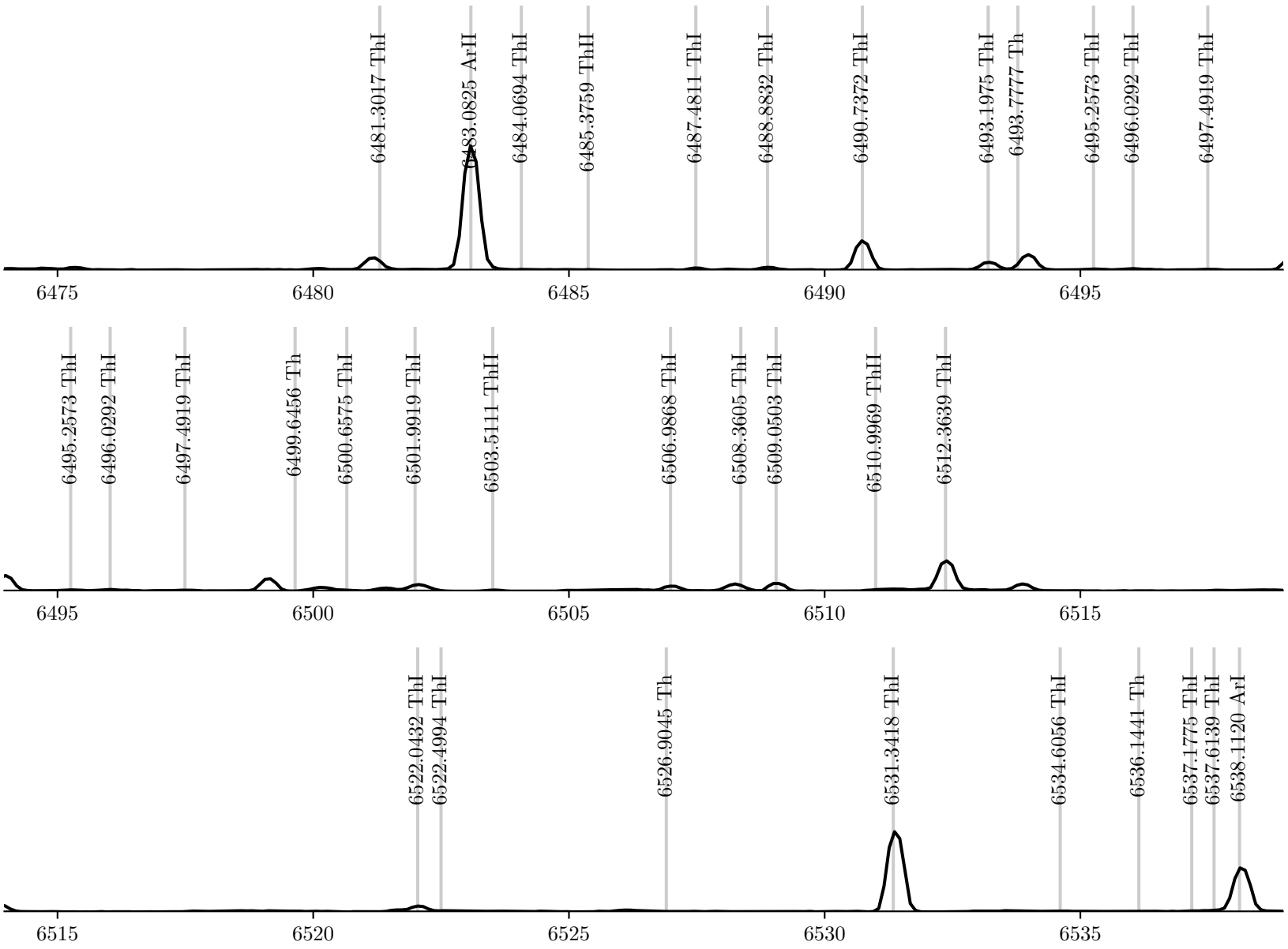
Order 35

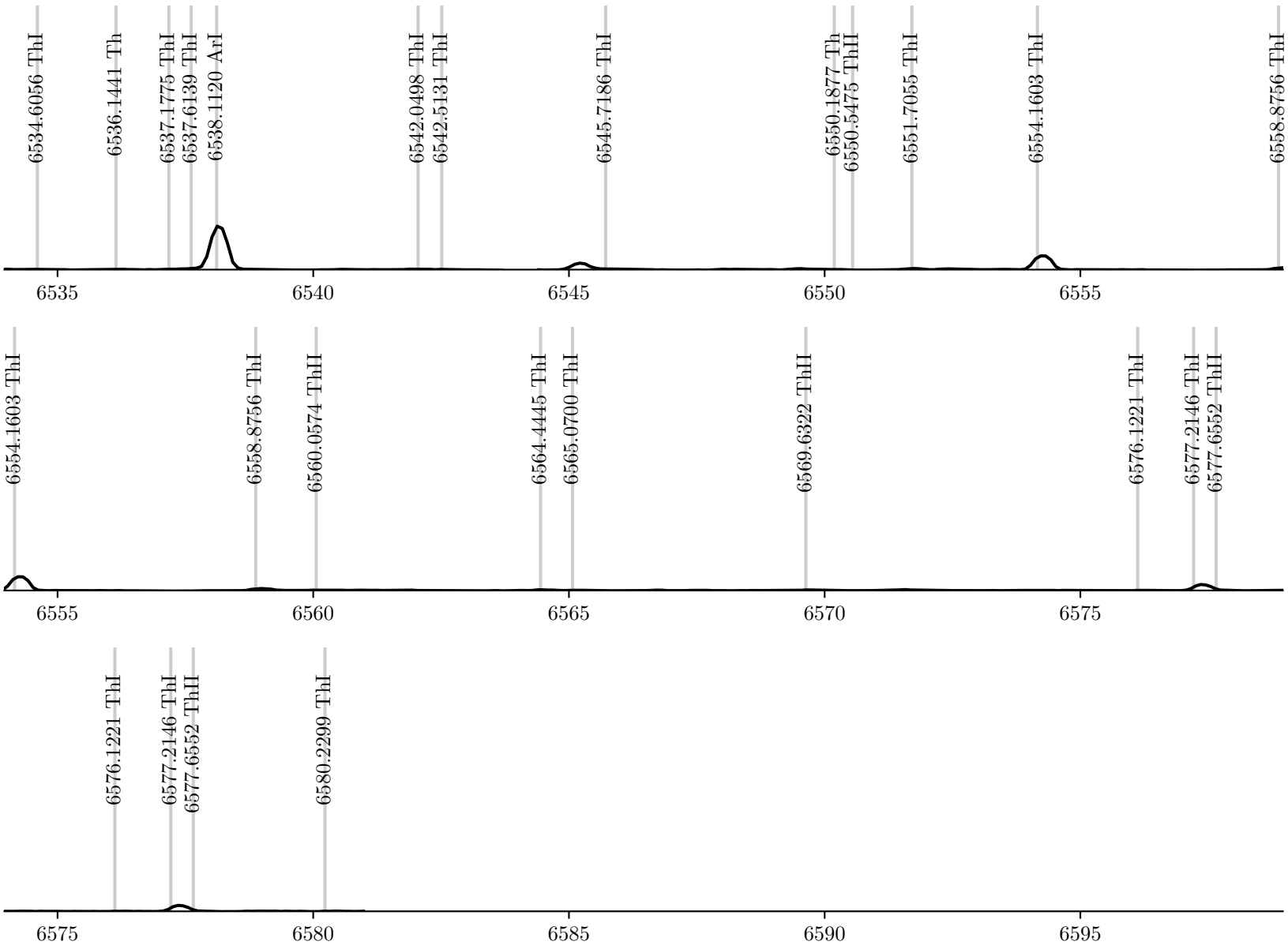


Approximate resolution:	17500
Dispersion:	0.111 Å pix^{-1}
Central wavelength:	6475 Å
Wavelength range:	6354 Å – 6581 Å

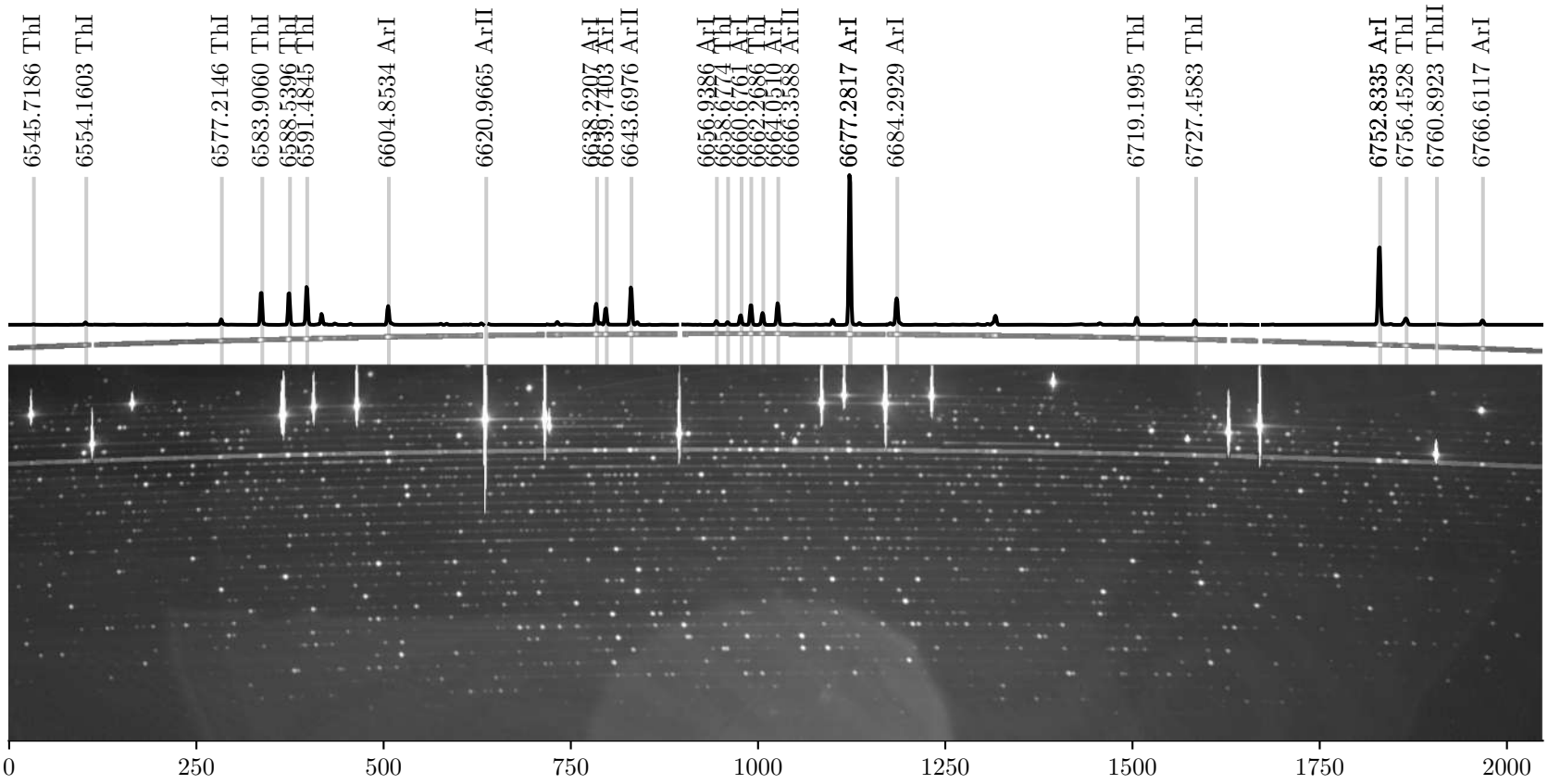

6355.6300 ThI
6355.9108 ThI
6357.0229 ArII
6357.6779 ArII
6358.6211 ThII
6359.1345 ThII
6359.6744 Th
6362.2518 ThI
6364.8937 ArI
6369.1394 ThI
6369.5748 ArI
6371.9436 ThI
6372.4602 ThI
6374.8255 Th
6376.9305 ThI
6378.7604 ThII
6355
6360
6365
6370
6375
6374.8255 Th
6376.9305 ThI
6378.7604 ThII
6379.6733 ThI
6381.3598 ThI
6381.7588 ThI
6384.7169 ArI
6387.3957 ThI
6388.4179 ThI
6389.3900 ThI
6390.1289 ThI
6391.1774 ThI
6392.3685 ThI
6393.7972 ArII
6394.0498 ThI
6396.9499 ThI
6397.4536 ThI
6375
6380
6385
6390
6395
6394.0498 ThI
6396.9499 ThI
6397.4536 ThI
6399.2065 ArII
6399.9498 ThI
6400.6961 ThI
6403.0128 ArII
6406.4462 ThI
6408.9035 ArII
6411.8991 ThI
6413.6145 ThI
6414.7040 ThI
6415.5374 ThI
6416.3071 ArI
6418.3703 ArII
6395
6400
6405
6410
6415


6414.7040 ThI
6415.5374 ThI
6416.3071 ArI
6418.3703 ArII
6422.1075 ThI
6422.8969 ArII
6424.8127 ThII
6427.5044 ThI
6428.7733 ThI
6431.5550 ArI
6436.6699 ThI
6437.7613 ThI
6438.3095 ThI
6438.9160 ThI
6415
6420
6425
6430
6435
6436.6699 ThI
6437.7613 ThI
6438.3095 ThI
6441.8994 ArII
6443.8598 ArII
6445.6407 ThI
6446.7712 ThI
6450.9557 ThI
6452.0591 ThI
6455.2647 ThI
6455.5964 Th
6457.2824 ThI
6435
6440
6445
6450
6455
6455.2647 ThI
6455.5964 Th
6457.2824 ThI
6460.4750 ThI
6461.0596 Th
6467.6754 ThI
6471.2131 ThII
6471.6577 Th
6455
6460
6465
6470
6475

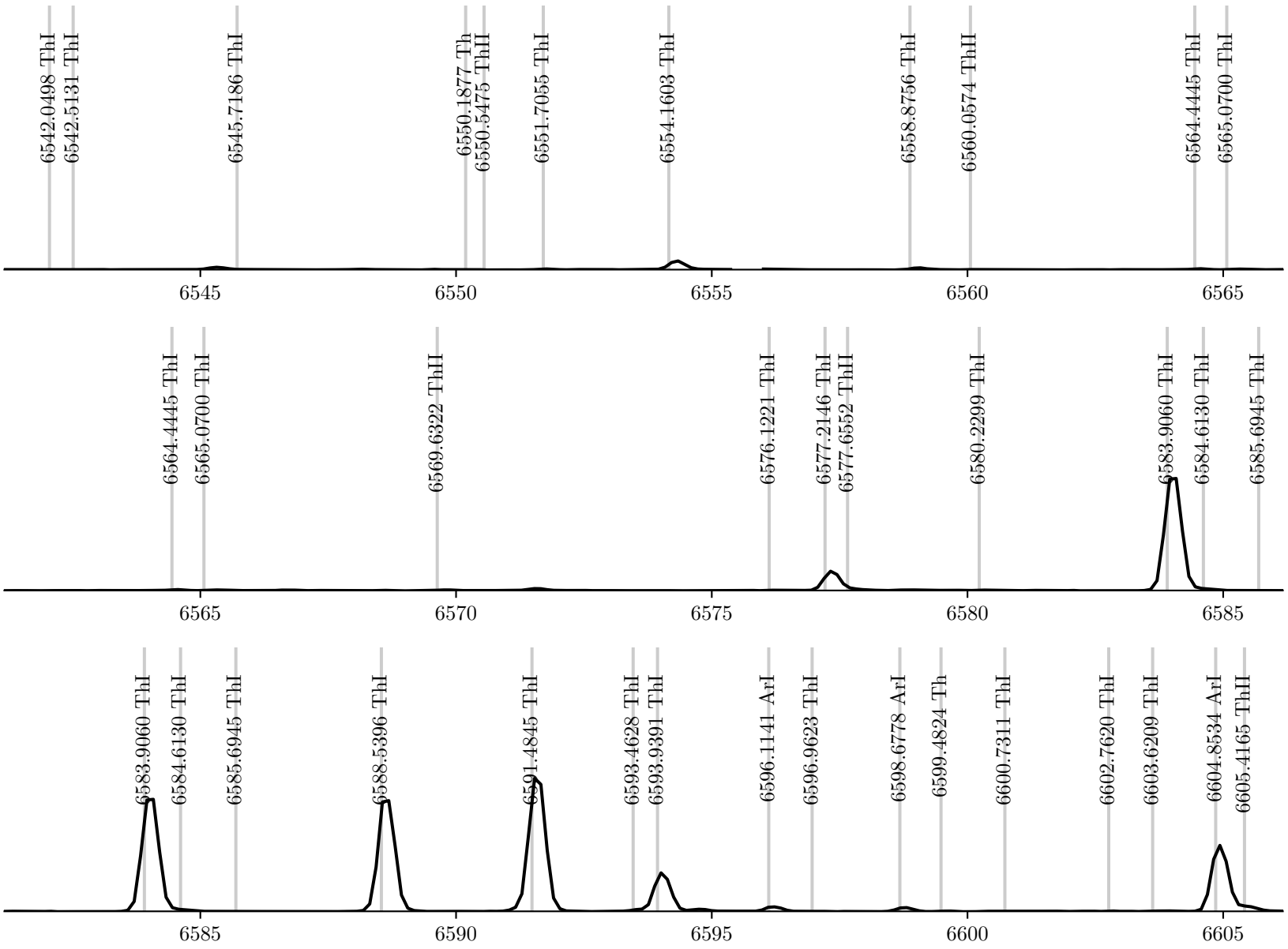

6481.3017 ThI
6483.0825 ArII
6484.0694 ThI
6485.3759 ThII
6487.4811 ThI
6488.8832 ThI
6490.7372 ThI
6493.1975 ThI
6493.7777 Th
6495.2573 ThI
6496.0292 ThI
6497.4919 ThI
6475
6480
6485
6490
6495
6495.2573 ThI
6496.0292 ThI
6497.4919 ThI
6499.6456 Th
6500.6575 ThI
6501.9919 ThI
6503.5111 ThII
6506.9868 ThI
6508.3605 ThI
6509.0503 ThI
6510.9969 ThII
6512.3639 ThI
6495
6500
6505
6510
6515
6522.0432 ThI
6522.4994 ThI
6526.9045 Th
6531.3418 ThI
6534.6056 ThI
6536.1441 Th
6537.1775 ThI
6537.6139 ThI
6538.1120 ArI
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6520
6525
6530
6535

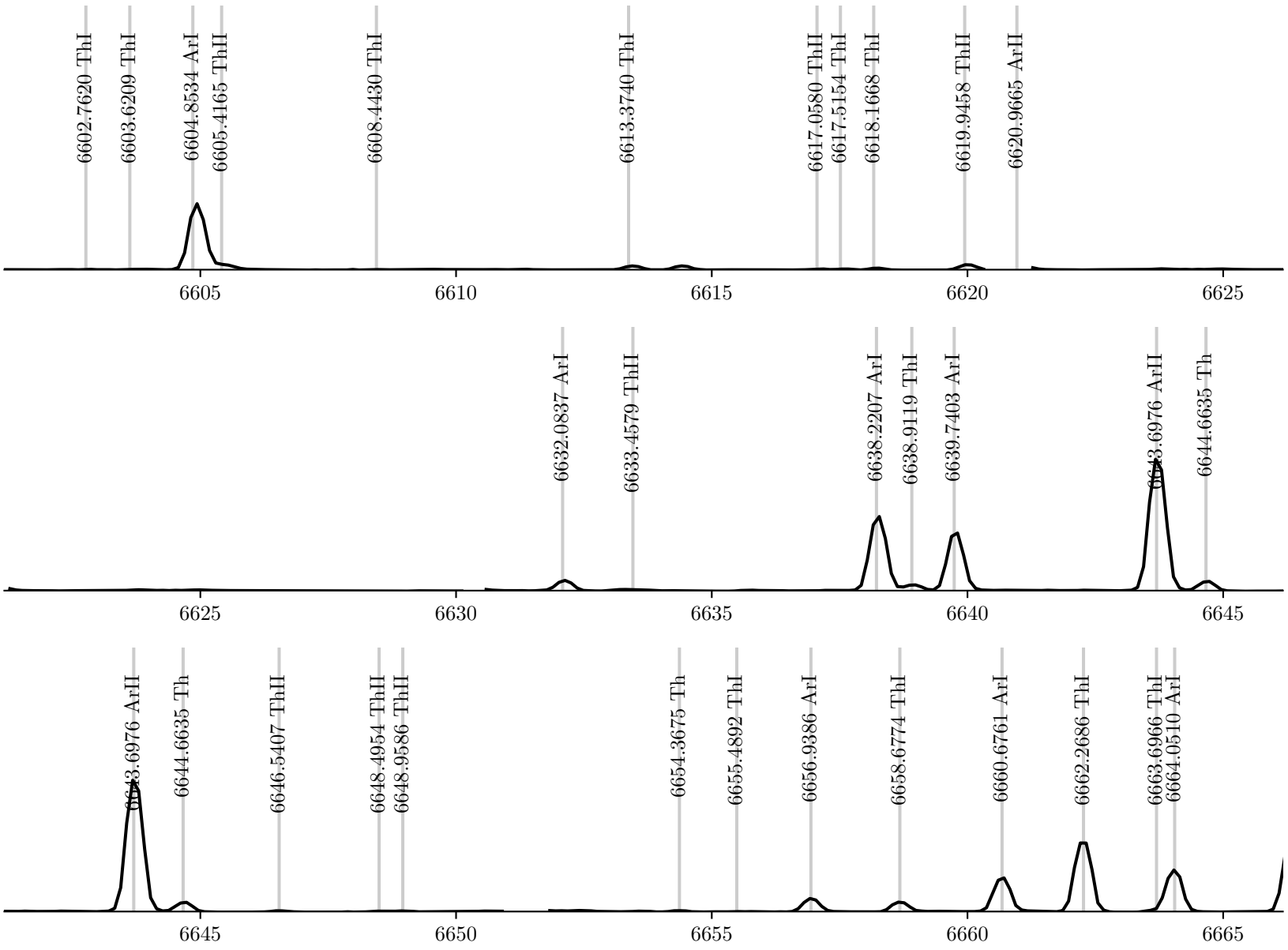

6534.6056 ThI
6536.1441 Th
6537.1775 ThI
6537.6139 ThI
6538.1120 ArI
6542.0498 ThI
6542.5131 ThI
6545.7186 ThI
6550.1877 Th
6550.5475 ThII
6551.7055 ThI
6554.1603 ThI
6558.8756 ThI
6535
6540
6545
6550
6555
6554.1603 ThI
6558.8756 ThI
6560.0574 ThII
6564.4445 ThI
6565.0700 ThI
6569.6322 ThII
6576.1221 ThI
6577.2146 ThI
6577.6552 ThII
6555
6560
6565
6570
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6575
6580
6585
6590
6595

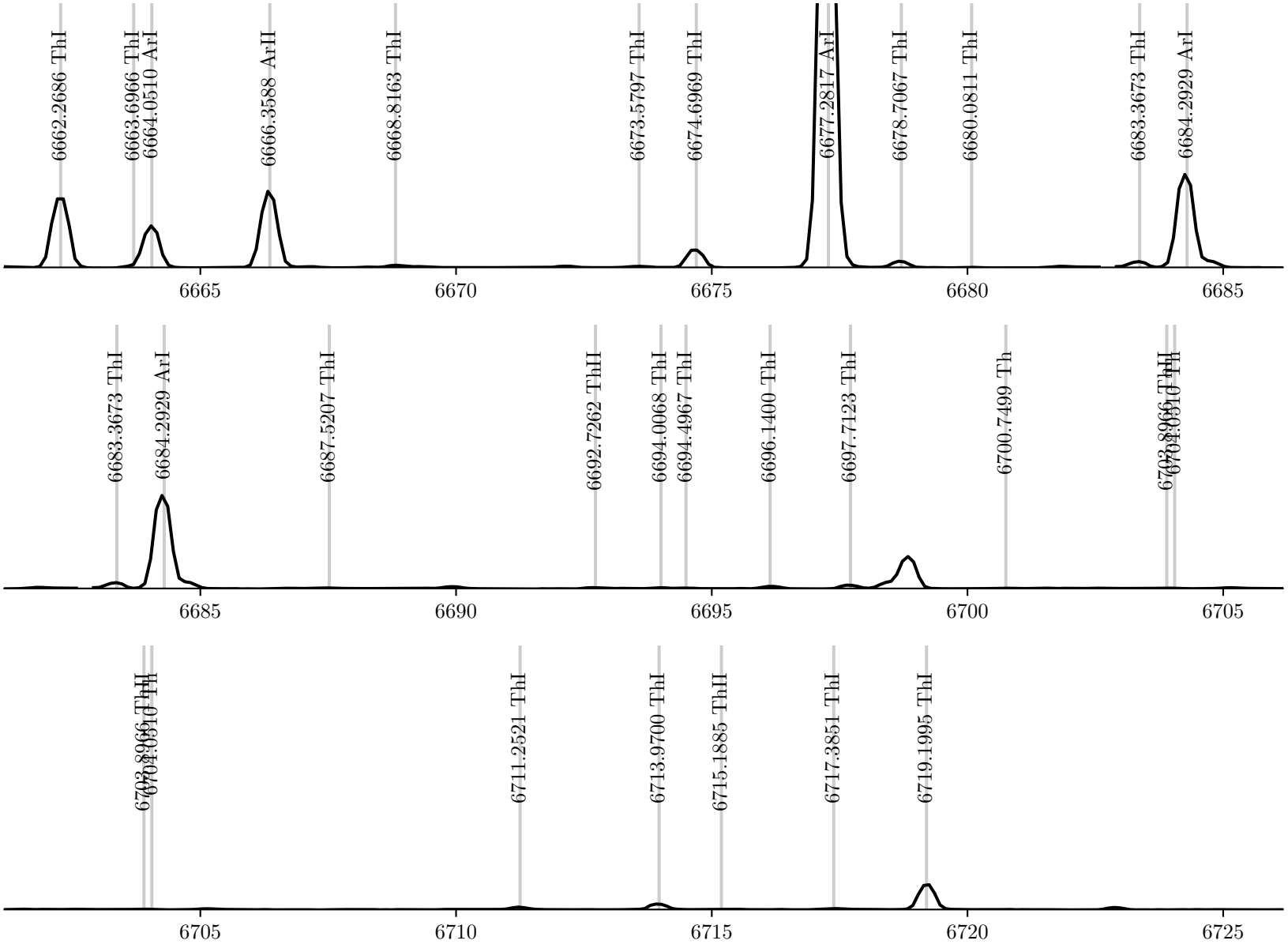
Order 34

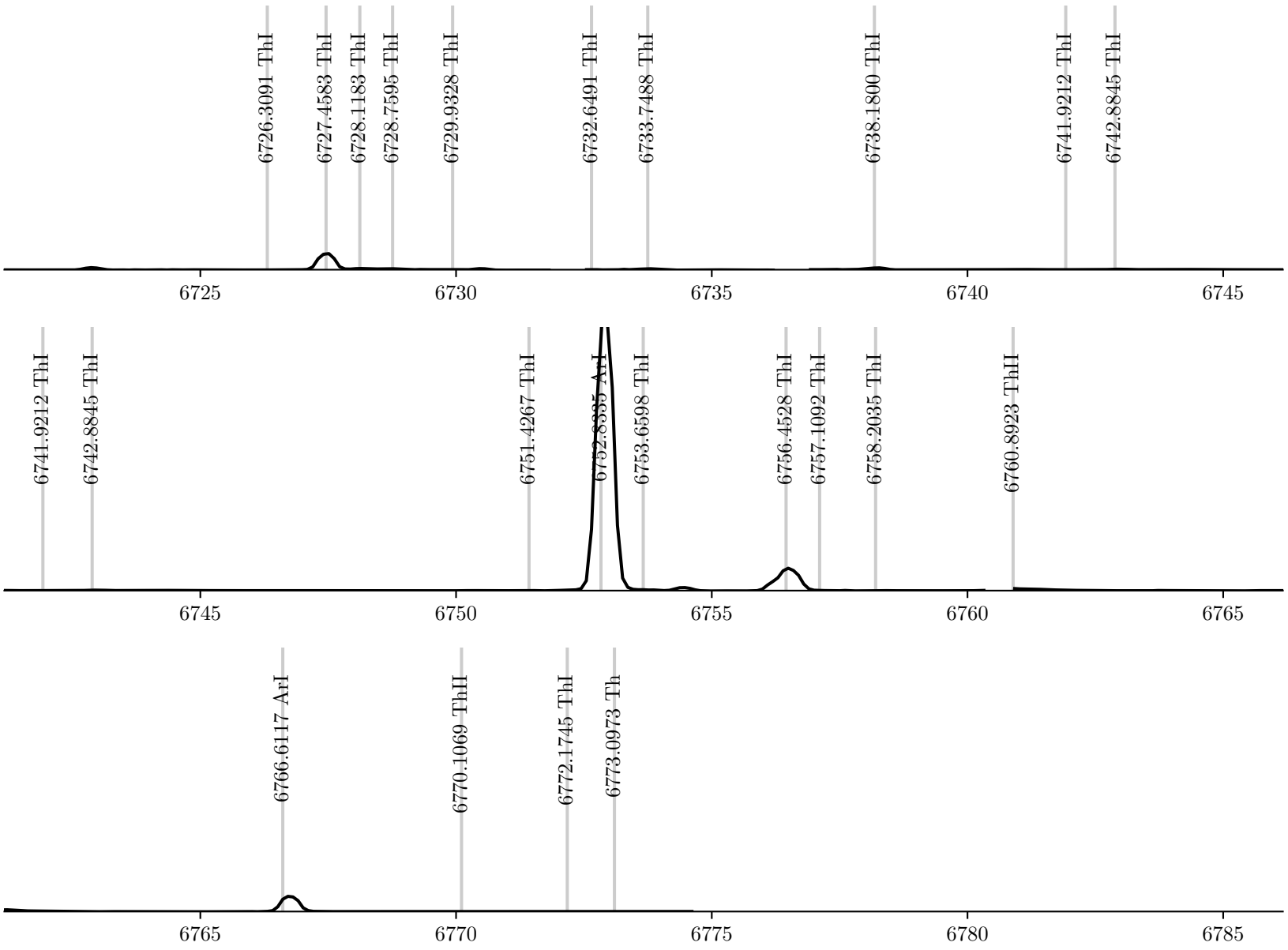


Approximate resolution:	17600
Dispersion:	0.114 Å pix^{-1}
Central wavelength:	6666 Å
Wavelength range:	6541 Å – 6775 Å

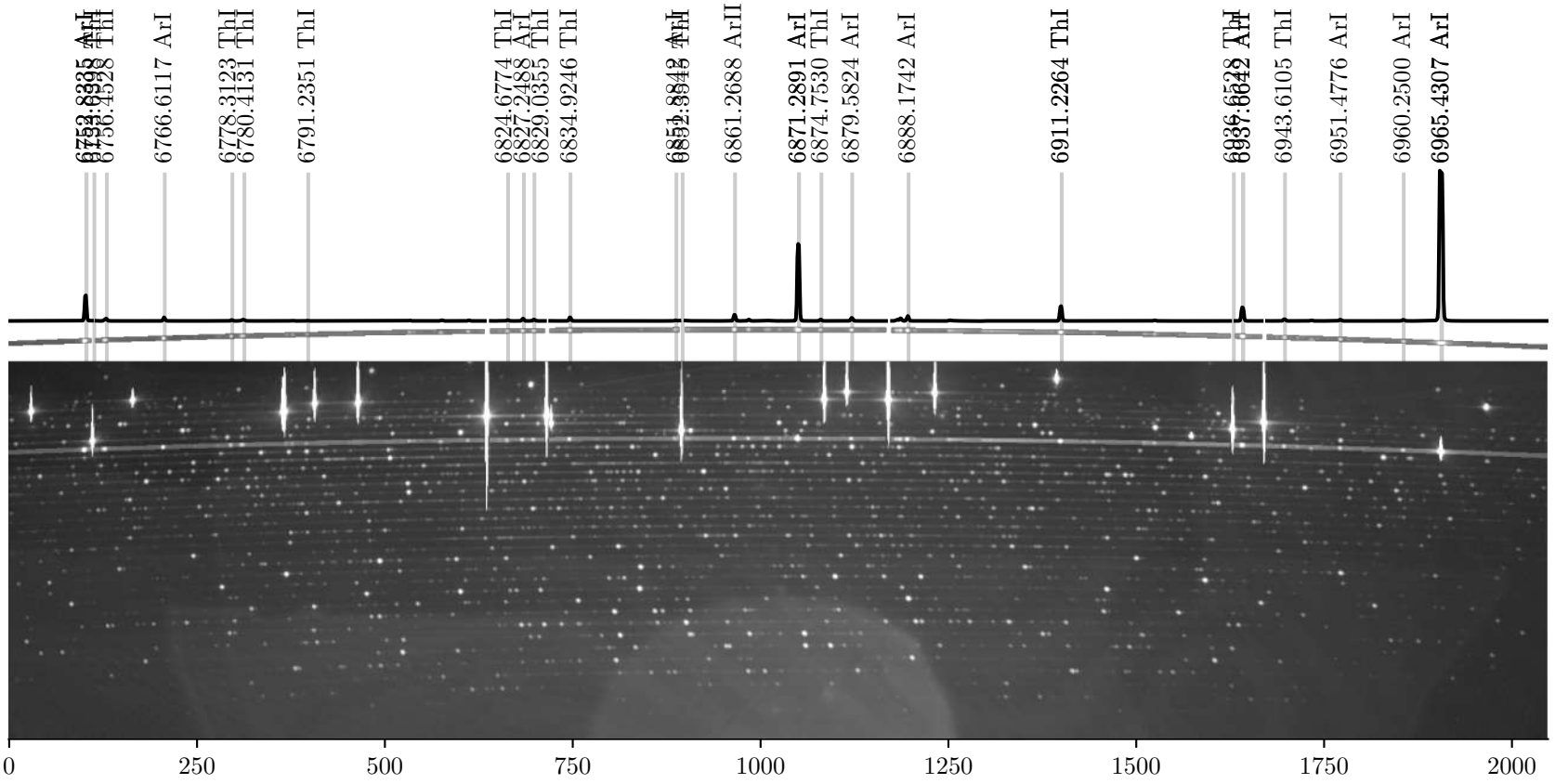

6542.0498 ThI
6542.5131 ThI
6545.7186 ThI
6550.1877 Th
6550.5475 ThII
6551.7055 ThI
6554.1603 ThI
6558.8756 ThI
6560.0574 ThII
6564.4445 ThI
6565.0700 ThI
6545
6550
6555
6560
6565
6564.4445 ThI
6565.0700 ThI
6569.6322 ThII
6576.1221 ThI
6577.2146 ThI
6577.6552 ThII
6580.2299 ThI
6583.9060 ThI
6584.6130 ThI
6585.6945 ThI
6565
6570
6575
6580
6585
6583.9060 ThI
6584.6130 ThI
6585.6945 ThI
6588.5396 ThI
6591.4845 ThI
6593.4628 ThI
6593.9391 ThI
6596.1141 ArI
6596.9623 ThI
6598.6778 ArI
6599.4824 Th
6600.7311 ThI
6602.7620 ThI
6603.6209 ThI
6604.8534 ArI
6605.4165 ThII
6585
6590
6595
6600
6605


6602.7620 ThI
6603.6209 ThI
6604.8534 ArI
6605.4165 ThII
6608.4430 ThI
6613.3740 ThI
6617.0580 ThII
6617.5154 ThI
6618.1668 ThI
6619.9458 ThII
6620.9665 ArII
6605
6610
6615
6620
6625
6632.0837 ArI
6633.4579 ThII
6638.2207 ArI
6638.9119 ThI
6639.7403 ArI
6643.6976 ArII
6644.6635 Th
6625
6630
6635
6640
6645
6643.6976 ArII
6644.6635 Th
6646.5407 ThII
6648.4954 ThII
6648.9586 ThII
6654.3675 Th
6655.4892 ThI
6656.9386 ArI
6658.6774 ThI
6660.6761 ArI
6662.2686 ThI
6663.6966 ThI
6664.0510 ArI
6645
6650
6655
6660
6665

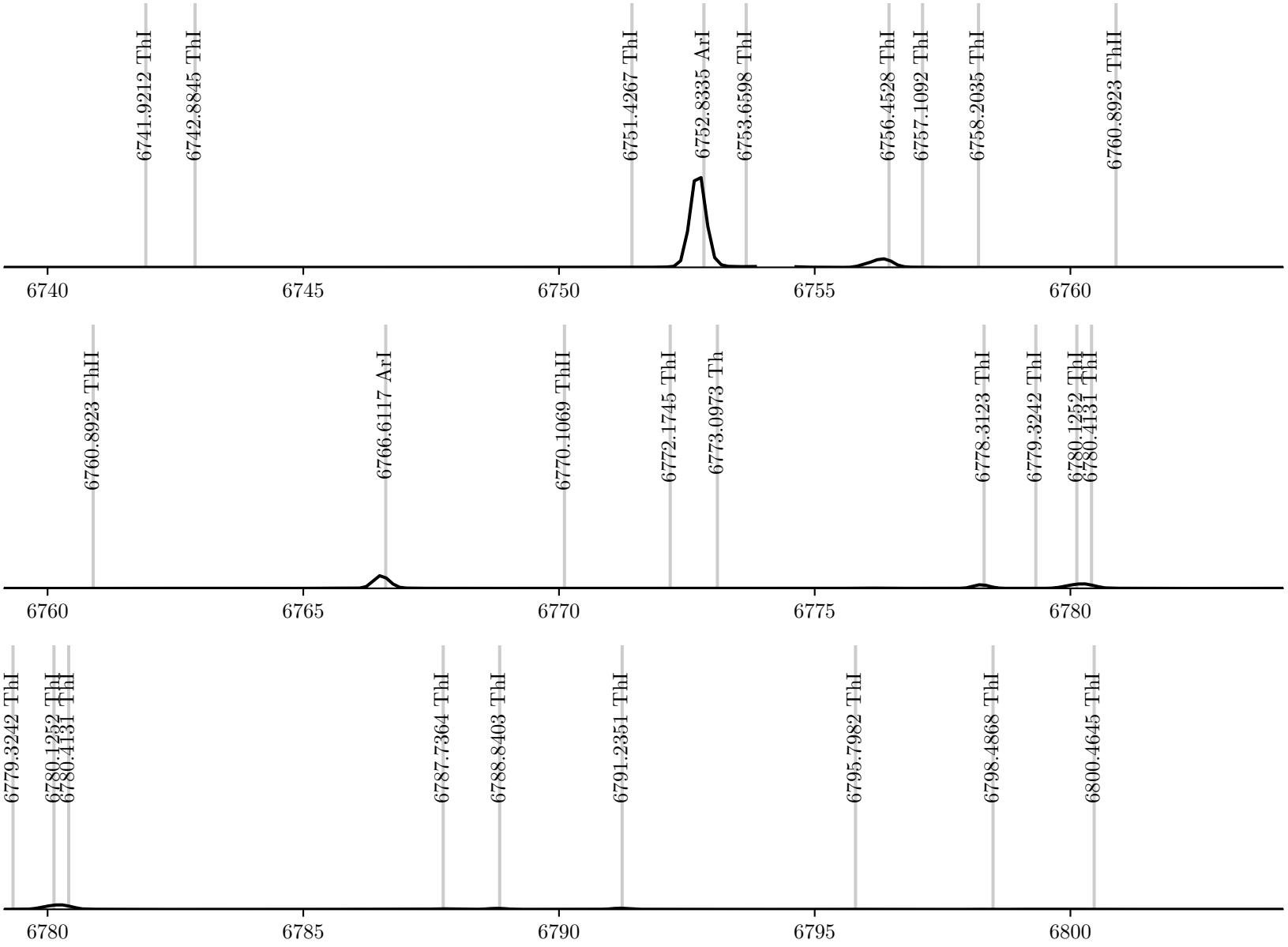

6662.2686 ThI
6663.6966 ThI
6664.0510 ArI
6666.3588 ArII
6668.8163 ThI
6673.5797 ThI
6674.6969 ThI
6677.2817 ArI
6678.7067 ThI
6680.0811 ThI
6683.3673 ThI
6684.2929 ArI
6665
6670
6675
6680
6685
6683.3673 ThI
6684.2929 ArI
6687.5207 ThI
6692.7262 ThII
6694.0068 ThI
6694.4967 ThI
6696.1400 ThI
6697.7123 ThI
6700.7499 Th
6703.8966 ThI
6704.0910 ThI
6685
6690
6695
6700
6705
6703.8966 ThI
6704.0910 ThI
6711.2521 ThI
6713.9700 ThI
6715.1885 ThII
6717.3851 ThI
6719.1995 ThI
6705
6710
6715
6720
6725

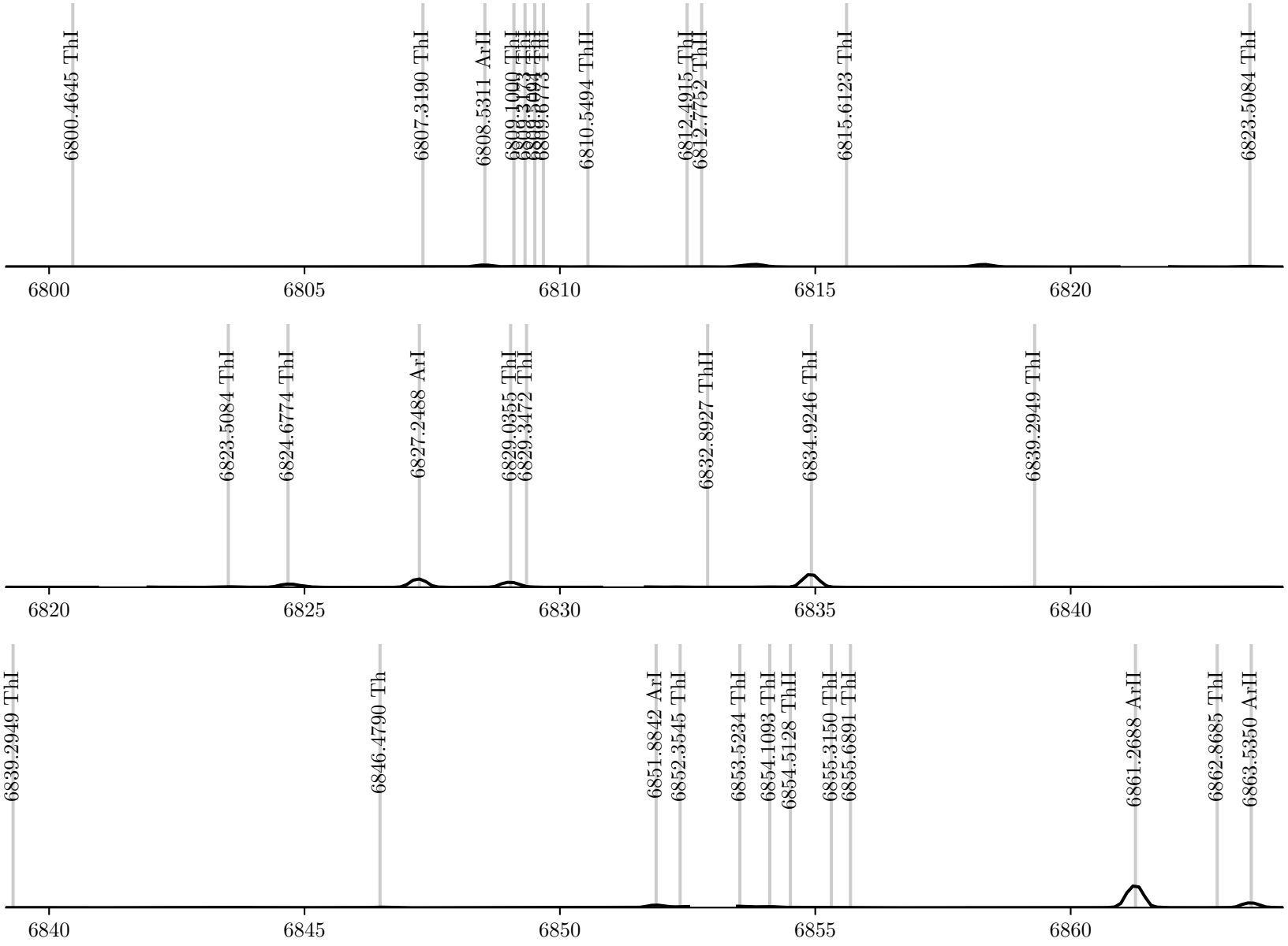

6726.3091 ThI
6727.4583 ThI
6728.1183 ThI
6728.7595 ThI
6729.9328 ThI
6732.6491 ThI
6733.7488 ThI
6738.1800 ThI
6741.9212 ThI
6742.8845 ThI
6725
6730
6735
6740
6745
6741.9212 ThI
6742.8845 ThI
6751.4267 ThI
6752.8335 ArI
6753.6598 ThI
6756.4528 ThI
6757.1092 ThI
6758.2035 ThI
6760.8923 ThII
6745
6750
6755
6760
6765
6766.6117 ArI
6770.1069 ThII
6772.1745 ThI
6773.0973 Th
6765
6770
6775
6780
6785

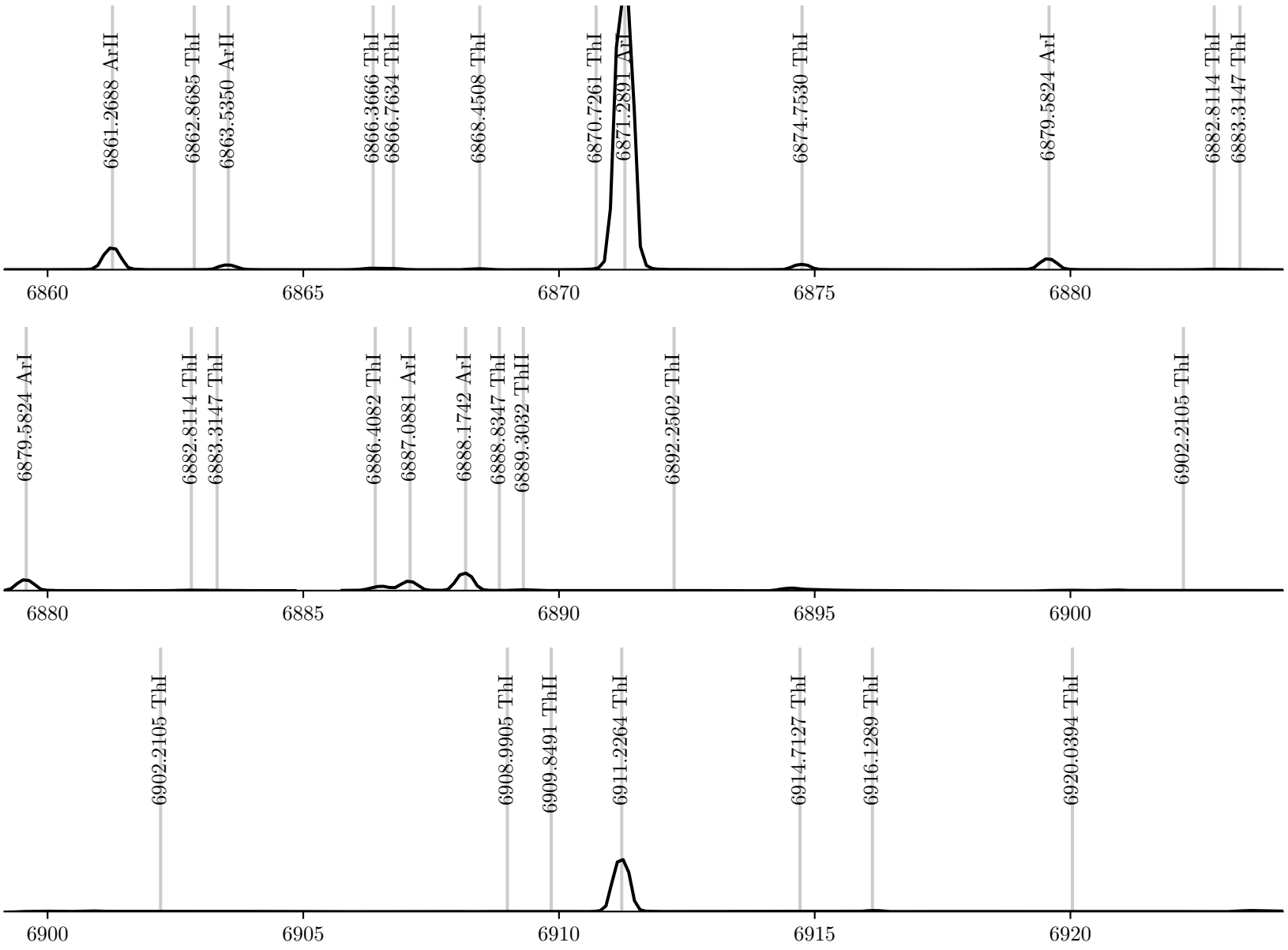
Order 33

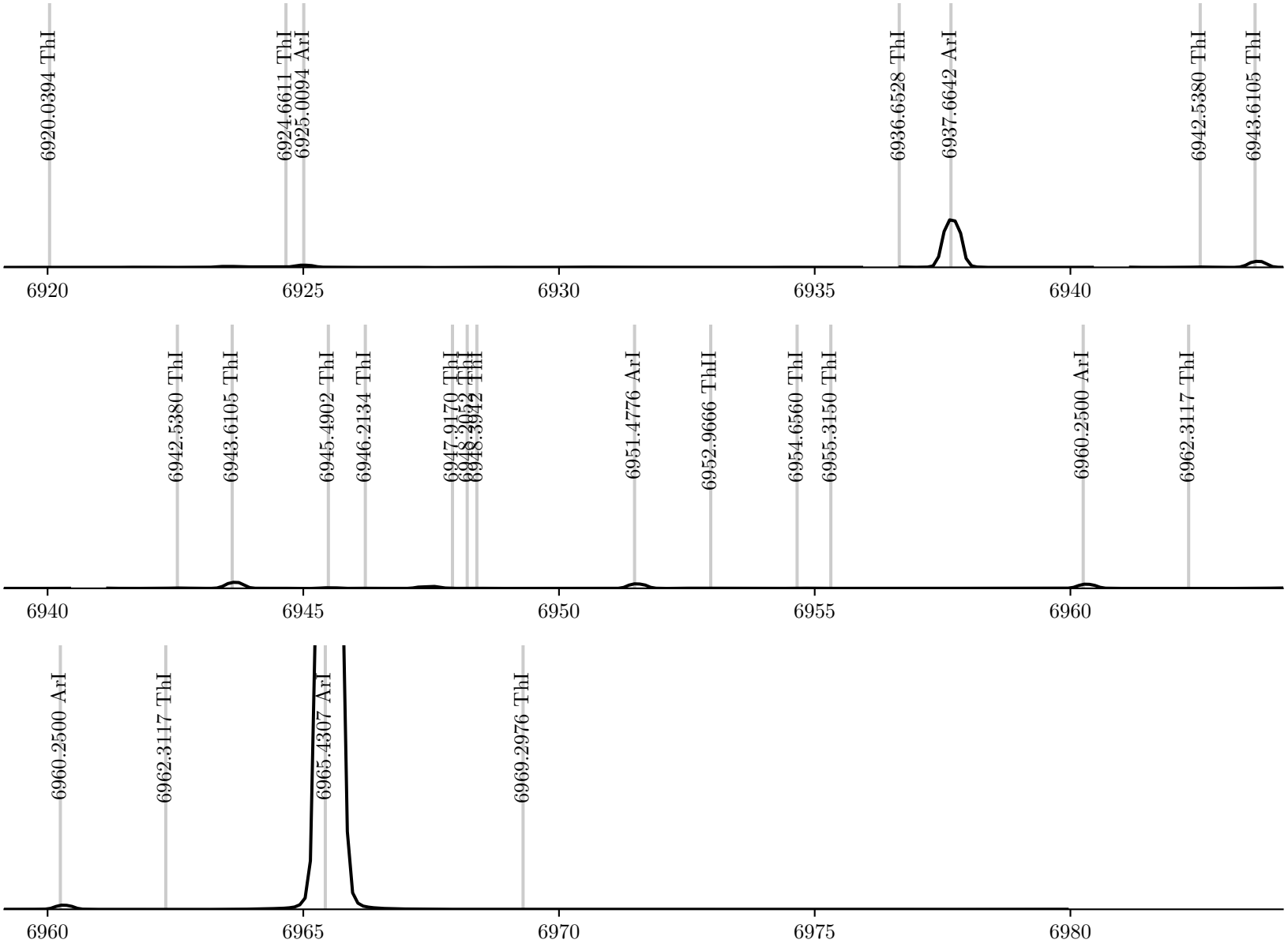


Approximate resolution:	18400
Dispersion:	0.118 Å pix^{-1}
Central wavelength:	6868 Å
Wavelength range:	6739 Å – 6980 Å

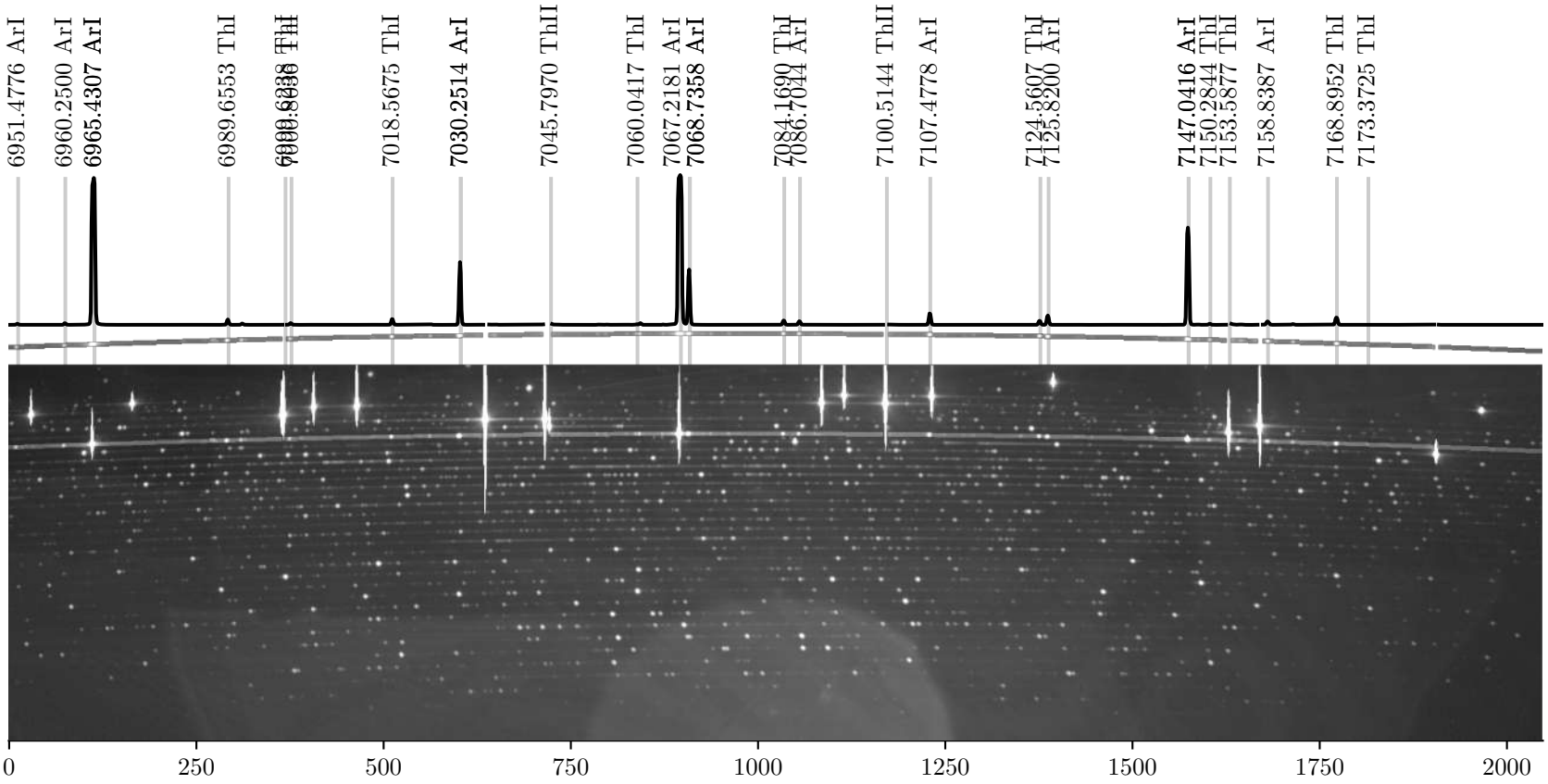

6741.9212 ThI
6742.8845 ThI
6751.4267 ThI
6752.8335 ArI
6753.6598 ThI
6756.4528 ThI
6757.1092 ThI
6758.2035 ThI
6760.8923 ThII
6740
6745
6750
6755
6760
6760.8923 ThII
6766.6117 ArI
6770.1069 ThII
6772.1745 ThI
6773.0973 Th
6778.3123 ThI
6779.3242 ThI
6780.1252 ThI
6780.4131 ThI
6760
6765
6770
6775
6780
6779.3242 ThI
6780.1252 ThI
6780.4131 ThI
6787.7364 ThI
6788.8403 ThI
6791.2351 ThI
6795.7982 ThI
6798.4868 ThI
6800.4645 ThI
6780
6785
6790
6795
6800


6800.4645 ThI
6807.3190 ThI
6808.5311 ArII
6810.5494 ThII
6812.4915 ThI
6812.7752 ThII
6815.6123 ThI
6823.5084 ThI
6800
6805
6810
6815
6820
6823.5084 ThI
6824.6774 ThI
6827.2488 ArI
6829.0355 ThI
6829.3472 ThI
6832.8927 ThII
6834.9246 ThI
6839.2949 ThI
6820
6825
6830
6835
6840
6839.2949 ThI
6846.4790 Th
6851.8842 ArI
6852.3545 ThI
6853.5234 ThI
6854.1093 ThI
6854.5128 ThII
6855.3150 ThI
6855.6891 ThI
6861.2688 ArII
6862.8685 ThI
6863.5350 ArII
6840
6845
6850
6855
6860

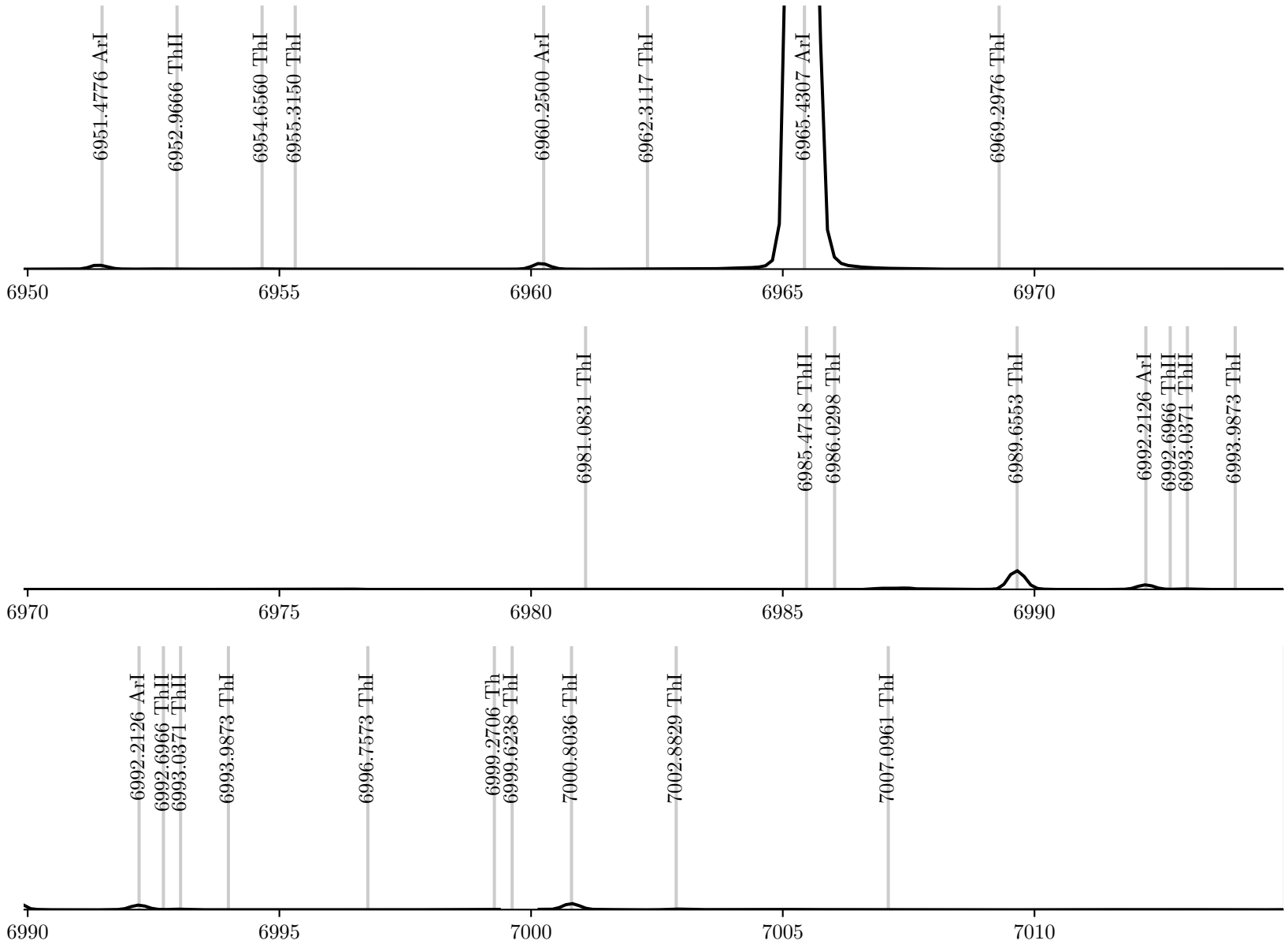

6861.2688 ArII
6862.8685 ThI
6863.5350 ArII
6866.3666 ThI
6866.7634 ThI
6868.4508 ThI
6870.7261 ThI
6871.2891 ArI
6874.7530 ThI
6879.5824 ArI
6882.8114 ThI
6883.3147 ThI
6860
6865
6870
6875
6880
6879.5824 ArI
6882.8114 ThI
6883.3147 ThI
6886.4082 ThI
6887.0881 ArI
6888.1742 ArI
6888.8347 ThI
6889.3032 ThII
6892.2502 ThI
6902.2105 ThI
6880
6885
6890
6895
6900
6902.2105 ThI
6908.9905 ThI
6909.8491 ThII
6911.2264 ThI
6914.7127 ThI
6916.1289 ThI
6920.0394 ThI
6900
6905
6910
6915
6920

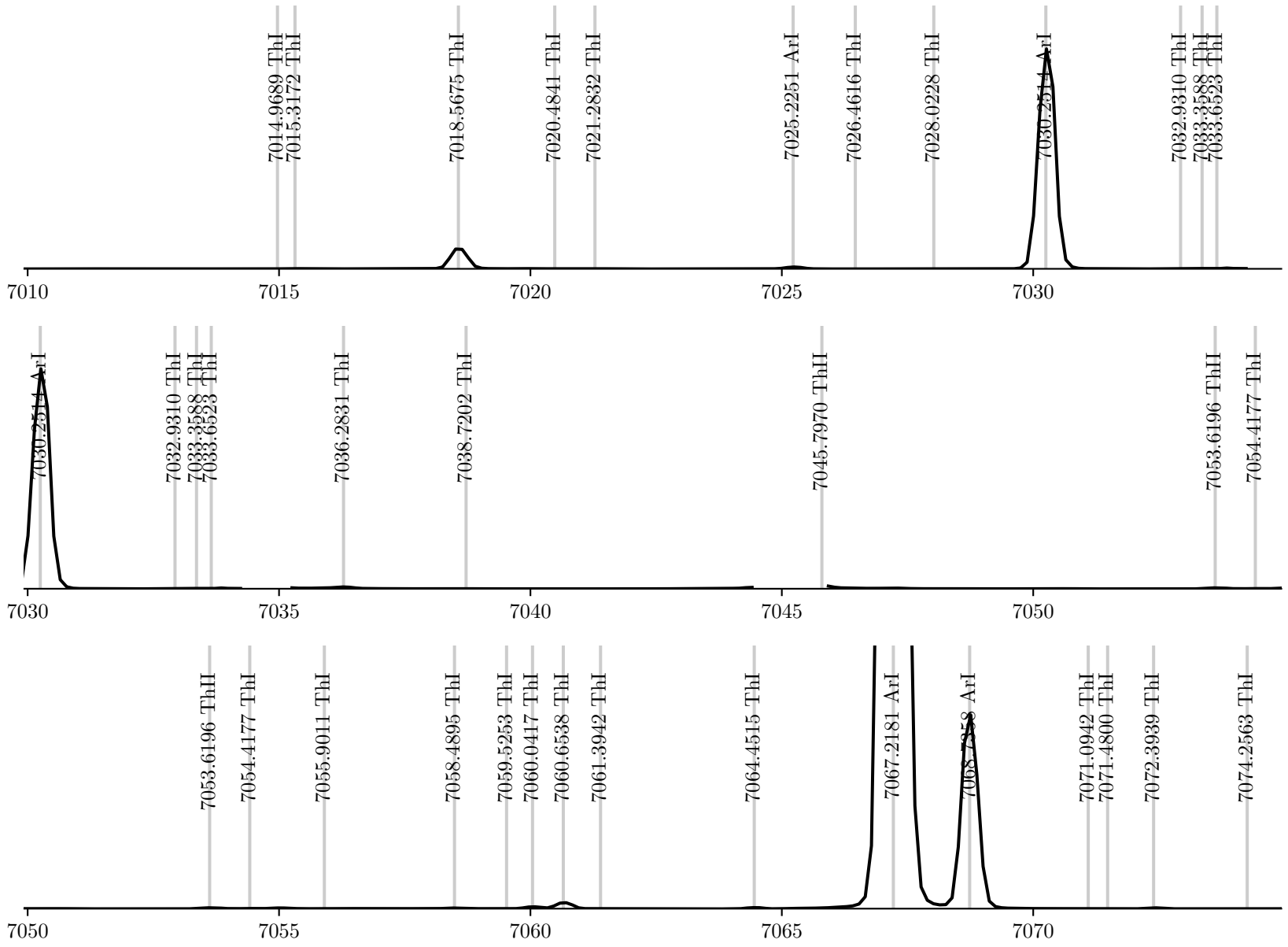

6920.0394 ThI
6924.6611 ThI
6925.0094 ArI
6936.6528 ThI
6937.6642 ArI
6942.5380 ThI
6943.6105 ThI
6920
6925
6930
6935
6940
6942.5380 ThI
6943.6105 ThI
6945.4902 ThI
6946.2134 ThI
6947.9170 ThI
6951.4776 ArI
6952.9666 ThII
6954.6560 ThI
6955.3150 ThI
6960.2500 ArI
6962.3117 ThI
6940
6945
6950
6955
6960
6960.2500 ArI
6962.3117 ThI
6965.4307 ArI
6969.2976 ThI
6960
6965
6970
6975
6980

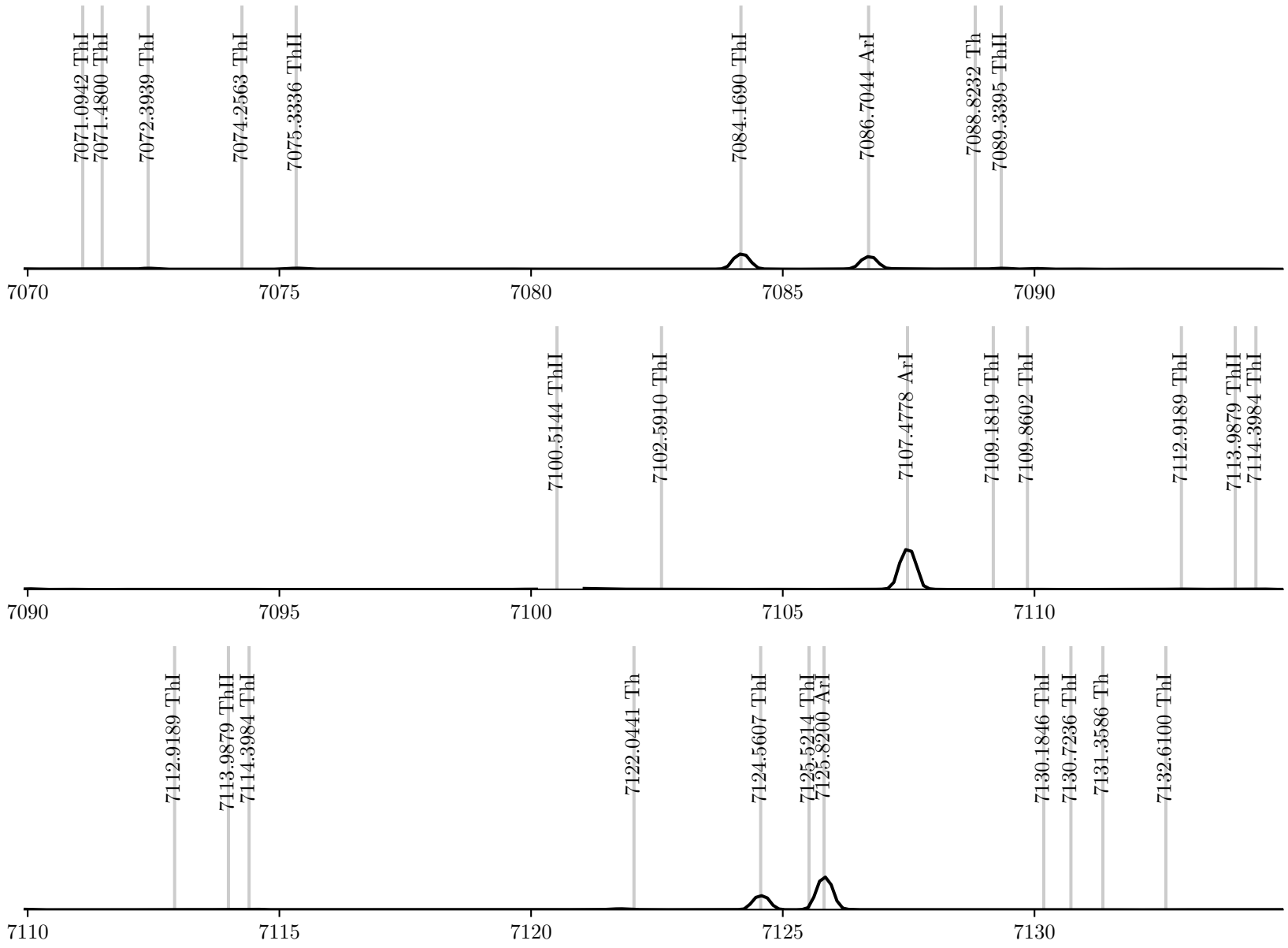
Order 32

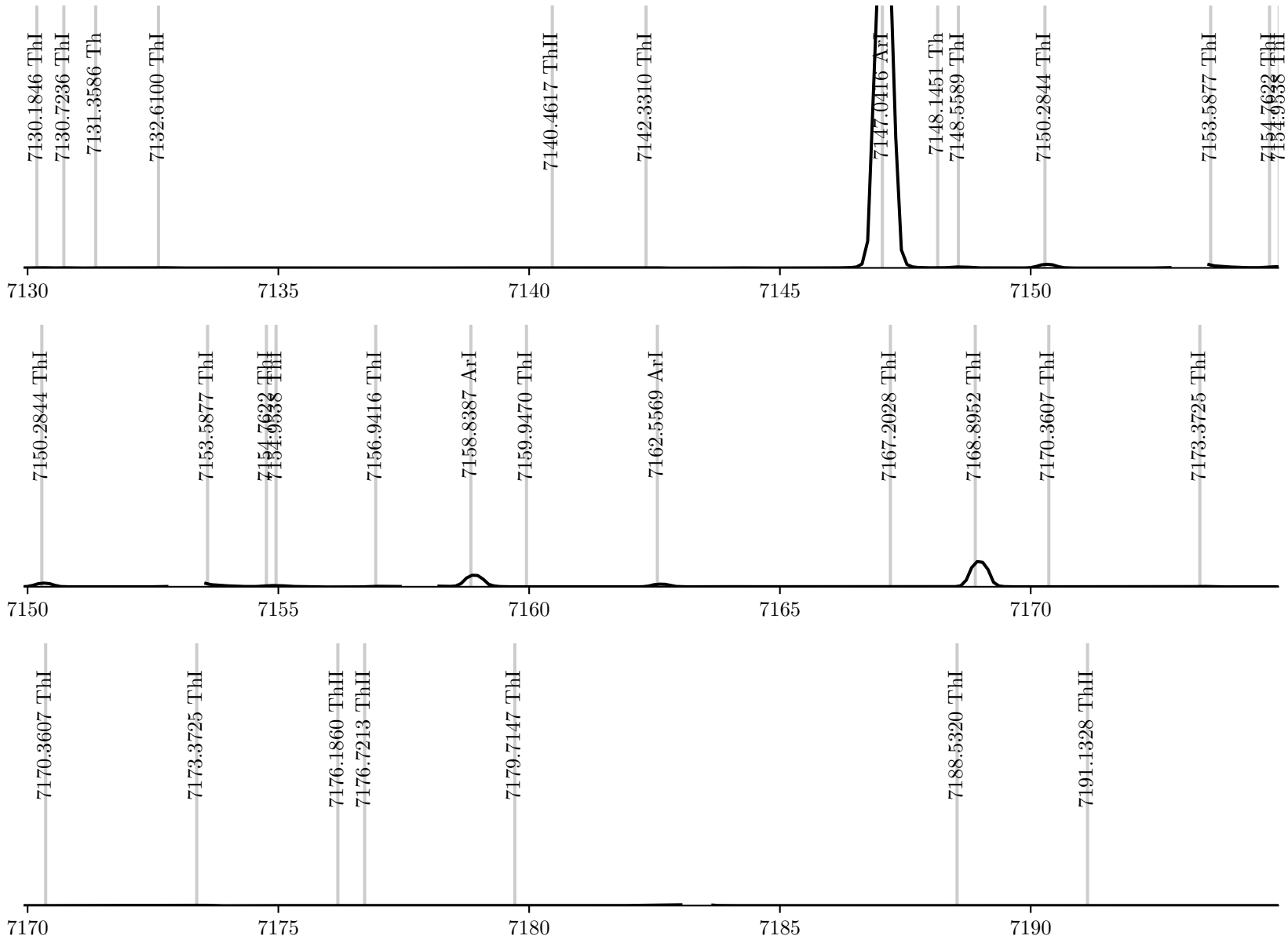


Approximate resolution:	17800
Dispersion:	0.121 Å pix^{-1}
Central wavelength:	7083 Å
Wavelength range:	6950 Å – 7198 Å

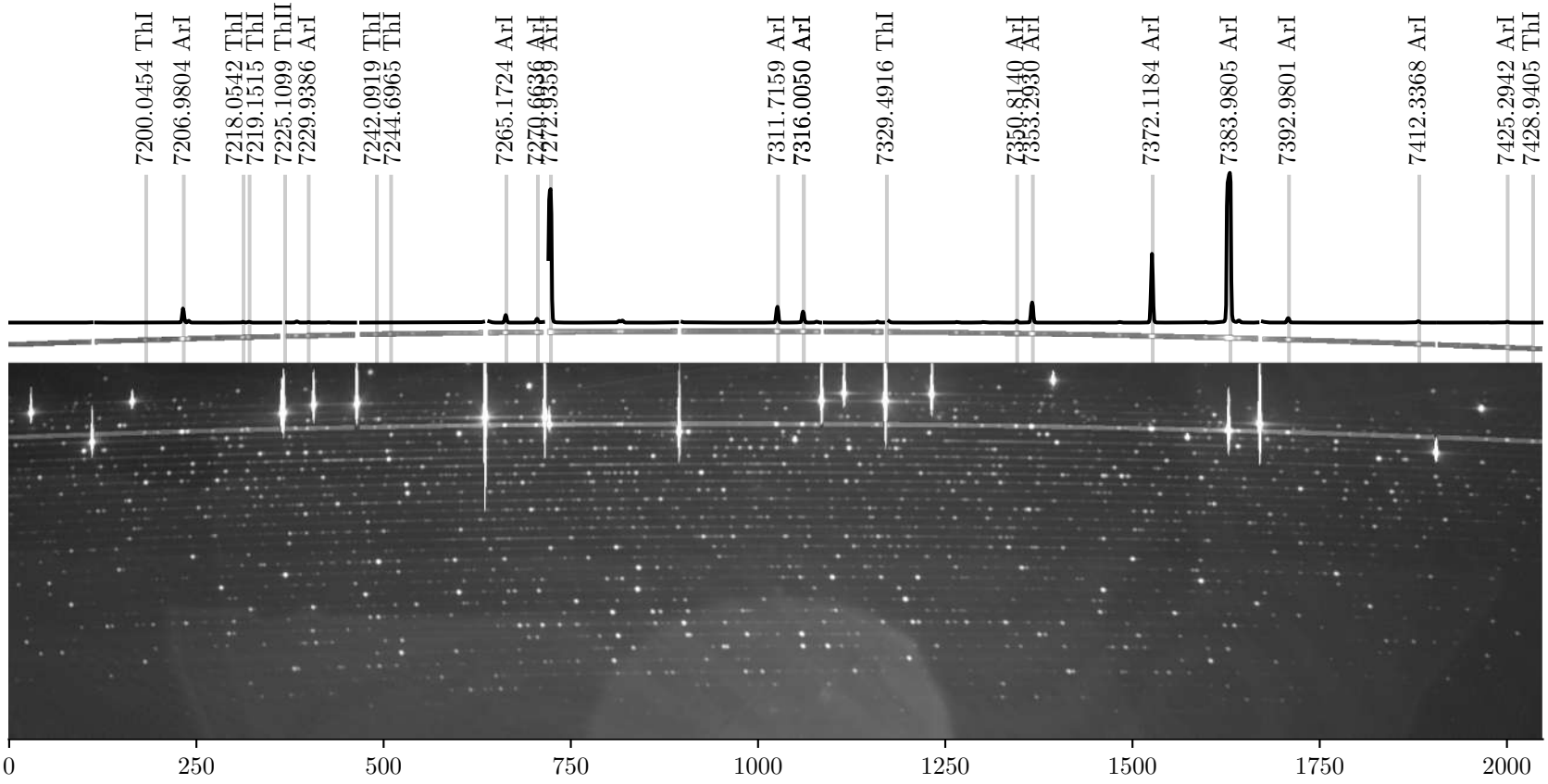

6951.4776 ArI
6952.9666 ThII
6954.6560 ThI
6955.3150 ThI
6960.2500 ArI
6962.3117 ThI
6965.4307 ArI
6969.2976 ThI
6950
6955
6960
6965
6970
6981.0831 ThI
6985.4718 ThII
6986.0298 ThI
6989.6553 ThI
6992.2126 ArI
6992.6966 ThII
6993.0371 ThII
6993.9873 ThI
6970
6975
6980
6985
6990
6992.2126 ArI
6992.6966 ThII
6993.0371 ThII
6993.9873 ThI
6996.7573 ThI
6999.2706 Th
6999.6238 ThI
7000.8036 ThI
7002.8829 ThI
7007.0961 ThI
6990
6995
7000
7005
7010


7014.9689 ThI
7015.3172 ThI
7018.5675 ThI
7020.4841 ThI
7021.2832 ThI
7025.2251 ArI
7026.4616 ThI
7028.0228 ThI
7030.2514 ArI
7032.9310 ThI
7033.3588 ThI
7033.6523 ThI
7010
7015
7020
7025
7030
7030.2514 ArI
7032.9310 ThI
7033.3588 ThI
7033.6523 ThI
7036.2831 ThI
7038.7202 ThI
7045.7970 ThII
7053.6196 ThII
7054.4177 ThI
7030
7035
7040
7045
7050
7053.6196 ThII
7054.4177 ThI
7055.9011 ThI
7058.4895 ThI
7059.5253 ThI
7060.0417 ThI
7060.6538 ThI
7061.3942 ThI
7064.4515 ThI
7067.2181 ArI
7068.7358 ArI
7071.0942 ThI
7071.4800 ThI
7072.3939 ThI
7074.2563 ThI
7050
7055
7060
7065
7070

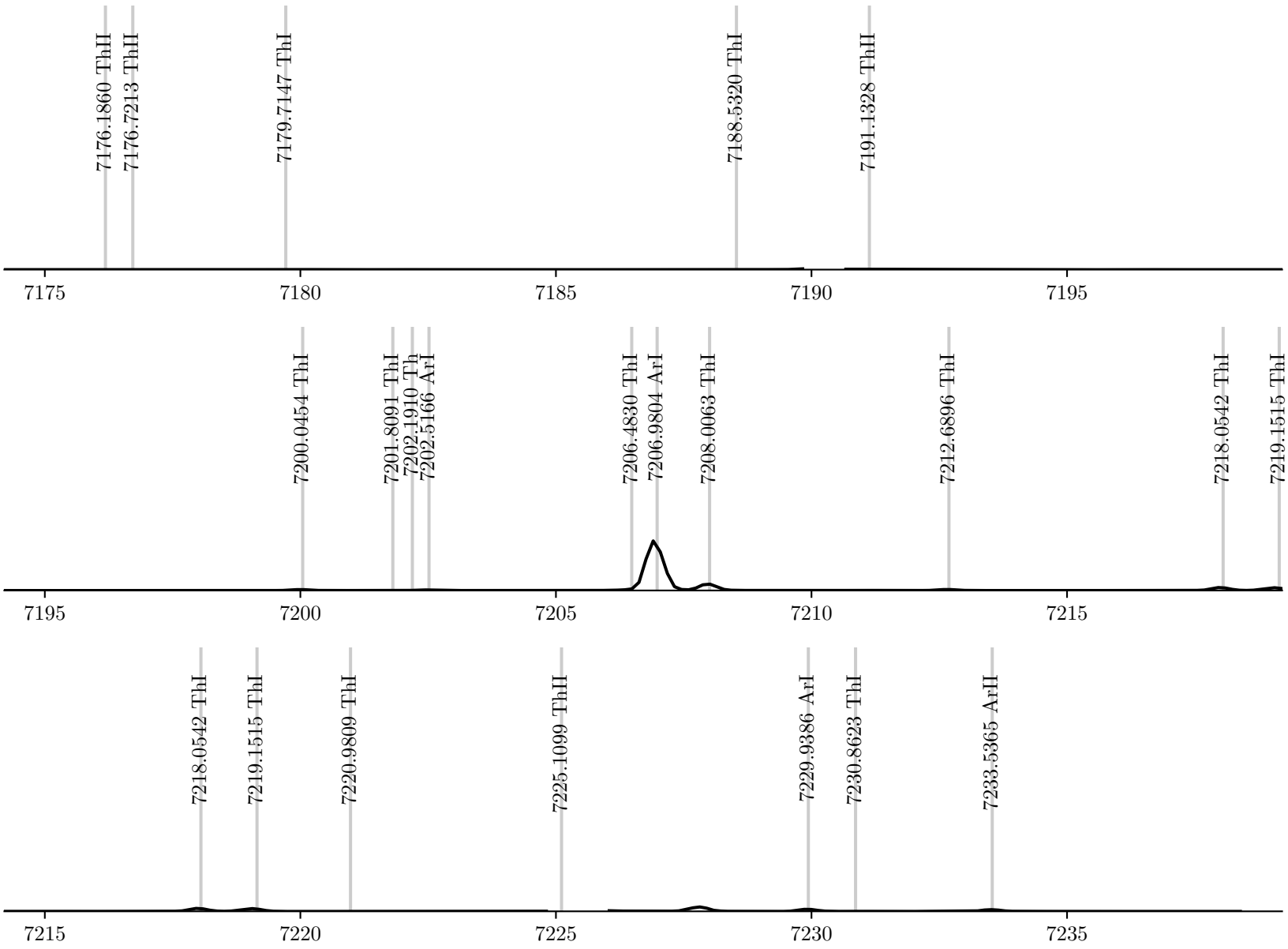

7071.0942 ThI
7071.4800 ThI
7072.3939 ThI
7074.2563 ThI
7075.3336 ThII
7084.1690 ThI
7086.7044 ArI
7088.8232 Th
7089.3395 ThII
7070
7075
7080
7085
7090
7100.5144 ThII
7102.5910 ThI
7107.4778 ArI
7109.1819 ThI
7109.8602 ThI
7112.9189 ThI
7113.9879 ThII
7114.3984 ThI
7090
7095
7100
7105
7110
7112.9189 ThI
7113.9879 ThII
7114.3984 ThI
7122.0441 Th
7124.5607 ThI
7125.5214 ThI
7125.8200 ArI
7130.1846 ThI
7130.7236 ThI
7131.3586 Th
7132.6100 ThI
7110
7115
7120
7125
7130

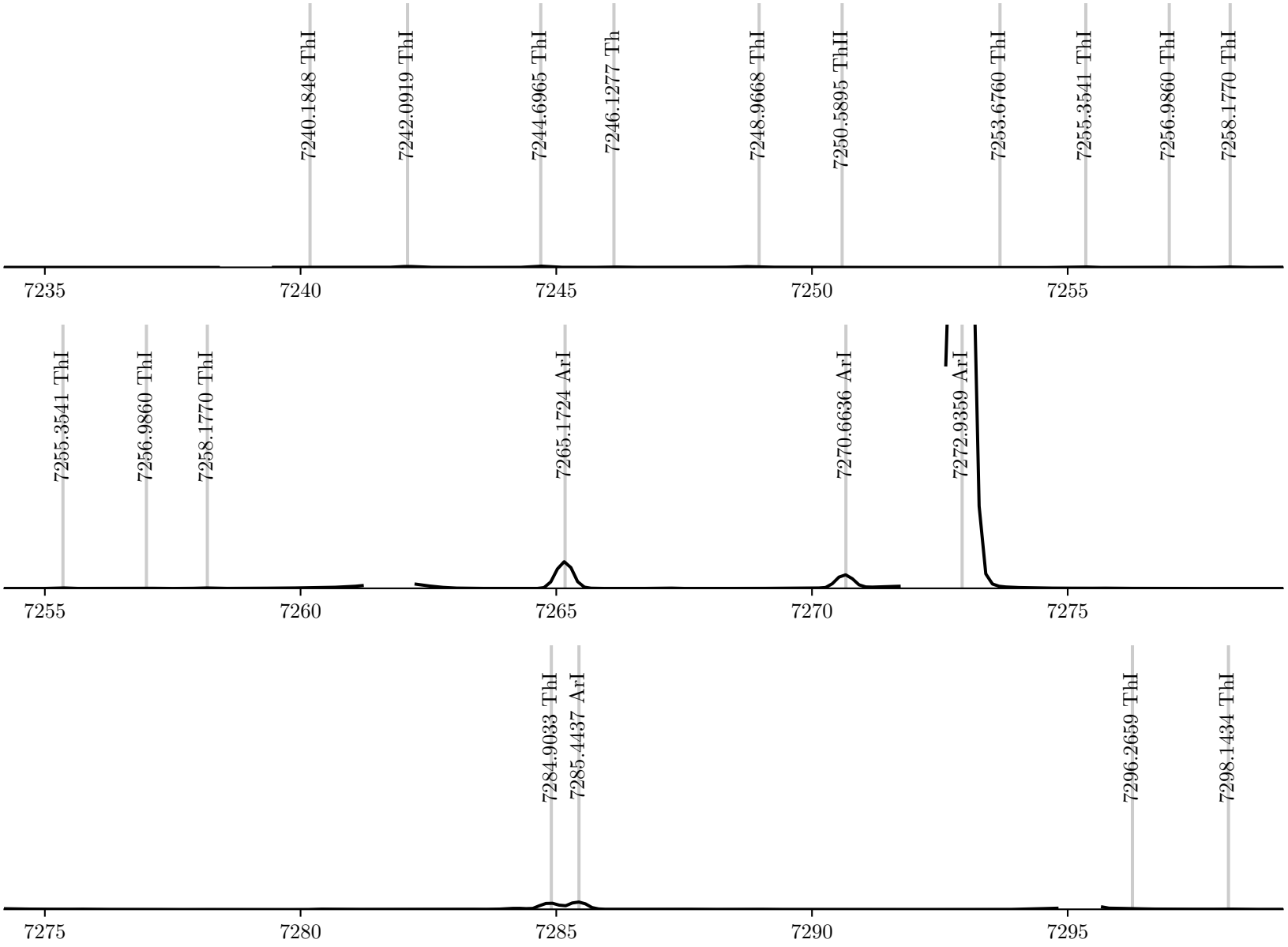

7130.1846 ThI
7130.7236 ThI
7131.3586 Th
7132.6100 ThI
7140.4617 ThII
7142.3310 ThI
7147.0416 ArI
7148.1451 Th
7148.5589 ThI
7150.2844 ThI
7153.5877 ThI
7154.7622 ThI
7154.9338 ThI
7130
7135
7140
7145
7150
7150.2844 ThI
7153.5877 ThI
7154.7622 ThI
7154.9338 ThI
7156.9416 ThI
7158.8387 ArI
7159.9470 ThI
7162.5569 ArI
7167.2028 ThI
7168.8952 ThI
7170.3607 ThI
7173.3725 ThI
7150
7155
7160
7165
7170
7170.3607 ThI
7173.3725 ThI
7176.1860 ThII
7176.7213 ThII
7179.7147 ThI
7188.5320 ThI
7191.1328 ThII
7170
7175
7180
7185
7190

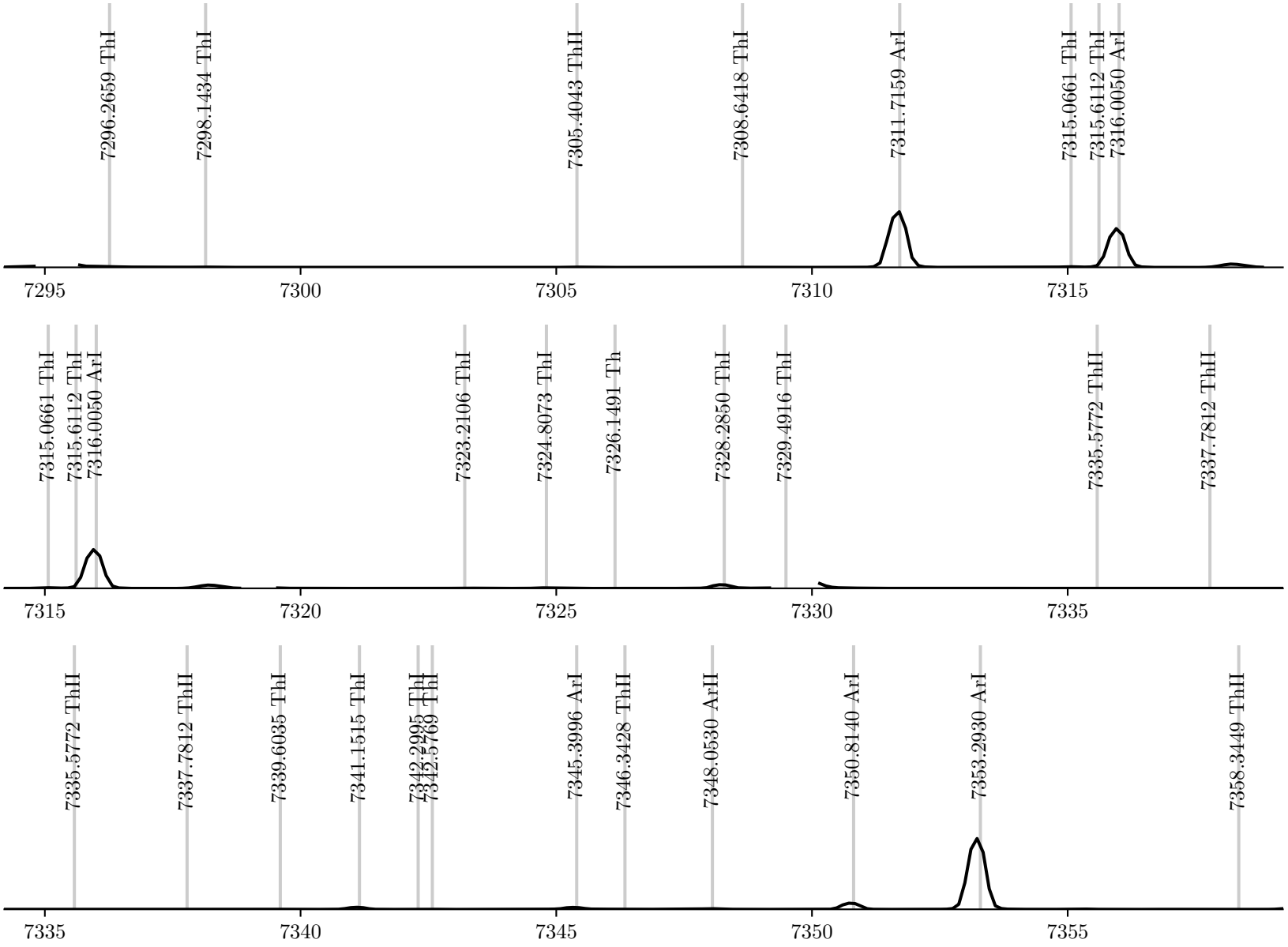
Order 31

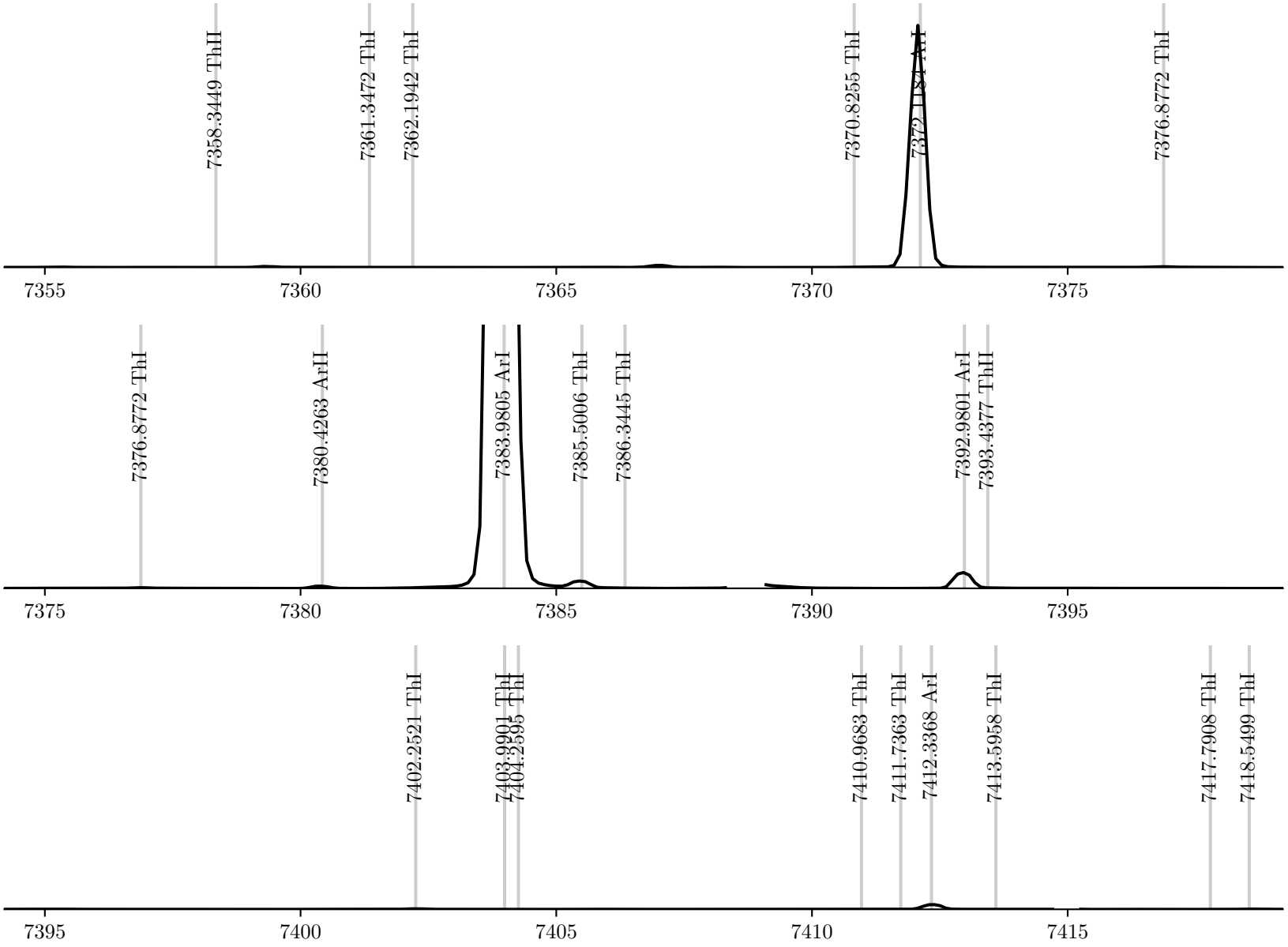


Approximate resolution:	18000
Dispersion:	0.125 Å pix^{-1}
Central wavelength:	7311 Å
Wavelength range:	7174 Å – 7430 Å

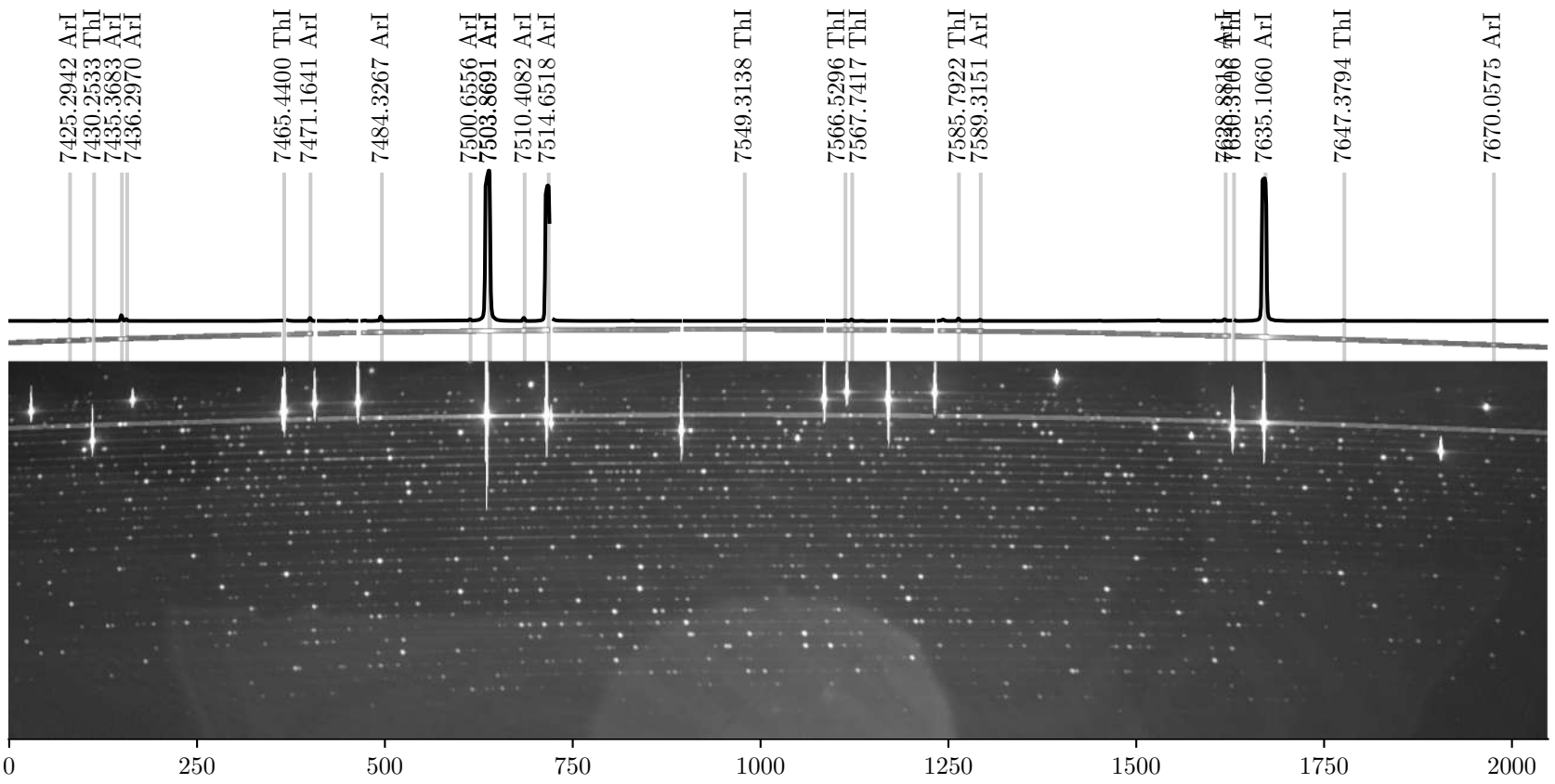

7176.1860 ThII
7176.7213 ThII
7179.7147 ThI
7188.5320 ThI
7191.1328 ThII
7175
7180
7185
7190
7195
7200.0454 ThI
7201.8091 ThI
7202.1910 Th
7202.5166 ArI
7206.4830 ThI
7206.9804 ArI
7208.0063 ThI
7212.6896 ThI
7218.0542 ThI
7219.1515 ThI
7195
7200
7205
7210
7215
7218.0542 ThI
7219.1515 ThI
7220.9809 ThI
7225.1099 ThII
7229.9386 ArI
7230.8623 ThI
7233.5365 ArII
7215
7220
7225
7230
7235


7240.1848 ThI
7242.0919 ThI
7244.6965 ThI
7246.1277 Th
7248.9668 ThI
7250.5895 ThII
7253.6760 ThI
7255.3541 ThI
7256.9860 ThI
7258.1770 ThI
7235
7240
7245
7250
7255
7255.3541 ThI
7256.9860 ThI
7258.1770 ThI
7265.1724 ArI
7270.6636 ArI
7272.9359 ArI
7255
7260
7265
7270
7275
7284.9033 ThI
7285.4437 ArI
7296.2659 ThI
7298.1434 ThI
7275
7280
7285
7290
7295

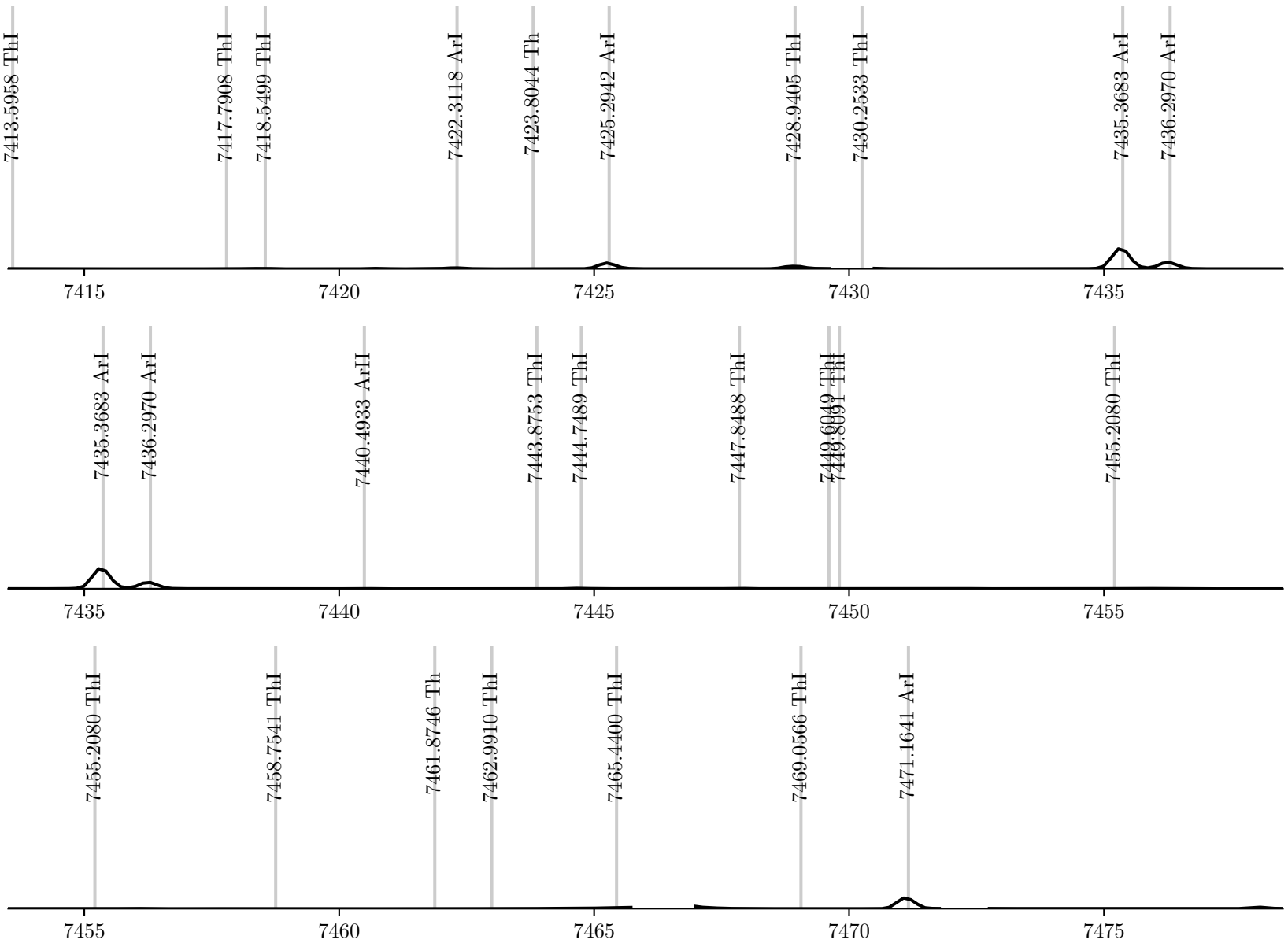

7296.2659 ThI
7298.1434 ThI
7305.4043 ThII
7308.6418 ThI
7311.7159 ArI
7315.0661 ThI
7315.6112 ThI
7316.0050 ArI
7295
7300
7305
7310
7315
7315.0661 ThI
7315.6112 ThI
7316.0050 ArI
7323.2106 ThI
7324.8073 ThI
7326.1491 Th
7328.2850 ThI
7329.4916 ThI
7335.5772 ThII
7337.7812 ThII
7315
7320
7325
7330
7335
7335.5772 ThII
7337.7812 ThII
7339.6035 ThI
7341.1515 ThI
7345.3996 ArI
7346.3428 ThII
7348.0530 ArII
7350.8140 ArI
7353.2930 ArI
7358.3449 ThII
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7345
7350
7355

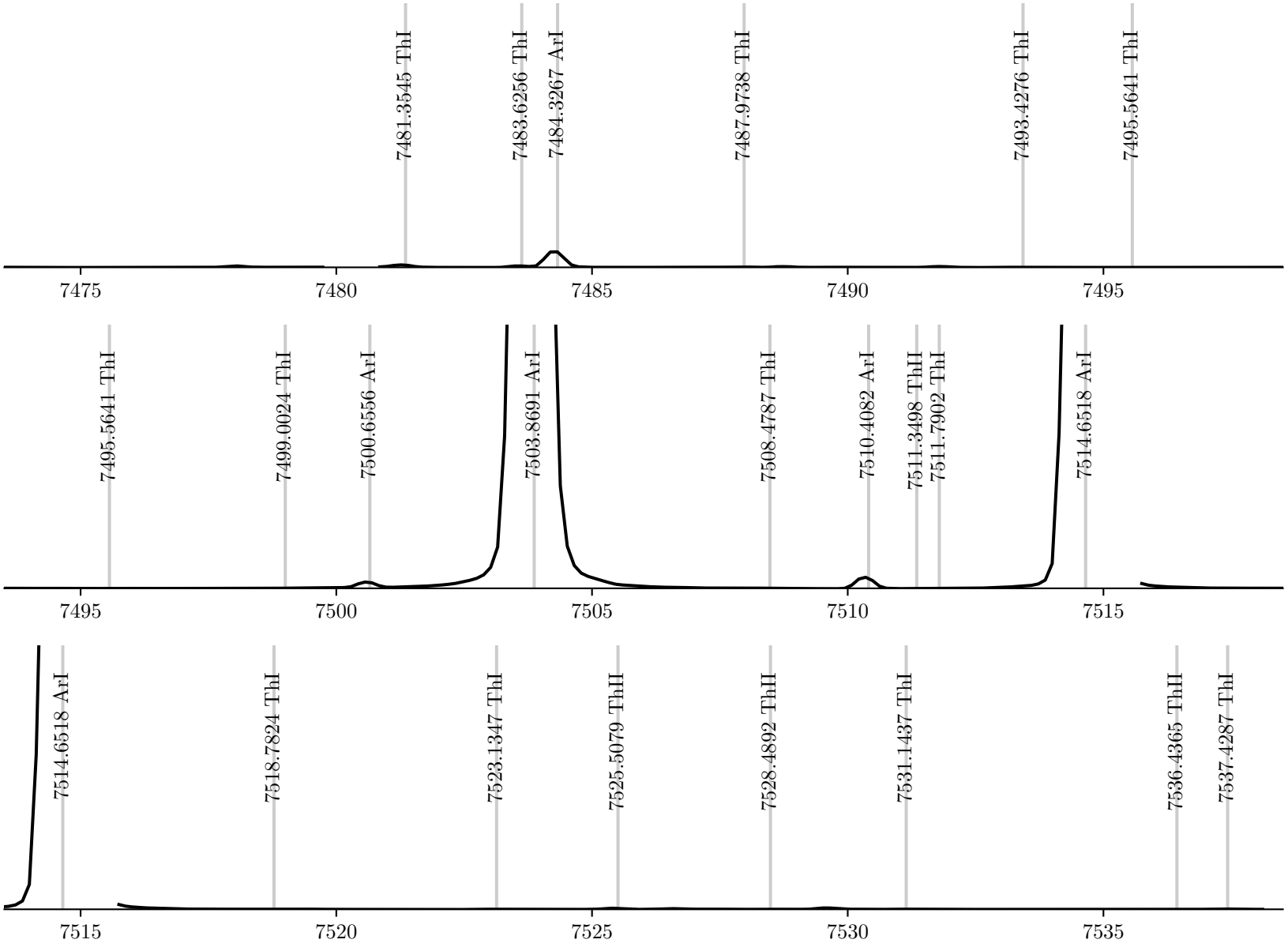

7358.3449 ThII
7361.3472 ThI
7362.1942 ThI
7370.8255 ThI
7372.1184 ArI
7376.8772 ThI
7355
7360
7365
7370
7375
7376.8772 ThI
7380.4263 ArII
7383.9805 ArI
7385.5006 ThI
7386.3445 ThI
7392.9801 ArI
7393.4377 ThII
7375
7380
7385
7390
7395
7402.2521 ThI
7403.3901 ThI
7404.2595 ThI
7410.9683 ThI
7411.7363 ThI
7412.3368 ArI
7413.5958 ThI
7417.7908 ThI
7418.5499 ThI
7395
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7405
7410
7415

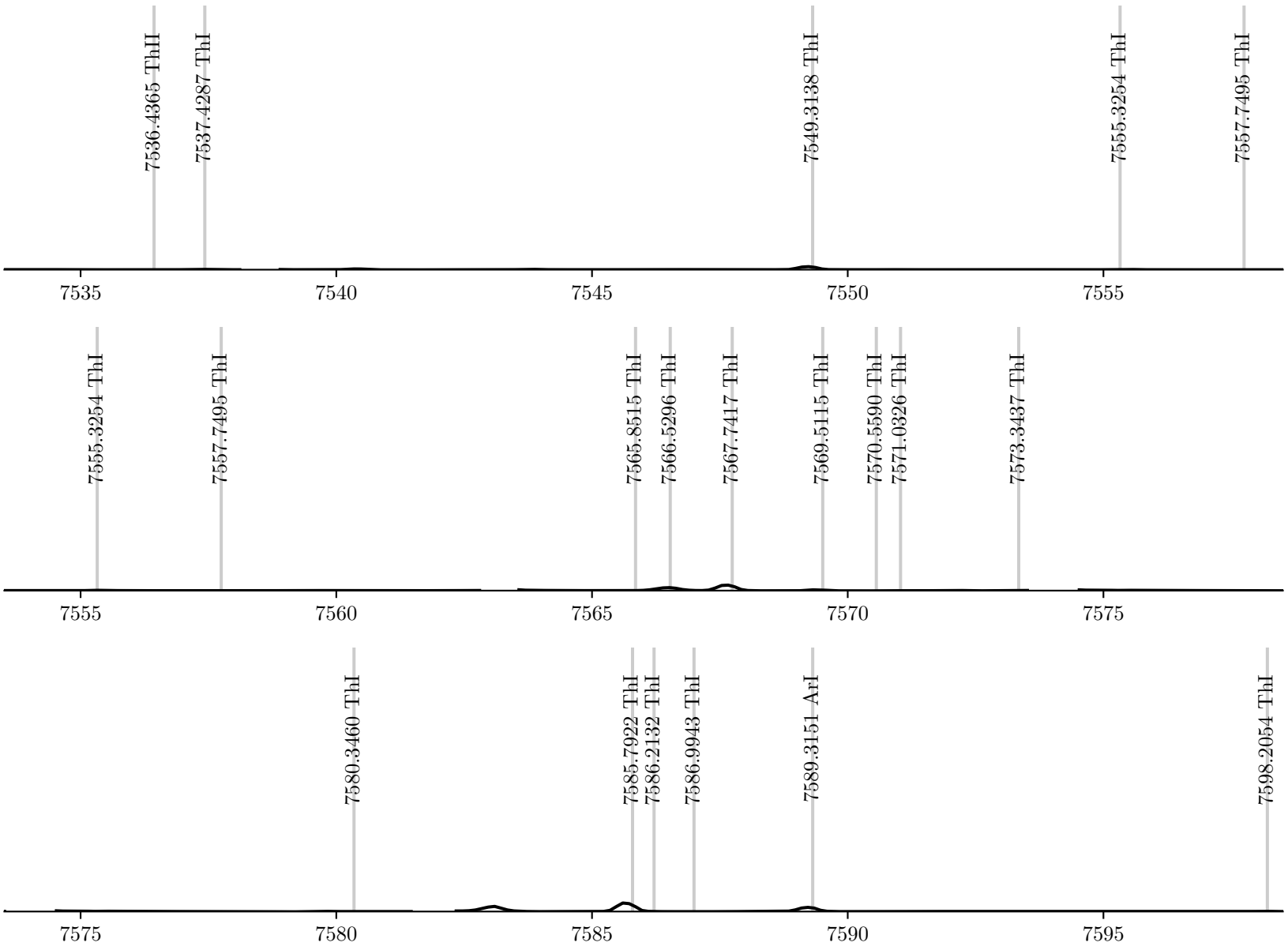
Order 30

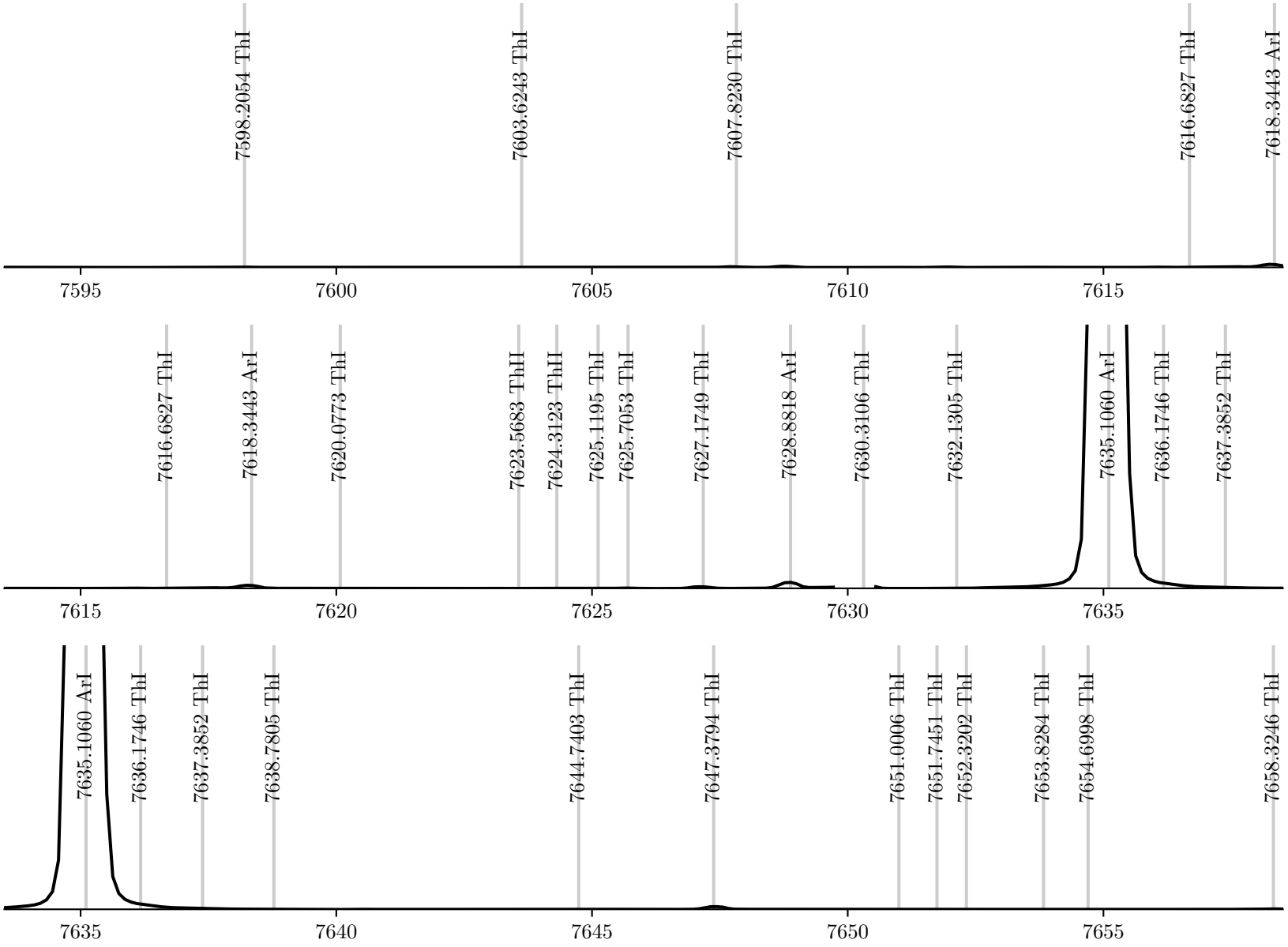


Approximate resolution:	17700
Dispersion:	0.129 Å pix^{-1}
Central wavelength:	7555 Å
Wavelength range:	7414 Å – 7678 Å

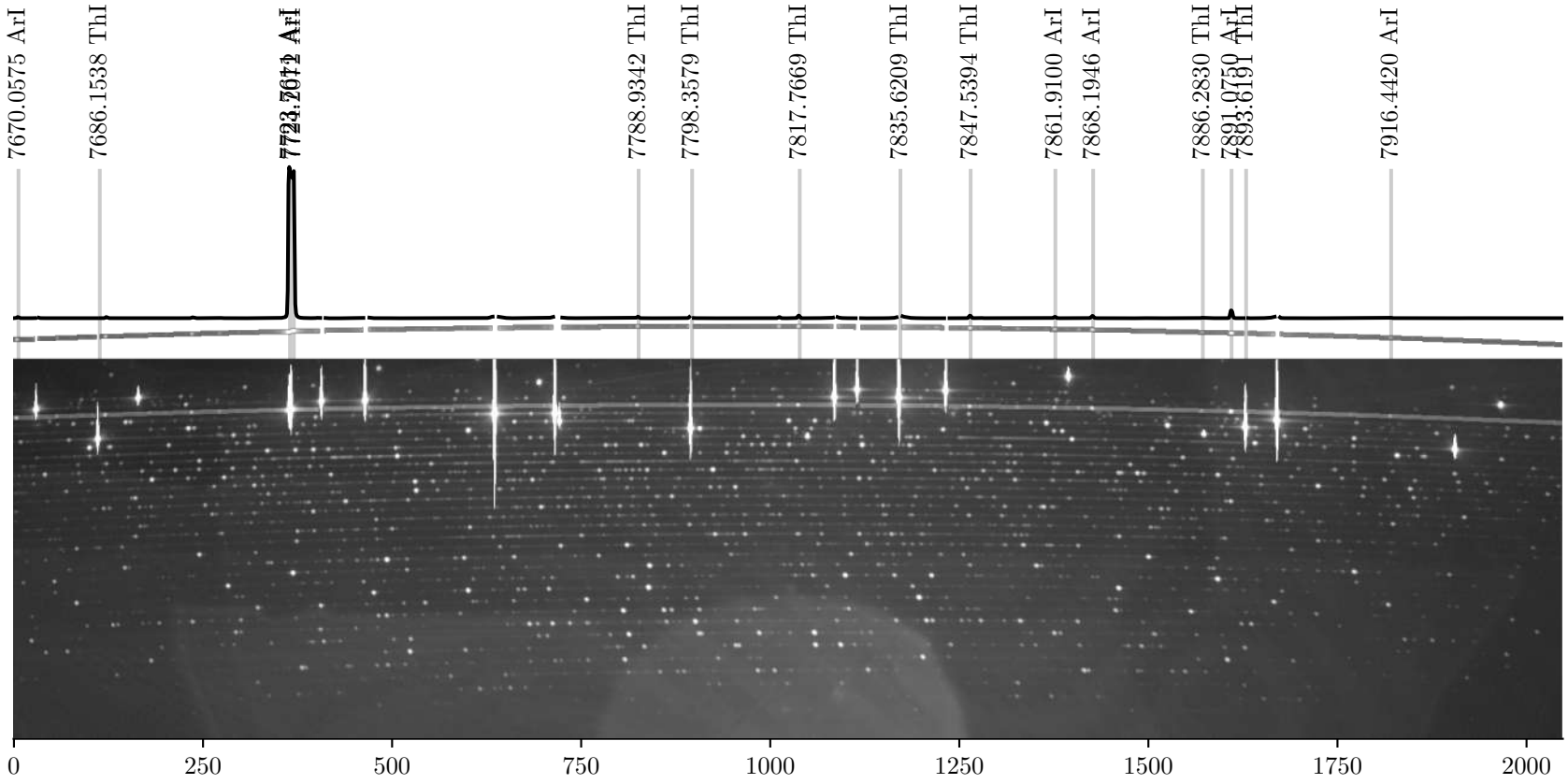

7413.5958 ThI
7417.7908 ThI
7418.5499 ThI
7422.3118 ArI
7423.8044 Th
7425.2942 ArI
7428.9405 ThI
7430.2533 ThI
7435.3683 ArI
7436.2970 ArI
7415
7420
7425
7430
7435
7435.3683 ArI
7436.2970 ArI
7440.4933 ArII
7443.8753 ThI
7444.7489 ThI
7447.8488 ThI
7455.2080 ThI
7435
7440
7445
7450
7455
7455.2080 ThI
7458.7541 ThI
7461.8746 Th
7462.9910 ThI
7465.4400 ThI
7469.0566 ThI
7471.1641 ArI
7455
7460
7465
7470
7475


7481.3545 ThI
7483.6256 ThI
7484.3267 ArI
7487.9738 ThI
7493.4276 ThI
7495.5641 ThI
7475
7480
7485
7490
7495
7495.5641 ThI
7499.0024 ThI
7500.6556 ArI
7503.8691 ArI
7508.4787 ThI
7510.4082 ArI
7511.3498 ThII
7511.7902 ThI
7514.6518 ArI
7495
7500
7505
7510
7515
7514.6518 ArI
7518.7824 ThI
7523.1347 ThI
7525.5079 ThII
7528.4892 ThII
7531.1437 ThI
7536.4365 ThII
7537.4287 ThI
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7520
7525
7530
7535

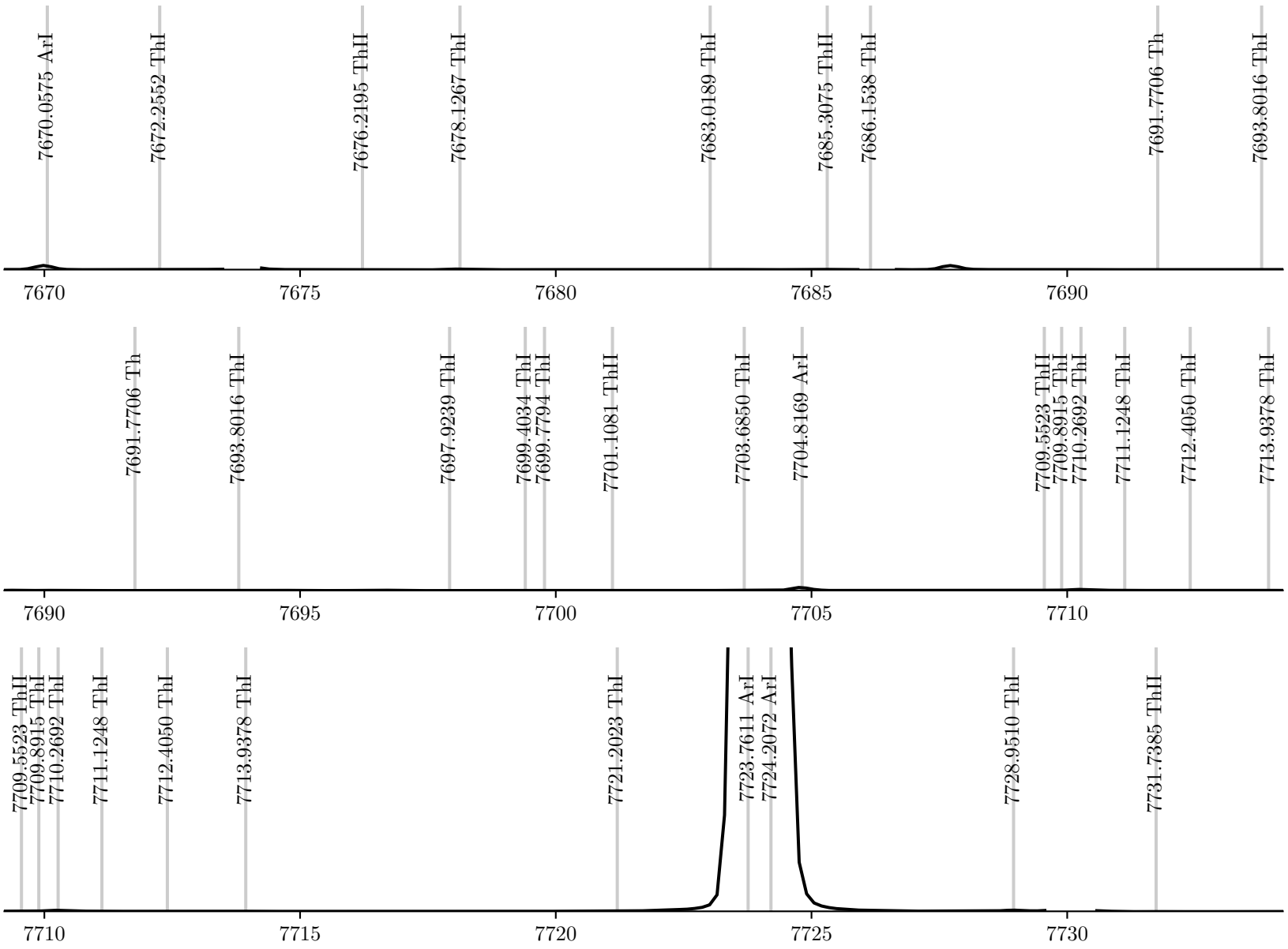

7536.4365 ThII
7537.4287 ThI
7549.3138 ThI
7555.3254 ThI
7557.7495 ThI
7535
7540
7545
7550
7555
7555.3254 ThI
7557.7495 ThI
7565.8515 ThI
7566.5296 ThI
7567.7417 ThI
7569.5115 ThI
7570.5590 ThI
7571.0326 ThI
7573.3437 ThI
7555
7560
7565
7570
7575
7580.3460 ThI
7585.7922 ThI
7586.2132 ThI
7586.9943 ThI
7589.3151 ArI
7598.2054 ThI
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7580
7585
7590
7595

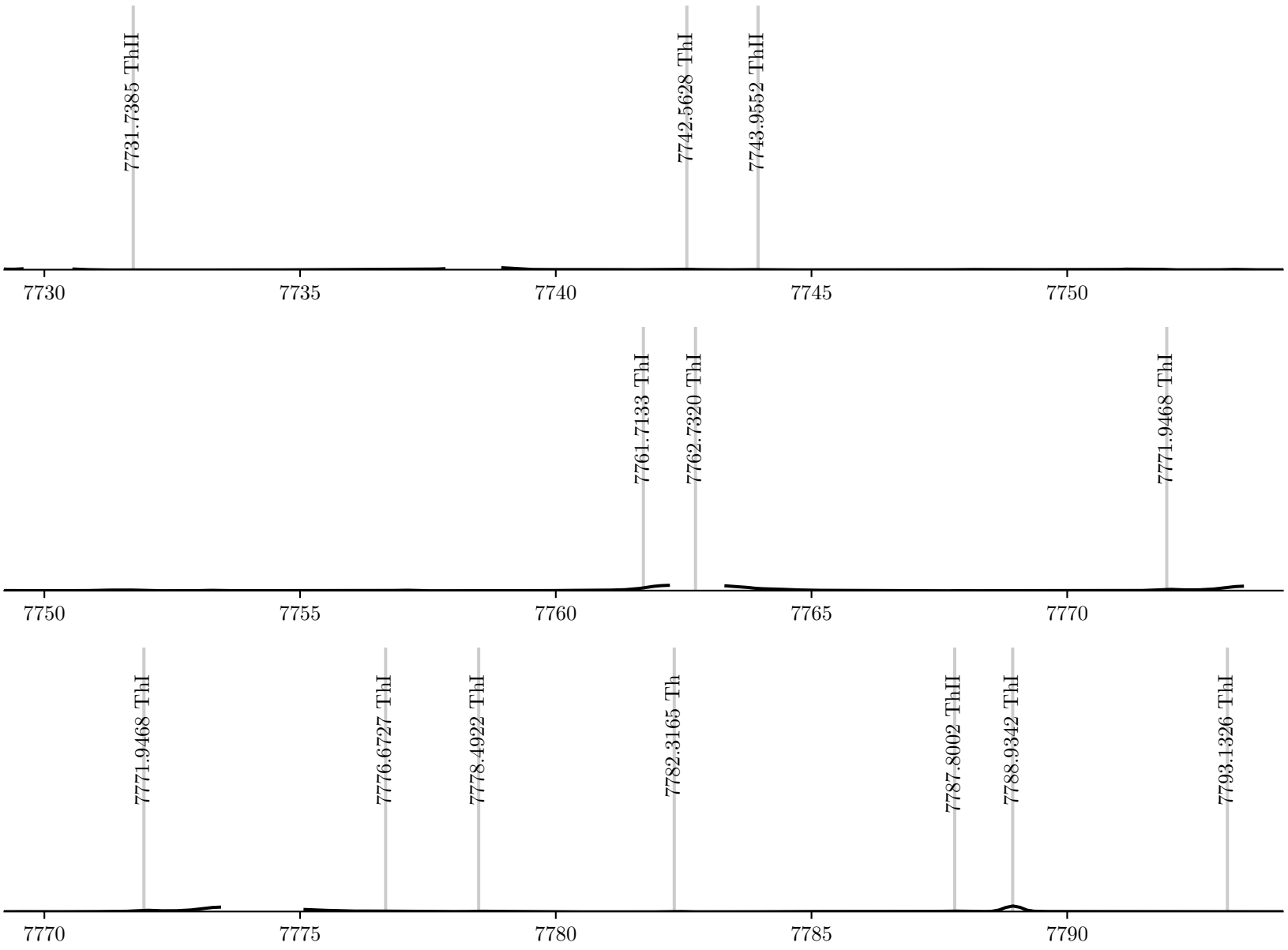

7598.2054 ThI
7603.6243 ThI
7607.8230 ThI
7616.6827 ThI
7618.3443 ArI
7595
7600
7605
7610
7615
7616.6827 ThI
7618.3443 ArI
7620.0773 ThI
7623.5683 ThII
7624.3123 ThII
7625.1195 ThI
7625.7053 ThI
7627.1749 ThI
7628.8818 ArI
7630.3106 ThI
7632.1305 ThI
7635.1060 ArI
7636.1746 ThI
7637.3852 ThI
7615
7620
7625
7630
7635
7635.1060 ArI
7636.1746 ThI
7637.3852 ThI
7638.7805 ThI
7644.7403 ThI
7647.3794 ThI
7651.0006 ThI
7651.7451 ThI
7652.3202 ThI
7653.8284 ThI
7654.6998 ThI
7658.3246 ThI
7635
7640
7645
7650
7655

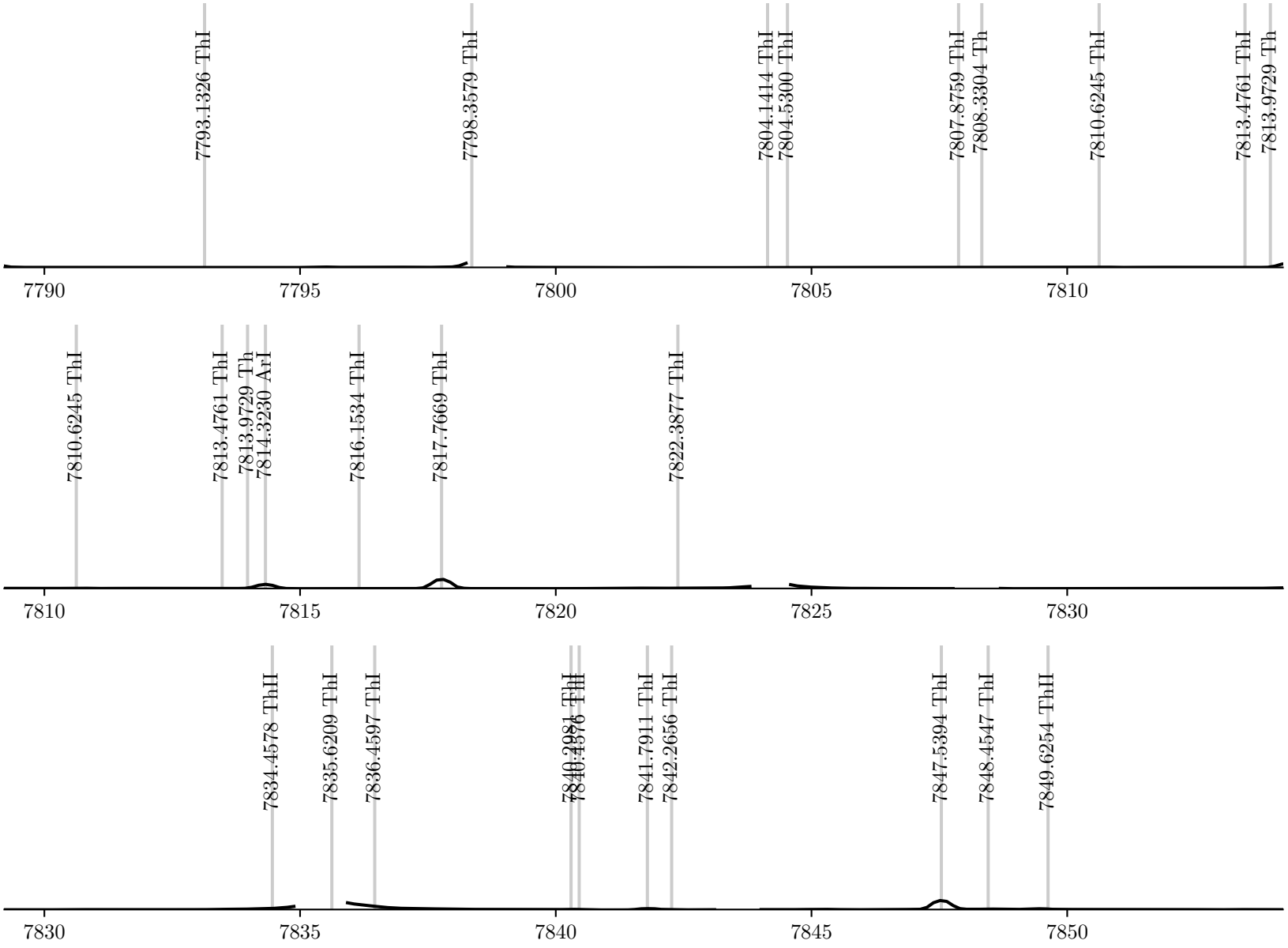
Order 29

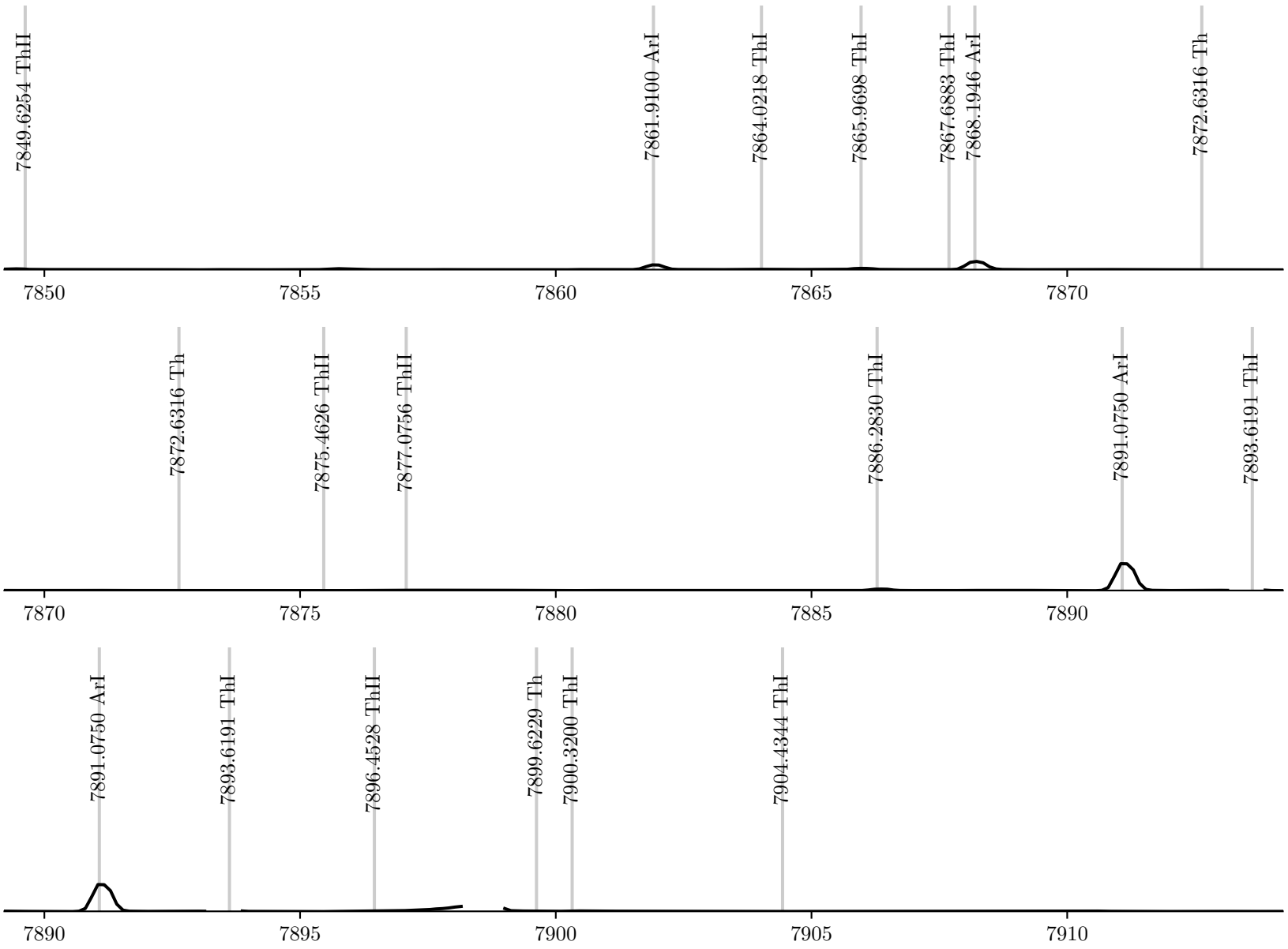


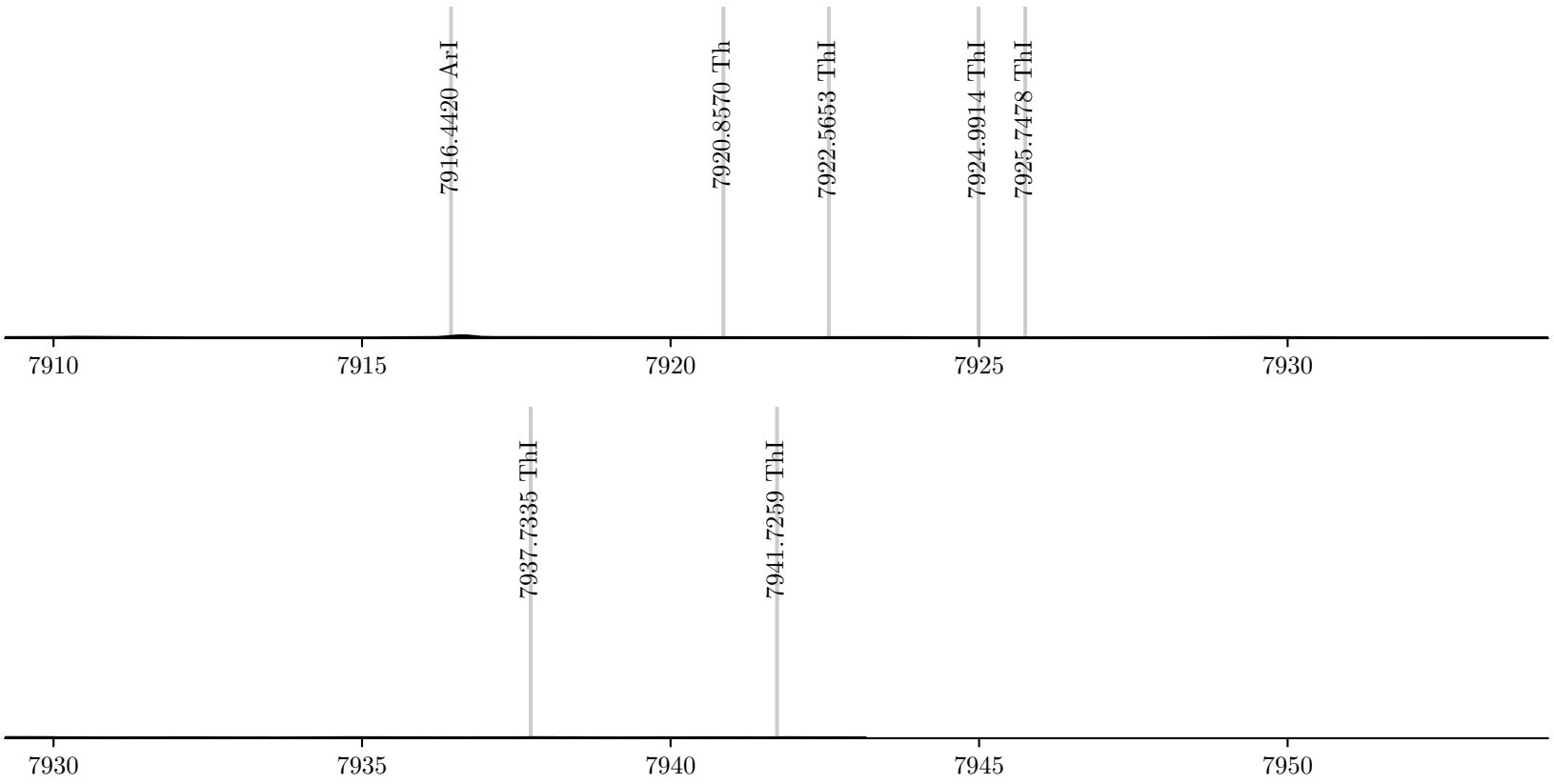
Approximate resolution:	18000
Dispersion:	0.134 Å pix^{-1}
Central wavelength:	7816 Å
Wavelength range:	7669 Å – 7943 Å


7670.0575 ArI
7672.2552 ThI
7676.2195 ThII
7678.1267 ThI
7683.0189 ThI
7685.3075 ThII
7686.1538 ThI
7691.7706 Th
7693.8016 ThI
7670
7675
7680
7685
7690
7691.7706 Th
7693.8016 ThI
7697.9239 ThI
7699.4034 ThI
7699.7794 ThI
7701.1081 ThII
7703.6850 ThI
7704.8169 ArI
7709.5523 ThII
7709.8915 ThI
7710.2692 ThI
7711.1248 ThI
7712.4050 ThI
7713.9378 ThI
7690
7695
7700
7705
7710
7709.5523 ThII
7709.8915 ThI
7710.2692 ThI
7711.1248 ThI
7712.4050 ThI
7713.9378 ThI
7721.2023 ThI
7723.7611 ArI
7724.2072 ArI
7728.9510 ThI
7731.7385 ThII
7710
7715
7720
7725
7730

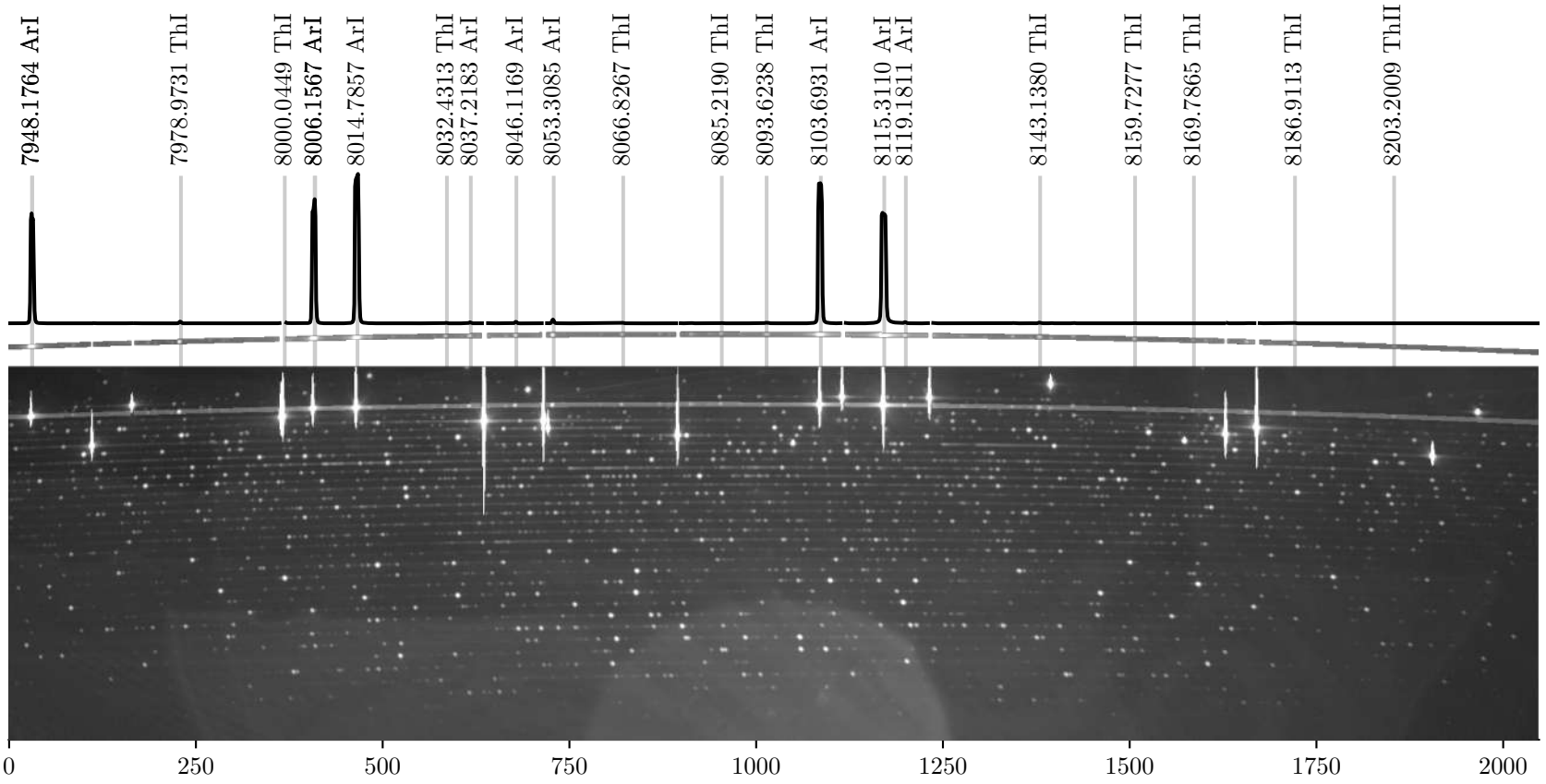

7731.7385 ThII
7742.5628 ThI
7743.9552 ThII
7730
7735
7740
7745
7750
7761.7133 ThI
7762.7320 ThI
7771.9468 ThI
7750
7755
7760
7765
7770
7771.9468 ThI
7776.6727 ThI
7778.4922 ThI
7782.3165 Th
7787.8002 ThII
7788.9342 ThI
7793.1326 ThI
7770
7775
7780
7785
7790


7793.1326 ThI
7798.3579 ThI
7804.1414 ThI
7804.5300 ThI
7807.8759 ThI
7808.3304 Th
7810.6245 ThI
7813.4761 ThI
7813.9729 Th
7790
7795
7800
7805
7810
7810.6245 ThI
7813.4761 ThI
7813.9729 Th
7814.3230 ArI
7816.1534 ThI
7817.7669 ThI
7822.3877 ThI
7810
7815
7820
7825
7830
7834.4578 ThII
7835.6209 ThI
7836.4597 ThI
7841.7911 ThI
7842.2656 ThI
7847.5394 ThI
7848.4547 ThI
7849.6254 ThII
7830
7835
7840
7845
7850

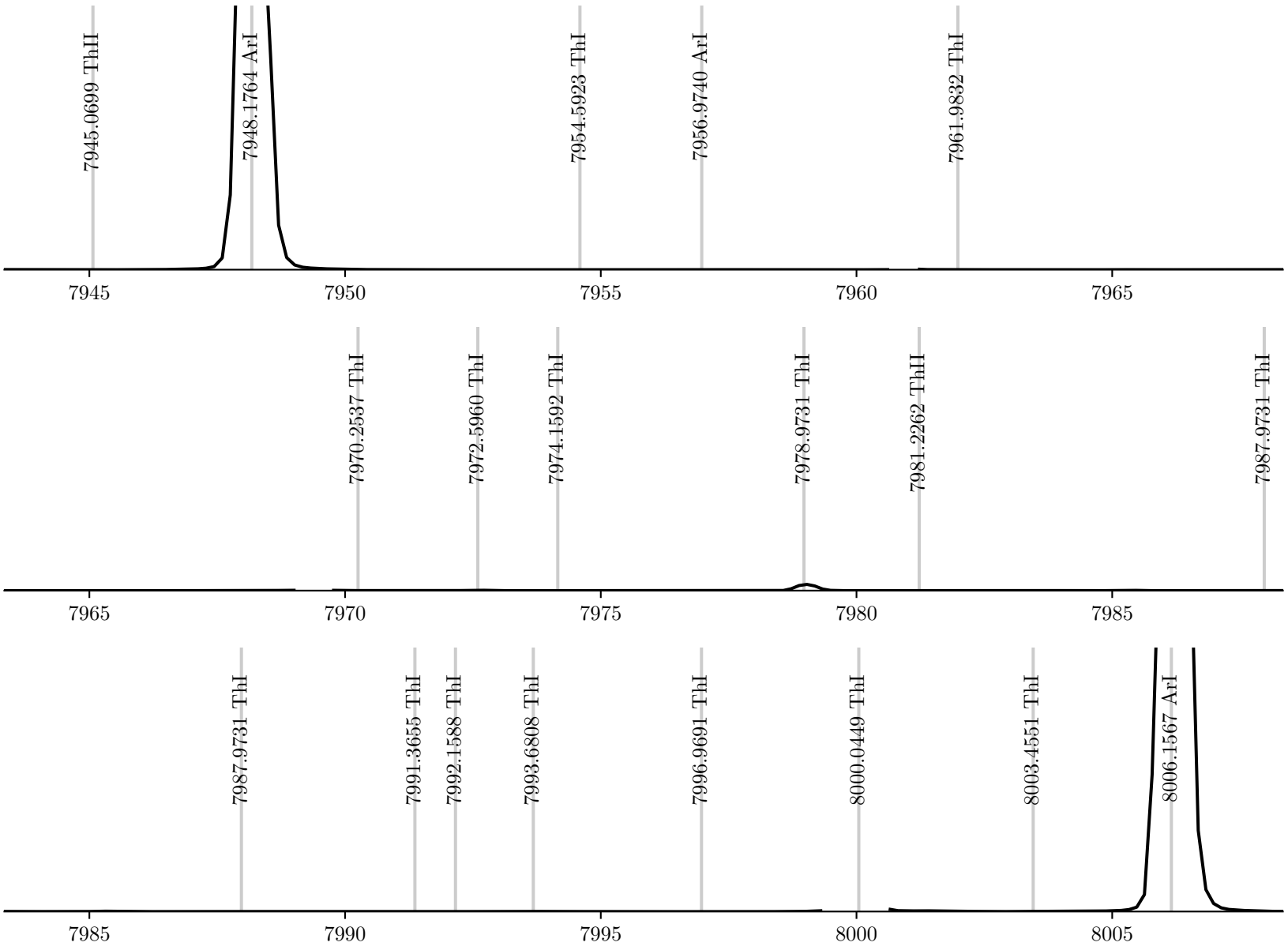

7849.6254 ThII
7861.9100 ArI
7864.0218 ThI
7865.9698 ThI
7867.6883 ThI
7868.1946 ArI
7872.6316 Th
7850
7855
7860
7865
7870
7872.6316 Th
7875.4626 ThII
7877.0756 ThII
7886.2830 ThI
7891.0750 ArI
7893.6191 ThI
7870
7875
7880
7885
7890
7891.0750 ArI
7893.6191 ThI
7896.4528 ThII
7899.6229 Th
7900.3200 ThI
7904.4344 ThI
7890
7895
7900
7905
7910

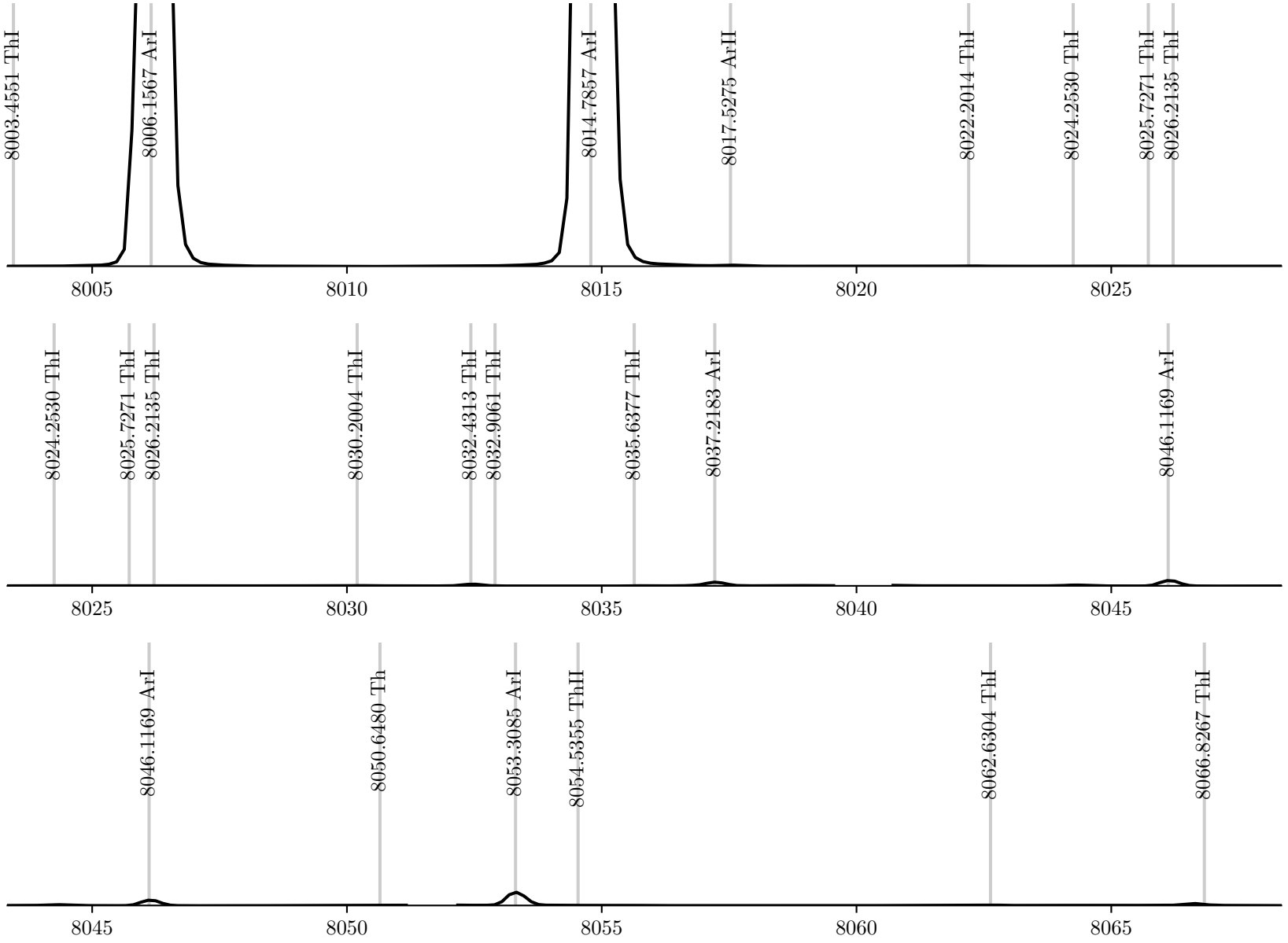

7916.4420 ArI
7920.8570 Th
7922.5653 ThI
7924.9914 ThI
7925.7478 ThI
7910
7915
7920
7925
7930
7937.7335 ThI
7941.7259 ThI
7930
7935
7940
7945
7950

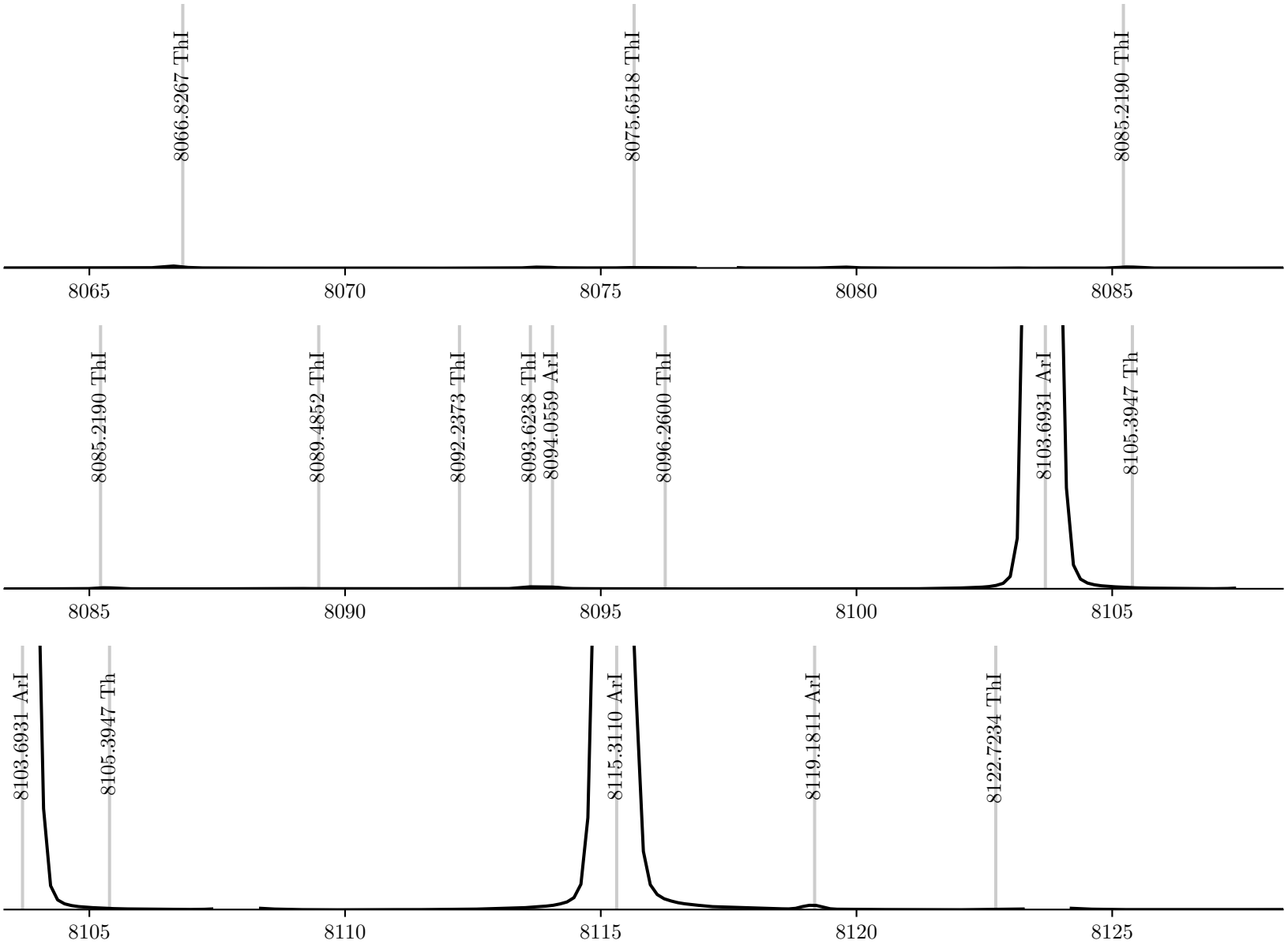
Order 28

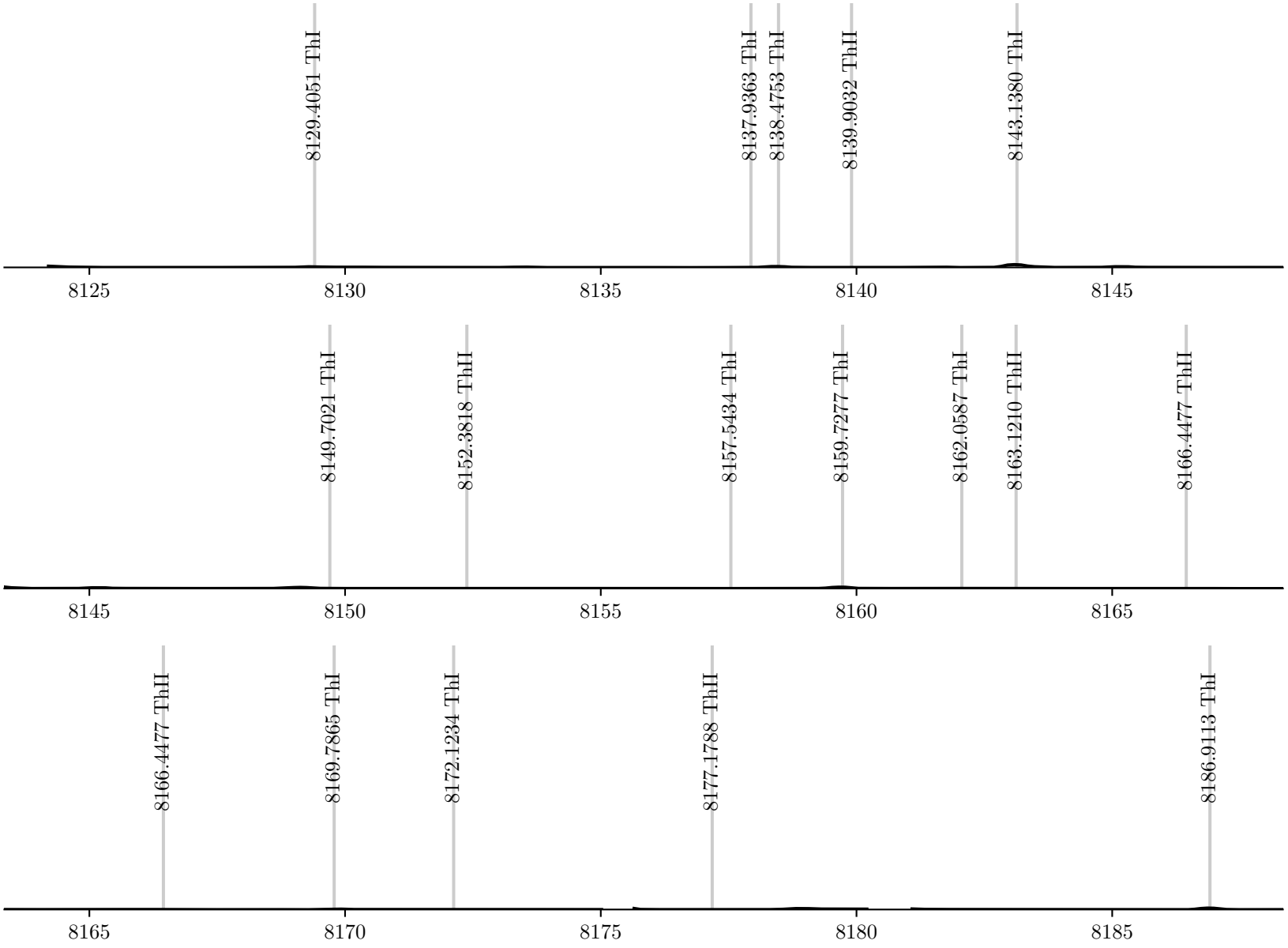


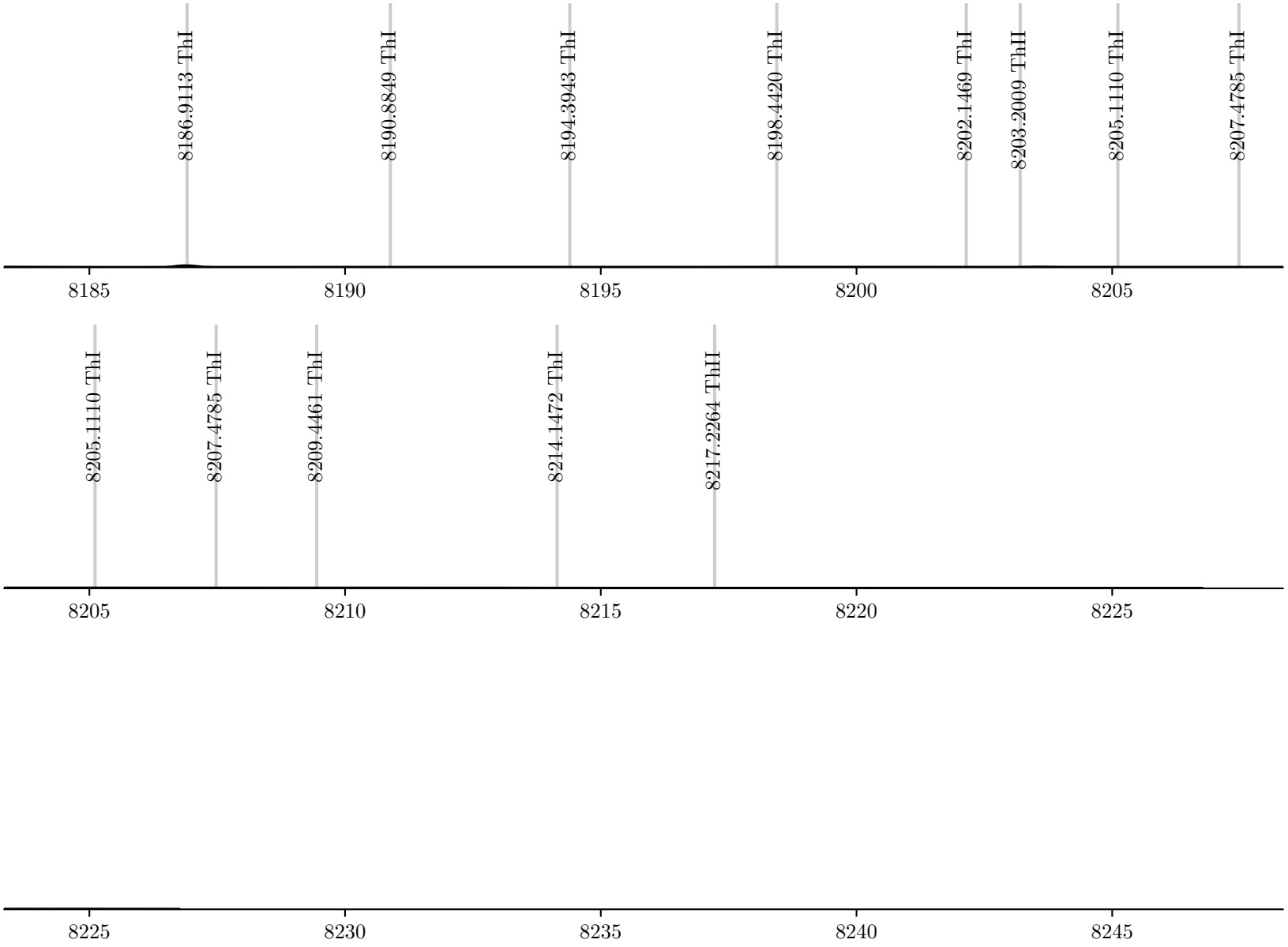
Approximate resolution:	17800
Dispersion:	0.138 Å pix^{-1}
Central wavelength:	8095 Å
Wavelength range:	7943 Å – 8227 Å


7945.0699 ThII
7948.1764 ArI
7954.5923 ThI
7956.9740 ArI
7961.9832 ThI
7945
7950
7955
7960
7965
7970.2537 ThI
7972.5960 ThI
7974.1592 ThI
7978.9731 ThI
7981.2262 ThII
7987.9731 ThI
7965
7970
7975
7980
7985
7987.9731 ThI
7991.3655 ThI
7992.1588 ThI
7993.6808 ThI
7996.9691 ThI
8000.0449 ThI
8003.4551 ThI
8006.1567 ArI
7985
7990
7995
8000
8005

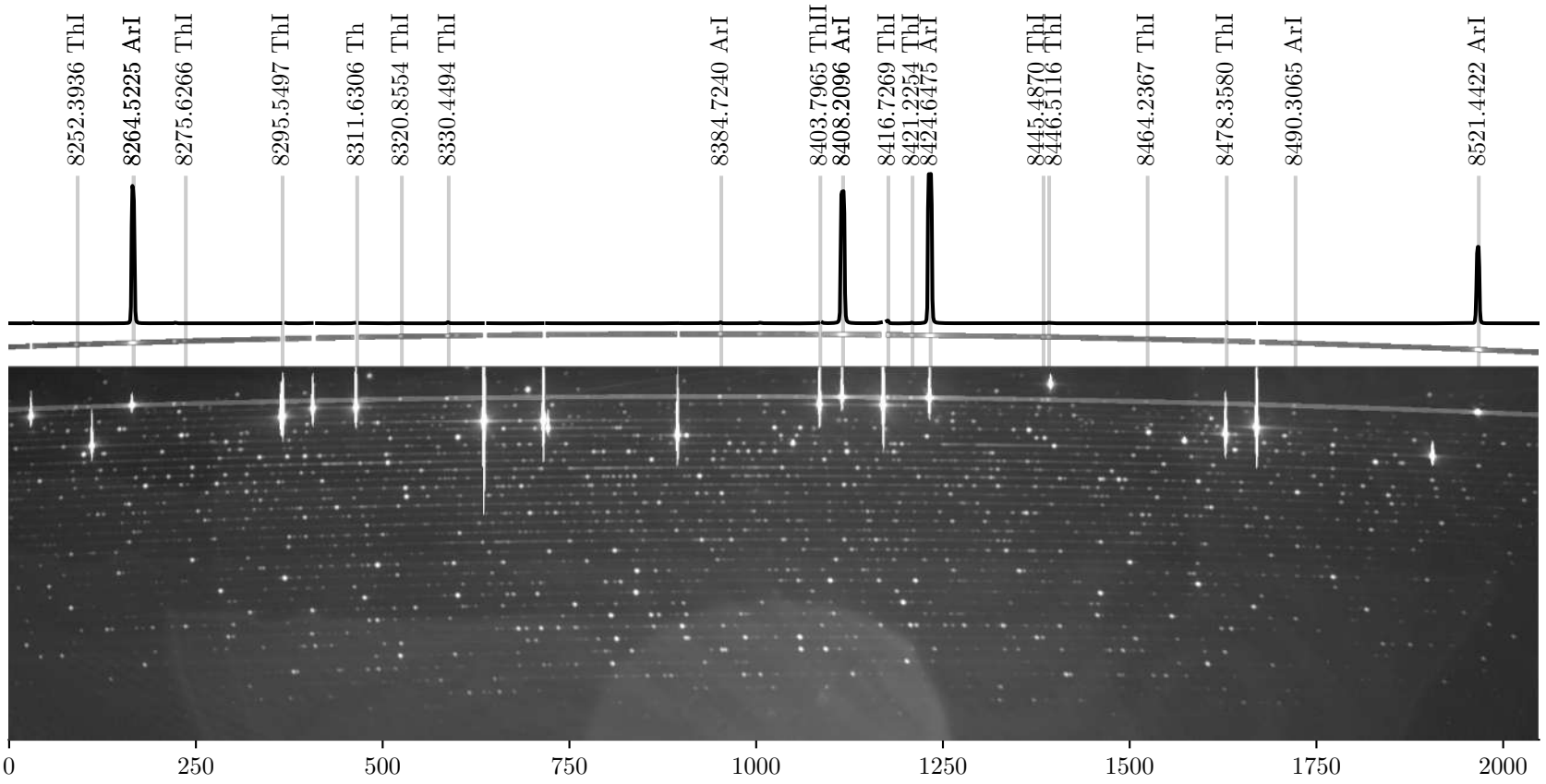

8003.4551 ThI
8006.1567 ArI
8014.7857 ArI
8017.5275 ArII
8022.2014 ThI
8024.2530 ThI
8025.7271 ThI
8026.2135 ThI
8005
8010
8015
8020
8025
8024.2530 ThI
8025.7271 ThI
8026.2135 ThI
8030.2004 ThI
8032.4313 ThI
8032.9061 ThI
8035.6377 ThI
8037.2183 ArI
8046.1169 ArI
8025
8030
8035
8040
8045
8046.1169 ArI
8050.6480 Th
8053.3085 ArI
8054.5355 ThII
8062.6304 ThI
8066.8267 ThI
8045
8050
8055
8060
8065


8066.8267 ThI
8075.6518 ThI
8085.2190 ThI
8065
8070
8075
8080
8085
8085.2190 ThI
8089.4852 ThI
8092.2373 ThI
8093.6238 ThI
8094.0559 ArI
8096.2600 ThI
8103.6931 ArI
8105.3947 Th
8085
8090
8095
8100
8105
8103.6931 ArI
8105.3947 Th
8115.3110 ArI
8119.1811 ArI
8122.7234 ThI
8105
8110
8115
8120
8125

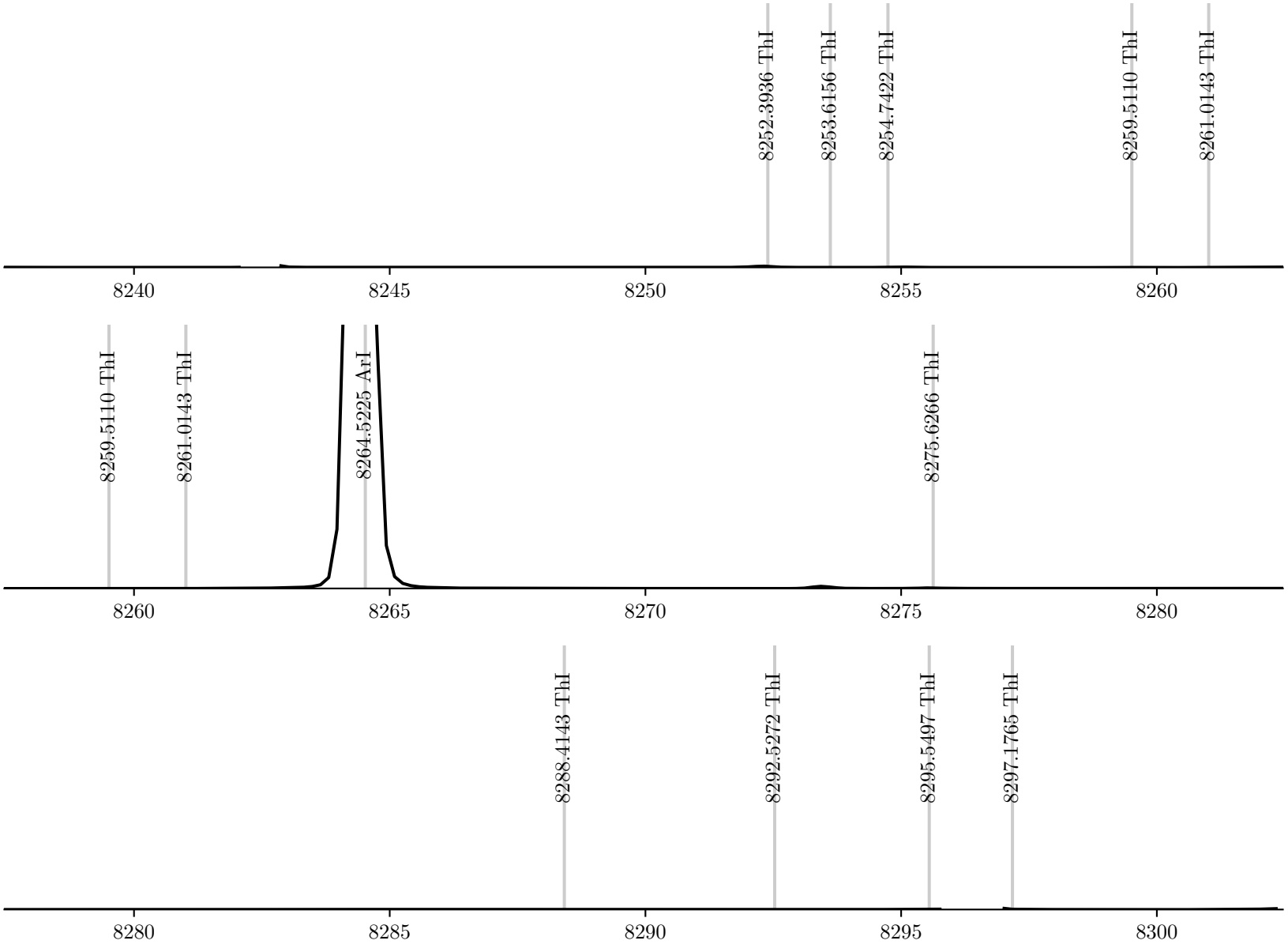

8129.4051 ThI
8137.9363 ThI
8138.4753 ThI
8139.9032 ThII
8143.1380 ThI
8125
8130
8135
8140
8145
8149.7021 ThI
8152.3818 ThII
8157.5434 ThI
8159.7277 ThI
8162.0587 ThI
8163.1210 ThII
8166.4477 ThII
8145
8150
8155
8160
8165
8166.4477 ThII
8169.7865 ThI
8172.1234 ThI
8177.1788 ThII
8186.9113 ThI
8165
8170
8175
8180
8185

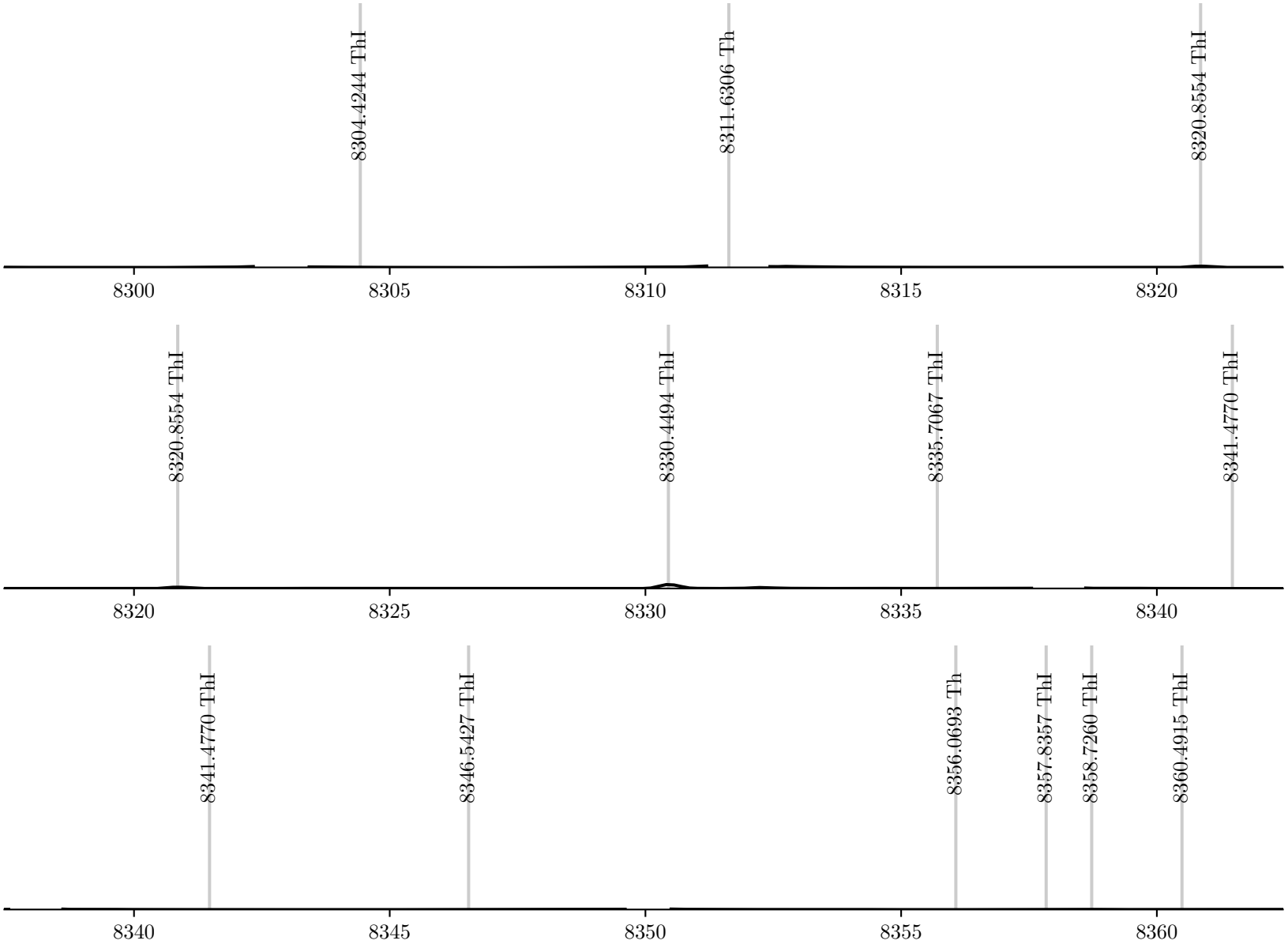

8186.9113 ThI
8190.8849 ThI
8194.3943 ThI
8198.4420 ThI
8202.1469 ThI
8203.2009 ThII
8205.1110 ThI
8207.4785 ThI
8185
8190
8195
8200
8205
8205.1110 ThI
8207.4785 ThI
8209.4461 ThI
8214.1472 ThI
8217.2264 ThII
8205
8210
8215
8220
8225
8225
8230
8235
8240
8245

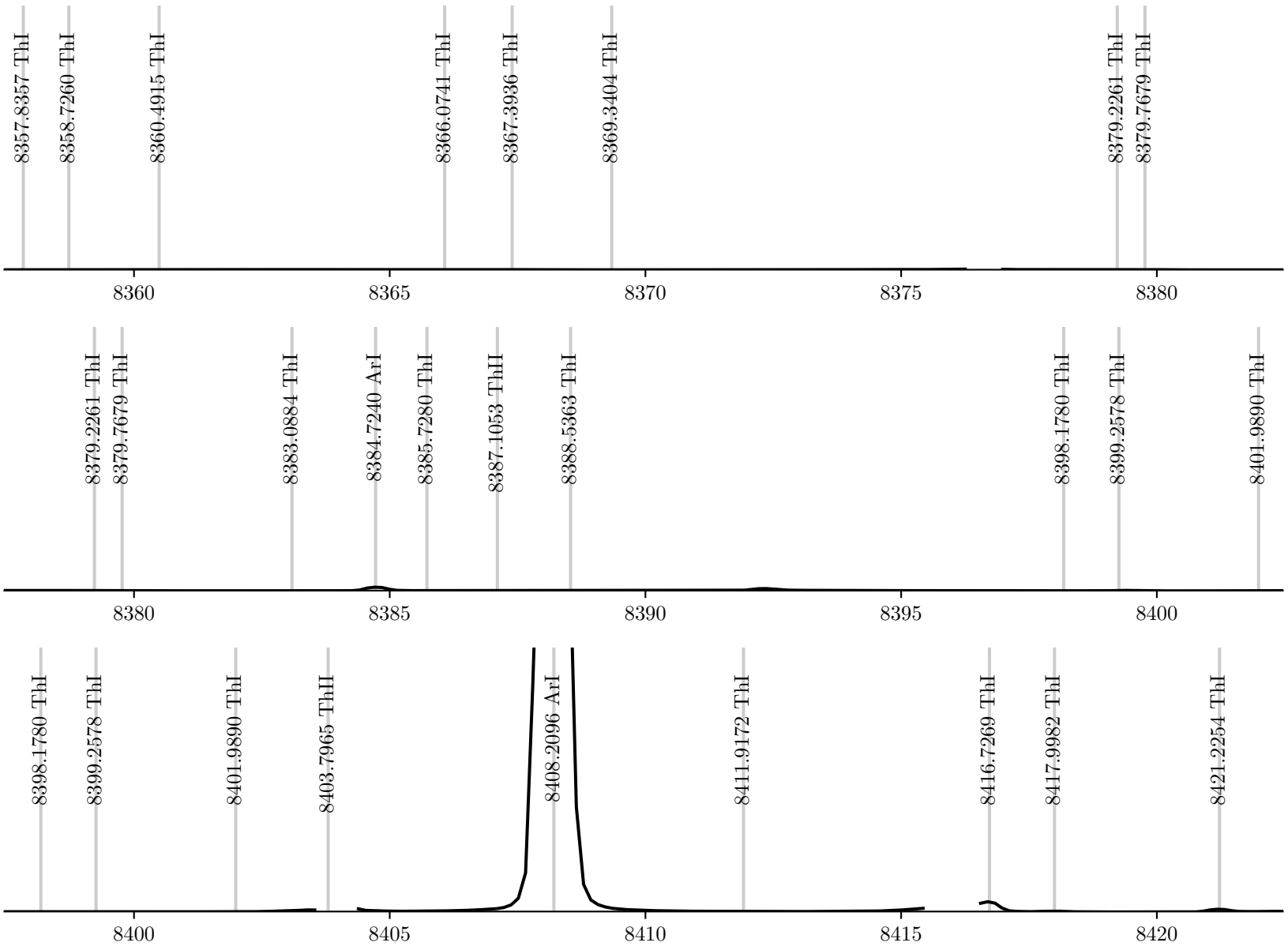
Order 27

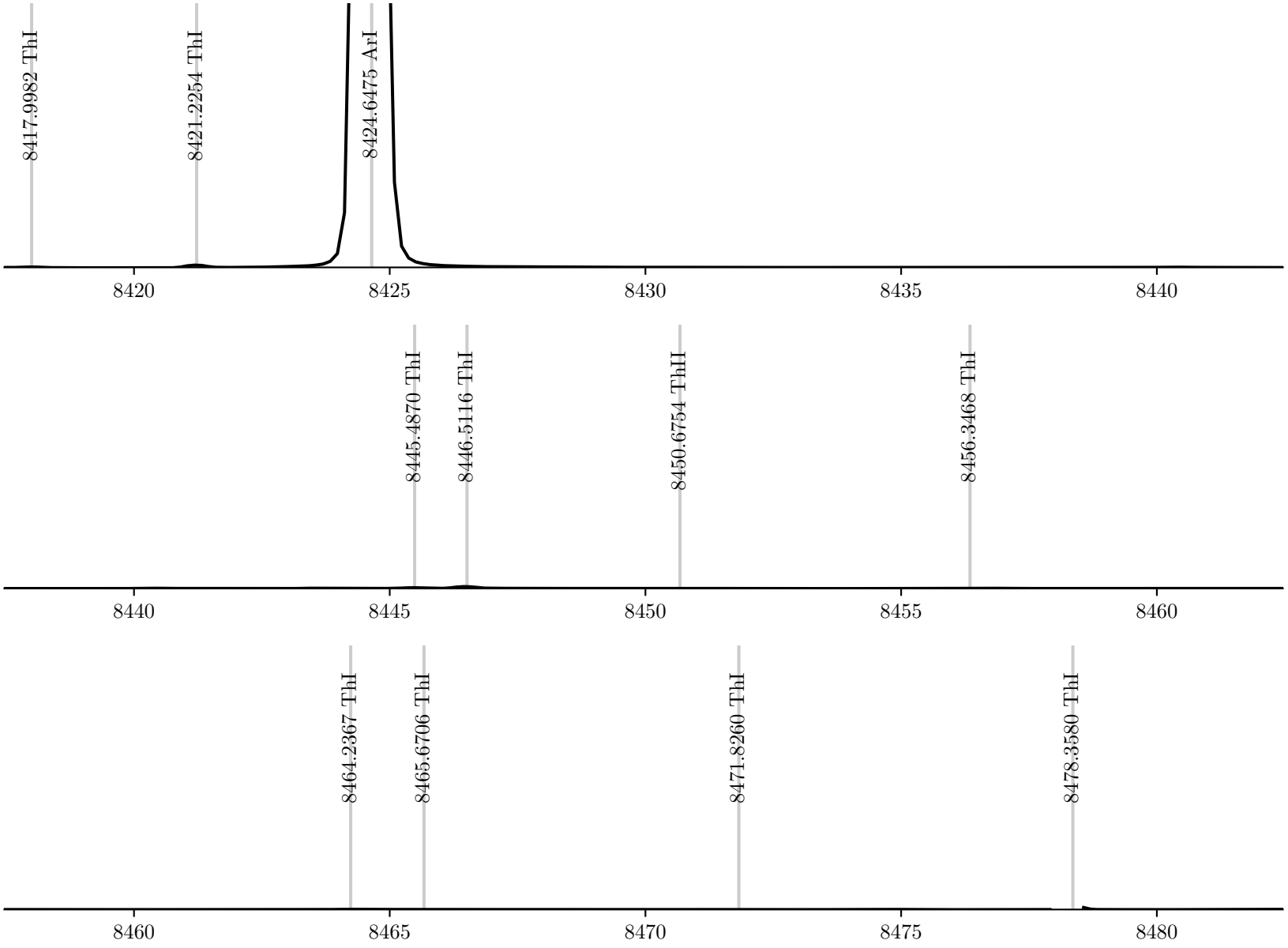


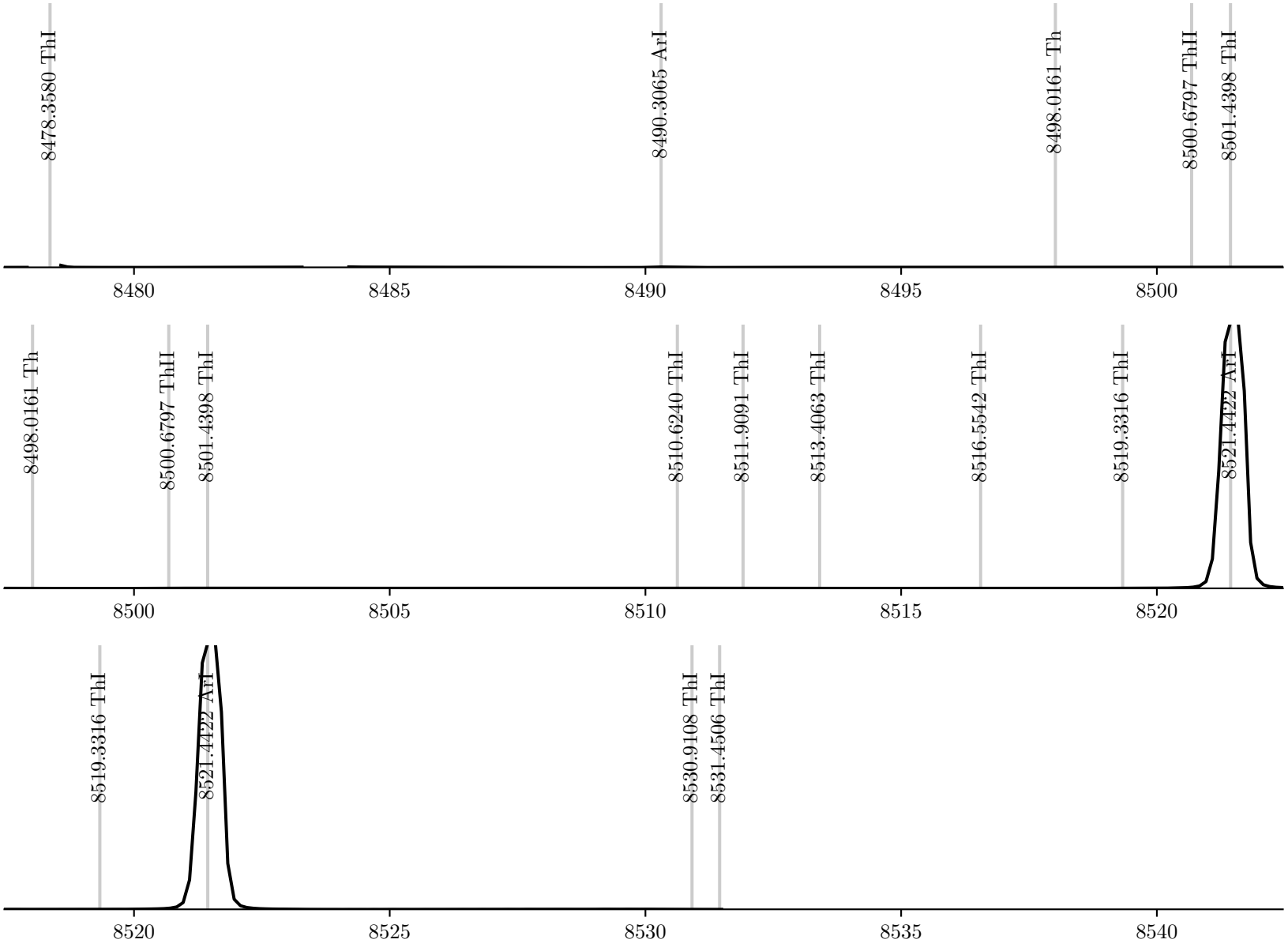
Approximate resolution:	18300
Dispersion:	0.144 Å pix^{-1}
Central wavelength:	8395 Å
Wavelength range:	8237 Å – 8532 Å


8252.3936 ThI
8253.6156 ThI
8254.7422 ThI
8259.5110 ThI
8261.0143 ThI
8240
8245
8250
8255
8260
8259.5110 ThI
8261.0143 ThI
8264.5225 ArI
8275.6266 ThI
8260
8265
8270
8275
8280
8288.4143 ThI
8292.5272 ThI
8295.5497 ThI
8297.1765 ThI
8280
8285
8290
8295
8300

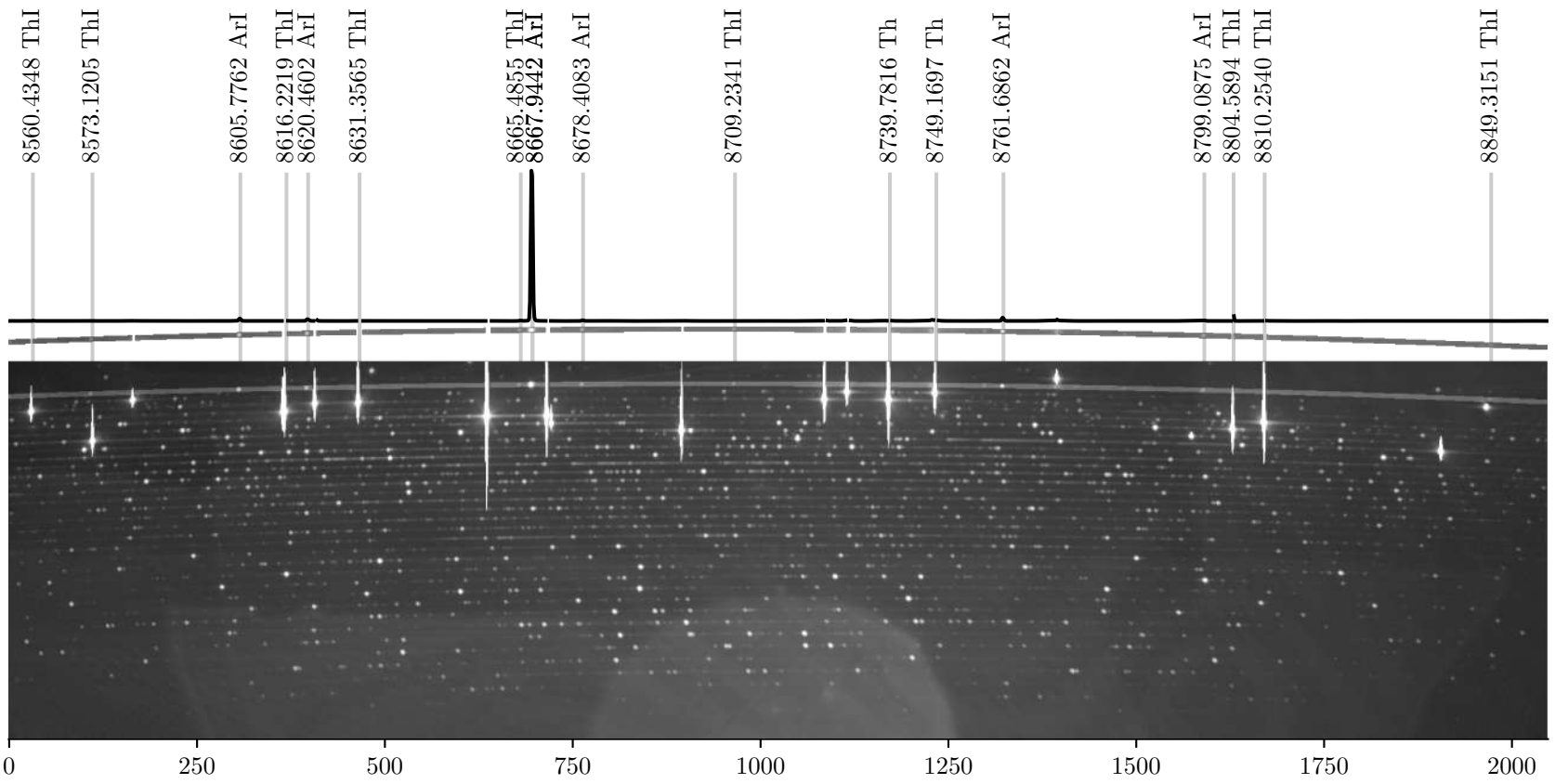

8304.4244 ThI
8311.6306 Th
8320.8554 ThI
8300
8305
8310
8315
8320
8320.8554 ThI
8330.4494 ThI
8335.7067 ThI
8341.4770 ThI
8320
8325
8330
8335
8340
8341.4770 ThI
8346.5427 ThI
8356.0693 Th
8357.8357 ThI
8358.7260 ThI
8360.4915 ThI
8340
8345
8350
8355
8360


8357.8357 ThI
8358.7260 ThI
8360.4915 ThI
8366.0741 ThI
8367.3936 ThI
8369.3404 ThI
8379.2261 ThI
8379.7679 ThI
8360
8365
8370
8375
8380
8379.2261 ThI
8379.7679 ThI
8383.0884 ThI
8384.7240 ArI
8385.7280 ThI
8387.1053 ThII
8388.5363 ThI
8398.1780 ThI
8399.2578 ThI
8401.9890 ThI
8380
8385
8390
8395
8400
8398.1780 ThI
8399.2578 ThI
8401.9890 ThI
8403.7965 ThII
8408.2096 ArI
8411.9172 ThI
8416.7269 ThI
8417.9982 ThI
8421.2254 ThI
8400
8405
8410
8415
8420

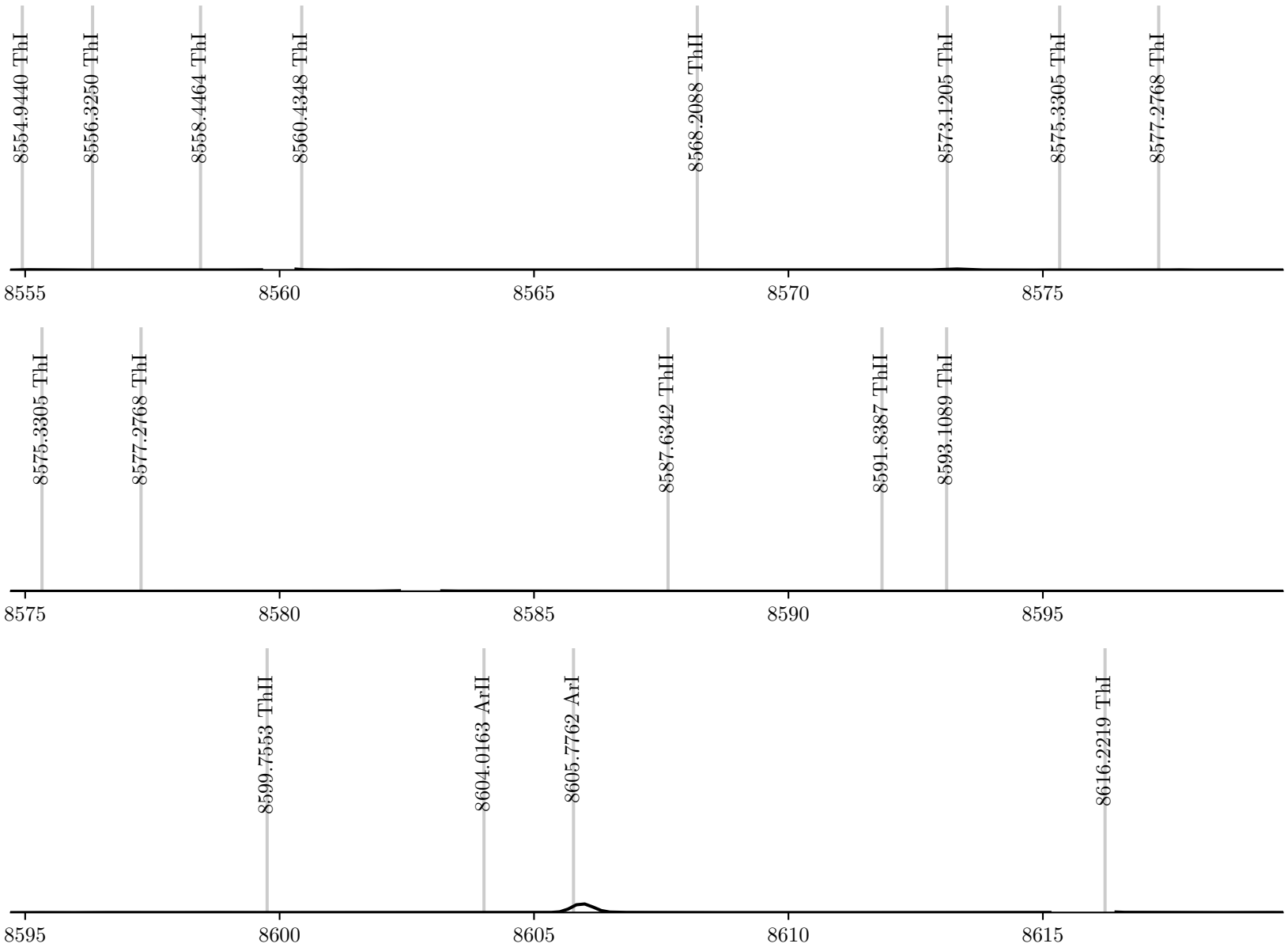

8417.9982 ThI
8421.2254 ThI
8424.6475 ArI
8420
8425
8430
8435
8440
8445.4870 ThI
8446.5116 ThI
8450.6754 ThII
8456.3468 ThI
8440
8445
8450
8455
8460
8464.2367 ThI
8465.6706 ThI
8471.8260 ThI
8478.3580 ThI
8460
8465
8470
8475
8480

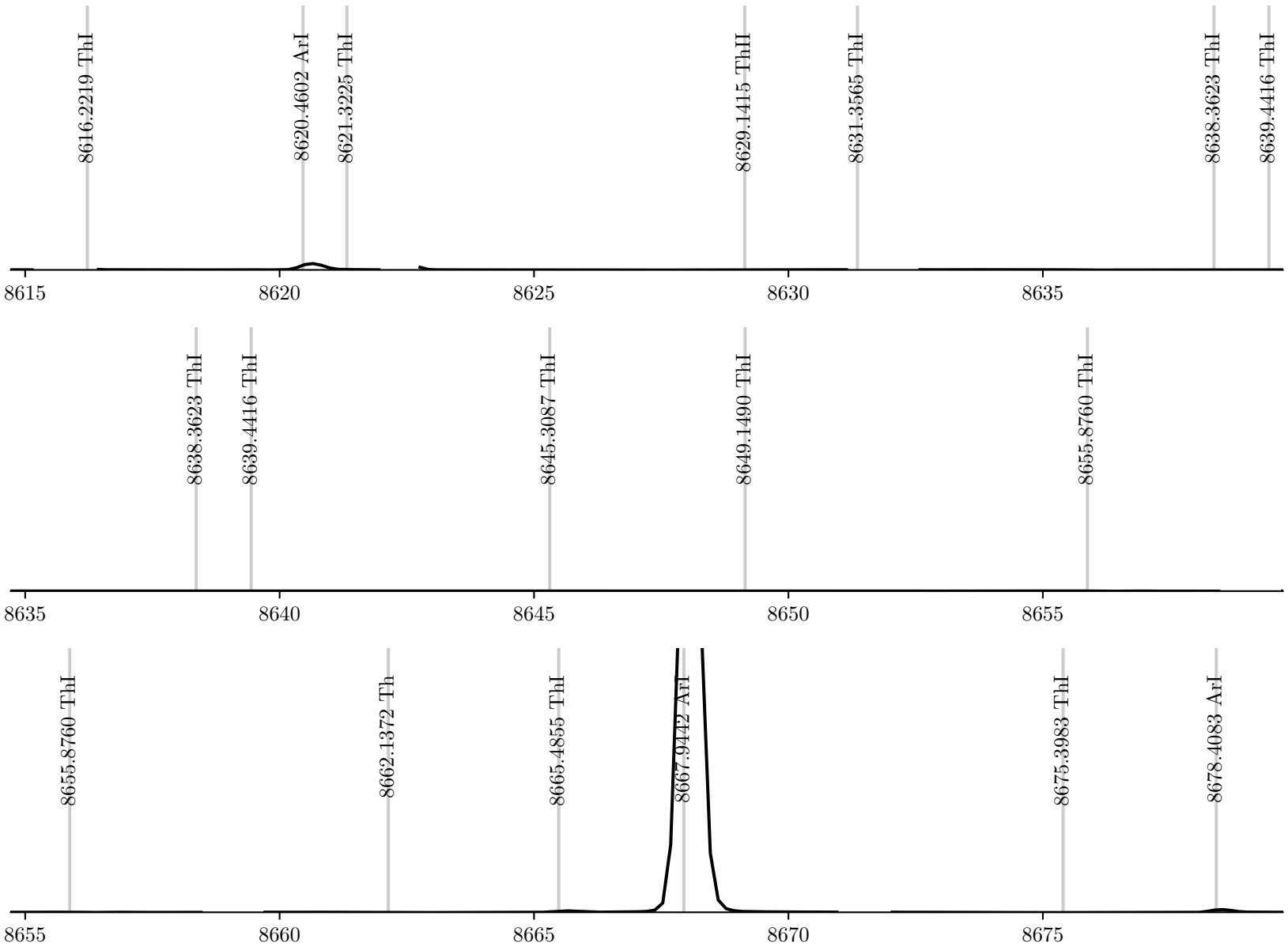

8478.3580 ThI
8490.3065 ArI
8498.0161 Th
8500.6797 ThII
8501.4398 ThI
8480
8485
8490
8495
8500
8498.0161 Th
8500.6797 ThII
8501.4398 ThI
8510.6240 ThI
8511.9091 ThI
8513.4063 ThI
8516.5542 ThI
8519.3316 ThI
8521.4422 ArI
8500
8505
8510
8515
8520
8519.3316 ThI
8521.4422 ArI
8530.9108 ThI
8531.4506 ThI
8520
8525
8530
8535
8540

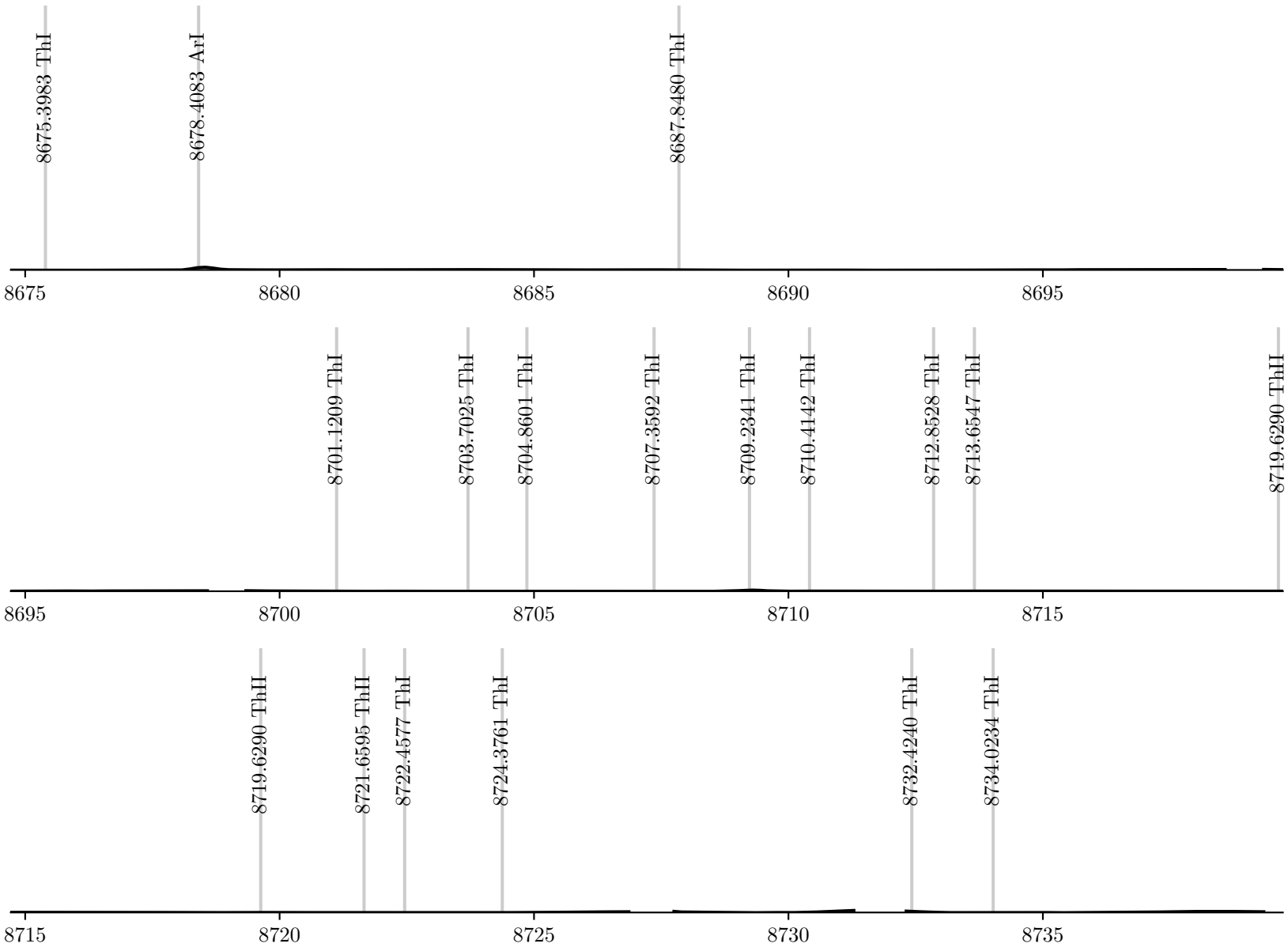
Order 26

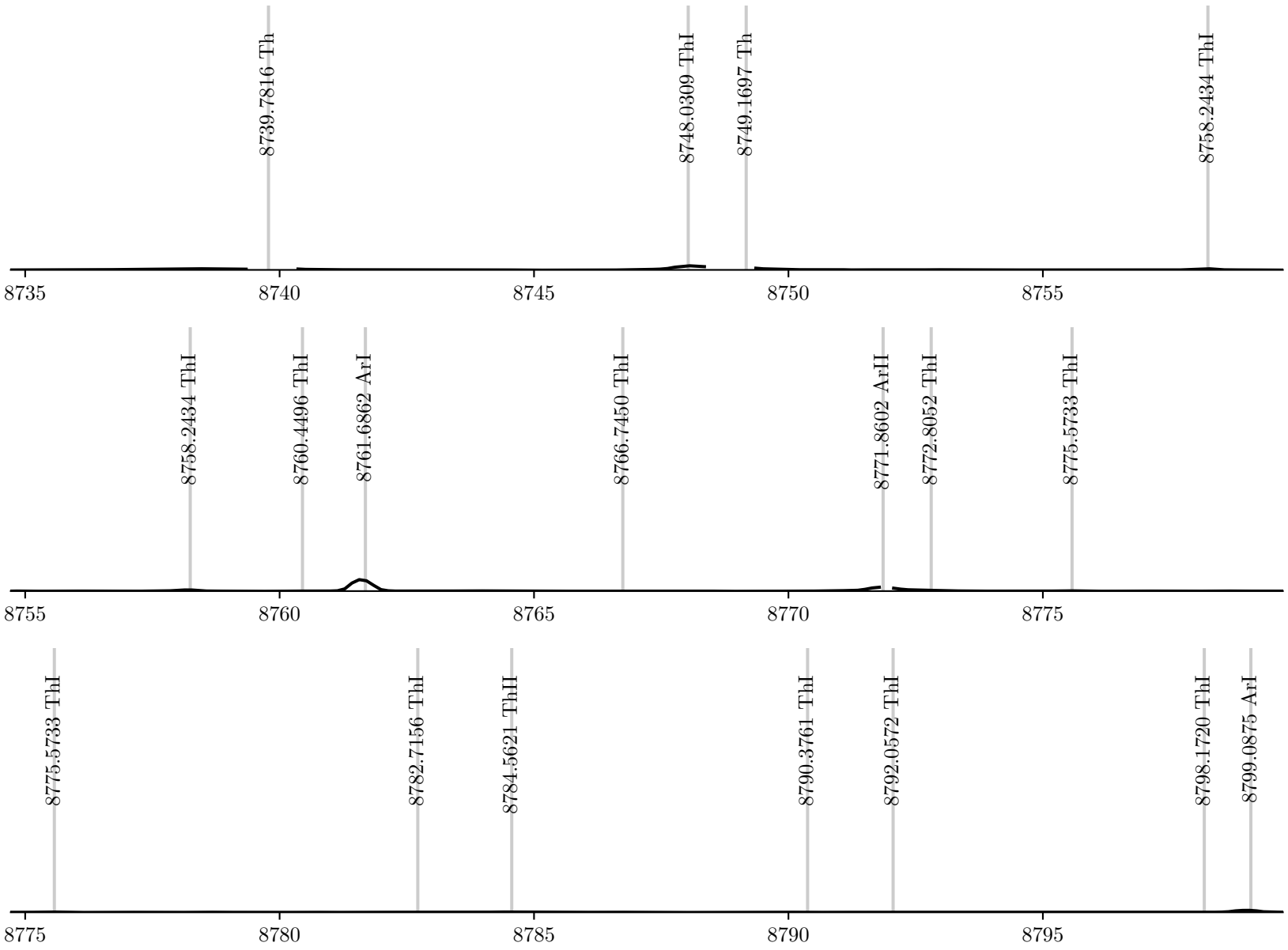


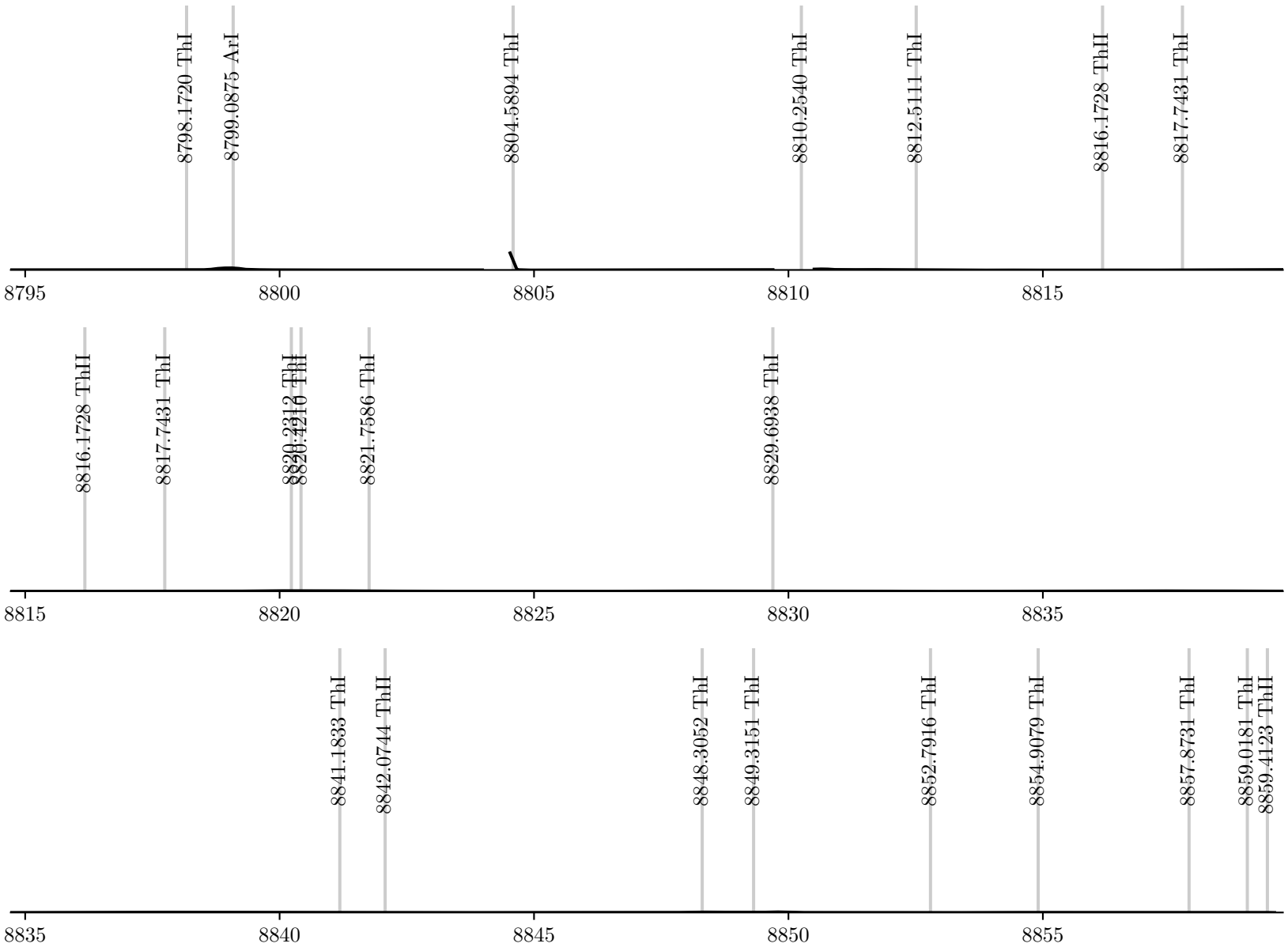
Approximate resolution:	17700
Dispersion:	0.149 Å pix^{-1}
Central wavelength:	8718 Å
Wavelength range:	8555 Å – 8860 Å


8554.9440 ThI
8556.3250 ThI
8558.4464 ThI
8560.4348 ThI
8568.2088 ThII
8573.1205 ThI
8575.3305 ThI
8577.2768 ThI
8555
8560
8565
8570
8575
8575.3305 ThI
8577.2768 ThI
8587.6342 ThII
8591.8387 ThII
8593.1089 ThI
8575
8580
8585
8590
8595
8599.7553 ThII
8604.0163 ArII
8605.7762 ArI
8616.2219 ThI
8595
8600
8605
8610
8615

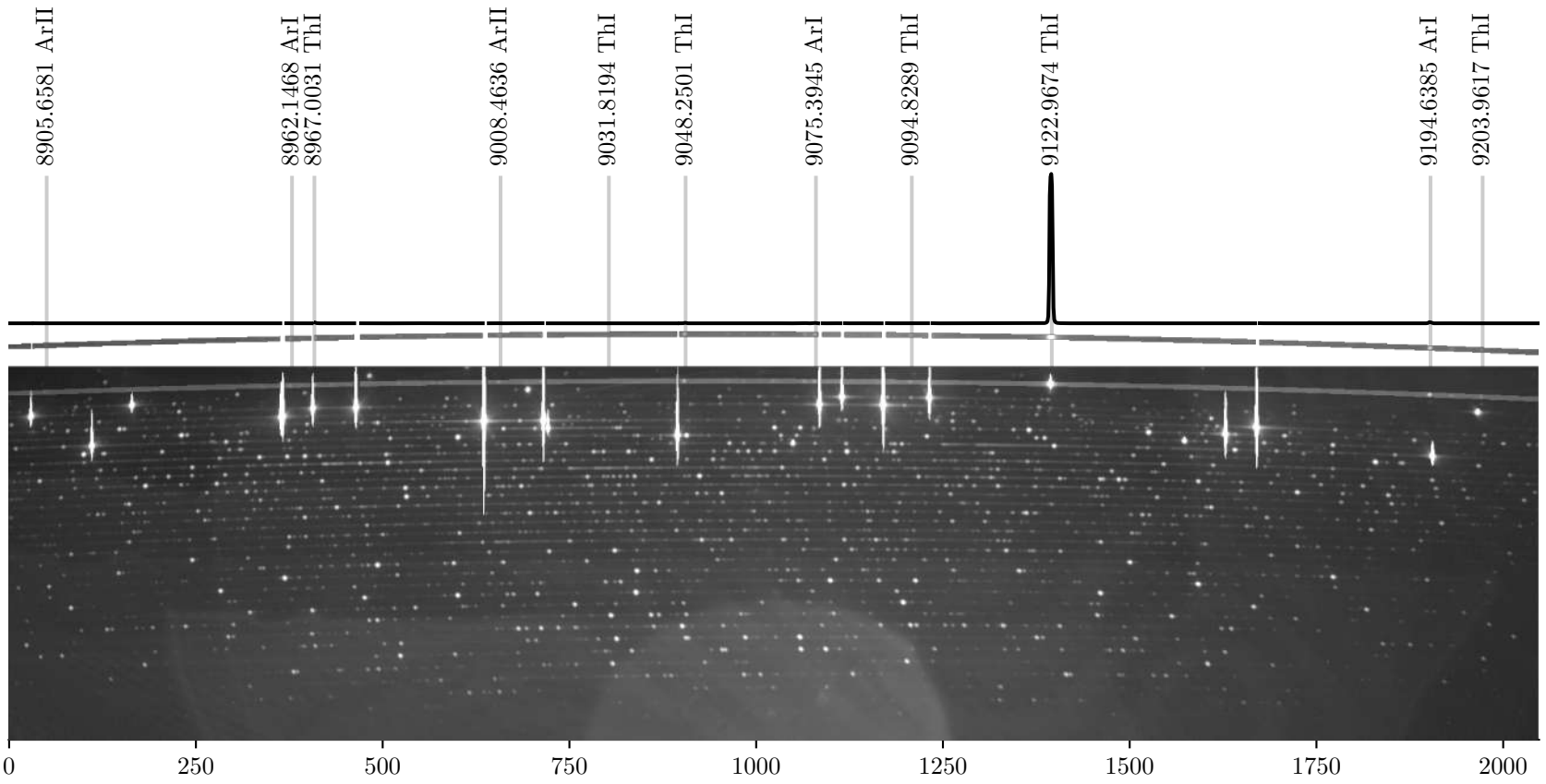

8616.2219 ThI
8620.4602 ArI
8621.3225 ThI
8629.1415 ThII
8631.3565 ThI
8638.3623 ThI
8639.4416 ThI
8615
8620
8625
8630
8635
8638.3623 ThI
8639.4416 ThI
8645.3087 ThI
8649.1490 ThI
8655.8760 ThI
8635
8640
8645
8650
8655
8655.8760 ThI
8662.1372 Th
8665.4855 ThI
8667.9442 ArI
8675.3983 ThI
8678.4083 ArI
8655
8660
8665
8670
8675


8675.3983 ThI
8678.4083 ArI
8687.8480 ThI
8675
8680
8685
8690
8695
8701.1209 ThI
8703.7025 ThI
8704.8601 ThI
8707.3592 ThI
8709.2341 ThI
8710.4142 ThI
8712.8528 ThI
8713.6547 ThI
8719.6290 ThII
8695
8700
8705
8710
8715
8719.6290 ThII
8721.6595 ThII
8722.4577 ThI
8724.3761 ThI
8732.4240 ThI
8734.0234 ThI
8715
8720
8725
8730
8735

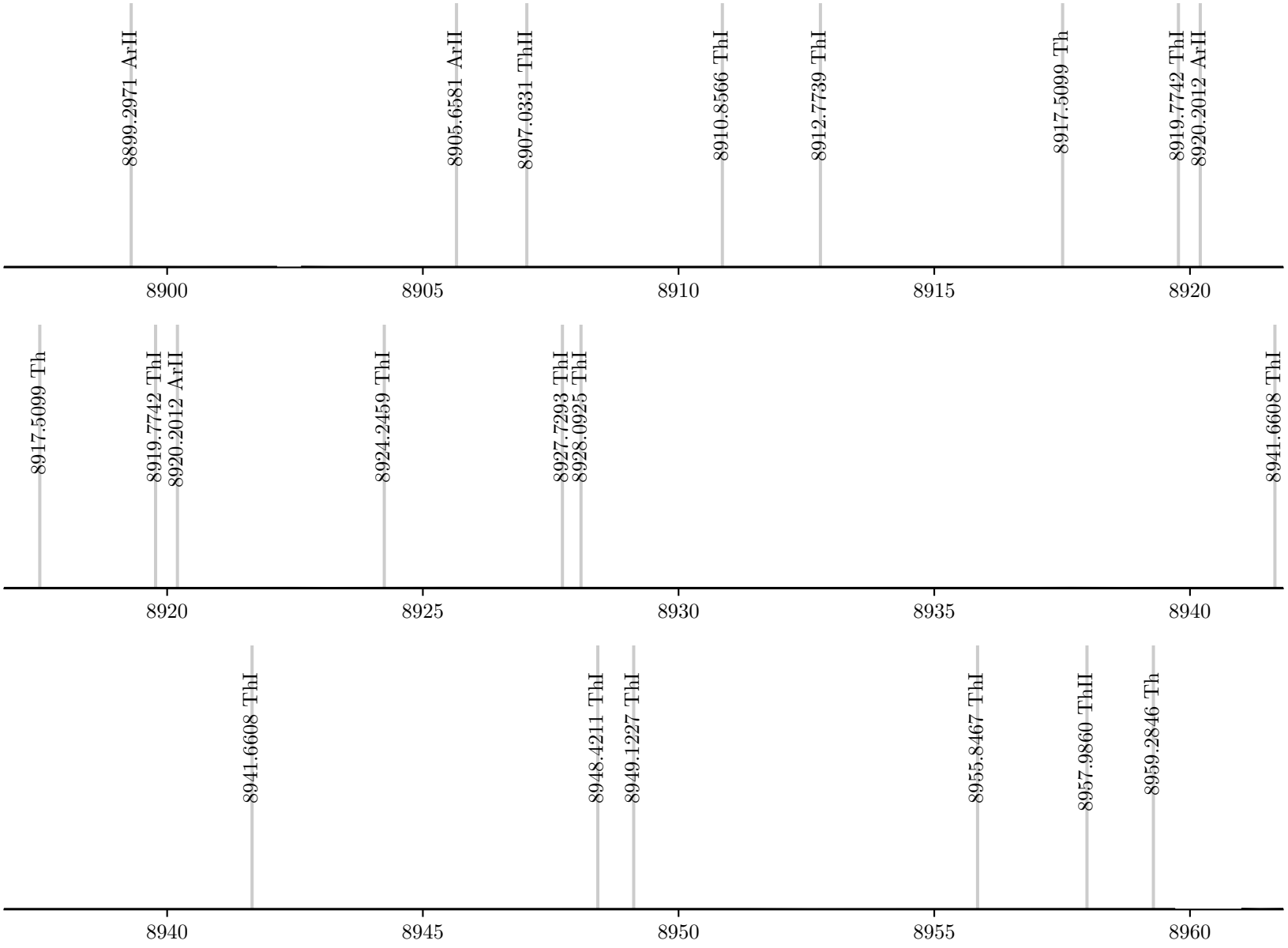

8739.7816 Th
8748.0309 ThI
8749.1697 Th
8758.2434 ThI
8735
8740
8745
8750
8755
8758.2434 ThI
8760.4496 ThI
8761.6862 ArI
8766.7450 ThI
8771.8602 ArII
8772.8052 ThI
8775.5733 ThI
8755
8760
8765
8770
8775
8775.5733 ThI
8782.7156 ThI
8784.5621 ThII
8790.3761 ThI
8792.0572 ThI
8798.1720 ThI
8799.0875 ArI
8775
8780
8785
8790
8795

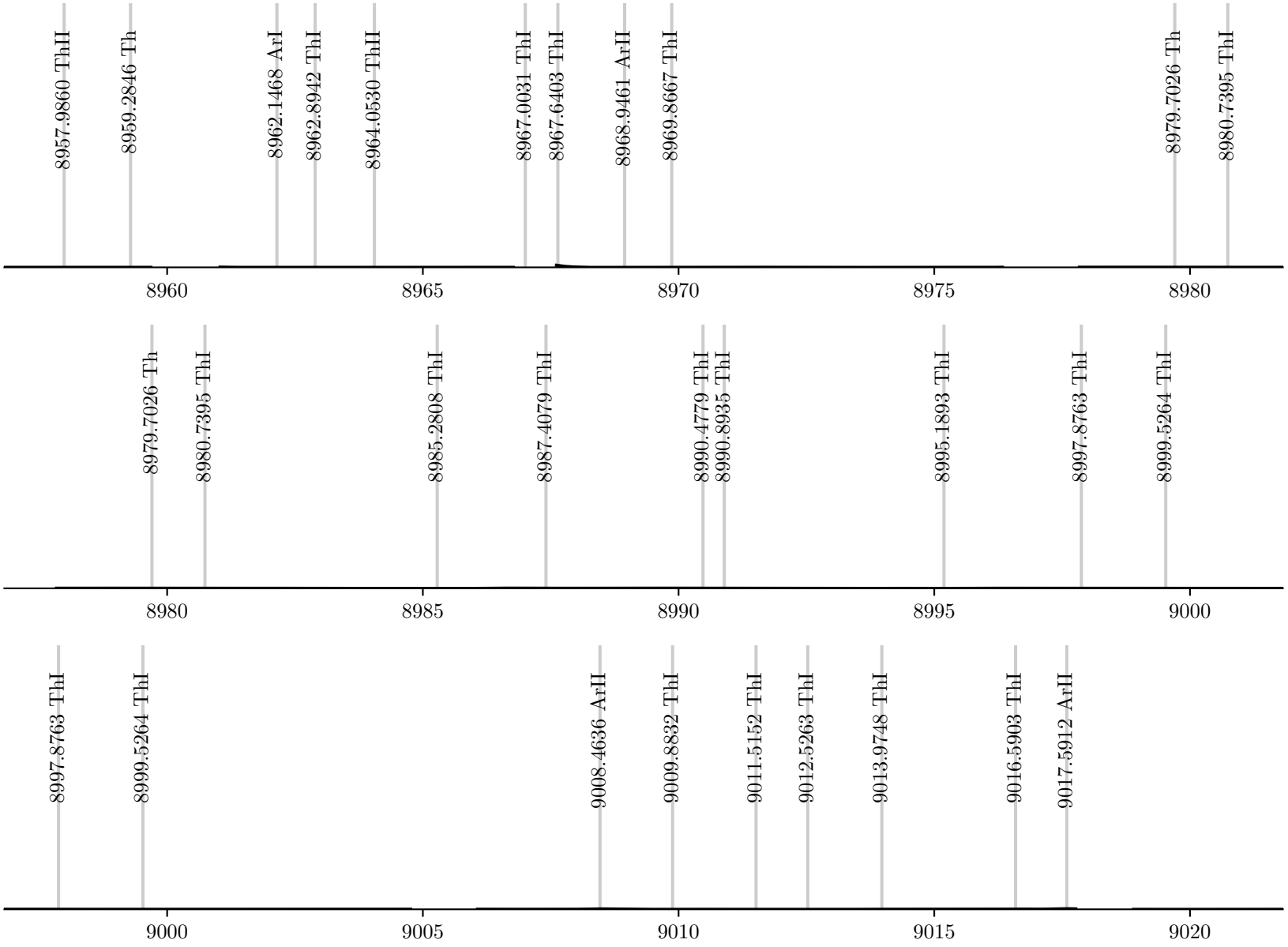

8798.1720 ThI
8799.0875 ArI
8804.5894 ThI
8810.2540 ThI
8812.5111 ThI
8816.1728 ThII
8817.7431 ThI
8795
8800
8805
8810
8815
8816.1728 ThII
8817.7431 ThI
8821.7586 ThI
8829.6938 ThI
8815
8820
8825
8830
8835
8841.1833 ThI
8842.0744 ThII
8848.3052 ThI
8849.3151 ThI
8852.7916 ThI
8854.9079 ThI
8857.8731 ThI
8859.0181 ThI
8859.4123 ThII
8835
8840
8845
8850
8855

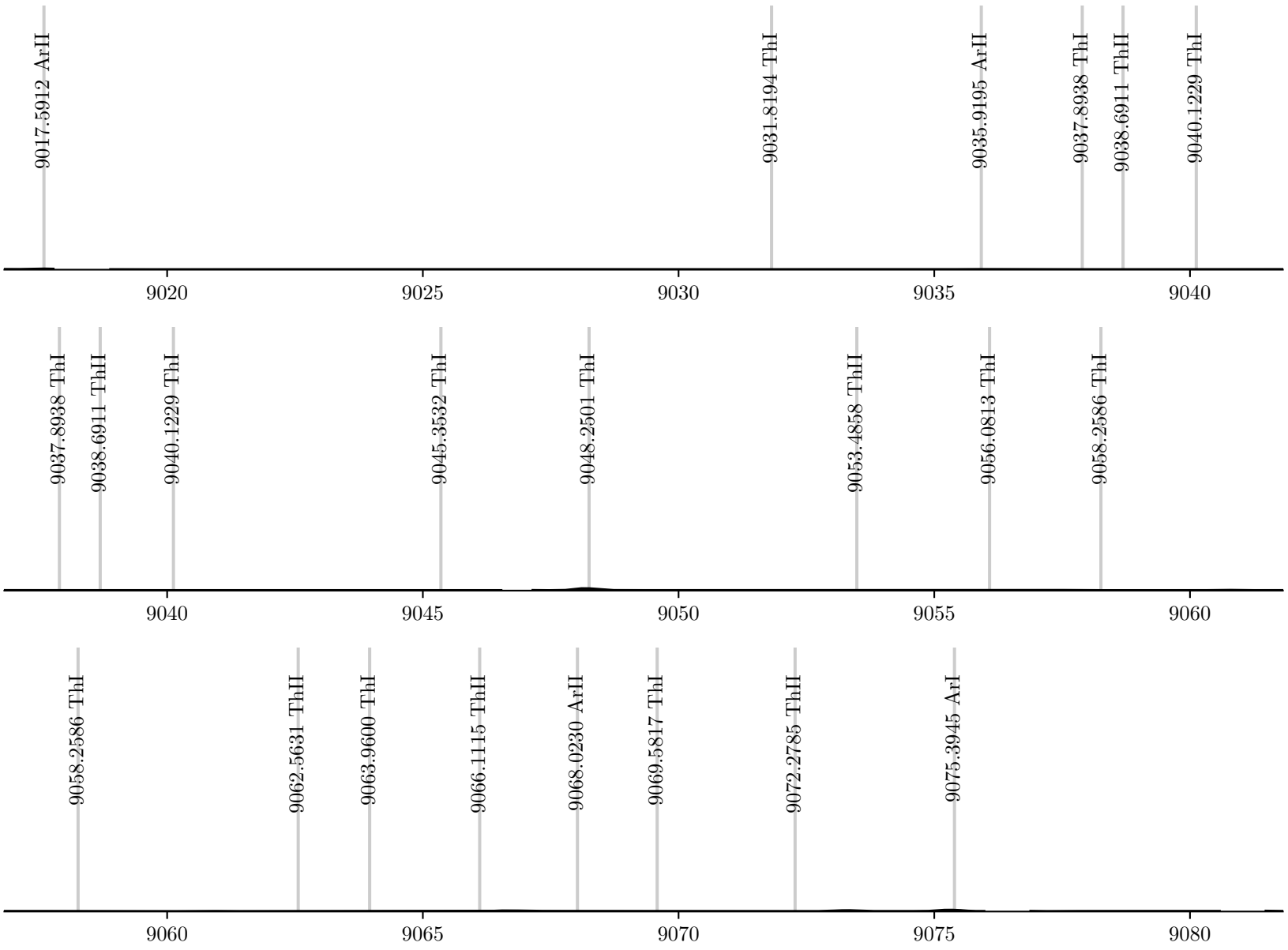
Order 25

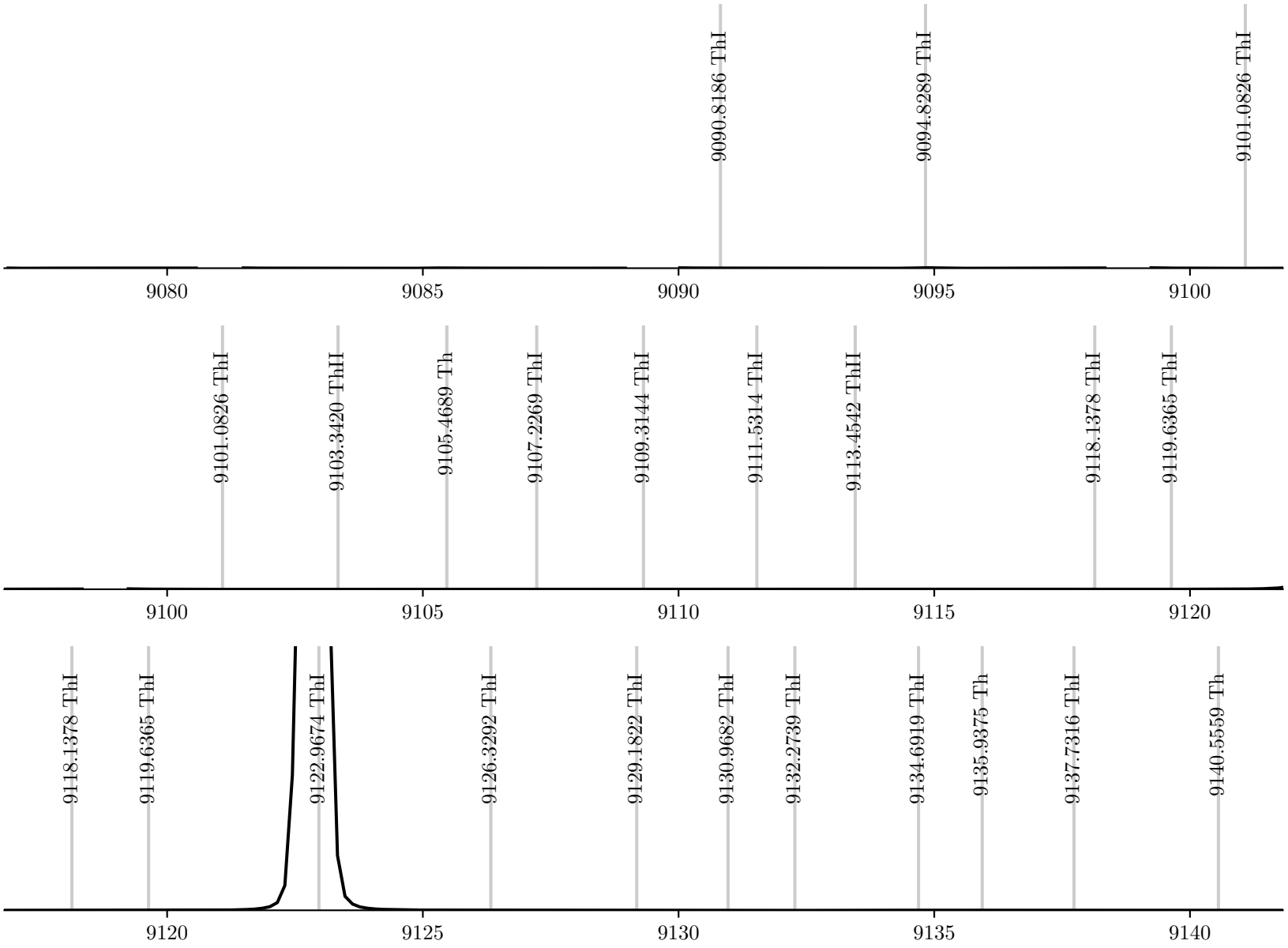


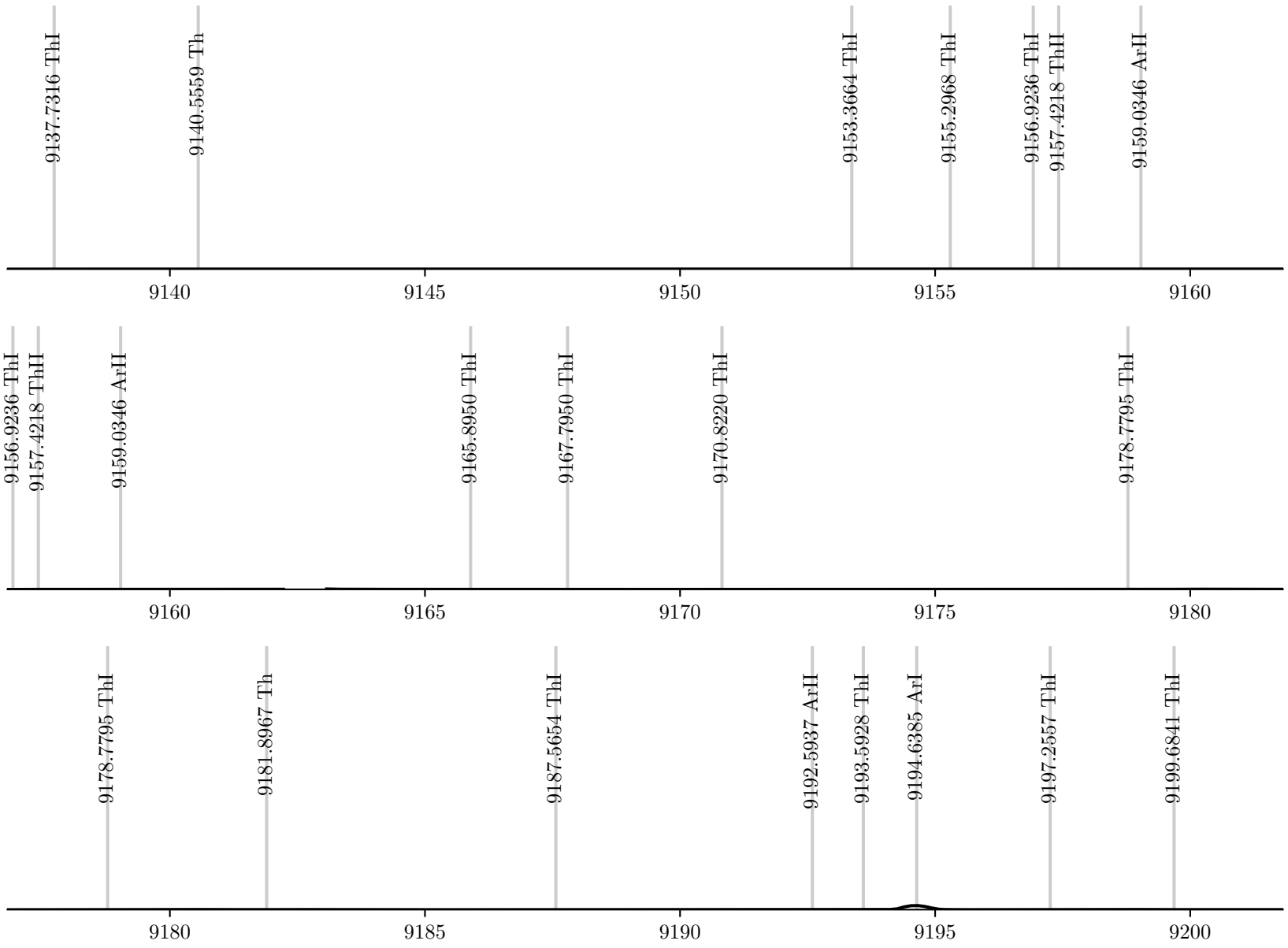
Approximate resolution:	18100
Dispersion:	0.155 Å pix^{-1}
Central wavelength:	9067 Å
Wavelength range:	8897 Å – 9214 Å


8899.2971 ArII
8905.6581 ArII
8907.0331 ThII
8910.8566 ThI
8912.7739 ThI
8917.5099 Th
8919.7742 ThI
8920.2012 ArII
8900
8905
8910
8915
8920
8917.5099 Th
8919.7742 ThI
8920.2012 ArII
8924.2459 ThI
8927.7293 ThI
8928.0925 ThI
8941.6608 ThI
8920
8925
8930
8935
8940
8941.6608 ThI
8948.4211 ThI
8949.1227 ThI
8955.8467 ThI
8957.9860 ThII
8959.2846 Th
8940
8945
8950
8955
8960

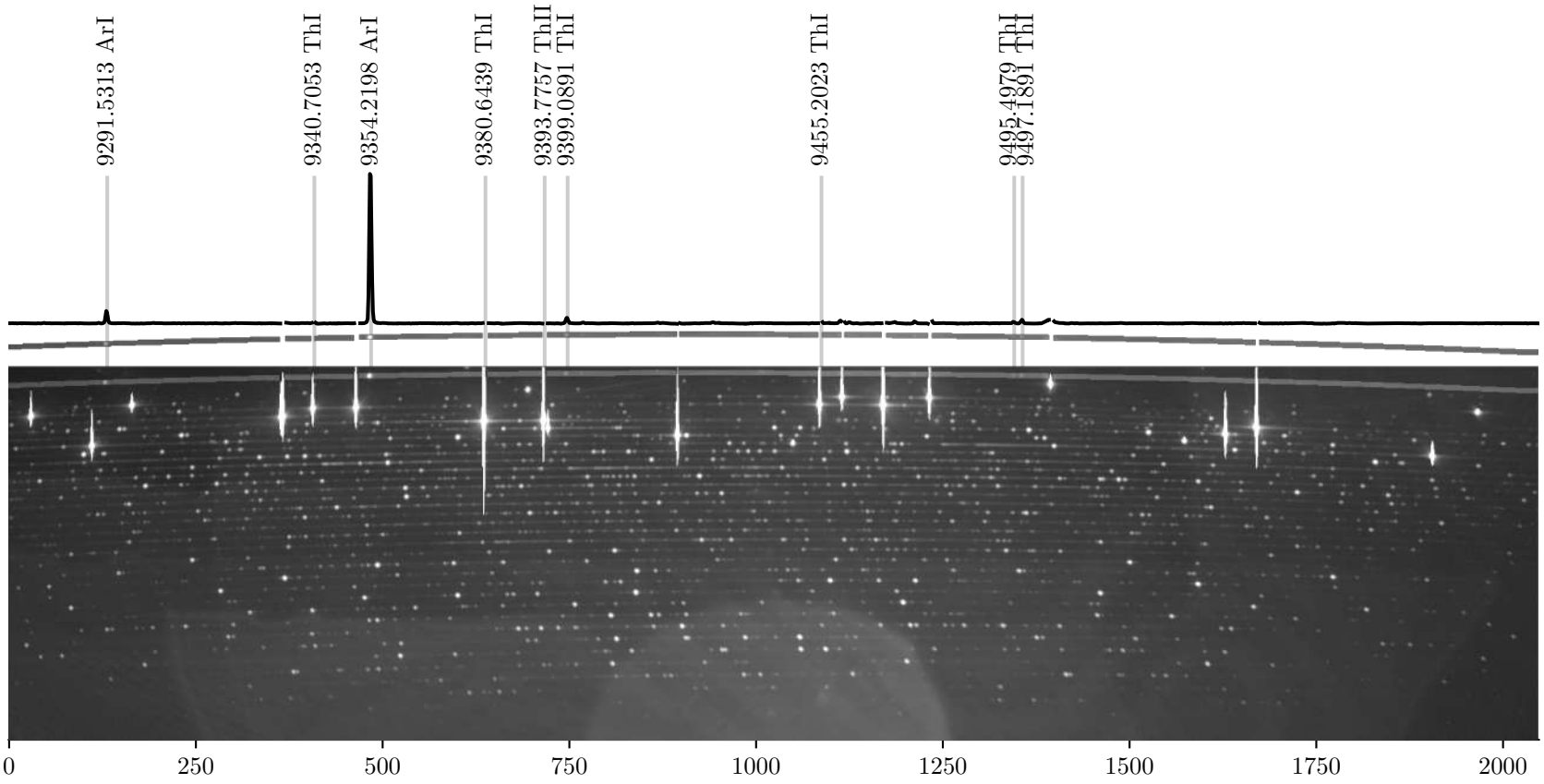

8957.9860 ThII
8959.2846 Th
8962.1468 ArI
8962.8942 ThI
8964.0530 ThII
8967.0031 ThI
8967.6403 ThI
8968.9461 ArII
8969.8667 ThI
8979.7026 Th
8980.7395 ThI
8960
8965
8970
8975
8980
8979.7026 Th
8980.7395 ThI
8985.2808 ThI
8987.4079 ThI
8990.4779 ThI
8990.8935 ThI
8995.1893 ThI
8997.8763 ThI
8999.5264 ThI
8980
8985
8990
8995
9000
8997.8763 ThI
8999.5264 ThI
9008.4636 ArII
9009.8832 ThI
9011.5152 ThI
9012.5263 ThI
9013.9748 ThI
9016.5903 ThI
9017.5912 ArII
9000
9005
9010
9015
9020


9017.5912 ArII
9031.8194 ThI
9035.9195 ArII
9037.8938 ThI
9038.6911 ThII
9040.1229 ThI
9020
9025
9030
9035
9040
9037.8938 ThI
9038.6911 ThII
9040.1229 ThI
9045.3532 ThI
9048.2501 ThI
9053.4858 ThII
9056.0813 ThI
9058.2586 ThI
9040
9045
9050
9055
9060
9058.2586 ThI
9062.5631 ThII
9063.9600 ThI
9066.1115 ThII
9068.0230 ArII
9069.5817 ThI
9072.2785 ThII
9075.3945 ArI
9060
9065
9070
9075
9080

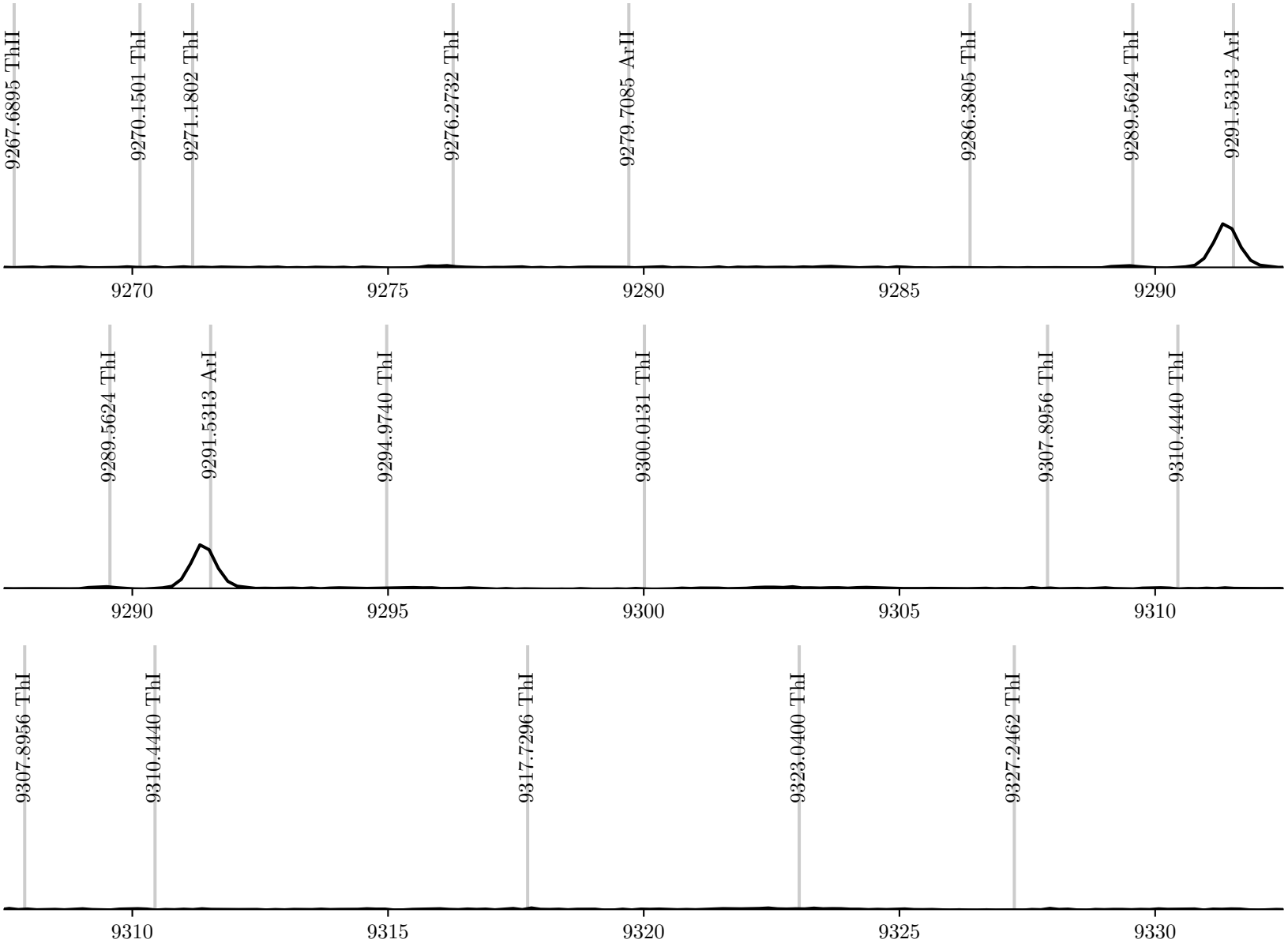

9090.8186 ThI
9094.8289 ThI
9101.0826 ThI
9080
9085
9090
9095
9100
9101.0826 ThI
9103.3420 ThII
9105.4689 Th
9107.2269 ThI
9109.3144 ThI
9111.5314 ThI
9113.4542 ThII
9118.1378 ThI
9119.6365 ThI
9100
9105
9110
9115
9120
9118.1378 ThI
9119.6365 ThI
9122.9674 ThI
9126.3292 ThI
9129.1822 ThI
9130.9682 ThI
9132.2739 ThI
9134.6919 ThI
9135.9375 Th
9137.7316 ThI
9140.5559 Th
9120
9125
9130
9135
9140

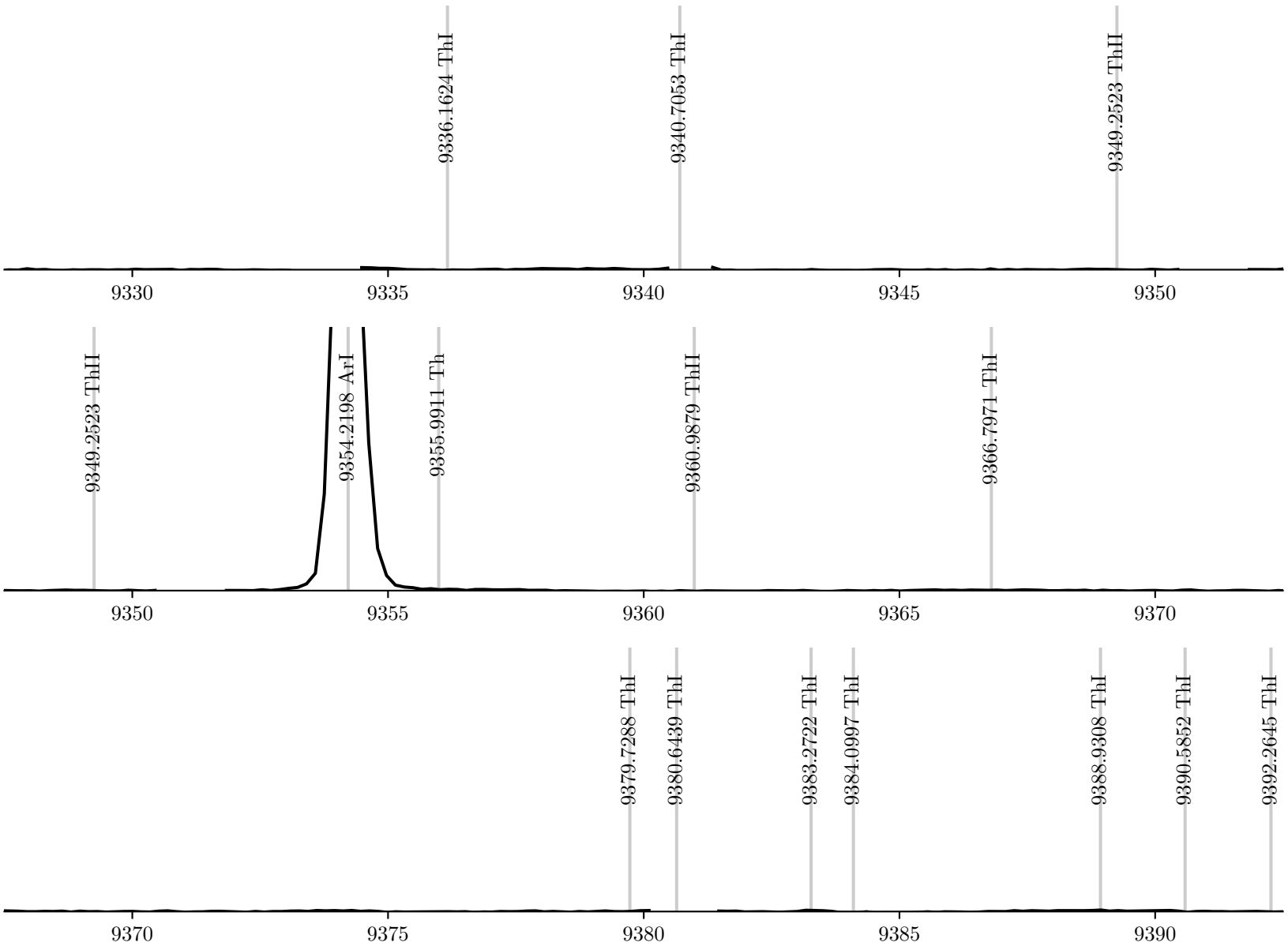

9137.7316 ThI
9140.5559 Th
9153.3664 ThI
9155.2968 ThI
9156.9236 ThI
9157.4218 ThII
9159.0346 ArII
9140
9145
9150
9155
9160
9156.9236 ThI
9157.4218 ThII
9159.0346 ArII
9165.8950 ThI
9167.7950 ThI
9170.8220 ThI
9178.7795 ThI
9160
9165
9170
9175
9180
9178.7795 ThI
9181.8967 Th
9187.5654 ThI
9192.5937 ArII
9193.5928 ThI
9194.6385 ArI
9197.2557 ThI
9199.6841 ThI
9180
9185
9190
9195
9200

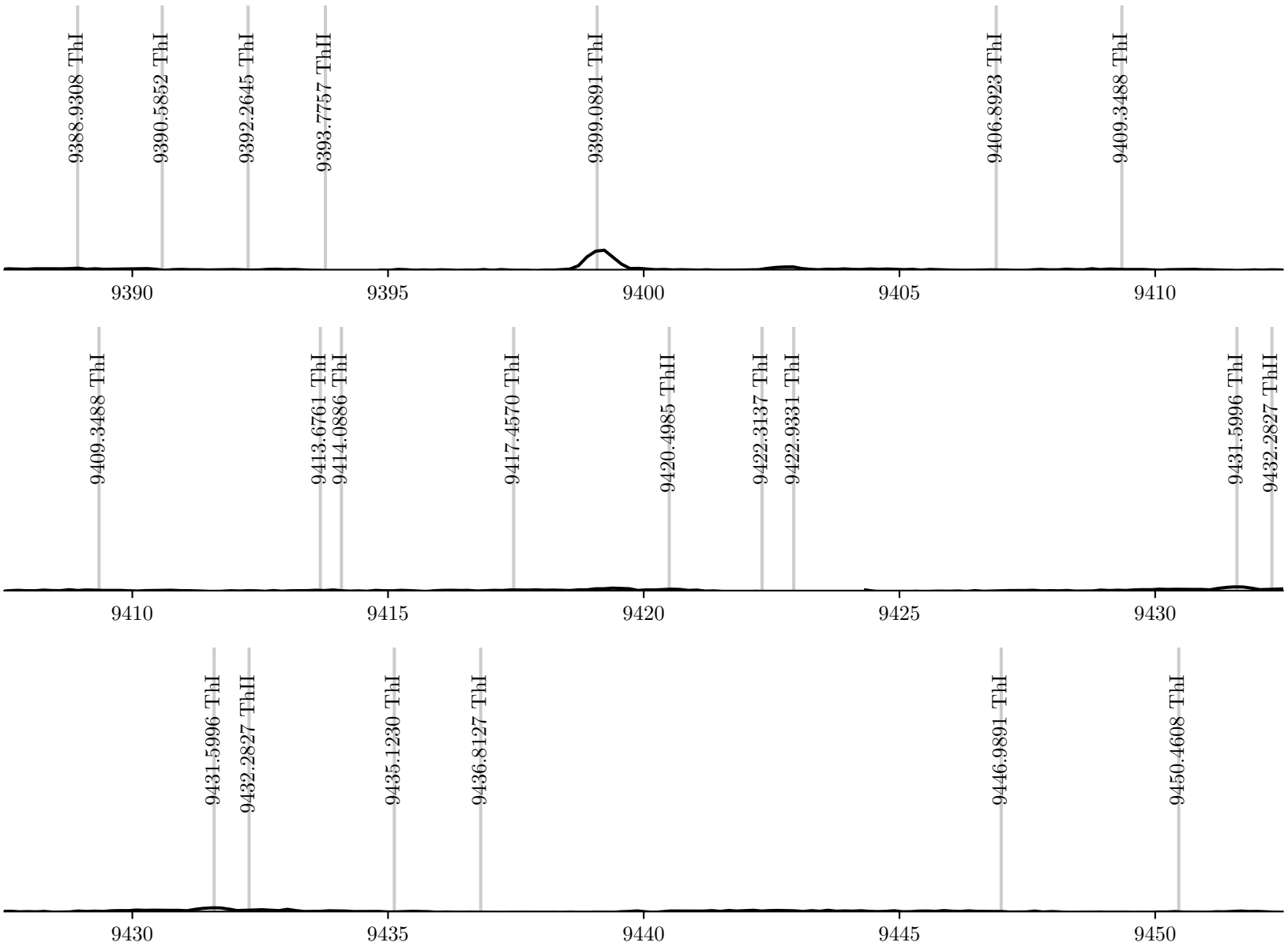
Order 24

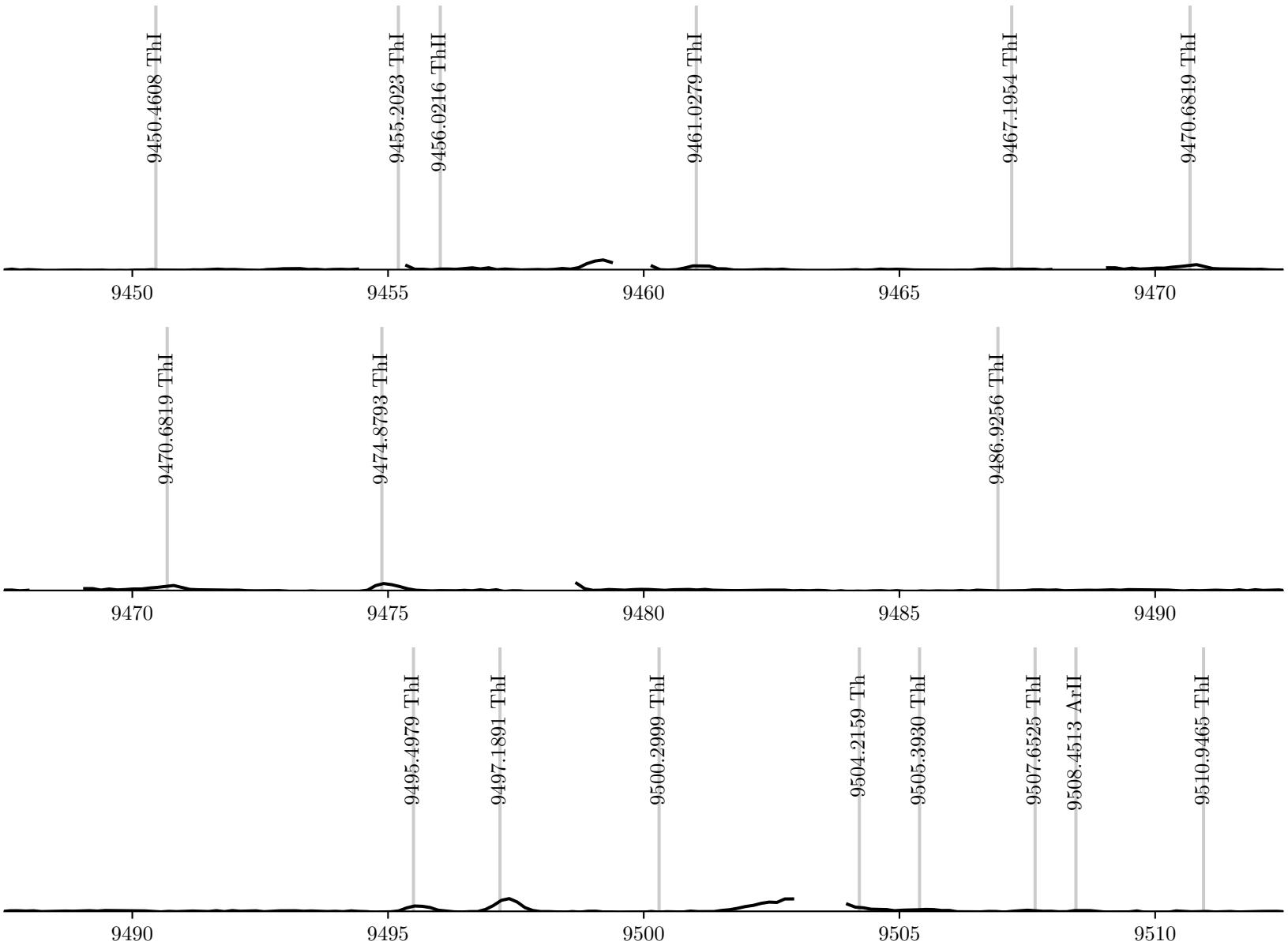


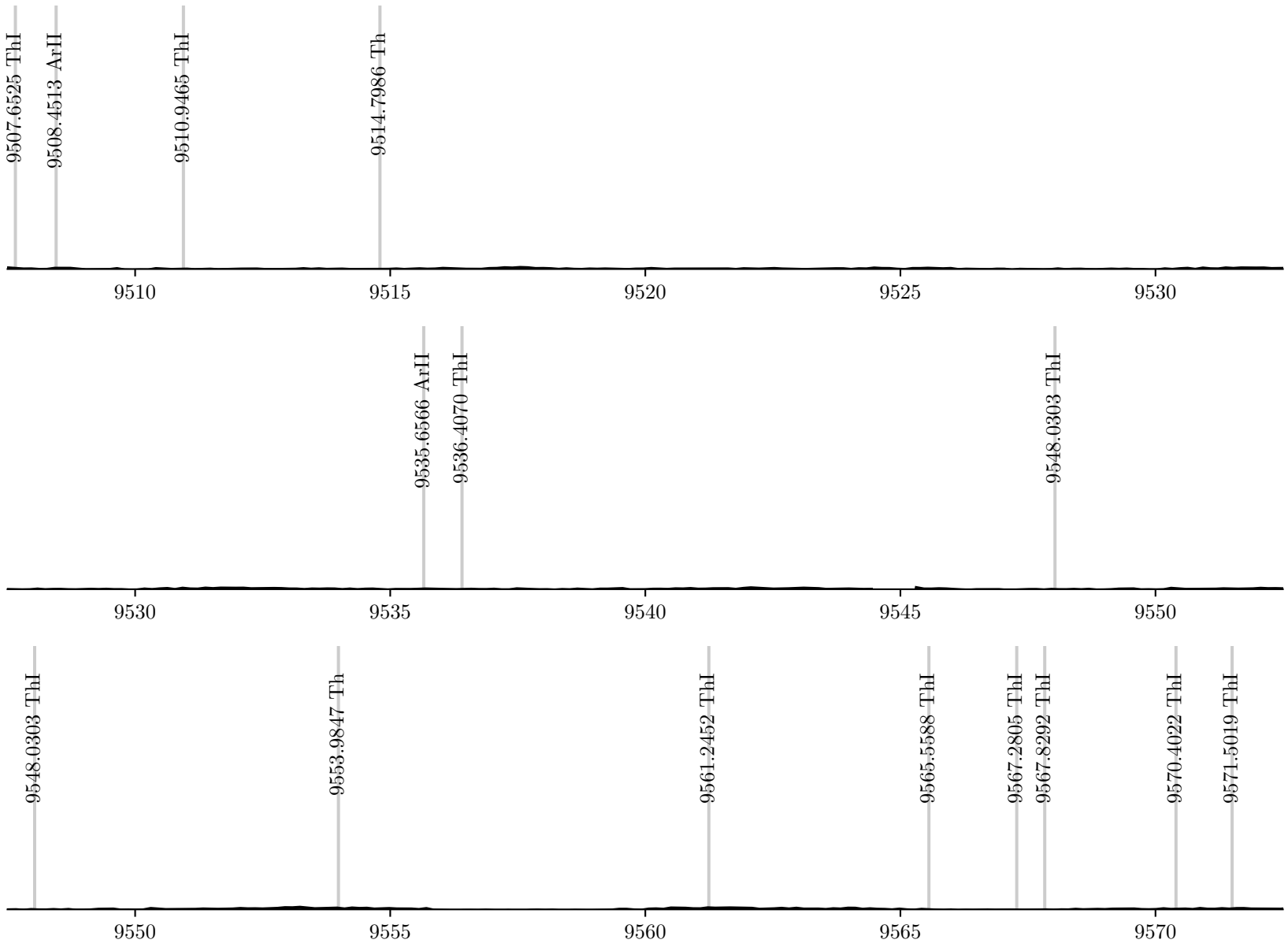
Approximate resolution:	17500
Dispersion:	0.162 Å pix^{-1}
Central wavelength:	9445 Å
Wavelength range:	9268 Å – 9599 Å

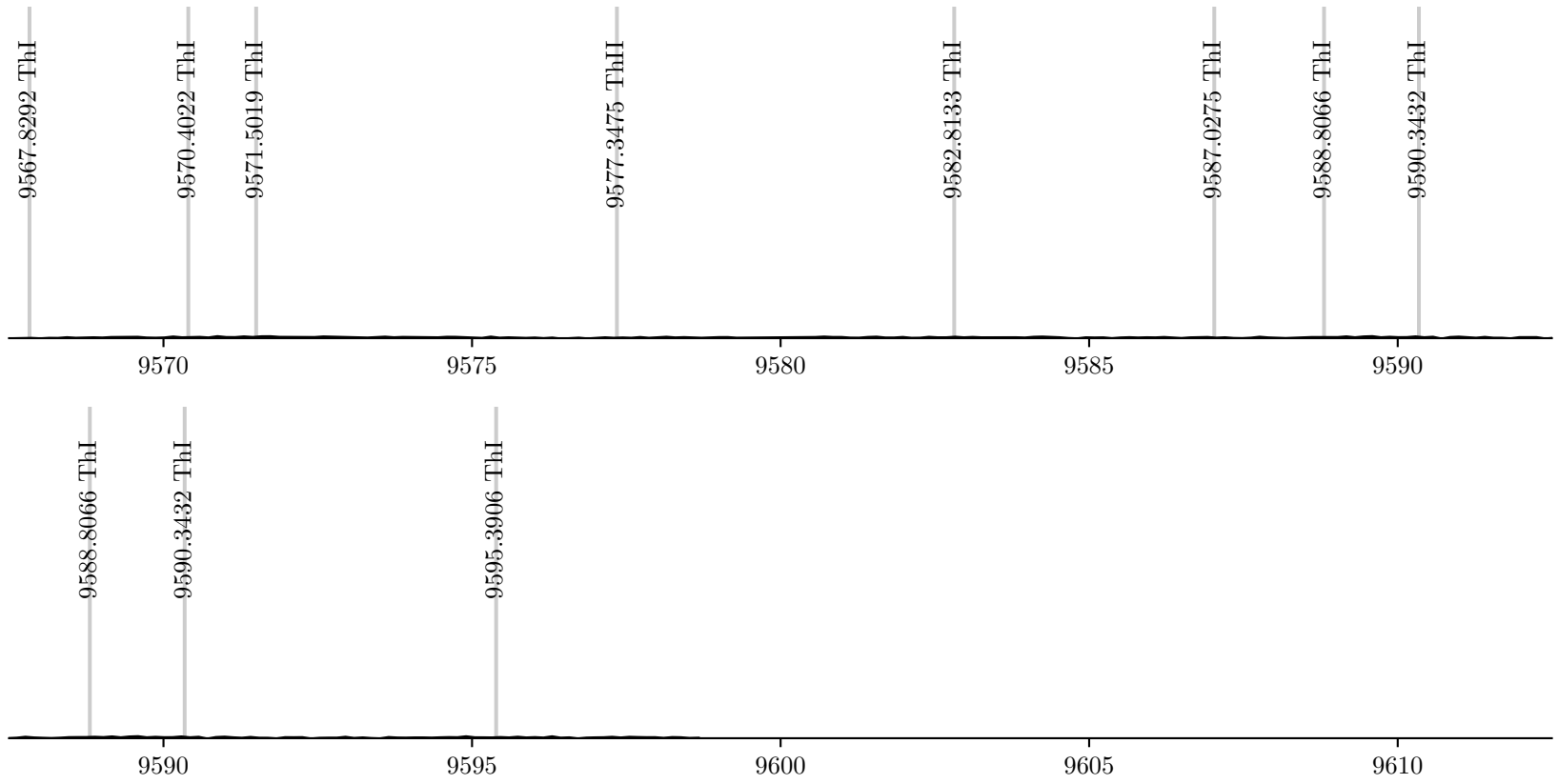

9267.6895 ThII
9270.1501 ThI
9271.1802 ThI
9276.2732 ThI
9279.7085 ArII
9286.3805 ThI
9289.5624 ThI
9291.5313 ArI
9270
9275
9280
9285
9290
9289.5624 ThI
9291.5313 ArI
9294.9740 ThI
9300.0131 ThI
9307.8956 ThI
9310.4440 ThI
9290
9295
9300
9305
9310
9307.8956 ThI
9310.4440 ThI
9317.7296 ThI
9323.0400 ThI
9327.2462 ThI
9310
9315
9320
9325
9330


9336.1624 ThI
9340.7053 ThI
9349.2523 ThII
9330
9335
9340
9345
9350
9349.2523 ThII
9354.2198 ArI
9355.9911 Th
9360.9879 ThII
9366.7971 ThI
9350
9355
9360
9365
9370
9379.7288 ThI
9380.6439 ThI
9383.2722 ThI
9384.0997 ThI
9388.9308 ThI
9390.5852 ThI
9392.2645 ThI
9370
9375
9380
9385
9390


9388.9308 ThI
9390.5852 ThI
9392.2645 ThI
9393.7757 ThII
9399.0891 ThI
9406.8923 ThI
9409.3488 ThI
9390
9395
9400
9405
9410
9409.3488 ThI
9413.6761 ThI
9414.0886 ThI
9417.4570 ThI
9420.4985 ThII
9422.3137 ThI
9422.9331 ThI
9431.5996 ThI
9432.2827 ThII
9410
9415
9420
9425
9430
9431.5996 ThI
9432.2827 ThII
9435.1230 ThI
9436.8127 ThI
9446.9891 ThI
9450.4608 ThI
9430
9435
9440
9445
9450


9450.4608 ThI
9455.2023 ThI
9456.0216 ThII
9461.0279 ThI
9467.1954 ThI
9470.6819 ThI
9450
9455
9460
9465
9470
9470.6819 ThI
9474.8793 ThI
9486.9256 ThI
9470
9475
9480
9485
9490
9495.4979 ThI
9497.1891 ThI
9500.2999 ThI
9504.2159 Th
9505.3930 ThI
9507.6525 ThI
9508.4513 ArII
9510.9465 ThI
9490
9495
9500
9505
9510


9507.6525 ThI
9508.4513 ArII
9510.9465 ThI
9514.7986 Th
9510
9515
9520
9525
9530
9535.6566 ArII
9536.4070 ThI
9548.0303 ThI
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9545
9550
9548.0303 ThI
9553.9847 Th
9561.2452 ThI
9565.5588 ThI
9567.2805 ThI
9567.8292 ThI
9570.4022 ThI
9571.5019 ThI
9550
9555
9560
9565
9570


9567.8292 ThI
9570.4022 ThI
9571.5019 ThI
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9587.0275 ThI
9588.8066 ThI
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9590
9588.8066 ThI
9590.3432 ThI
9595.3906 ThI
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9605
9610